Lie Groups and Their Representation Theory

Noncompact representations, geometry and physics

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Studying this course

This course has two parts of fourteen lectures. Each lecture is intended for about ninety minutes of study. The exercise sheet is separate. Allow another two to four hours for its core problems.

The prerequisites are *Linear Algebra for Physics*, *Finite Groups and Their Representations*, *Lie Algebras and Their Representations*, and basic category language. We use their algebraic results throughout. Ordinary multivariable calculus and elementary ordinary differential equations are also assumed.

Part I studies the global structure of Lie groups and the representations of compact groups. Part II studies selected noncompact groups and the geometry behind their representations. It includes the Heisenberg, Lorentz and Poincaré groups. The final lecture prepares for a later course on quantum groups.

Reading a lecture

Start with the objectives and outcomes. Work through the displayed calculations before reading the next paragraph. Then answer the self-checks without looking at their answers.

Attempt the core exercises before opening the worked solutions. Compare the reasoning as well as the final result. Some sheets include an optional extension. It is marked explicitly and is not needed for the next lecture.

The notes introduce the geometry and analysis needed at each stage. They do not replace courses on differential geometry, topology or functional analysis. Long existence theorems are stated with their hypotheses and a source. The main calculations are carried out here.

The course plan uses four labels. [P] means that the result is proved in the lecture. [S] means that the argument is sketched and the omitted step is identified. [C] means that a theorem is cited. [I] means that a result comes from a prerequisite course. A cited theorem is an input to the argument that follows it.

Conventions

Lie groups are finite-dimensional real Lie groups. Their manifolds are Hausdorff and second countable. Compactness and connectedness are stated when they are needed. Representations are continuous and act on complex vector spaces. A Hilbert-space unitary representation is strongly continuous. Inner products are antilinear in the first slot.

A real Lie algebra and its complexification are distinct objects. In particular, $\mathfrak{k} = \mathrm{Lie}(K)$ is real, while highest-weight theory applies to $\mathfrak{k}_{\mathbb{C}}$. For a torus with Lie algebra $\mathfrak{t}$, our period lattice is

$$\Lambda = \ker(\exp\colon \mathfrak{t} \to T).$$

We write a character as $\chi(\exp H) = e^{i\ell(H)}$. Thus its period condition is $\ell(\Lambda) \subseteq 2\pi\mathbb{Z}$.

The module V_m of $\mathrm{SU}(2)$ has dimension $m+1$ and spin $m/2$. The operator of a group representation is $\Pi(g)$, and its differential is $d\Pi(X)$. When a basis is chosen, D denotes an operator matrix. Every change of convention is explained where it is used.

Books used in the notes

The reading lists give a route into the books. The following abbreviations are used in some lectures.

H Brian C. Hall, *Lie Groups, Lie Algebras, and Representations*, second edition, Springer, 2015. This is the main route for matrix groups, compact groups and integration of representations.

BD Theodor Bröcker and Tammo tom Dieck, *Representations of Compact Lie Groups*, Springer, 1985. This supplies the general compact-group theory and reconstruction.

S Mark R. Sepanski, *Compact Lie Groups*, Springer, 2007. This is a second route through compact structure and Peter–Weyl theory.

Knapp Beyond
Anthony W. Knapp, *Lie Groups Beyond an Introduction*, second edition. This supplies general Lie-group structure.

Knapp Representation Theory
Anthony W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples*, Princeton University Press, 2001 edition. This supplies noncompact examples, smooth vectors and induced representations.

Kirillov Jr.
Alexander Kirillov Jr., *An Introduction to Lie Groups and Lie Algebras*, Cambridge University Press, 2008.

Kirillov Elements
A. A. Kirillov, *Elements of the Theory of Representations*, Springer, 1976.

Kirillov Orbit Method
A. A. Kirillov, *Lectures on the Orbit Method*, American Mathematical Society, 2004.

Warner
Frank W. Warner, *Foundations of Differentiable Manifolds and Lie Groups*, Springer edition, 1983.

Further primary sources are named in the relevant lectures. Knapp's two books serve different purposes; their titles are distinguished throughout.

Part II

Noncompact representations, geometry and physics

Lecture 15

Strong Continuity and Infinitesimal Generators

Objectives. *Part II studies unitary representations on infinite-dimensional Hilbert spaces. Their derivatives may exist only on a proper subspace. This lecture fixes the continuity hypothesis, defines the derivative on smooth vectors, and uses Stone's theorem to identify the skew-adjoint generator of each one-parameter subgroup, together with its domain. Translations on $L^2(\mathbb{R})$ are the model throughout.*

By the end of this lecture you will be able to:

- Prove strong continuity of a homomorphism into the unitary group from continuity at the identity on a dense set.
- Define smooth vectors and $d\Pi$, and prove that each $d\Pi(X)$ is skew-symmetric on smooth vectors.
- State Stone's theorem with its hypotheses, and identify the closure of $d\Pi(X)$ with the generator.
- Locate explicit vectors relative to the domains for translations: Schwartz functions, a non-Schwartz vector in the domain, and a vector outside it.

15.1 From matrices to Hilbert spaces

Let $\Pi\colon G \to \mathrm{GL}(V)$ be a continuous representation on a finite-dimensional complex space V. Continuous homomorphisms of Lie groups are smooth (Lecture 3). So every $X \in \mathfrak{g}$ gives an operator

$$d\Pi(X) = \frac{d}{dt}\Big|_{t=0} \Pi(\exp tX) \in \mathrm{End}(V), \qquad \Pi(\exp tX) = e^{t\,d\Pi(X)}.$$

If Π is unitary, then $d\Pi(X)^* = -d\Pi(X)$. Every vector can be differentiated.

For compact G, the Peter–Weyl theorem (Lecture 10) decomposes each unitary representation into finite-dimensional subrepresentations. Noncompact groups behave differently. The representations studied in Part II are infinite-dimensional, and the derivative may be defined only on a proper subspace. The basic example is the translation representation on $L^2(\mathbb{R})$, so we first construct that space.

Let $C_c(\mathbb{R})$ be the space of continuous functions $\mathbb{R} \to \mathbb{C}$ with compact support, with inner product

$$\langle f, g\rangle = \int_{\mathbb{R}} \overline{f(x)}\, g(x)\, dx.$$

Its Hilbert completion is $L^2(\mathbb{R})$.

We use one fact about $L^2(\mathbb{R})$. A *jump* of a function $f\colon \mathbb{R} \to \mathbb{C}$ is a point where f is discontinuous but has both one-sided limits. We call f *piecewise continuous* if it is continuous except at finitely many jumps in each bounded interval. If moreover the improper integral $\int_{\mathbb{R}} |f|^2\,dx$ is finite, we call f *square integrable*. Such a function f defines a vector of $L^2(\mathbb{R})$, and inner products of these vectors are given by the same integral. We first treat functions that vanish outside a bounded interval. Then we pass to square integrable functions by truncation.

Let f be piecewise continuous and vanish outside a bounded interval. Linear interpolation across the jumps of f gives functions $g_n \in C_c(\mathbb{R})$ with $\int_{\mathbb{R}} |f - g_n|^2\,dx \to 0$. The sequence (g_n) is Cauchy in $L^2(\mathbb{R})$. We call its limit the vector f. This vector does not depend on the approximating sequence, and it depends linearly on f. For two such functions f and $\tilde f$ we have $\|f\|^2 = \int_{\mathbb{R}} |f|^2\,dx$ and $\langle f, \tilde f\rangle = \int_{\mathbb{R}} \bar f \tilde f\,dx$. The preliminary note in the solution of Exercise 15.1 proves these statements. It constructs the functions g_n explicitly and uses the Cauchy–Schwarz inequality for Riemann integrals.

Now let f be square integrable. Each truncation $f \cdot 1_{[-n,n]}$ is piecewise continuous and vanishes outside $[-n, n]$, so it defines a vector. The difference of two truncations is $f \cdot 1_{\{n<|x|\le m\}}$. By linearity and the norm formula, the truncations form a Cauchy sequence, because

$$\left\| f \cdot 1_{[-m,m]} - f \cdot 1_{[-n,n]} \right\|^2 = \int_{n<|x|\le m} |f|^2\,dx \longrightarrow 0 \qquad (m > n \to \infty).$$

Their limit is the vector f. For square integrable f and $\tilde f$, letting $n \to \infty$ in the formulas for the truncations gives

$$\|f\|^2 = \int_{\mathbb{R}} |f|^2\,dx, \qquad \langle f, \tilde f\rangle = \int_{\mathbb{R}} \bar f \tilde f\,dx.$$

The second improper integral converges absolutely, since $\int_{-n}^{n} |\bar f \tilde f|\,dx \le \|f\|\,\|\tilde f\|$ by the Cauchy–Schwarz inequality.

Changing f at finitely many points does not change the vector. Indeed, the difference of the two functions is nonzero at only finitely many points, so its vector has norm 0.

For $t \in \mathbb{R}$ and $f \in C_c(\mathbb{R})$ put

$$(\lambda(t) f)(x) = f(x - t).$$

Then $\lambda(s)\lambda(t) = \lambda(s+t)$, and $\|\lambda(t) f\| = \|f\|$ by substitution. So each $\lambda(t)$ extends to a unitary operator on $L^2(\mathbb{R})$ with inverse $\lambda(-t)$. On the piecewise continuous functions above, $\lambda(t)$ is still translation.

We identify the Lie algebra of $\mathbb{R}$ with $\mathbb{R}$. With this identification the exponential map $\mathbb{R} \to \mathbb{R}$ is the identity, so $d\lambda(1) f$ is the derivative of $t \mapsto \lambda(t) f$ at $t = 0$. Formally $\frac{d}{dt}\lambda(t) f\big|_{t=0} = -f'$. This formula has no meaning for a general vector. In the domain example, Example 15.15, the difference quotients of $1_{[0,1]}$ are unbounded. So $d\lambda(1)$ needs a stated domain.

15.2 Strong continuity

Definition 15.1. Let $\mathcal{H}$ be a complex Hilbert space and $\mathrm{U}(\mathcal{H})$ its group of unitary operators. A *unitary representation* of a Lie group G on $\mathcal{H}$ is a homomorphism $\Pi\colon G \to \mathrm{U}(\mathcal{H})$ such that the map $G \to \mathcal{H}$, $g \mapsto \Pi(g)v$, is continuous for every $v \in \mathcal{H}$. This condition is *strong continuity*. The homomorphism is *norm continuous* if $g \mapsto \Pi(g)$ is continuous for the operator norm.

Lemma 15.2. *Let $\Pi\colon G \to \mathrm{U}(\mathcal{H})$ be a homomorphism. Let $D \subset \mathcal{H}$ be dense, and suppose $g \mapsto \Pi(g)w$ is continuous at e for every $w \in D$. Then Π is strongly continuous.*

Proof. Let $v \in \mathcal{H}$ and $w \in D$. Since $\Pi(g)$ is unitary,

$$\|\Pi(g)v - v\| \le \|\Pi(g)(v-w)\| + \|\Pi(g)w - w\| + \|w - v\| = 2\|v-w\| + \|\Pi(g)w - w\|.$$

Given $\varepsilon > 0$, choose w with $\|v - w\| < \varepsilon$. Then $\limsup_{g\to e} \|\Pi(g)v - v\| \le 2\varepsilon$. So $g \mapsto \Pi(g)v$ is continuous at e. At another point h we use

$$\|\Pi(g)v - \Pi(h)v\| = \|\Pi(h^{-1}g)v - v\|,$$

and $h^{-1}g \to e$ as $g \to h$. ∎

Proposition 15.3. *The translations λ form a unitary representation of $\mathbb{R}$ on $L^2(\mathbb{R})$.*

Proof. Only strong continuity remains. We apply the dense-set criterion, Lemma 15.2, with $D = C_c(\mathbb{R})$. Let $f \in C_c(\mathbb{R})$ vanish outside $[-R, R]$. For $|t| \le 1$ the function $\lambda(t)f - f$ vanishes outside $[-R-1, R+1]$. Hence

$$\|\lambda(t)f - f\|^2 \le (2R+2) \sup_{x\in\mathbb{R}} |f(x-t) - f(x)|^2.$$

Since f is continuous with compact support, it is uniformly continuous. Hence the right side tends to 0 as $t \to 0$, and the dense-set criterion gives strong continuity. ∎

Norm continuity would be too strong a requirement, since translations are not norm continuous. Indeed, Exercise 15.1 shows that $\|\lambda(t) - I\| = 2$ for every $t \neq 0$. Strong continuity requires continuity separately for each vector. It does not require this continuity to be uniform over unit vectors.

15.3 Smooth vectors and the derived representation

A map from an open subset of $\mathbb{R}^n$ to $\mathcal{H}$ is *smooth* if all its partial derivatives of all orders exist as limits in the norm of $\mathcal{H}$ and are continuous. A map $G \to \mathcal{H}$ is smooth if it is smooth in every chart.

Definition 15.4. Let Π be a unitary representation of G on $\mathcal{H}$. A vector $v \in \mathcal{H}$ is *smooth* if $g \mapsto \Pi(g)v$ is a smooth map $G \to \mathcal{H}$. We write $\mathcal{H}^\infty$ for the set of smooth vectors. For $v \in \mathcal{H}^\infty$ and $X \in \mathfrak{g}$ put

$$d\Pi(X)v = \frac{d}{dt}\Big|_{t=0} \Pi(\exp tX)v = \lim_{t \to 0} \frac{\Pi(\exp tX)v - v}{t}.$$

The set $\mathcal{H}^\infty$ is a subspace of $\mathcal{H}$, since linear combinations of smooth maps $G \to \mathcal{H}$ are smooth. The limit defining $d\Pi(X)v$ exists because $t \mapsto \Pi(\exp tX)v$ is the smooth curve $t \mapsto \exp tX$ followed by the smooth map in the definition of a smooth vector.

Proposition 15.5. *Let Π be a unitary representation of a Lie group G on $\mathcal{H}$, and let $g \in G$, $X \in \mathfrak{g}$.*

(a) $\Pi(g)\mathcal{H}^\infty \subset \mathcal{H}^\infty$ *and* $d\Pi(X)\mathcal{H}^\infty \subset \mathcal{H}^\infty$.

(b) $\mathcal{H}^\infty$ *is dense in* $\mathcal{H}$.

(c) $X \mapsto d\Pi(X)$ *is a representation of* $\mathfrak{g}$ *on* $\mathcal{H}^\infty$, *and* $d\Pi(\mathrm{Ad}(g)X) = \Pi(g)\, d\Pi(X)\, \Pi(g)^{-1}$ *on* $\mathcal{H}^\infty$.

(d) For $v, w \in \mathcal{H}^\infty$ *we have* $\langle d\Pi(X)v, w\rangle = -\langle v, d\Pi(X)w\rangle$.

Proof. We cite Knapp, *Representation Theory of Semisimple Groups*, Sections III.3–4, for the invariance under $d\Pi(X)$ in (a), the density in (b), and the representation property and Ad formula in (c). We prove the rest.

For the invariance under $\Pi(g)$ in (a), let $v \in \mathcal{H}^\infty$, $h \in G$ and $F(g) = \Pi(g)v$. Then $\Pi(g)\Pi(h)v = F(gh)$. Right translation by h is smooth on G, so $g \mapsto F(gh)$ is smooth.

For the skew-symmetry in (d), unitarity gives

$$\langle \Pi(\exp tX)v, w\rangle = \langle v, \Pi(\exp(-tX))w\rangle.$$

Subtract $\langle v, w\rangle$ from both sides and divide by $t \neq 0$:

$$\left\langle \frac{\Pi(\exp tX)v - v}{t},\, w \right\rangle = -\left\langle v,\, \frac{\Pi(\exp(-tX))w - w}{-t} \right\rangle.$$

The inner product is continuous, so letting $t \to 0$ gives (d). ∎

The omitted density step in (b) is the hard one, so we sketch it. The sketch uses a vector-valued integral against a left Haar measure dg on G. For compact G, Haar measure is introduced in Lecture 5. For noncompact G it is introduced in Lecture 16, so until then we cite Knapp for this integral. For $\varphi \in C_c^\infty(G)$ and $v \in \mathcal{H}$ form the *Gårding vector*

$$\Pi(\varphi)v = \int_G \varphi(g)\, \Pi(g)v \, dg.$$

Let $h \in G$. Since $\Pi(h)$ is bounded, it passes inside the integral. Left invariance of dg then allows the substitution $g \mapsto h^{-1}g$. It gives

$$\Pi(h)\Pi(\varphi)v = \int_G \varphi(g)\, \Pi(hg)v \, dg = \int_G \varphi(h^{-1}g)\, \Pi(g)v \, dg.$$

Now h appears only inside the smooth scalar function $(h,g)\mapsto \varphi(h^{-1}g)$. Differentiating under the integral in local coordinates of h shows that $h\mapsto \Pi(h)\Pi(\varphi)v$ is smooth on G. So $\Pi(\varphi)v$ is a smooth vector. If $\varphi_n\ge 0$, $\int_G \varphi_n\,dg=1$ and the supports shrink to e, then

$$\|\Pi(\varphi_n)v-v\|\le \sup_{g\in \operatorname{supp}\varphi_n}\|\Pi(g)v-v\|\longrightarrow 0$$

by strong continuity. Exercise 15.4 carries out the argument for $G=\mathbb{R}$ with Riemann integrals.

15.4 Unbounded operators and Stone's theorem

On $L^2(\mathbb{R})$ the operator $-d/dx$ is defined only on a subspace. We need language for such operators.

Definition 15.6. An *operator* A on $\mathcal{H}$ is a linear map $A\colon D(A)\to\mathcal{H}$ defined on a subspace $D(A)\subset\mathcal{H}$, its *domain*. It is *densely defined* if $D(A)$ is dense. We write $B\subset A$ if $D(B)\subset D(A)$ and $Bv=Av$ for $v\in D(B)$.

Definition 15.7. Let A be densely defined. The domain of the *adjoint* A^* is the set of $w\in\mathcal{H}$ for which some $u\in\mathcal{H}$ satisfies

$$\langle w, Av\rangle=\langle u,v\rangle\qquad\text{for all } v\in D(A),$$

and then $A^*w=u$. The operator A is *skew-symmetric* if $\langle Av,w\rangle=-\langle v,Aw\rangle$ for all $v,w\in D(A)$. It is *skew-adjoint* if $A^*=-A$, including $D(A^*)=D(A)$, and *self-adjoint* if $A^*=A$, including $D(A^*)=D(A)$.

Since $D(A)$ is dense, the vector u is unique. Skew-symmetry of A means exactly that $-A\subset A^*$.

Suppose $B\subset A$ and B is densely defined. Then $D(A)\supset D(B)$ is dense too, so both adjoints exist. In this case $A^*\subset B^*$. Indeed, the defining identity for A^*w holds in particular for $v\in D(B)$.

Lemma 15.8. *Let B be densely defined and skew-symmetric. If $v_n\in D(B)$, $v_n\to 0$ and $Bv_n\to u$, then $u=0$.*

Proof. For every $w\in D(B)$, continuity of the inner product and skew-symmetry of B give

$$\langle w,u\rangle=\lim_n\langle w,Bv_n\rangle=-\lim_n\langle Bw,v_n\rangle=0.$$

So u is orthogonal to the dense subspace $D(B)$, and $u=0$. ∎

Definition 15.9. Let B be densely defined and skew-symmetric. The domain of its *closure* $\bar B$ is the set of $v\in\mathcal{H}$ for which there are $v_n\in D(B)$ with $v_n\to v$ and (Bv_n) convergent. Then $\bar Bv=\lim_n Bv_n$.

The value $\bar Bv$ does not depend on the sequence. Indeed, the difference of two such sequences satisfies the hypotheses of the limit lemma for skew-symmetric operators, Lemma 15.8.

Definition 15.10. Let $U\colon \mathbb{R} \to \mathrm{U}(\mathcal{H})$ be a unitary representation. Its *generator* A has domain

$$D(A) = \left\{ v \in \mathcal{H} : \lim_{t\to 0} \frac{U(t)v - v}{t} \text{ exists in } \mathcal{H} \right\},$$

and Av is this limit.

Theorem 15.11 (Stone). *Let $U\colon \mathbb{R} \to \mathrm{U}(\mathcal{H})$ be a strongly continuous unitary representation with generator A.*

(a) A is densely defined and skew-adjoint. Moreover $U(t)D(A) \subset D(A)$ and $AU(t)v = U(t)Av$ for $v \in D(A)$.

(b) Let $D \subset D(A)$ be a dense subspace with $U(t)D \subset D$ for all t. Then $A|_D$ is skew-symmetric, and its closure is A.

(c) Every skew-adjoint operator is the generator of exactly one strongly continuous unitary representation of $\mathbb{R}$, written $t \mapsto e^{tA}$.

We cite most of this theorem and prove only the invariance statement in (a). For $v \in D(A)$, since $U(t)$ is bounded and commutes with $U(h)$,

$$\frac{U(h)U(t)v - U(t)v}{h} = U(t)\,\frac{U(h)v - v}{h} \longrightarrow U(t)Av \qquad (h \to 0).$$

So $U(t)v \in D(A)$ and $AU(t)v = U(t)Av$.

Hall, *Quantum Theory for Mathematicians*, Section 10.2, Theorem 10.15, gives Stone's theorem. Hall writes $U(t) = e^{itH_{\mathrm{Hall}}}$ with a self-adjoint operator H_{Hall}, so $H_{\mathrm{Hall}} = -iA$. Reed–Simon, *Methods of Modern Mathematical Physics I*, Section VIII.4, Theorems VIII.10–11, gives the dense invariant-domain criterion in (b).

Our physical convention writes $U(t) = e^{-itH}$ with a self-adjoint operator H, so $H = iA$ and $H_{\mathrm{Hall}} = -H$.

Corollary 15.12. *Let Π be a unitary representation of a Lie group G on $\mathcal{H}$. Let $X \in \mathfrak{g}$, and let B_X be the generator of $t \mapsto \Pi(\exp tX)$. Then $\mathcal{H}^\infty \subset D(B_X)$ and $B_X = d\Pi(X)$ on $\mathcal{H}^\infty$. The closure of $d\Pi(X)$ with domain $\mathcal{H}^\infty$ is the skew-adjoint operator B_X, and $\Pi(\exp tX) = e^{t\,\overline{d\Pi(X)}}$.*

Proof. The map $t \mapsto \exp tX$ is a continuous homomorphism $\mathbb{R} \to G$. Composing it with Π gives a homomorphism $\mathbb{R} \to \mathrm{U}(\mathcal{H})$. For each v, the map $t \mapsto \Pi(\exp tX)v$ is a continuous map followed by the continuous map $g \mapsto \Pi(g)v$. So $t \mapsto \Pi(\exp tX)$ is a strongly continuous unitary representation of $\mathbb{R}$, and its generator B_X is defined.

The first two claims restate the definition of $d\Pi(X)$. By the invariance and density statements of Proposition 15.5, $\mathcal{H}^\infty$ is dense and invariant under every $\Pi(\exp tX)$. The dense invariant-domain criterion of Theorem 15.11(b), applied with $D = \mathcal{H}^\infty$, shows that the closure of $d\Pi(X)$ is B_X. The uniqueness statement of Theorem 15.11(c) then gives $\Pi(\exp tX) = e^{t\,\overline{d\Pi(X)}}$. ∎

In the physical convention above, the *physical generator* of $t \mapsto \Pi(\exp tX)$ is $A_X = iB_X$, so $\Pi(\exp tX) = e^{-itA_X}$. Since B_X is skew-adjoint, A_X is self-adjoint. For translations and $X = 1 \in \mathbb{R}$, we have $B_1 = A$. With $\hbar = 1$, the operator $A_1 = iA$ is the momentum operator P, formally $-i\,d/dx$.

15.5 Translations on $L^2(\mathbb{R})$

The *Schwartz space* $\mathcal{S}(\mathbb{R})$ is the space of smooth $f\colon \mathbb{R}\to\mathbb{C}$ such that

$$\sup_{x\in\mathbb{R}} |x^k f^{(j)}(x)| < \infty \qquad \text{for all } j,k\ge 0.$$

It contains $C_c^\infty(\mathbb{R})$ and $e^{-x^2/2}$. Let A be the generator of λ.

Lemma 15.13. *Let $f\in C^2(\mathbb{R})$ with f and f' square integrable. Suppose $|f''(y)|\le C(1+y^2)^{-1}$ for all y. Then $f\in D(A)$ and $Af=-f'$.*

Proof. Fix $0<|h|\le 1$. Taylor's formula with integral remainder, for $s\mapsto f(x-sh)$ on $[0,1]$, gives

$$f(x-h)=f(x)-hf'(x)+h^2\int_0^1 (1-s)f''(x-sh)\,ds.$$

So the error function is

$$q_h(x)=\frac{f(x-h)-f(x)}{h}+f'(x)=h\int_0^1(1-s)f''(x-sh)\,ds,$$

and $|q_h(x)|\le \frac{|h|}{2}\sup_{|y-x|\le 1}|f''(y)|$. If $|y-x|\le 1$, then $x^2\le (|y|+1)^2\le 2y^2+2$, so $1+x^2\le 3(1+y^2)$. Therefore

$$|q_h(x)|\le \frac{3C|h|}{2(1+x^2)}, \qquad \|q_h\|^2\le \frac{9C^2h^2}{4}\int_{\mathbb{R}}\frac{dx}{(1+x^2)^2}=\frac{9\pi C^2h^2}{8}.$$

Thus $\left\|\frac{\lambda(h)f-f}{h}-(-f')\right\|=\|q_h\|\to 0$. ∎

The essential hypothesis is the weighted bound $|f''(y)|\le C(1+y^2)^{-1}$. The difference quotient in the proof converges pointwise for every C^1 function f. Pointwise convergence alone does not give convergence in norm. The weighted bound makes the error q_h at most $|h|$ times a fixed square-integrable function, so the norm of q_h is $O(|h|)$.

Proposition 15.14. *The space $\mathcal{S}(\mathbb{R})$ is dense in $L^2(\mathbb{R})$, invariant under every $\lambda(t)$, and contained in $L^2(\mathbb{R})^\infty$. For $f\in\mathcal{S}(\mathbb{R})$ we have $d\lambda(1)f=-f'$. The closure of $B=-d/dx$ with domain $\mathcal{S}(\mathbb{R})$ is the skew-adjoint generator A.*

Proof. Invariance. Put $y=x-t$. Expanding $x^k=(y+t)^k$ bounds $|x^kf^{(j)}(x-t)|$ by a finite sum of terms $\binom{k}{i}|t|^{k-i}|y^if^{(j)}(y)|$.

Smoothness. Each $f\in\mathcal{S}(\mathbb{R})$ satisfies the hypotheses of the Taylor-remainder lemma, Lemma 15.13. Indeed $(1+y^2)f$, $(1+y^2)f'$ and $(1+y^2)f''$ are bounded. So f and f' are square integrable, and $|f''|\le C(1+y^2)^{-1}$. In particular $\mathcal{S}(\mathbb{R})\subset L^2(\mathbb{R})$. By the lemma, $t\mapsto\lambda(t)f$ is differentiable at 0 with derivative $-f'$. At any t_0,

$$\frac{\lambda(t_0+h)f-\lambda(t_0)f}{h}=\lambda(t_0)\,\frac{\lambda(h)f-f}{h}\longrightarrow\lambda(t_0)(-f'),$$

because $\lambda(t_0)$ is bounded. The derivative $t\mapsto\lambda(t)(-f')$ is continuous by the strong continuity of translations, Proposition 15.3. Since $-f'\in\mathcal{S}(\mathbb{R})$, we may repeat. The k-th derivative is $t\mapsto\lambda(t)\big((-1)^kf^{(k)}\big)$, so f is a smooth vector.

Density. Fix $\varphi \in C_c^\infty(\mathbb{R})$ with $\varphi \geq 0$, support in $[-1,1]$ and $\int \varphi = 1$. Put $\varphi_n(t) = n\varphi(nt)$. For $f \in C_c(\mathbb{R})$ vanishing outside $[-R,R]$, let

$$f_n(x) = \int_{\mathbb{R}} \varphi_n(x-y) f(y)\, dy = \int_{\mathbb{R}} \varphi_n(t) f(x-t)\, dt.$$

The first form shows $f_n \in C_c^\infty(\mathbb{R}) \subset \mathcal{S}(\mathbb{R})$, with support in $[-R-1, R+1]$. The second form gives

$$|f_n(x) - f(x)| \leq \sup_{|t| \leq 1/n} |f(x-t) - f(x)|,$$

which tends to 0 uniformly in x because f is uniformly continuous. Since f_n and f vanish outside $[-R-1, R+1]$, it follows that $f_n \to f$ in $L^2(\mathbb{R})$. Since $C_c(\mathbb{R})$ is dense in $L^2(\mathbb{R})$, so is $\mathcal{S}(\mathbb{R})$.

Closure. By the three steps above, $D = \mathcal{S}(\mathbb{R})$ is a dense invariant subspace of $D(A)$. So the dense invariant-domain criterion of Theorem 15.11(b) shows that the closure of B is A. ∎

The function f_n is the Gårding vector $\lambda(\varphi_n)f$ written pointwise.

Example 15.15. The vectors w and h, and the operator $B = -d/dx$ on $\mathcal{S}(\mathbb{R})$, show where the domains lie.

(i) Let $w(x) = (1+x^2)^{-1}$. Then

$$w'(x) = \frac{-2x}{(1+x^2)^2}, \qquad w''(x) = \frac{6x^2 - 2}{(1+x^2)^3}.$$

Since $|6x^2 - 2| \leq 6(1+x^2)$, we get $|w''(x)| \leq 6(1+x^2)^{-2} \leq 6(1+x^2)^{-1}$. Both w and w' are square integrable. By the Taylor-remainder lemma, Lemma 15.13, $w \in D(A)$ and

$$(Aw)(x) = \frac{2x}{(1+x^2)^2}.$$

But $x^2 w(x) \to 1$, so $x^4 w(x)$ is unbounded and $w \notin \mathcal{S}(\mathbb{R})$.

(ii) The operator $B = -d/dx$ on $\mathcal{S}(\mathbb{R})$ is the restriction of $d\lambda(1)$ to $\mathcal{S}(\mathbb{R})$ by Proposition 15.14. So B is skew-symmetric by the skew-symmetry of $d\lambda(1)$ on smooth vectors, Proposition 15.5(d). It is not skew-adjoint. Indeed $B \subset A$, the domain $\mathcal{S}(\mathbb{R})$ is dense, and $A^* = -A$. So $-A = A^* \subset B^*$. Hence $w \in D(A) \subset D(B^*)$, while $w \notin D(B)$. So $D(B^*) \neq D(B)$, and therefore $B^* \neq -B$.

(iii) Let $h = 1_{[0,1]}$. For $0 < t \leq 1$, up to finitely many points,

$$\lambda(t)h - h = 1_{[t,1+t]} - 1_{[0,1]} = 1_{(1,1+t]} - 1_{[0,t)}, \qquad \|\lambda(t)h - h\|^2 = 2t.$$

For $-1 \leq t < 0$ the same computation gives $2|t|$. Therefore

$$\left\| \frac{\lambda(t)h - h}{t} \right\| = \sqrt{\frac{2}{|t|}} \longrightarrow \infty.$$

Convergent quotients would stay bounded near 0. So $h \notin D(A)$. Smooth vectors lie in $D(A)$ by Corollary 15.12, so also $h \notin L^2(\mathbb{R})^\infty$.

Together these give

$$\mathcal{S}(\mathbb{R}) \subset L^2(\mathbb{R})^\infty \subset D(A), \qquad w \in D(A) \setminus \mathcal{S}(\mathbb{R}), \qquad h \in L^2(\mathbb{R}) \setminus D(A).$$

Exercise 15.2 gives a tent function in $D(A)$ that is not a smooth vector. The momentum operator $P = iA$ has domain $D(A)$, not $\mathcal{S}(\mathbb{R})$.

15.6 Self-checks

1. Let $\dim \mathcal{H} < \infty$. What are $\mathcal{H}^\infty$ and $D(B_X)$?
   Answer. Both equal $\mathcal{H}$. The homomorphism $\Pi\colon G \to \mathrm{U}(\mathcal{H})$ is continuous, hence smooth (Lecture 3), so every vector is smooth. Since $d\Pi(X)$ is then defined on all of $\mathcal{H}$, its closure B_X also has domain $\mathcal{H}$.
2. Let $f(x) = xe^{-x^2/2}$. Compute Af and $\langle f, Af\rangle$.
   Answer. Since $f \in \mathcal{S}(\mathbb{R})$, the Schwartz-space proposition, Proposition 15.14, gives $Af = -f' = (x^2 - 1)e^{-x^2/2}$. Then $\langle f, Af\rangle = \int_{\mathbb{R}} x(x^2 - 1)e^{-x^2}\,dx = 0$, because the integrand is odd. This is consistent with skew-symmetry, which forces $\langle f, Af\rangle$ to be purely imaginary.
3. Let D be the space of smooth functions with compact support in $(0, \infty)$. Then $D \subset D(A)$. Does the dense invariant-domain criterion of Theorem 15.11(b) apply to D?
   Answer. No. The space D is not dense, since every element is orthogonal to $1_{[-1,0]} \neq 0$. It is not invariant either. Given $t < 0$, choose a nonzero $f \in D$ supported in $(0, |t|)$. Then $\lambda(t)f$ is supported in $(t, 0)$, so $\lambda(t)f \notin D$.

15.7 Summary

Unitary representations are required to be strongly continuous, not norm continuous. Smooth vectors form a dense invariant subspace $\mathcal{H}^\infty$. On it, $d\Pi$ is a representation of $\mathfrak{g}$ by skew-symmetric operators. Stone's theorem attaches a skew-adjoint generator to each strongly continuous unitary representation of $\mathbb{R}$. The closure of $d\Pi(X)$ on $\mathcal{H}^\infty$ is this generator. For translations, $-d/dx$ on $\mathcal{S}(\mathbb{R})$ is skew-symmetric but not skew-adjoint. Its closure has the strictly larger domain $D(A)$, and $D(A)$ is still not all of $L^2(\mathbb{R})$.

15.8 Reading

- Hall, *Quantum Theory for Mathematicians* (2013): unbounded self-adjoint operators and Stone's theorem, Section 10.2.
- Reed–Simon, *Methods of Modern Mathematical Physics I: Functional Analysis*, Section VIII.4, especially Theorems VIII.10–11 on dense invariant domains.
- Knapp, *Representation Theory of Semisimple Groups* (2001), Sections III.3–4: smooth vectors, Gårding vectors and the action of $\mathfrak{g}$.
- Etingof, MIT 18.757 lecture notes, Lectures 1–7: an alternative introduction to continuous representations of Lie groups.

Lecture 16

Noncompact Haar Measure and the Modular Function

Objectives. *Lecture 5 integrated over compact groups. This lecture drops compactness. It asks which translation-invariant integrals a Lie group carries and how left and right integration differ. Every Lie group has a left-invariant and a right-invariant density. With equal normalisations at the identity, the left density divided by the right density is the modular function. It controls right translation, inversion and the right regular representation.*

By the end of this lecture you will be able to:

- Construct the left and right Haar densities of a Lie group and prove that each is unique up to a positive factor.
- Prove the right-translation and inversion identities for $\Delta(g) = |\det \mathrm{Ad}(g)|^{-1}$.
- Compute both Haar densities and Δ for the affine group $ax + b$.
- Construct the left and right regular representations as strongly continuous unitary representations on $L^2(G)$.

Throughout, G is a Lie group of dimension n with Lie algebra $\mathfrak{g} = T_eG$. We write left translation, right translation, conjugation and inversion as

$$L_g(x) = gx, \qquad R_g(x) = xg, \qquad C_g = L_g \circ R_{g^{-1}}, \qquad \iota(x) = x^{-1}.$$

Then $R_g \circ R_h = R_{hg}$ and $d(C_g)_e = \mathrm{Ad}(g)$.

16.1 Densities and integration

Under a linear map T of $\mathbb{R}^n$, Lebesgue measure scales by $|\det T|$. A density imposes this rule on each tangent space.

Definition 16.1. Let V be a real vector space of dimension n. A *positive density* on V is a nonzero function $\delta\colon V^n \to [0,\infty)$ such that

$$\delta(Tv_1,\dots,Tv_n) = |\det T|\,\delta(v_1,\dots,v_n) \qquad (T \in \mathrm{End}\,V).$$

Let $e_1,\dots,e_n$ be a basis and $v_j = Te_j$. Then $\delta(v_1,\dots,v_n) = |\det T|\,\delta(e_1,\dots,e_n)$. So δ is fixed by the number $\delta(e_1,\dots,e_n)$, which is positive since $\delta \neq 0$. Two positive densities on V therefore differ by a positive factor.

Definition 16.2. A *smooth positive density* μ on G assigns a positive density μ_x on T_xG to each $x \in G$ such that, in every chart with coordinates y, the coefficient $m(y) = \mu_x(\partial_1, \dots, \partial_n)$ is smooth.

For a diffeomorphism F of G and a smooth function $\varphi > 0$ put

$$(F^*\mu)_x(v_1, \dots, v_n) = \mu_{F(x)}(dF_x v_1, \dots, dF_x v_n), \qquad (\varphi\mu)_x = \varphi(x)\,\mu_x.$$

The chain rule gives $(F \circ H)^* = H^*F^*$. The definitions give $F^*(\varphi\mu) = (\varphi \circ F)\,F^*\mu$.

Let $C_c(G)$ be the space of continuous functions $f\colon G \to \mathbb{C}$ with compact support.

If $\operatorname{supp} f$ lies in one chart, set $\int_G f\mu = \int_{\mathbb{R}^n} f(y)\,m(y)\,dy$. A change of chart multiplies the coefficient m by the absolute Jacobian determinant. Hence, by the change of variables formula, the integral is well defined.

For general $f \in C_c(G)$ we use a partition of unity, adapted from Lecture 5. Here a *partition of unity* on $\operatorname{supp} f$ is a finite family of smooth functions $\psi_i \geq 0$ on G, each with compact support in one chart, such that $\sum_i \psi_i = 1$ on $\operatorname{supp} f$. On a compact manifold, Lecture 5 requires the sum to equal 1 everywhere. Since f vanishes outside $\operatorname{supp} f$, we have $f = \sum_i \psi_i f$. We add the chart integrals of the functions $\psi_i f$.

We cite this construction and the following three facts. Knapp, *Lie Groups Beyond an Introduction*, Chapter VIII, carries out the same construction with differential forms.

(I1) $f \mapsto \int_G f\mu$ is linear, and $\int_G f\mu > 0$ if $f \geq 0$ and $f \neq 0$.
(I2) $\int_G (f \circ F)\,F^*\mu = \int_G f\mu$ for every diffeomorphism F of G.
(I3) $\int_G f\,(\varphi\mu) = \int_G (f\varphi)\,\mu$.

In a chart, $F^*\mu$ has coefficient $m(F(y))\,|\det DF(y)|$. So (I2) is the change of variables formula.

16.2 Left and right Haar densities

Theorem 16.3 (Haar densities)**.** *Fix a positive density μ_e on $\mathfrak{g}$.*

(a) There is exactly one smooth positive density μ_L with $(\mu_L)_e = \mu_e$ and $L_g^*\mu_L = \mu_L$ for all $g \in G$. Every left-invariant smooth positive density is a positive multiple of μ_L.

(b) There is exactly one smooth positive density μ_R with $(\mu_R)_e = \mu_e$ and $R_g^*\mu_R = \mu_R$ for all $g \in G$. Every right-invariant smooth positive density is a positive multiple of μ_R.

(c) Write $\int_G f(x)\,d_Lx$ for $\int_G f\mu_L$, and similarly d_Rx. For $f \in C_c(G)$ and $g \in G$,

$$\int_G f(gx)\,d_Lx = \int_G f(x)\,d_Lx, \qquad \int_G f(xg)\,d_Rx = \int_G f(x)\,d_Rx.$$

Proof. Uniqueness. Let ν be a left-invariant smooth positive density. Evaluate $L_{x^{-1}}^*\nu = \nu$ at x:

$$\nu_x(v_1, \dots, v_n) = \nu_e\big(d(L_{x^{-1}})_x v_1, \dots, d(L_{x^{-1}})_x v_n\big). \tag{16.1}$$

So ν is determined by ν_e. Two positive densities on $\mathfrak{g}$ differ by a positive factor. Thus $\nu_e = c\mu_e$ for some $c > 0$. Once μ_L is constructed below, the formula gives $\nu = c\mu_L$.

Existence. Define μ_L by (16.1) with $\nu_e = \mu_e$. Linear maps act entrywise on tuples $v = (v_1, \dots, v_n)$. Since $L_{(gx)^{-1}} \circ L_g = L_{x^{-1}}$, the chain rule gives

$$(L_g^* \mu_L)_x(v) = \mu_e\big(d(L_{(gx)^{-1}})_{gx}\, d(L_g)_x v\big) = \mu_e\big(d(L_{x^{-1}})_x v\big) = (\mu_L)_x(v).$$

Smoothness. This step uses the smooth group law. Choose a basis $X_1, \dots, X_n$ of $\mathfrak{g}$. The left-invariant fields $\tilde{X}_i(x) = d(L_x)_e X_i$ are smooth (Lecture 1). Since $L_{x^{-1}} \circ L_x = \mathrm{id}$, we have $d(L_{x^{-1}})_x \tilde{X}_i(x) = X_i$. Hence

$$(\mu_L)_x\big(\tilde{X}_1(x), \dots, \tilde{X}_n(x)\big) = \mu_e(X_1, \dots, X_n) =: c_0 > 0.$$

In a chart write $\tilde{X}_i = \sum_j P_{ji}(y)\, \partial_j$. The matrix $P(y)$ is smooth. It is invertible because $d(L_x)_e$ is an isomorphism. The density rule gives $c_0 = |\det P(y)|\, m(y)$. So $m = c_0\, |\det P|^{-1}$ is smooth.

Right translations. Use $\nu_x(v) = \nu_e(d(R_{x^{-1}})_x v)$. The key identity is $R_{(xg)^{-1}} \circ R_g = R_{x^{-1}}$, since $z\, g\, g^{-1} x^{-1} = z x^{-1}$. The fields $d(R_x)_e X_i$ replace $\tilde{X}_i$.

Integrals. Apply the change of variables (I2) with $F = L_g$ or $F = R_g$. ∎

On a compact group one normalises the total mass to 1, as in Lecture 5. For noncompact G there is no canonical normalisation. We use the same density μ_e for both μ_L and μ_R.

16.3 The modular function

Definition 16.4. The *modular function* of G is $\Delta(g) = |\det \mathrm{Ad}(g)|^{-1}$. The group G is *unimodular* if $\Delta \equiv 1$.

Since $\mathrm{Ad}\colon G \to \mathrm{GL}(\mathfrak{g})$ is a smooth homomorphism (Lecture 2), Δ is a smooth homomorphism into the multiplicative group $(0, \infty)$.

Theorem 16.5 (Integration identities). *Let $f \in C_c(G)$ and $g \in G$.*

(i) $R_g^* \mu_L = \Delta(g)\, \mu_L$. *Hence* $\int_G f(xg)\, d_L x = \Delta(g)^{-1} \int_G f(x)\, d_L x$.

(ii) $\mu_R = \Delta^{-1} \mu_L$, *that is,* $\int_G f(x)\, d_R x = \int_G f(x)\, \Delta(x)^{-1}\, d_L x$.

(iii) $\int_G f(x^{-1})\, d_L x = \int_G f(x)\, d_R x$ *and* $\int_G f(x^{-1})\, \Delta(x^{-1})\, d_L x = \int_G f(x)\, d_L x$.

Proof. (i) Left and right translations commute. Hence

$$L_h^* R_g^* \mu_L = (R_g \circ L_h)^* \mu_L = (L_h \circ R_g)^* \mu_L = R_g^* L_h^* \mu_L = R_g^* \mu_L.$$

So $R_g^* \mu_L$ is left-invariant. By the uniqueness in Theorem 16.3(a), it equals $c(g) \mu_L$ for some $c(g) > 0$. We find $c(g)$ at e. Since $L_{g^{-1}} \circ R_g = C_{g^{-1}}$,

$$(R_g^* \mu_L)_e(v) = \mu_e\big(d(L_{g^{-1}})_g\, d(R_g)_e v\big) = \mu_e\big(\mathrm{Ad}(g^{-1}) v\big) = |\det \mathrm{Ad}(g)|^{-1} \mu_e(v).$$

Thus $c(g) = \Delta(g)$. Now the change of variables (I2) with $F = R_g$ and the multiplication rule (I3) give

$$\int_G f \mu_L = \int_G (f \circ R_g)\, R_g^* \mu_L = \Delta(g) \int_G f(xg)\, d_L x.$$

(ii) Use $F^*(\varphi\mu) = (\varphi \circ F)F^*\mu$, the right-translation identity (i) and the homomorphism property of Δ:

$$\big(R_g^*(\Delta^{-1}\mu_L)\big)_x = \Delta(xg)^{-1}\Delta(g)\,(\mu_L)_x = \Delta(x)^{-1}(\mu_L)_x.$$

So $\Delta^{-1}\mu_L$ is right-invariant with value μ_e at e. By uniqueness in Theorem 16.3(b) it equals μ_R. Then the multiplication rule (I3) gives the integral form.

(iii) First, $d\iota_e = -\mathrm{id}$. Indeed, the multiplication map $\mathrm{mult}\colon G\times G \to G$ satisfies $\mathrm{mult}(x,e) = x$ and $\mathrm{mult}(e,x) = x$ for $x \in G$. So $d(\mathrm{mult})_{(e,e)}(v,w) = v + w$ for $v, w \in \mathfrak{g}$. Differentiating $x\,\iota(x) = e$ at e gives $v + d\iota_e v = 0$. Therefore

$$(\iota^*\mu_L)_e(v_1,\dots,v_n) = \mu_e(-v_1,\dots,-v_n) = |\det(-\mathrm{id})|\,\mu_e(v_1,\dots,v_n) = \mu_e(v_1,\dots,v_n).$$

Next, $(xg)^{-1} = g^{-1}x^{-1}$ says $\iota \circ R_g = L_{g^{-1}} \circ \iota$. Hence

$$R_g^*\iota^*\mu_L = (\iota \circ R_g)^*\mu_L = (L_{g^{-1}} \circ \iota)^*\mu_L = \iota^* L_{g^{-1}}^*\mu_L = \iota^*\mu_L.$$

By the uniqueness in Theorem 16.3(b), $\iota^*\mu_L = \mu_R$. Apply the change of variables (I2) with $F = \iota$ to $f \circ \iota$. This gives $\int_G f\,d_R x = \int_G f(x^{-1})\,d_L x$. For the second identity apply the first to $h = f\Delta \in C_c(G)$ and use the density relation (ii):

$$\int_G f(x^{-1})\Delta(x^{-1})\,d_L x = \int_G h(x)\,d_R x = \int_G f(x)\Delta(x)\Delta(x)^{-1}\,d_L x. \qquad \blacksquare$$

Remark 16.6. The right-translation identity (i) is equivalent to $\int_G f(xg^{-1})\,d_L x = \Delta(g)\int_G f\,d_L x$. For Borel sets E, using the Radon measure of § 16.6, it reads $\mu_L(Eg) = \Delta(g)\mu_L(E)$. This is Folland's convention. Some texts call $|\det \mathrm{Ad}(g)|$ the modular function. When comparing sources, check which defining identity is used.

Corollary 16.7. *(a) Every compact Lie group is unimodular.*

(b) If G is connected, then $\Delta(\exp X) = e^{-\operatorname{tr}\operatorname{ad} X}$. So G is unimodular if and only if $\operatorname{tr}\operatorname{ad} X = 0$ for all $X \in \mathfrak{g}$. This holds when $\mathfrak{g}$ is abelian, nilpotent or semisimple.

Proof. (a) Since G is compact and Δ is continuous, the image $\Delta(G)$ is a compact subgroup of $(0,\infty)$. If $t \neq 1$ lies in it, then so does the unbounded set $\{t^k : k \in \mathbb{Z}\}$. So $\Delta(G) = \{1\}$.

(b) We have $\mathrm{Ad}(\exp X) = e^{\operatorname{ad} X}$ (Lecture 2) and $\det e^A = e^{\operatorname{tr} A}$. This gives the formula. If all traces vanish, then $\Delta = 1$ on $\exp\mathfrak{g}$. This set contains a neighbourhood U of e (Lecture 2). The subgroup H generated by U is open, since $hU \subset H$ for $h \in H$. It is closed, since its complement is a union of open cosets. By connectedness $H = G$, so $\Delta \equiv 1$. Conversely, $\Delta \equiv 1$ gives $e^{-t\operatorname{tr}\operatorname{ad} X} = 1$ for all t, so $\operatorname{tr}\operatorname{ad} X = 0$. For nilpotent $\mathfrak{g}$ each $\operatorname{ad} X$ is nilpotent. For semisimple $\mathfrak{g}$ we have $\mathfrak{g} = [\mathfrak{g},\mathfrak{g}]$ and $\operatorname{tr}\operatorname{ad}[X,Y] = \operatorname{tr}[\operatorname{ad} X, \operatorname{ad} Y] = 0$. $\blacksquare$

16.4 The affine group

The *$ax + b$ group* is

$$G = \left\{ g(a,b) = \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} :\ a > 0,\ b \in \mathbb{R} \right\}.$$

It acts on $\mathbb{R}$ by $x \mapsto ax+b$. The pair (a,b) is a global chart onto $(0,\infty)\times\mathbb{R}$. So integrals over G are ordinary double integrals. Matrix multiplication gives

$$g(a,b)\,g(a',b') = g(aa',\ ab'+b), \qquad g(a,b)^{-1} = g(a^{-1}, -a^{-1}b).$$

A tangent vector at g is a matrix $\begin{pmatrix}\alpha & \beta\\ 0 & 0\end{pmatrix}$, which is $\alpha\partial_a + \beta\partial_b$ in the chart. The Lie algebra has basis

$$X = \begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix}, \qquad Y = \begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix}, \qquad [X,Y] = Y.$$

Translations are restrictions of linear maps on matrices. Hence $d(L_g)_e Z = gZ$ and $d(R_g)_e Z = Zg$. We fix the density μ_e on $\mathfrak{g}$ by $\mu_e(X,Y) = 1$.

Left density. For $g = g(a,b)$,

$$gX = \begin{pmatrix}a & 0\\ 0 & 0\end{pmatrix} = a\,\partial_a, \qquad gY = \begin{pmatrix}0 & a\\ 0 & 0\end{pmatrix} = a\,\partial_b.$$

So the matrix P from the smoothness step in the proof of Theorem 16.3 is

$$P = \begin{pmatrix}a & 0\\ 0 & a\end{pmatrix}.$$

Then $m_L = |\det P|^{-1} = a^{-2}$.

Right density. Similarly,

$$Xg = \begin{pmatrix}a & b\\ 0 & 0\end{pmatrix} = a\,\partial_a + b\,\partial_b, \qquad Yg = \begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix} = \partial_b, \qquad P = \begin{pmatrix}a & 0\\ b & 1\end{pmatrix}.$$

Then $m_R = a^{-1}$. In summary,

$$d_L x = a^{-2}\,da\,db, \qquad d_R x = a^{-1}\,da\,db.$$

Modular function. With $g^{-1} = \begin{pmatrix}a^{-1} & -a^{-1}b\\ 0 & 1\end{pmatrix}$ we compute

$$gXg^{-1} = \begin{pmatrix}1 & -b\\ 0 & 0\end{pmatrix} = X - bY, \qquad gYg^{-1} = \begin{pmatrix}0 & a\\ 0 & 0\end{pmatrix} = aY.$$

In the basis (X,Y),

$$\mathrm{Ad}(g) = \begin{pmatrix}1 & 0\\ -b & a\end{pmatrix}, \qquad \det \mathrm{Ad}(g) = a, \qquad \Delta(g(a,b)) = a^{-1}.$$

This matches the density relation in Theorem 16.5(ii): $\Delta^{-1} m_L = a \cdot a^{-2} = a^{-1} = m_R$. It also matches the trace formula in Corollary 16.7(b). Indeed $\operatorname{ad} X$ fixes Y and kills X, so $\operatorname{tr}\operatorname{ad} X = 1$. And $\exp tX = g(e^t, 0)$ has $\Delta = e^{-t}$. Hence the $ax+b$ group is not unimodular.

The connectedness hypothesis in Corollary 16.7(b) is needed. Let H be the subgroup of elements $g(a,b)$ with $a \in 2^{\mathbb{Z}}$ and $b \in \mathbb{R}$. It is closed, since $2^{\mathbb{Z}}$ is closed in $(0,\infty)$. Its Lie algebra $\mathbb{R}Y$ is abelian. Yet $\mathrm{Ad}(g)Y = gYg^{-1} = aY$ for $g = g(a,b) \in H$. So the modular function Δ_H of H satisfies $\Delta_H(g(a,b)) = a^{-1}$, which is not identically 1.

Direct check of the right-translation identity in Theorem 16.5(i). Let $x = g(a', b')$ and $g = g(a, b)$. Then $xg = g(a'a,\ a'b + b')$. Substitute $u = a'a$, $v = a'b + b'$. Then $du\,dv = a\,da'\,db'$ and $a'^{-2} = a^2 u^{-2}$. So

$$\int_G f(xg)\,d_L x = \iint f(a'a,\ a'b+b')\,\frac{da'\,db'}{a'^2} = \iint f(u,v)\,\frac{a^2}{u^2}\,\frac{du\,dv}{a} = a\int_G f\,d_L x.$$

This is $\Delta(g)^{-1}\int_G f\,d_L x$.

Direct check of the second inversion identity in Theorem 16.5(iii). Here $x^{-1} = g(1/a, -b/a)$ and $\Delta(x^{-1}) = a$. Substitute $u = 1/a$, $v = -b/a$. The Jacobian determinant of $(a,b) \mapsto (u,v)$ is

$$\det\begin{pmatrix} -a^{-2} & 0 \\ ba^{-2} & -a^{-1}\end{pmatrix} = a^{-3}, \qquad \text{so} \qquad da\,db = u^{-3}\,du\,dv.$$

Since $a \cdot a^{-2} = a^{-1} = u$,

$$\int_G f(x^{-1})\Delta(x^{-1})\,d_L x = \iint f\!\left(\tfrac{1}{a}, -\tfrac{b}{a}\right)\frac{a\,da\,db}{a^2} = \iint f(u,v)\,u\,u^{-3}\,du\,dv = \int_G f\,d_L x.$$

16.5 Regular representations

Definition 16.8. The space $L^2(G)$ is the Hilbert completion of $C_c(G)$ for the inner product

$$\langle f, h\rangle = \int_G \overline{f(x)}\,h(x)\,d_L x.$$

By the positivity statement (I1), this inner product is positive definite. For $f \in C_c(G)$ define

$$(\lambda(g)f)(x) = f(g^{-1}x), \qquad (\rho(g)f)(x) = \Delta(g)^{1/2} f(xg).$$

Lemma 16.9. *Let $f \in C_c(G)$ and $\varepsilon > 0$. There is a neighbourhood $V = V^{-1}$ of e with $|f(yx) - f(x)| < \varepsilon$ and $|f(xy) - f(x)| < \varepsilon$ for all $x \in G$ and $y \in V$.*

Proof. We treat $f(yx)$. The other case is symmetric. If U_L works for $f(yx)$ and U_R for $f(xy)$, put $V' = U_L \cap U_R$ and $V = V' \cap V'^{-1}$.

Fix a compact neighbourhood W of e. Choose a decreasing neighbourhood basis $V_1 \supset V_2 \supset \cdots$ of e inside W. Suppose no V_k works. Then there are $y_k \in V_k$ and x_k with $|f(y_k x_k) - f(x_k)| \geq \varepsilon$. One of x_k and $y_k x_k$ lies in $K = \operatorname{supp} f$. So x_k lies in the compact set $K \cup W^{-1}K$. Since G is second countable and Hausdorff, compact subsets are sequentially compact. So a subsequence has $x_k \to x$. As $y_k \to e$, also $y_k x_k \to x$. Continuity gives $f(y_k x_k) - f(x_k) \to 0$, which is a contradiction. $\blacksquare$

Proposition 16.10. *The maps λ and ρ extend to strongly continuous unitary representations of G on $L^2(G)$, and $\lambda(g)\rho(h) = \rho(h)\lambda(g)$ for all $g, h \in G$.*

Proof. Both are homomorphisms on $C_c(G)$. For example,

$$\rho(g)\rho(h)f(x) = \Delta(g)^{1/2}\Delta(h)^{1/2} f(xgh) = \rho(gh)f(x).$$

Both operators preserve norms. By the left invariance in Theorem 16.3(c) and the right-translation identity in Theorem 16.5(i), applied to $|f|^2$,

$$\|\lambda(g)f\|^2 = \int_G |f(g^{-1}x)|^2 \, d_L x = \|f\|^2, \qquad \|\rho(g)f\|^2 = \Delta(g) \int_G |f(xg)|^2 \, d_L x = \|f\|^2.$$

Each operator is a bijection of $C_c(G)$, with inverse $\lambda(g^{-1})$ or $\rho(g^{-1})$. An isometric bijection of a dense subspace extends to an isometry with closed, dense range, hence to a unitary operator. The homomorphism property passes to the extensions by continuity. The operators commute, by the computation

$$\lambda(g)\rho(h)f(x) = \Delta(h)^{1/2} f(g^{-1}xh) = \rho(h)\lambda(g)f(x).$$

Strong continuity. Fix $f \in C_c(G)$ with support K, and a compact neighbourhood $W = W^{-1}$ of e. For $g \in W$, the functions $\lambda(g)f - f$ and $f(\cdot\, g) - f$ vanish outside $WK \cup KW$. A finite sum of chart bump functions gives $\varphi \in C_c(G)$ with $\varphi \geq 0$ and $\varphi \geq 1$ on $WK \cup KW$. Fix $\varepsilon > 0$. Take the neighbourhood supplied by the uniform-continuity statement of Lemma 16.9 and intersect it with W to obtain $V \subset W$. For $g \in V$,

$$\|\lambda(g)f - f\|^2 \leq \varepsilon^2 \int_G \varphi \, d_L x.$$

The function φ was fixed before ε, and $\varepsilon > 0$ is arbitrary. Hence $\lambda(g)f \to f$ as $g \to e$. For ρ use

$$\|\rho(g)f - f\| \leq \Delta(g)^{1/2} \, \|f(\cdot\, g) - f\| + |\Delta(g)^{1/2} - 1| \, \|f\|$$

and the continuity of Δ. The squared norm $\|f(\cdot\, g) - f\|^2$ obeys the same bound as $\|\lambda(g)f - f\|^2$, by the second estimate of Lemma 16.9. Hence $\rho(g)f \to f$ as $g \to e$.

For general $h \in L^2(G)$, choose $f \in C_c(G)$ close to h. Then

$$\|\lambda(g)h - h\| \leq 2\|h - f\| + \|\lambda(g)f - f\|.$$

This gives continuity at e. Continuity at g_0 follows from $\lambda(g)h - \lambda(g_0)h = \lambda(g_0)\big(\lambda(g_0^{-1}g)h - h\big)$. The same holds for ρ. ∎

Without the factor $\Delta(g)^{1/2}$, right translation multiplies squared norms by $\Delta(g)^{-1}$. On the $ax + b$ group the factor $\Delta(g)^{-1}$ equals a for $g = g(a, b)$. Exercise 16.5 shows that λ and ρ are unitarily equivalent.

Right translation of a rectangle in the $ax + b$ group

In the $ax + b$ group the modular function can be computed from weighted areas, without differentiating a matrix. For a region E in the (a, b)-chart we write

$$\mu_L(E) = \iint_E a^{-2} \, da \, db$$

and call it the *left Haar measure* of E. Now let

$$E = \{(a, b) : 1 \le a \le 2,\ 0 \le b \le 1\}.$$

Its left Haar measure is

$$\mu_L(E) = \int_1^2 \int_0^1 a^{-2}\, db\, da = \left[-a^{-1}\right]_1^2 = \frac{1}{2}.$$

For $h = (s, t)$, right translation sends (a, b) to $(as, b + at)$. Its Jacobian in (a, b) is s, while the new density coefficient is $(as)^{-2}$. Hence

$$\mu_L(Eh) = \int_E (as)^{-2} s\, da\, db = s^{-1}\mu_L(E) = \Delta(h)\mu_L(E).$$

The shear parameter t changes the shape of the region but not this scale factor. Left translation by h has Jacobian s^2 and density coefficient $(sa)^{-2}$, so it preserves $\mu_L(E)$.

For $f \in C_c(G)$, the right-translation identity in Theorem 16.5(i), applied to $|f|^2$, gives $\|f(\cdot\, h)\|_2^2 = \Delta(h)^{-1}\|f\|_2^2 = s\,\|f\|_2^2$.

The indicator $\mathbf{1}_E$ of E is not continuous, so it is not in $C_c(G)$. It lies in $L^2(G)$ through the Radon-measure interpretation of § 16.6, where $\mu_L(E)$ is the Radon measure of E. The uncorrected right translation of $\mathbf{1}_E$ is $\mathbf{1}_E(xh) = \mathbf{1}_{Eh^{-1}}(x)$. Therefore

$$\|\mathbf{1}_E(\,\cdot\, h)\|_2^2 = \mu_L(Eh^{-1}) = s\,\mu_L(E).$$

Multiplication by $\Delta(h)^{1/2} = s^{-1/2}$ restores the norm in both cases. The inverse in the translated set Eh^{-1} explains why the integral identity has $\Delta(h)^{-1}$, whereas the region identity has $\Delta(h)$.

16.6 Locally compact groups

A topological group is *locally compact* if e has a compact neighbourhood. We cite the following result (Folland, *A Course in Abstract Harmonic Analysis*, Chapter 2). Let G be a locally compact Hausdorff group. A linear functional I on $C_c(G)$ is *positive* if $I(f) \ge 0$ whenever $f \ge 0$. There is a nonzero positive linear functional I on $C_c(G)$ with $I(f(g\,\cdot)) = I(f)$ for all $f \in C_c(G)$ and $g \in G$. It is unique up to a positive factor. There is a unique continuous homomorphism $\Delta\colon G \to (0, \infty)$ satisfying $I(f(\cdot\, g^{-1})) = \Delta(g)I(f)$.

A general locally compact group has no normalisation at e that compares left and right integrals. The cited result instead contains two identities. They replace the density relation in Theorem 16.5(ii) and the inversion identities in part (iii). First, the functional $f \mapsto I(f\Delta^{-1})$ is right-invariant. Second, I takes the same value on $f \in C_c(G)$ and on the function $x \mapsto f(x^{-1})\Delta(x^{-1})$.

The Riesz representation theorem realises I as a Radon measure. We cite Folland, *Real Analysis*, second edition, Theorem 7.2 in §7.1.

For a Lie group, uniqueness makes I a multiple of $f \mapsto \int_G f\, d_L x$. So the two modular functions agree. Since $C_c(G)$ is dense in L^2 of a Radon measure (Folland, *Real Analysis*, Proposition 7.9 in §7.2), our $L^2(G)$ is the usual one.

Self-checks

1. In the $ax+b$ group let $g = g(2,5)$. Suppose $\int_G f\, d_L x = 7$. Find $\Delta(g)$ and $\int_G f(xg)\, d_L x$.

   Answer. $\Delta(g) = 1/2$. By the right-translation identity in Theorem 16.5(i), the integral is $\Delta(g)^{-1} \cdot 7 = 14$.

2. Suppose some left-invariant smooth positive density on G is also right-invariant. Must G be unimodular?

   Answer. Yes. By the uniqueness in Theorem 16.3(a), the density is $c\mu_L$ for some $c > 0$, so $R_g^* \mu_L = \mu_L$ for all $g \in G$. By the right-translation identity in Theorem 16.5(i), this means $\Delta(g) = 1$ for all $g \in G$.

3. On the $ax+b$ group, by what factor does $f \mapsto f(\cdot\, g(3,1))$ multiply $\|f\|^2$ in $L^2(G)$?

   Answer. By the right-translation identity in Theorem 16.5(i), applied to $|f|^2$, the factor is $\Delta(g(3,1))^{-1} = 3$. Hence the map is not unitary without the factor $\Delta^{1/2}$.

Summary

Translating a fixed positive density on $\mathfrak{g}$ gives the left and right Haar densities. Each is unique up to a positive factor. Right translation scales μ_L by $\Delta(g) = |\det \mathrm{Ad}(g)|^{-1}$. When μ_L and μ_R have the same value at e, we have $\mu_R = \Delta^{-1}\mu_L$, and inversion exchanges μ_L and μ_R. For the $ax+b$ group, $d_L x = a^{-2}\, da\, db$, $d_R x = a^{-1}\, da\, db$ and $\Delta = a^{-1}$. The factor $\Delta^{1/2}$ makes right translation unitary on $L^2(G)$.

Reading

- Knapp, *Lie Groups Beyond an Introduction*, Chapter VIII "Integration", sections "Differential Forms and Measure Zero" and "Haar Measure for Lie Groups". Compare its normalisation of the modular function with ours.
- Kirillov, *Elements of the Theory of Representations*, §9, from the course reference route. Check its modular-function convention.
- Folland, *A Course in Abstract Harmonic Analysis*, Chapter 2, sections "Haar Measure" and "The Modular Function", for general locally compact groups.

Lecture 17

Unitary induction

Objectives. *Lecture 12 constructed induced representations of compact groups using an invariant measure on K/H. For a noncompact group G, the quotient G/H may carry no invariant measure. We replace it by a* quasi-invariant *measure, which translation by an element of G multiplies by a positive continuous function. We then correct the action by a square-root factor. The model is the $ax+b$ group, with representations induced from characters of its translation subgroup.*

By the end of this lecture you will be able to:

- State the quotient-measure theorem and derive the quasi-invariance factor and its cocycle identity.
- Define the normalised induced representation and prove that it is unitary.
- Compute the $ax+b$ representations induced from translation characters in two measure pictures.
- Explain why the unitary equivalence class of the induced representation depends only on the scaling orbit of the frequency.

17.1 Equivariant functions

Throughout, G is a Lie group, H is a closed subgroup and $q(x) = xH$ is the quotient map $G \to G/H$. By Lecture 3, G/H is a manifold on which G acts by $g \cdot xH = gxH$. The map q is continuous and open. Let (σ, V) be a unitary representation of H on a Hilbert space V.

Definition 17.1. An *equivariant function of type σ* is a continuous map $f : G \to V$ with

$$f(xh) = \sigma(h)^{-1} f(x) \qquad (x \in G,\ h \in H). \tag{17.1}$$

Let $\mathcal{F}_c(\sigma)$ be the space of those f for which $q(\operatorname{supp} f)$ is compact.

This is the convention of Lecture 12. Let $s : U \to G$ be a continuous section of q over an open set $U \subseteq G/H$. Then $f(s(p)h) = \sigma(h)^{-1} f(s(p))$. So over U the function f is just the ordinary function $f \circ s : U \to V$. Continuous local sections exist near every point; they are part of the construction of G/H in Lecture 3. Geometrically f is a section of a vector bundle over G/H. We use only the function description.

Since $\sigma(h)$ is unitary, $n_f(xH) = \|f(x)\|^2$ is well defined. It is continuous on G/H because q is open, and its support is compact.

Averaging over H turns compactly supported continuous functions $G \to V$ into elements of $\mathcal{F}_c(\sigma)$. Fix a left Haar measure dh on H. For continuous compactly supported $F\colon G \to V$ define $P_\sigma F$ by

$$P_\sigma F(x) = \int_H \sigma(h) F(xh)\, dh.$$

The integral is defined weakly by the Riesz argument of Lecture 10, applied to a continuous Hilbert-space-valued function with compact support. For fixed x, the integrand vanishes outside $H \cap x^{-1} \operatorname{supp} F$.

Left invariance of dh under $h' = h_0 h$ gives

$$P_\sigma F(xh_0) = \int_H \sigma(h_0^{-1}h') F(xh')\, dh' = \sigma(h_0)^{-1} P_\sigma F(x).$$

For x in a compact neighbourhood, all the integrands have support in one compact subset of H. Uniform continuity there makes $P_\sigma F$ continuous. Also $q(\operatorname{supp} P_\sigma F) \subseteq q(\operatorname{supp} F)$, which is compact. Thus $P_\sigma F \in \mathcal{F}_c(\sigma)$.

17.2 Quotient measures

As in Lecture 16, a *measure* on a locally compact space X here means a positive linear functional $\varphi \mapsto \int_X \varphi\, d\mu$ on $C_c(X)$ which is strictly positive on nonzero $\varphi \geq 0$. Then $L^2(X, \mu)$ is the completion of $C_c(X)$ for $\langle \varphi, \psi \rangle = \int_X \bar{\varphi}\psi\, d\mu$.

Fix left Haar measures dx on G and dh on H. Modular functions follow Lecture 16:

$$\int_G F(xg)\, dx = \Delta_G(g)^{-1} \int_G F(x)\, dx,$$

and likewise for Δ_H. For $F \in C_c(G)$ define $F^H \in C_c(G/H)$ by

$$F^H(xH) = \int_H F(xh)\, dh. \tag{17.2}$$

It is well defined by left invariance of dh.

> **Lemma 17.2.** *If a measure μ on G/H satisfies $\int_G F\, dx = \int_{G/H} F^H\, d\mu$ for all $F \in C_c(G)$, then $\Delta_G(h) = \Delta_H(h)$ for all $h \in H$.*

Proof. Fix $h_0 \in H$ and apply the identity to $F_{h_0}(x) = F(xh_0)$. The left side becomes $\Delta_G(h_0)^{-1} \int_G F\, dx$. On the right,

$$(F_{h_0})^H(xH) = \int_H F(xhh_0)\, dh = \Delta_H(h_0)^{-1} F^H(xH).$$

Hence $\Delta_G(h_0)^{-1} \int_G F\, dx = \Delta_H(h_0)^{-1} \int_G F\, dx$. Choose $F \geq 0$ nonzero, so that $\int_G F\, dx > 0$. Dividing both sides by $\int_G F\, dx$ gives $\Delta_G(h_0)^{-1} = \Delta_H(h_0)^{-1}$, so $\Delta_G(h_0) = \Delta_H(h_0)$. ∎

For the $ax + b$ group and its scaling subgroup A, we will see $\Delta_G(a, 0) = 1/a$ while $\Delta_A = 1$. Hence, by the modular compatibility lemma, Lemma 17.2, G/A carries no such measure. Instead, we insert a weight function on G into the integral over G.

Theorem 17.3 (Quotient measures; cited)**.** *Let G be a Lie group and H a closed subgroup.*

(a) There is a continuous function $\rho\colon G \to (0,\infty)$ with

$$\rho(xh) = \frac{\Delta_H(h)}{\Delta_G(h)}\,\rho(x) \qquad (x \in G,\ h \in H). \tag{17.3}$$

(b) The map $F \mapsto F^H$ sends $C_c(G)$ onto $C_c(G/H)$. If $\varphi \geq 0$, a lift F with $F^H = \varphi$ can be chosen with $F \geq 0$.

(c) For each ρ as in (a) there is a unique measure μ_ρ on G/H with

$$\int_G F(x)\rho(x)\,dx = \int_{G/H} F^H\,d\mu_\rho \qquad (F \in C_c(G)). \tag{17.4}$$

For a general closed subgroup, see G. B. Folland, *A Course in Abstract Harmonic Analysis*, second edition (2016), Chapter 2, "Measures on Homogeneous Spaces", cited below as Folland, Abstract Harmonic Analysis. Its modular convention agrees with ours. Uniqueness of μ_ρ follows from the onto property of the averaging map. A. A. Kirillov, *Elements of the Theory of Representations* (1976), Section 9.1, cited below as Kirillov, Elements, treats quasi-invariant measures and the modular criterion. Its conventions must be translated before use.

A continuous positive function ρ with the transformation law (17.3) is called a *rho-function*. Repeating the computation in the modular compatibility lemma, Lemma 17.2, with $F\rho$ in place of F confirms that the transformation law (17.3) is compatible with our modular convention. If $\Delta_G|_H = \Delta_H$ we may take $\rho = 1$.

Proposition 17.4. *Let ρ be a rho-function and $\mu = \mu_\rho$. Define λ by $\lambda(g, xH) = \rho(gx)/\rho(x)$. Then λ is well defined. For $g, g_1, g_2 \in G$, $p \in G/H$ and $\varphi \in C_c(G/H)$,*

$$\lambda(g_1g_2, p) = \lambda(g_1, g_2p)\,\lambda(g_2, p), \qquad \lambda(e, p) = 1, \tag{17.5}$$

$$\int_{G/H} \varphi(g^{-1}p)\,d\mu(p) = \int_{G/H} \varphi(p)\,\lambda(g, p)\,d\mu(p). \tag{17.6}$$

Proof. By the transformation law (17.3), replacing x by xh multiplies $\rho(gx)$ and $\rho(x)$ by the same factor. So λ is well defined. The function $(g, x) \mapsto \rho(gx)/\rho(x)$ is continuous and constant on fibres of $\mathrm{id} \times q$. Since q is open, $\mathrm{id} \times q$ is an open surjection, hence a quotient map. Thus λ is jointly continuous in (g, p). For $p = xH$,

$$\lambda(g_1g_2, p) = \frac{\rho(g_1g_2x)}{\rho(g_2x)} \cdot \frac{\rho(g_2x)}{\rho(x)} = \lambda(g_1, g_2p)\,\lambda(g_2, p).$$

Also $\lambda(e, p) = \rho(x)/\rho(x) = 1$.

For the quasi-invariance identity (17.6), use the onto property of averaging, Theorem 17.3(b), to write $\varphi = F^H$ with $F \in C_c(G)$. The function $x \mapsto F(g^{-1}x)$ has average $\varphi(g^{-1}\,\cdot)$. By the weighted quotient integral (17.4) and left invariance of dx, with $y = g^{-1}x$,

$$\int \varphi(g^{-1}p)\,d\mu(p) = \int_G F(g^{-1}x)\rho(x)\,dx = \int_G F(y)\rho(gy)\,dy = \int_G F(y)\,\lambda(g, yH)\,\rho(y)\,dy.$$

Since $\lambda(g, \cdot)$ is constant on cosets, $y \mapsto F(y)\lambda(g, yH)$ has average $\varphi\,\lambda(g, \cdot)$. Applying the weighted quotient integral (17.4) again gives the right side of (17.6). ∎

We say μ is *quasi-invariant*: translation by g changes it by the positive continuous factor $\lambda(g,\cdot)$. If $\rho = 1$, then $\lambda = 1$ and μ is G-invariant.

17.3 The normalised induced representation

Fix a rho-function ρ, and let μ and λ be as in the quasi-invariance proposition, Proposition 17.4. For $f_1, f_2 \in \mathcal{F}_c(\sigma)$ put

$$\langle f_1, f_2 \rangle = \int_{G/H} \langle f_1(x), f_2(x) \rangle \, d\mu(xH), \qquad \|f\|^2 = \int_{G/H} n_f \, d\mu.$$

The integrand is well defined because σ is unitary. If $\|f\| = 0$, then $n_f = 0$ by strict positivity, so $f = 0$.

Theorem 17.5. *For $f \in \mathcal{F}_c(\sigma)$ and $g \in G$ define*

$$(\Pi(g)f)(x) = \left(\frac{\rho(g^{-1}x)}{\rho(x)}\right)^{1/2} f(g^{-1}x) = \lambda(g^{-1}, xH)^{1/2} f(g^{-1}x). \tag{17.7}$$

Then (a) $\Pi(g)f \in \mathcal{F}_c(\sigma)$; (b) $\Pi(e) = I$ *and* $\Pi(g_1 g_2) = \Pi(g_1)\Pi(g_2)$; (c) $\|\Pi(g)f\| = \|f\|$.

Proof. (a) The factor depends only on xH. So replacing x by xh only changes $f(g^{-1}x)$ into $\sigma(h)^{-1} f(g^{-1}x)$. Also $q(\operatorname{supp} \Pi(g)f) = g \cdot q(\operatorname{supp} f)$ is compact.

(b) Let $p = xH$. Applying the normalised action (17.7) twice,

$$(\Pi(g_1)\Pi(g_2)f)(x) = \lambda(g_1^{-1}, p)^{1/2} \, \lambda(g_2^{-1}, g_1^{-1}p)^{1/2} \, f(g_2^{-1} g_1^{-1} x).$$

The cocycle identity for the pair g_2^{-1}, g_1^{-1} turns the product of factors into $\lambda((g_1 g_2)^{-1}, p)^{1/2}$. Also $\lambda(e, p) = 1$ gives $\Pi(e) = I$.

(c) Translation by g changes μ by the factor $\lambda(g,\cdot)$. The square-root factor in $\Pi(g)$ is chosen to cancel this change. We have $n_{\Pi(g)f}(p) = \lambda(g^{-1}, p)\, n_f(g^{-1}p)$. Put $\varphi(p) = n_f(p)\, \lambda(g^{-1}, gp)$, so that $\varphi(g^{-1}p) = n_{\Pi(g)f}(p)$. Then the quasi-invariance identity (17.6) gives

$$\|\Pi(g)f\|^2 = \int \varphi(g^{-1}p)\, d\mu = \int n_f(p)\, \lambda(g^{-1}, gp)\, \lambda(g, p)\, d\mu(p) = \int n_f \, d\mu.$$

The last step is the cocycle identity (17.5) with $g_1 = g^{-1}$ and $g_2 = g$. ∎

By the representation law and the isometry property in the normalised-induction theorem, Theorem 17.5, each $\Pi(g)$ is an isometry of $\mathcal{F}_c(\sigma)$ with inverse $\Pi(g^{-1})$. So $\Pi(g)$ extends to a unitary operator on the completion. The representation law persists by continuity.

Definition 17.6. The *normalised induced representation* $\operatorname{Ind}_H^G \sigma$ is the completion of $\mathcal{F}_c(\sigma)$, for the inner product defined by the chosen rho-function ρ, together with these unitary operators $\Pi(g)$.

Proposition 17.7. *The induced representation $\operatorname{Ind}_H^G \sigma$ is strongly continuous.*

Proof. First take $f \in \mathcal{F}_c(\sigma)$ and let $C = q(\operatorname{supp} f)$, a compact set. Choose a compact neighbourhood B of e in G. For $g \in B$, the quotient supports of both $\Pi(g)f$ and f lie in the compact set

$$S = B \cdot C \subset G/H.$$

It suffices to show that their difference tends uniformly to zero in fibre norm on S. This norm is independent of the representative in G because σ is unitary.

By compactness of S and the manifold structure of G/H, we can cover S by finitely many compact subsets S_i. Each S_i lies in an open set with a continuous local section s_i of q. The lifted sets $s_i(S_i)$ are compact. At these representatives,

$$(\Pi(g)f - f)(s_i(p)) = \lambda(g^{-1}, p)^{1/2} f(g^{-1}s_i(p)) - f(s_i(p)).$$

The function $(g, p) \mapsto \lambda(g^{-1}, p)$ is jointly continuous and equals one at $g = e$. Also $(g, p) \mapsto f(g^{-1}s_i(p))$ is jointly continuous as a Hilbert-space-valued function. On the compact sets $B \times S_i$, this continuity is uniform. Thus the displayed difference tends uniformly to zero for $p \in S_i$ as $g \to e$. There are only finitely many S_i, so the same holds on S.

The measure μ is finite on compact sets, by its Radon-measure interpretation in Lecture 16. So if the uniform bound is ε, then

$$\|\Pi(g)f - f\|^2 \le \varepsilon^2 \mu(S).$$

This proves norm continuity on $\mathcal{F}_c(\sigma)$.

For a vector v in the completion, choose f with $\|v - f\| < \varepsilon$. Unitarity gives

$$\|\Pi(g)v - v\| \le 2\varepsilon + \|\Pi(g)f - f\|.$$

Hence continuity holds at e for every v. By the representation law and unitarity, continuity at e implies continuity at every g_0. ∎

Remark 17.8. Without the square root, the map $f \mapsto f(g^{-1}\,\cdot)$ still preserves $\mathcal{F}_c(\sigma)$ and satisfies the representation law. By the quasi-invariance identity (17.6), this map sends $\|f\|^2$ to $\int n_f\, \lambda(g, \cdot)\, d\mu$. So in general it is not unitary; in the $ax+b$ example it multiplies $\|\psi\|^2$ by a_0 (Proposition 17.9(a)). For compact G and finite-dimensional σ, we take $\rho = 1$ and recover the construction of Lecture 12. Different rho-functions give unitarily equivalent representations (Exercise 17.3). Finite-group Frobenius reciprocity does not transfer verbatim, and we do not use it.

17.4 Representations of the $ax + b$ group induced from translations

Let $G = \{(a, b) : a > 0,\ b \in \mathbb{R}\}$ with

$$(a, b)(a', b') = (aa', b + ab'), \qquad (a, b)^{-1} = (a^{-1}, -a^{-1}b), \qquad (a, b) = \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}.$$

It acts on $\mathbb{R}$ by $x \mapsto ax + b$. By Lecture 16, $dx = a^{-2}\, da\, db$ is a left Haar measure and $\Delta_G(a, b) = 1/a$. Let $N = \{n_c = (1, c)\}$ be the translations and $A = \{(a, 0)\}$ the scalings. Then $N \cong \mathbb{R}$ has Haar measure dc and $\Delta_N = 1$. For a *frequency* $\xi \in \mathbb{R}$, the map $\chi_\xi(n_c) = e^{i\xi c}$ is a unitary character of N.

17.4.1 The quotient and two measures

Since $(a,b)n_c = (a, b+ac)$, the coset $(a,b)N$ is $\{a\}\times\mathbb{R}$. So $(a,b)N \mapsto a$ identifies G/N with $(0,\infty)$, and $s(a) = (a,0)$ is a global section. The action is $(a_0,b_0)\cdot a = a_0 a$.

Since $\Delta_G(n_c) = 1 = \Delta_N(n_c)$, the transformation law (17.3) says $\rho(xn) = \rho(x)$. Thus rho-functions are positive continuous functions of a alone. The substitution $b = ac$ gives

$$F^N(a) = \int_{\mathbb{R}} F(a,ac)\,dc = \frac{1}{a}\int_{\mathbb{R}} F(a,b)\,db.$$

Hence for $\rho = \rho(a)$,

$$\int_G F\rho\,dx = \int_0^\infty \Big(\int_{\mathbb{R}} F(a,b)\,db\Big)\frac{\rho(a)\,da}{a^2} = \int_0^\infty F^N(a)\,\rho(a)\,\frac{da}{a}.$$

By the uniqueness in the quotient-measure theorem, Theorem 17.3, $d\mu_\rho = \rho(a)\,da/a$.

We use two choices. The rho-function $\rho_1 = 1$ gives the invariant measure, and the rho-function $\rho_L(a,b) = a$ gives Lebesgue measure:

$$d\mu_{\rho_1} = \frac{da}{a}, \qquad d\mu_{\rho_L} = da.$$

For the Lebesgue choice, take $g = (a_0,b_0)$ and $x = (a,0)$. Then $gx = (a_0 a, b_0)$, so $\lambda_L(g,a) = a_0$. In this case the quasi-invariance identity (17.6) is the substitution rule $\int_0^\infty \varphi(a/a_0)\,da = a_0\int_0^\infty \varphi(u)\,du$. The cocycle identity reads $a_1a_2 = a_1\cdot a_2$, because g_1g_2 has first entry a_1a_2.

17.4.2 Equivariant functions and the action

For $\sigma = \chi_\xi$, the equivariance condition (17.1) says $f(a,b+ac) = e^{-i\xi c}f(a,b)$. Take $b=0$ and then $c = b/a$:

$$f(a,b) = e^{-i\xi b/a}\varphi(a), \qquad \varphi = f\circ s. \tag{17.8}$$

Conversely, for $\varphi \in C_c((0,\infty))$ this f lies in $\mathcal{F}_c(\chi_\xi)$, since $e^{-i\xi(b+ac)/a} = e^{-i\xi c}e^{-i\xi b/a}$. So $f\mapsto\varphi$ is a bijection onto $C_c((0,\infty))$. Moreover $n_f = |\varphi|^2$ and $\|f\|^2 = \int_0^\infty |\varphi|^2\rho(a)\,da/a$.

Let $g = (a_0,b_0)$. Then $g^{-1}(a,0) = (a/a_0,\ -b_0/a_0)$, and the equivariant form (17.8) gives

$$f(g^{-1}(a,0)) = \exp\Big(-i\xi\,\frac{-b_0/a_0}{a/a_0}\Big)\varphi\Big(\frac{a}{a_0}\Big) = e^{i\xi b_0/a}\,\varphi\Big(\frac{a}{a_0}\Big).$$

The factor at $x=(a,0)$ is $(\rho(a/a_0)/\rho(a))^{1/2}$. This yields two pictures:

$$(\Pi_\xi(a_0,b_0)\varphi)(a) = e^{i\xi b_0/a}\,\varphi(a/a_0) \qquad \text{on } L^2((0,\infty),da/a), \tag{17.9}$$

$$(\Pi^L_\xi(a_0,b_0)\psi)(a) = a_0^{-1/2}e^{i\xi b_0/a}\,\psi(a/a_0) \qquad \text{on } L^2((0,\infty),da). \tag{17.10}$$

The normalised-induction theorem, Theorem 17.5, already gives unitarity. We now verify unitarity and the representation law directly in this model.

Proposition 17.9. *Let $g=(a_0,b_0)$, $g_j=(a_j,b_j)$ and $\varphi,\psi\in C_c((0,\infty))$.*

(a) $\|\Pi^L_\xi(g)\psi\| = \|\psi\|$, *while* $\psi\mapsto e^{i\xi b_0/a}\psi(a/a_0)$ *multiplies* $\|\psi\|^2$ *by* a_0.

(b) $\Pi^L_\xi(g_1)\Pi^L_\xi(g_2) = \Pi^L_\xi(g_1 g_2)$.

(c) $(W\varphi)(a) = a^{-1/2}\varphi(a)$ *extends to a unitary map* $L^2(da/a) \to L^2(da)$ *with* $W\Pi_\xi(g) = \Pi^L_\xi(g)W$.

Proof. (a) Substitute $a = a_0 u$:

$$\int_0^\infty a_0^{-1}|\psi(a/a_0)|^2\,da = \int_0^\infty |\psi(u)|^2\,du.$$

Without the factor a_0^{-1} the result is $a_0\|\psi\|^2$.

(b) Since $g_1 g_2 = (a_1 a_2,\ b_1 + a_1 b_2)$,

$$\begin{aligned}(\Pi^L_\xi(g_1)\Pi^L_\xi(g_2)\psi)(a) &= a_1^{-1/2} e^{i\xi b_1/a}\,(\Pi^L_\xi(g_2)\psi)(a/a_1)\\ &= a_1^{-1/2}a_2^{-1/2}\,e^{i\xi b_1/a}e^{i\xi a_1 b_2/a}\,\psi\Big(\frac{a}{a_1 a_2}\Big)\\ &= (a_1 a_2)^{-1/2} e^{i\xi(b_1+a_1 b_2)/a}\,\psi\Big(\frac{a}{a_1 a_2}\Big) = (\Pi^L_\xi(g_1 g_2)\psi)(a).\end{aligned}$$

The square roots combine because λ_L is multiplicative; this is the cocycle identity. The phases combine into the second entry of $g_1 g_2$; this is where equivariance enters.

(c) We have $\int |a^{-1/2}\varphi|^2\,da = \int |\varphi|^2\,da/a$, and W has inverse multiplication by $a^{1/2}$. Finally,

$$(\Pi^L_\xi(g)W\varphi)(a) = a_0^{-1/2}e^{i\xi b_0/a}\Big(\frac{a}{a_0}\Big)^{-1/2}\varphi\Big(\frac{a}{a_0}\Big) = a^{-1/2}e^{i\xi b_0/a}\varphi\Big(\frac{a}{a_0}\Big) = (W\Pi_\xi(g)\varphi)(a). \qquad \blacksquare$$

Note that W is multiplication by $(\rho_1/\rho_L)^{1/2}$.

17.5 The scaling orbit of a frequency

On translations, the first picture (17.9) gives $(\Pi_\xi(n_c)\varphi)(a) = e^{i(\xi/a)c}\varphi(a)$. So at the point $a \in G/N$ the group N acts by the character $\chi_{\xi/a}$. Let A act on characters of N by $(t\cdot\chi)(n) = \chi(t^{-1}nt)$. For $t = (a, 0)$,

$$t^{-1}n_c t = (a^{-1}, a^{-1}c)(a, 0) = (1, c/a), \qquad \text{so} \qquad t\cdot\chi_\xi = \chi_{\xi/a}.$$

Thus the character at aN is $s(a)\cdot\chi_\xi$. As a runs through $(0,\infty)$, the frequency ξ/a fills the orbit $\mathbb{R}_{>0}\,\xi$. There are three orbits: $(0,\infty)$, $(-\infty, 0)$ and $\{0\}$.

Proposition 17.10. *Let* $\kappa > 0$. *The operator* $(U\varphi)(a) = \varphi(a/\kappa)$ *is unitary on* $L^2((0,\infty), da/a)$, *and* $U\Pi_\xi(g) = \Pi_{\kappa\xi}(g)U$ *for all* $g \in G$.

Proof. The substitution $a = \kappa u$ preserves da/a, so U is isometric on $C_c((0,\infty))$. Its inverse is $\varphi \mapsto \varphi(\kappa\,\cdot)$. For $g = (a_0, b_0)$,

$$\begin{aligned}(U\Pi_\xi(g)\varphi)(a) &= (\Pi_\xi(g)\varphi)(a/\kappa) = e^{i\xi b_0\kappa/a}\,\varphi\Big(\frac{a}{\kappa a_0}\Big),\\ (\Pi_{\kappa\xi}(g)U\varphi)(a) &= e^{i\kappa\xi b_0/a}\,(U\varphi)(a/a_0) = e^{i\kappa\xi b_0/a}\,\varphi\Big(\frac{a}{a_0\kappa}\Big).\end{aligned}$$

The two expressions agree on the dense subspace $C_c((0,\infty))$ of $L^2((0,\infty), da/a)$. Both sides are bounded operators, so they agree on the whole space. $\blacksquare$

Thus, by Proposition 17.10, the unitary equivalence class of Π_ξ depends only on the scaling orbit of ξ. We write $\Pi_+ = \Pi_1$ and $\Pi_- = \Pi_{-1}$. By Exercise 17.5, the three representations Π_+, Π_- and Π_0 are pairwise inequivalent.

In Π_0 we have $(\Pi_0(a_0, b_0)\varphi)(a) = \varphi(a/a_0)$. So N acts trivially. The representation Π_0 is the left regular representation of $(0, \infty) \cong G/N$, composed with the quotient map $G \to G/N$. Irreducibility is not discussed here.

A quotient with no invariant Radon measure

For comparison, give the scaling subgroup $A = \{(a, 0) : a > 0\}$ the Haar measure da/a and induce its trivial character. Identify G/A with the line by $(a, b)A \mapsto b$. Left multiplication gives the familiar affine action

$$(a_0, b_0) \cdot t = b_0 + a_0 t.$$

Lebesgue measure dt is quasi-invariant, but a dilation multiplies it by a_0. Any invariant Radon measure would be translation invariant. By Haar uniqueness on $\mathbb{R}$, it would be a constant multiple of dt. Dilation invariance then forces this constant to be zero.

Here $\Delta_A = 1$ and $\Delta_G(a, 0) = 1/a$, so a rho-function is $\rho(a, b) = a$. The quotient identity follows directly:

$$\int_G F(a, b)\rho(a, b)\,\frac{da\,db}{a^2} = \int_{\mathbb{R}} \left(\int_0^\infty F(a, t)\,\frac{da}{a} \right) dt.$$

Since $(1, t)(a, 0) = (a, t)$ and A carries the Haar measure da/a, the inner integral is the coset average $F^A((1, t)A)$ of (17.2) with $H = A$. By the uniqueness in the quotient-measure theorem, Theorem 17.3, dt is exactly the measure μ_ρ supplied by this rho-function.

The square-root factor in the normalised action (17.7) becomes $a_0^{-1/2}$, and the induced action is

$$(\Pi(a_0, b_0)u)(t) = a_0^{-1/2} u\left(\frac{t - b_0}{a_0} \right).$$

To verify its norm, set $r = (t - b_0)/a_0$. Then $dt = a_0\,dr$, so

$$\|\Pi(a_0, b_0)u\|_2^2 = a_0^{-1} \int_{\mathbb{R}} |u((t - b_0)/a_0)|^2 dt = \int_{\mathbb{R}} |u(r)|^2 dr.$$

Without the square root, the squared norm would be multiplied by a_0. This quotient makes the measure correction visible in one variable. In the earlier translation-subgroup model, the quotient was the positive scaling line and an invariant quotient measure was available. The subgroup, not just the ambient group, controls the answer.

Self-checks

1. In the $ax + b$ group, derive the transformation law of a rho-function for $H = A$ and check that $\rho(a, b) = a$ satisfies it. Is $\rho = 1$ possible?
   Answer. For $h = (t, 0)$ we have $\Delta_A(h)/\Delta_G(h) = 1/(1/t) = t$, so we need $\rho(xh) = t\rho(x)$. Since $(a, b)(t, 0) = (at, b)$, the function $\rho(a, b) = a$ works. The choice $\rho = 1$ fails for $t \neq 1$.

2. Is $\rho(a,b) = a + b^2$ a rho-function for $H = N$?
   Answer. No. A rho-function for N must satisfy $\rho(xn) = \rho(x)$. But $\rho((a,0)n_1) = \rho(a,a) = a + a^2 \neq a = \rho(a,0)$.
3. Give $\Pi_\xi(g)^*$ for $g = (a_0, b_0)$ in the first picture (17.9).
   Answer. Since Π_ξ is unitary, the adjoint is $\Pi_\xi(g^{-1})$, with $g^{-1} = (a_0^{-1}, -a_0^{-1}b_0)$. Hence

$$(\Pi_\xi(g)^*\varphi)(a) = e^{-i\xi b_0/(a_0 a)}\varphi(a_0 a).$$

   Substituting $a = a_0 u$ in $\langle \Pi_\xi(g)\varphi, \psi\rangle$ confirms this formula.

Summary

A rho-function gives a quasi-invariant measure μ_ρ on G/H. Its translation factor $\lambda(g, xH) = \rho(gx)/\rho(x)$ is a cocycle. The action $f \mapsto \lambda(g^{-1}, \cdot)^{1/2} f(g^{-1}\,\cdot)$ on equivariant functions is unitary, because the cocycle cancels the change of measure. For the representation Π_ξ of the $ax + b$ group induced from the character χ_ξ of the translation subgroup N, the point $a \in G/N$ carries the frequency ξ/a. The unitary equivalence class of Π_ξ depends only on the scaling orbit of ξ.

Reading

Folland, Abstract Harmonic Analysis, Chapter 2, “Measures on Homogeneous Spaces”: quotient measures and rho-functions for a general closed subgroup. Kirillov, Elements, Section 9.1: quasi-invariant measures on homogeneous spaces. Kirillov, Elements, Section 13: models of induced representations. A. W. Knapp, *Representation Theory of Semisimple Groups* (2001), Sections VII.1–VII.2: pictures of representations induced from a parabolic subgroup of a linear connected reductive group.

Lecture 18

Heisenberg group and the Weyl relations

Objectives. *The Heisenberg group is the simplest noncommutative nilpotent Lie group. This lecture builds its Schrödinger representation on $L^2(\mathbb{R})$ from translations and modulations, and proves the Weyl relations, which describe how translations and modulations commute up to a phase. We then state the Stone–von Neumann theorem. It says that a fixed nonzero central character, which records the scalars by which the centre acts, determines one irreducible unitary representation up to unitary equivalence. For central character zero, the irreducibles are one-dimensional characters, and the translation representation on $L^2(\mathbb{R})$ contains none of them.*

By the end of this lecture you will be able to:

- Compute products, inverses, commutators, the exponential map and Ad in the Heisenberg group.
- Prove that the Schrödinger formula is a strongly continuous unitary representation satisfying the Weyl relations, and compute its derivative on $C_c^\infty(\mathbb{R})$.
- State the Stone–von Neumann theorem with its hypotheses, and turn a Weyl pair into a representation to which it applies.
- Compute a Gaussian matrix coefficient and use it to contrast nonzero and zero central character.

18.1 The group law

Definition 18.1. The *Heisenberg group* H is the set of real matrices

$$(a,b,c) := \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}, \qquad a,b,c \in \mathbb{R},$$

with matrix multiplication.

Multiplying the matrices gives

$$(a,b,c)(a',b',c') = (a+a',\ b+b',\ c+c'+ab'). \tag{18.1}$$

The identity is $(0,0,0)$, and

$$(a,b,c)^{-1} = (-a,-b,-c+ab).$$

Indeed, the third coordinate of $(a,b,c)(-a,-b,-c+ab)$ is $c-c+ab-ab=0$. The group H is a closed subgroup of $\mathrm{GL}_3(\mathbb{R})$, hence a Lie group by the closed-subgroup theorem of Lecture 3. The coordinates identify H with $\mathbb{R}^3$, so H is connected and simply connected.

Swapping the factors in the group law (18.1) changes only the last coordinate. For $g=(a,b,c)$ and $g'=(a',b',c')$,

$$gg' = (0,0,ab'-a'b)\,g'g. \tag{18.2}$$

By the group law (18.1), each $(0,0,d)$ commutes with every element. So the commutation formula (18.2) shows that g is central exactly when $ab'-a'b=0$ for all a',b', that is, when $a=b=0$. The centre is therefore

$$Z(H)=\{(0,0,c)\,:\,c\in\mathbb{R}\}\cong\mathbb{R},$$

and $(a,b,c)Z(H)\mapsto(a,b)$ identifies $H/Z(H)$ with $\mathbb{R}^2$.

The Lie algebra $\mathfrak{h}$ has basis $X=E_{12}$, $Y=E_{23}$, $Z=E_{13}$, where E_{ij} are matrix units. Then

$$[X,Y]=Z,\qquad [X,Z]=[Y,Z]=0.$$

Let $W=xX+yY+zZ$. Multiplying matrix units gives $W^2=xyZ$ and $W^3=0$. So the exponential series of W stops after the quadratic term:

$$\exp(xX+yY+zZ)=I+W+\tfrac12 W^2=\big(x,\ y,\ z+\tfrac12 xy\big). \tag{18.3}$$

So $\exp\colon\mathfrak{h}\to H$ is a diffeomorphism with inverse $(a,b,c)\mapsto aX+bY+(c-\frac12 ab)Z$. In particular $\exp(sX)=(s,0,0)$, $\exp(sY)=(0,s,0)$ and $\exp(sZ)=(0,0,s)$.

For the adjoint action we show one representative step. By the group law (18.1),

$$g\exp(sX)g^{-1}=(a+s,b,c)(-a,-b,-c+ab)=(s,0,-sb)=\exp\big(s(X-bZ)\big).$$

Thus $\mathrm{Ad}(g)X=X-bZ$. In the same way $\mathrm{Ad}(g)Y=Y+aZ$ and $\mathrm{Ad}(g)Z=Z$. In the ordered basis (X,Y,Z),

$$D_{\mathrm{Ad}}(g)=\begin{pmatrix}1&0&0\\0&1&0\\-b&a&1\end{pmatrix},\qquad \det D_{\mathrm{Ad}}(g)=1.$$

With the convention $\Delta(g)=|\det\mathrm{Ad}(g)|^{-1}$ of Lecture 16, H is unimodular. Directly, left and right translations are polynomial maps of $\mathbb{R}^3$ with triangular Jacobian matrices and ones on the diagonal. So Lebesgue measure $da\,db\,dc$ is a left and right Haar measure.

18.2 The Schrödinger representation

We use $L^2(\mathbb{R})$ with $\langle f,h\rangle=\int_{\mathbb{R}}\overline{f(t)}\,h(t)\,dt$. We use the Lebesgue-integration inputs from Lecture 16: completeness of L^2, Fubini–Tonelli, dominated convergence and change of variables. In particular, dominated convergence justifies differentiation under the integrals below. We also use two standard facts. First, Lebesgue integrals are invariant under $t\mapsto t+a$. Second, the space $C_c(\mathbb{R})$ of continuous compactly supported functions, and its subspace $C_c^\infty(\mathbb{R})$ of smooth ones, are dense in $L^2(\mathbb{R})$.

Definition 18.2. Fix $\lambda \in \mathbb{R}$. For $g = (a, b, c) \in H$ and $f \in L^2(\mathbb{R})$, the *Schrödinger operator* $\Pi_\lambda(g)$ is defined by

$$(\Pi_\lambda(a,b,c)f)(t) = e^{i\lambda(c+bt)} f(t+a). \tag{18.4}$$

The *translations* $U(a)$ and *modulations* $V_\lambda(b)$ are defined by

$$(U(a)f)(t) = f(t+a), \qquad (V_\lambda(b)f)(t) = e^{i\lambda bt} f(t).$$

By the definitions, the Schrödinger operators factor as

$$\Pi_\lambda(a,b,c) = e^{i\lambda c}\, V_\lambda(b)\, U(a).$$

This matches the factorisation of group elements

$$(a,b,c) = (0,0,c)(0,b,0)(a,0,0).$$

Our $U(a)$ is the translation by $-a$ in the convention of Lecture 15.

Proposition 18.3. *For every $\lambda \in \mathbb{R}$, Π_λ is a strongly continuous unitary representation of H on $L^2(\mathbb{R})$, and $\Pi_\lambda(0,0,c) = e^{i\lambda c} I$. The* Weyl relations *hold:*

$$U(a)V_\lambda(b) = e^{i\lambda ab} V_\lambda(b)U(a), \qquad a, b \in \mathbb{R}. \tag{18.5}$$

Proof. Homomorphism. Let $g = (a,b,c)$ and $g' = (a',b',c')$. Applying the defining formula (18.4) twice,

$$\begin{aligned}(\Pi_\lambda(g)\Pi_\lambda(g')f)(t) &= e^{i\lambda(c+bt)}(\Pi_\lambda(g')f)(t+a)\\ &= e^{i\lambda(c+bt)} e^{i\lambda(c'+b'(t+a))} f(t+a+a')\\ &= e^{i\lambda(c+c'+ab'+(b+b')t)} f(t+a+a').\end{aligned}$$

By the group law (18.1), this is $(\Pi_\lambda(gg')f)(t)$. Also $\Pi_\lambda(0,0,0) = I$.

Unitarity. Since $|e^{i\lambda(c+bt)}| = 1$, translation invariance gives $\|\Pi_\lambda(g)f\|^2 = \int |f(t+a)|^2 dt = \|f\|^2$. Each $\Pi_\lambda(g)$ is an isometry with inverse $\Pi_\lambda(g^{-1})$, hence unitary.

Centre and Weyl relations. The first claim is the defining formula (18.4) with $a = b = 0$. For the second,

$$(U(a)V_\lambda(b)f)(t) = (V_\lambda(b)f)(t+a) = e^{i\lambda b(t+a)} f(t+a) = e^{i\lambda ab}(V_\lambda(b)U(a)f)(t).$$

Strong continuity. This is the only analytic step. First let $f \in C_c(\mathbb{R})$ vanish outside $[-R, R]$, and let $|a| \le 1$. Then $\Pi_\lambda(g)f - f$ vanishes outside $[-R-1, R+1]$. On that interval, $|e^{i\theta} - 1| \le |\theta|$ gives

$$|(\Pi_\lambda(g)f)(t) - f(t)| \le |\lambda|\big(|c| + (R+1)|b|\big)\|f\|_\infty + \omega_f(|a|),$$

where $\omega_f(\delta) = \sup\{|f(t+s) - f(t)| : t \in \mathbb{R},\ |s| \le \delta\}$. Uniform continuity of f gives $\omega_f(\delta) \to 0$ as $\delta \to 0$. Integrating the square over an interval of length $2R + 2$,

$$\|\Pi_\lambda(g)f - f\| \le (2R+2)^{1/2}\Big(|\lambda|\big(|c| + (R+1)|b|\big)\|f\|_\infty + \omega_f(|a|)\Big) \longrightarrow 0 \quad (g \to e).$$

Now let $f \in L^2(\mathbb{R})$ and $\varepsilon > 0$. Choose $h \in C_c(\mathbb{R})$ with $\|f - h\| < \varepsilon$. Since $\Pi_\lambda(g)$ is unitary,

$$\|\Pi_\lambda(g)f - f\| \le \|\Pi_\lambda(g)(f-h)\| + \|\Pi_\lambda(g)h - h\| + \|h - f\| < 2\varepsilon + \|\Pi_\lambda(g)h - h\|.$$

So $\limsup_{g\to e} \|\Pi_\lambda(g)f - f\| \le 2\varepsilon$ for every ε. Finally, $\|\Pi_\lambda(g)f - \Pi_\lambda(g_0)f\| = \|\Pi_\lambda(g_0^{-1}g)f - f\|$, and $g_0^{-1}g \to e$ as $g \to g_0$. ∎

Combining the commutation formula (18.2) with the scalar action of the centre in Proposition 18.3 gives

$$\Pi_\lambda(g)\Pi_\lambda(g') = e^{i\lambda(ab'-a'b)}\,\Pi_\lambda(g')\Pi_\lambda(g).$$

For $\lambda \neq 0$ the operators need not commute. For $\lambda = 0$ they commute.

Remark 18.4. Let $A = \{(0,b,c)\}$, a closed abelian normal subgroup, with character $\chi_\lambda(0,b,c) = e^{i\lambda c}$. Exercise 18.5 shows that the representation unitarily induced from χ_λ, as in Lecture 17, is equivalent to Π_λ.

Next we differentiate. Write $\mathcal{D} = C_c^\infty(\mathbb{R})$ and tf for the function $t \mapsto tf(t)$.

Proposition 18.5. *Each $\Pi_\lambda(g)$ maps $\mathcal{D}$ into $\mathcal{D}$. For $f \in \mathcal{D}$ and $W \in \{X, Y, Z\}$, the limit*

$$d\Pi_\lambda(W)f = \lim_{s\to 0} s^{-1}(\Pi_\lambda(\exp sW)f - f)$$

exists in $L^2(\mathbb{R})$, and

$$d\Pi_\lambda(X)f = f', \qquad d\Pi_\lambda(Y)f = i\lambda t f, \qquad d\Pi_\lambda(Z)f = i\lambda f.$$

On $\mathcal{D}$ we have $[d\Pi_\lambda(X), d\Pi_\lambda(Y)] = d\Pi_\lambda(Z) = d\Pi_\lambda([X,Y])$.

Proof. Translation and multiplication by the smooth function $e^{i\lambda(c+bt)}$ preserve smoothness and compact support. Let f vanish outside $[-R,R]$ and let $0 < |s| \le 1$. For X, Taylor's theorem gives

$$\left|\frac{f(t+s)-f(t)}{s} - f'(t)\right| \le \tfrac{1}{2}|s|\,\|f''\|_\infty,$$

and the left side vanishes outside $[-R-1, R+1]$. Its L^2 norm is at most $(2R+2)^{1/2}\tfrac{1}{2}|s|\|f''\|_\infty \to 0$. For Y, the bound $|e^{i\theta} - 1 - i\theta| \le \theta^2/2$ gives

$$\left|\frac{e^{i\lambda st}-1}{s} - i\lambda t\right| |f(t)| \le \tfrac{1}{2}\lambda^2|s|R^2|f(t)|,$$

whose L^2 norm tends to 0. For Z, $(e^{i\lambda s}-1)/s \to i\lambda$. Finally,

$$[d\Pi_\lambda(X), d\Pi_\lambda(Y)]f = (i\lambda t f)' - i\lambda t f' = i\lambda f = d\Pi_\lambda(Z)f. \qquad \blacksquare$$

Remark 18.6. Since Π_λ is strongly continuous (Proposition 18.3), $s \mapsto \Pi_\lambda(\exp sW)$ is a strongly continuous one-parameter unitary group. Its generator B_W, as defined in Lecture 15, has domain exactly the vectors f at which $s \mapsto \Pi_\lambda(\exp sW)f$ is differentiable at $s = 0$ in $L^2(\mathbb{R})$. Stone's theorem (Lecture 15) shows that B_W is skew-adjoint. So the physical generator $A_W = iB_W$ is self-adjoint and satisfies $\Pi_\lambda(\exp sW) = \exp(-isA_W)$. The derivative formulas of Proposition 18.5 show that $\mathcal{D}$ lies in the domain of A_W and that on $\mathcal{D}$

$$A_X f = if', \qquad A_Y f = -\lambda t f, \qquad A_Z f = -\lambda f.$$

In quantum mechanics a particle on a line has state space $L^2(\mathbb{R})$. Take $\lambda = 1/\hbar$, where $\hbar > 0$ is the reduced Planck constant. Put $P = -\hbar A_X$ (momentum) and $Q = -\hbar A_Y$ (position). On $\mathcal{D}$, $Qf = tf$, $Pf = -i\hbar f'$ and $[Q,P] = i\hbar$.

In terms of these operators, $U(a) = \exp(iaP/\hbar)$ and $V_\lambda(b) = \exp(ibQ/\hbar)$. So the Weyl relations (18.5) are the bounded, exponentiated form of the commutation relation.

We never use the commutation relation as a hypothesis. We state without proof that a commutation relation of unbounded operators on a dense domain does not by itself imply the Weyl relations. Exercise 18.3 shows that no bounded operators on a nonzero Hilbert space satisfy $[Q,P] = i\hbar I$.

18.3 The Stone–von Neumann theorem

Definition 18.7. A unitary representation ρ of H on a Hilbert space $\mathcal{H}$ has *central character* $\lambda \in \mathbb{R}$ if $\rho(0,0,c) = e^{i\lambda c} I$ for all $c \in \mathbb{R}$.

By the centre statement in Proposition 18.3, the representation Π_λ has central character λ. Now let ρ be an irreducible unitary representation of H. Schur's lemma for unitary representations (Kirillov, *Lectures on the Orbit Method*, 2004, Appendix V) makes each central operator $\rho(0,0,c)$ a scalar. These scalars form a continuous homomorphism $\mathbb{R} \to \mathrm{U}(1)$, hence have the form $e^{i\lambda c}$. So every irreducible unitary representation has a central character. Reducible ones need not, for example $\Pi_1 \oplus \Pi_2$.

Lemma 18.8. *Let σ and τ be strongly continuous one-parameter unitary groups on a Hilbert space $\mathcal{H}$, and let $\lambda \in \mathbb{R}$. Suppose $\sigma(a)\tau(b) = e^{i\lambda ab}\tau(b)\sigma(a)$ for all $a, b \in \mathbb{R}$. Then $\rho(a,b,c) = e^{i\lambda c}\tau(b)\sigma(a)$ defines a strongly continuous unitary representation ρ of H on $\mathcal{H}$ with central character λ.*

Proof. Move $\sigma(a)$ past $\tau(b')$ using the hypothesis:

$$\rho(g)\rho(g') = e^{i\lambda(c+c')}\tau(b)\sigma(a)\tau(b')\sigma(a') = e^{i\lambda(c+c'+ab')}\tau(b+b')\sigma(a+a') = \rho(gg').$$

Each $\rho(g)$ is unitary, and $\rho(0,0,c) = e^{i\lambda c}I$. For continuity at (a_0,b_0,c_0), let $f \in \mathcal{H}$ and put $h = \sigma(a_0)f$. Then

$$\|\tau(b)\sigma(a)f - \tau(b_0)\sigma(a_0)f\| \le \|\sigma(a)f - \sigma(a_0)f\| + \|\tau(b)h - \tau(b_0)h\| \to 0,$$

and the scalar factor $e^{i\lambda c}$ is continuous. ∎

We call a pair (σ,τ) satisfying the hypotheses of Lemma 18.8 a *Weyl pair* with parameter λ. By the Weyl relations (18.5) and the strong continuity in Proposition 18.3, the translations and modulations form a Weyl pair (U,V_λ) with parameter λ. Two Weyl pairs are *equivalent* when a single unitary operator from the first space to the second conjugates each group of the first pair into the corresponding group of the second.

Theorem 18.9 (Stone–von Neumann). *Fix $\lambda \neq 0$.*

(a) The Schrödinger representation Π_λ is irreducible.

(b) Let ρ be a strongly continuous unitary representation of H on a nonzero separable Hilbert space $\mathcal{H}$ with central character λ. Then $\mathcal{H}$ is a Hilbert direct sum of mutually orthogonal closed ρ-invariant subspaces. On each summand ρ is unitarily equivalent to Π_λ.

We cite the Stone–von Neumann theorem, Theorem 18.9, from Kirillov, *Lectures on the Orbit Method*, Chapter 2.

The Hilbert space of an irreducible representation of H is automatically separable. By irreducibility, every nonzero vector is cyclic. By strong continuity, the closed span of the orbit of such a vector is already generated by its orbit under a countable dense subset of H. This closed span is the whole space, and it is separable.

The Stone–von Neumann theorem, Theorem 18.9, has the following consequences.

- Every irreducible unitary representation of H with central character $\lambda \neq 0$ is unitarily equivalent to Π_λ.
- By the lemma on Weyl pairs, Lemma 18.8, every Weyl pair with parameter $\lambda \neq 0$ on a nonzero separable space is equivalent to a direct sum of copies of (U, V_λ). In particular, every Weyl pair with parameter $1/\hbar$ on a nonzero separable space is unitarily equivalent to a direct sum of copies of $(\exp(iaP/\hbar), \exp(ibQ/\hbar))$.
- Distinct values of λ give inequivalent representations Π_λ (Exercise 18.6(c) on inequivalence).

A standard proof, going back to von Neumann, turns on one step. Let ρ be a representation as in the uniqueness statement (b) of Theorem 18.9. For $f, h \in \mathcal{H}$, consider the sesquilinear form

$$B(f,h) = \frac{|\lambda|}{2\pi} \int_{\mathbb{R}^2} e^{-(a^2+\lambda^2 b^2)/4} \left\langle f, \rho(\exp(aX+bY))h \right\rangle \, da\, db.$$

The Gaussian weight has integral $4\pi/|\lambda|$, so $|B(f,h)| \leq 2\|f\|\|h\|$. Hence the Riesz representation theorem for bounded sesquilinear forms gives a bounded operator E on $\mathcal{H}$ with $\|E\| \leq 2$ and $B(f,h) = \langle f, Eh\rangle$ for all $f, h \in \mathcal{H}$.

That proof shows, from the Weyl relations and strong continuity, that E is a nonzero orthogonal projection. Each unit vector $v \in E\mathcal{H}$ then generates one copy of Π_λ. In that copy, v corresponds to the Gaussian of the next section, § 18.4. The weight is that Gaussian's matrix coefficient, given by (18.6) at $g = \exp(aX + bY)$.

18.4 Gaussian matrix coefficients and central character zero

Let $\varphi(t) = \pi^{-1/4} e^{-t^2/2}$. Since $\int e^{-t^2} dt = \sqrt{\pi}$, $\|\varphi\| = 1$. We compute $\langle \varphi, \Pi_\lambda(g)\varphi\rangle$ for $g = (a,b,c)$. We use the calculus fact

$$\int_{\mathbb{R}} e^{-(t+w)^2} dt = \sqrt{\pi} \qquad (w \in \mathbb{C}).$$

For real w this is a translation. Write $w = u + iv$. The v-derivative of the integrand is $i\,\frac{d}{dt} e^{-(t+w)^2}$, which integrates to 0. Differentiation under the integral is justified on bounded v-intervals by an integrable Gaussian bound, including the derivative. So the integral does not depend on v.

By the defining formula (18.4), and since φ is real,

$$\langle \varphi, \Pi_\lambda(g)\varphi\rangle = \pi^{-1/2} e^{i\lambda c} \int_{\mathbb{R}} e^{-t^2/2 - (t+a)^2/2 + i\lambda b t}\, dt.$$

Complete the square with $w = \frac{1}{2}(a - i\lambda b)$:

$$-\tfrac{t^2}{2} - \tfrac{(t+a)^2}{2} + i\lambda b t = -t^2 - (a - i\lambda b)t - \tfrac{a^2}{2} = -(t+w)^2 + w^2 - \tfrac{a^2}{2},$$

$$w^2 - \tfrac{a^2}{2} = \tfrac{1}{4}\big(a^2 - 2i\lambda ab - \lambda^2 b^2\big) - \tfrac{a^2}{2} = -\tfrac{1}{4}\big(a^2 + \lambda^2 b^2\big) - \tfrac{i\lambda ab}{2}.$$

The integral is $\sqrt{\pi}$ times $e^{w^2 - a^2/2}$. Therefore

$$\langle \varphi, \Pi_\lambda(a,b,c)\varphi\rangle = e^{i\lambda(c - ab/2)}\, e^{-(a^2+\lambda^2 b^2)/4}. \tag{18.6}$$

We make two checks. First, at $g = e$ the value is $1 = \|\varphi\|^2$. Second, replacing g by $g^{-1} = (-a, -b, -c + ab)$ conjugates the value, as unitarity requires.

By the exponential formula (18.3), $c - ab/2$ is the Z-coordinate of $\log g$, where log is the inverse of the exponential map. Hence at $\exp(xX + yY + zZ)$ the coefficient is $e^{i\lambda z} e^{-(x^2+\lambda^2 y^2)/4}$.

The modulus depends only on the image (a, b) in $H/Z(H) \cong \mathbb{R}^2$. For $\lambda \neq 0$ it decays in every direction, and

$$\int_{\mathbb{R}^2} \big|\langle \varphi, \Pi_\lambda(a,b,c)\varphi\rangle\big|^2 da\, db = \int e^{-a^2/2} da \int e^{-\lambda^2 b^2/2} db = \sqrt{2\pi} \cdot \frac{\sqrt{2\pi}}{|\lambda|} = \frac{2\pi}{|\lambda|}.$$

Its reciprocal $|\lambda|/(2\pi)$ is the normalising constant in the form B above.

The operator E defined after the Stone–von Neumann theorem can be computed explicitly in this model. This computation verifies its normalisation. It is not a proof of the general theorem. Write E_λ for that operator with $\rho = \Pi_\lambda$. Since

$$\Pi_\lambda(\exp(aX + bY))f(t) = e^{i\lambda(bt + ab/2)} f(t + a),$$

we integrate first over b. For $f, h \in C_c(\mathbb{R})$, Cauchy–Schwarz in t and translation invariance give

$$\int_{\mathbb{R}^2} e^{-(a^2+\lambda^2 b^2)/4} \int_{\mathbb{R}} |h(t)|\, |f(t+a)|\, dt\, da\, db \le \frac{4\pi}{|\lambda|} \|f\| \|h\|.$$

Thus the Fubini–Tonelli theorem identifies the following kernel calculation with the form B defining E_λ. The Gaussian Fourier integral, obtained by the same square completion as above, gives

$$\int_{\mathbb{R}} e^{-\lambda^2 b^2/4} e^{i\lambda b(t + a/2)}\, db = \frac{2\sqrt{\pi}}{|\lambda|} e^{-(t+a/2)^2}.$$

Consequently, for $f \in C_c(\mathbb{R})$,

$$(E_\lambda f)(t) = \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} e^{-a^2/4 - (t+a/2)^2} f(t + a)\, da.$$

Set $u = t + a$. The exponent simplifies to

$$-\frac{(u - t)^2}{4} - \frac{(t + u)^2}{4} = -\frac{t^2 + u^2}{2}.$$

We therefore obtain

$$(E_\lambda f)(t) = \varphi(t) \int_{\mathbb{R}} \overline{\varphi(u)} f(u)\, du = \varphi(t) \langle \varphi, f\rangle.$$

Both operators are bounded, so density extends the identity to all $L^2(\mathbb{R})$. Thus E_λ is exactly the orthogonal projection onto the Gaussian line. For a general representation ρ with central

character $\lambda \neq 0$, proving that the operator E is a nonzero projection is the key step of the standard proof. Here its rank-one integral kernel is visible.

Now set $\lambda = 0$. The defining formula (18.4) becomes pure translation, $(\Pi_0(a,b,c)f)(t) = f(t+a)$. The modulations become trivial, and the Gaussian coefficient (18.6) becomes $e^{-a^2/4}$. This does not decay in b, and the integral above diverges.

The irreducible unitary representations of H with central character 0 factor through $H/Z(H) \cong \mathbb{R}^2$. By Schur's lemma (Kirillov, Appendix V), irreducible unitary representations of an abelian group are one-dimensional. Hence the irreducible unitary representations of H with central character 0 are the characters

$$\chi_{\alpha,\beta}(a,b,c) = e^{i(\alpha a+\beta b)}, \qquad \alpha,\beta \in \mathbb{R}.$$

Kirillov's Chapter 2 records this together with the Stone–von Neumann theorem, Theorem 18.9, as the full list of irreducibles. Lecture 19 explains the list through coadjoint orbits.

Proposition 18.10. *Π_0 has no one-dimensional invariant subspace. Hence it has no irreducible closed invariant subspace.*

Proof. Suppose $f \neq 0$ spans an invariant line. Then $U(a)f = \mu(a)f$ with $|\mu(a)| = 1$. For each integer n,

$$\int_n^{n+1} |f(t)|^2 dt = \int_0^1 |f(t+n)|^2 dt = \int_0^1 |\mu(n)f(t)|^2 dt = \int_0^1 |f(t)|^2 dt.$$

Summing over $n \in \mathbb{Z}$ gives $\|f\|^2 = \sum_n \int_0^1 |f|^2$. This is finite only if $\int_0^1 |f|^2 = 0$, and then $\|f\| = 0$, a contradiction. An irreducible closed invariant subspace would carry an irreducible representation with central character 0. By the classification above it would be a line. ∎

The following table compares central character $\lambda \neq 0$ with central character 0.

	central character $\lambda \neq 0$	central character 0
irreducibles	one class, Π_λ	$\chi_{\alpha,\beta}$, $\alpha,\beta \in \mathbb{R}$
dimension	infinite	one
the $L^2(\mathbb{R})$ model	Π_λ itself, irreducible	Π_0 has no irreducible piece
Gaussian coefficient	decays on $H/Z(H)$	constant in b

Self-checks

1. Compute gg' and $g'g$ for $g = (1,2,3)$ and $g' = (4,5,6)$, and check the commutation formula (18.2).
   Answer. $gg' = (5, 7, 3+6+5) = (5,7,14)$ and $g'g = (5,7,9+8) = (5,7,17)$. Here $ab' - a'b = 5 - 8 = -3$, and $(0,0,-3)(5,7,17) = (5,7,14)$.
2. Why are Π_2 and Π_3 inequivalent?
   Answer. An intertwiner T satisfies $Te^{2ic} = e^{3ic}T$ for all c. At $c = \pi$ this reads $T = -T$, so $T = 0$.

3. Compute $\langle \varphi, \Pi_1(2,2,2)\varphi\rangle$ and $\langle \varphi, \Pi_0(0,b,0)\varphi\rangle$.
   Answer. By the Gaussian coefficient (18.6), $c - ab/2 = 0$ and $(a^2+b^2)/4 = 2$, so the first is e^{-2}. The second is 1 for every b.

Summary

The Heisenberg group has centre $Z(H) \cong \mathbb{R}$, and every commutator lies in it. Translations and modulations on $L^2(\mathbb{R})$ form the Schrödinger representation Π_λ, which satisfies the Weyl relations. The Stone–von Neumann theorem, cited, says that for $\lambda \neq 0$ every strongly continuous unitary representation on a nonzero separable space with central character λ is a Hilbert direct sum of copies of Π_λ, up to unitary equivalence. At $\lambda = 0$ the irreducibles are the characters $\chi_{\alpha,\beta}$, and the translation representation Π_0 contains none of them.

Reading

- Kirillov, *Lectures on the Orbit Method* (2004), Chapter 2: the Heisenberg group, its irreducible unitary representations and the Stone–von Neumann theorem.
- Kirillov, *Lectures on the Orbit Method*, Appendix V: background on unitary representations, including Schur's lemma and induced representations.
- Hall, *Lie Groups, Lie Algebras, and Representations*, second edition (2015), Chapter 1: the Heisenberg group among the examples of matrix Lie groups.

Lecture 19

Coadjoint orbits and the nilpotent orbit method

Objectives. *A representation acts on vectors. The coadjoint action of a Lie group instead acts on linear functionals on its Lie algebra. Each orbit of this action has a natural symplectic form. For connected simply connected nilpotent groups, these orbits label all irreducible strongly continuous unitary representations up to unitary equivalence. We prove the geometric construction and work through the Heisenberg example. The general classification is an explicit cited theorem.*

By the end of this lecture you will be able to:

- Compute a coadjoint action and its stabilisers.
- Construct the orbit symplectic form and check its signs.
- State the nilpotent orbit theorem with its global hypotheses.
- Choose a polarisation for a functional on a two-dimensional Heisenberg orbit and recover the Schrödinger representation by induction.

19.1 The dual action

Throughout, G is a finite-dimensional real Lie group with Lie algebra $\mathfrak{g}$. Its real dual is $\mathfrak{g}^* = \mathrm{Hom}_{\mathbb{R}}(\mathfrak{g}, \mathbb{R})$. We pair a functional ℓ and a vector X by $\ell(X)$. This dual is different from the complex dual of a representation space.

Definition 19.1. The *coadjoint action* of G on $\mathfrak{g}^*$ is

$$(g \cdot \ell)(X) = \ell(\mathrm{Ad}(g^{-1})X).$$

The *coadjoint orbit* through ℓ is $O_\ell = \{g \cdot \ell : g \in G\}$. Its stabiliser is $G_\ell = \{g : g \cdot \ell = \ell\}$.

The inverse is needed for the action law. Indeed,

$$(g \cdot (h \cdot \ell))(X) = \ell(\mathrm{Ad}(h^{-1})\,\mathrm{Ad}(g^{-1})X) = \ell(\mathrm{Ad}((gh)^{-1})X) = ((gh) \cdot \ell)(X).$$

The stabiliser is closed by continuity. It is therefore an embedded Lie subgroup by the closed-subgroup theorem of Lecture 3. Differentiating the action gives

$$(\mathrm{ad}^*(X)\ell)(Y) = -\ell([X,Y]), \qquad \mathfrak{g}_\ell = \{X : \ell([X,Y]) = 0 \text{ for every } Y \in \mathfrak{g}\}. \tag{19.1}$$

The identity $\exp(tX)\cdot\ell = e^{t\,\mathrm{ad}^*(X)}\ell$ shows that $\exp(tX)\in G_\ell$ for every t exactly when $\mathrm{ad}^*(X)\ell = 0$. Thus $\mathrm{Lie}(G_\ell) = \mathfrak{g}_\ell$. For example, a central X lies in every $\mathfrak{g}_\ell$. Central directions therefore do not generate motion on an orbit.

We give $\mathcal{O}_\ell$ the smooth structure of G/G_ℓ. This uses the homogeneous-space theorem from Lecture 3. The natural map

$$G/G_\ell \longrightarrow \mathfrak{g}^*, \qquad gG_\ell \longmapsto g\cdot\ell$$

is injective and an immersion. At the identity coset, its derivative is induced by $X\mapsto \mathrm{ad}^*(X)\ell$, whose kernel is precisely $\mathfrak{g}_\ell$. At other points, equivariance gives the same rank. The orbit dimension is consequently $\dim\mathfrak{g} - \dim\mathfrak{g}_\ell$.

For a general Lie group, an orbit need not be an embedded submanifold of $\mathfrak{g}^*$. All tangent-space statements below use the quotient smooth structure. The Heisenberg orbits will be ordinary affine planes and points, so this distinction disappears there.

Write

$$\xi_X(\eta) = \left.\frac{d}{dt}\right|_0 \exp(tX)\cdot\eta = \mathrm{ad}^*(X)\eta, \qquad \eta\in\mathcal{O}_\ell.$$

The vectors $\xi_X(\eta)$ span the tangent space of $\mathcal{O}_\ell$ at η. Two choices of X give the same vector exactly when their difference belongs to $\mathfrak{g}_\eta$.

19.2 A symplectic form on each orbit

We need a small amount of differential-form language. A smooth 2-form ω assigns an alternating bilinear form ω_x to every tangent space, smoothly in local coordinates. On $\mathbb{R}^2$, for example,

$$(dp\wedge dq)((u_1,u_2),(v_1,v_2)) = u_1v_2 - u_2v_1.$$

A form ω is *nondegenerate* if, at every point x, the only tangent vector u at x with $\omega_x(u,v)=0$ for every tangent vector v at x is $u=0$.

In coordinates $x_1,\dots,x_d$, put $\omega_{ij} = \omega(\partial_i,\partial_j) = -\omega_{ji}$. Then $\omega = \frac{1}{2}\sum_{i,j}\omega_{ij}\,dx_i\wedge dx_j$. The form ω is *closed* when

$$\partial_i\omega_{jk} + \partial_j\omega_{ki} + \partial_k\omega_{ij} = 0 \quad \text{for all } i,j,k.$$

Equivalently, closedness means $d\omega = 0$. A closed nondegenerate 2-form is called a *symplectic form*. These are coordinate-independent conditions. We import this elementary differential-form calculus, including pullbacks and the exterior derivative d, from Warner, Chapter 2. We will state the identities used in the proof.

Theorem 19.2 (Orbit symplectic form)**.** *Every coadjoint orbit has a G-invariant symplectic form ω defined by*

$$\omega_\eta(\xi_X(\eta),\xi_Y(\eta)) = \eta([X,Y]). \tag{19.2}$$

This is the Kirillov–Kostant–Souriau form, with the sign convention of the infinitesimal action (19.1).

Proof. Well-definedness. If X changes by $A \in \mathfrak{g}_\eta$, then $\eta([A,Y]) = 0$. Changing Y has the same effect by antisymmetry. Thus the defining formula (19.2) depends only on the two tangent vectors. It is alternating and bilinear.

Nondegeneracy. If $\omega_\eta(\xi_X, \xi_Y) = 0$ for every Y, then $X \in \mathfrak{g}_\eta$ by the stabiliser formula (19.1). Hence $\xi_X(\eta) = 0$.

Smoothness. The fields ξ_X are smooth and span every tangent space. Choose a local frame from these fields. The defining formula (19.2) gives smooth coefficients in that frame, so ω is smooth.

Invariance. For $g \in G$, let g_* be the differential of the map $\eta \mapsto g \cdot \eta$ on the orbit. Since $g \exp(tX) = \exp(t \operatorname{Ad}(g)X)g$, differentiating $g \cdot (\exp(tX) \cdot \eta)$ at $t = 0$ gives

$$g_* \xi_X(\eta) = \xi_{\operatorname{Ad}(g)X}(g \cdot \eta).$$

Using that $\operatorname{Ad}(g)$ preserves brackets, we get

$$\begin{aligned} \omega_{g\cdot\eta}(g_*\xi_X, g_*\xi_Y) &= (g \cdot \eta)([\operatorname{Ad}(g)X, \operatorname{Ad}(g)Y]) \\ &= \eta([X,Y]). \end{aligned}$$

So the form is invariant.

Closedness. Let $Q : G \to \mathcal{O}_\ell$ be the orbit map, given by $Q(g) = g \cdot \ell$. Put $\beta = Q^*\omega$. Invariance makes β a left-invariant form on G, and $\beta_e(X,Y) = \ell([X,Y])$. Exterior differentiation commutes with pullback. For a left-invariant 2-form, the differential at e is

$$(d\beta)_e(X,Y,Z) = -\beta_e([X,Y],Z) + \beta_e([X,Z],Y) - \beta_e([Y,Z],X).$$

We have now stated the two identities imported from Warner, Chapter 2. They are the commutation of d with pullback and the preceding formula for a left-invariant 2-form. Substitution gives

$$(d\beta)_e(X,Y,Z) = -\ell([[X,Y],Z]) + \ell([[X,Z],Y]) - \ell([[Y,Z],X]) = 0$$

by the Jacobi identity. Left invariance gives $d\beta = 0$ everywhere. Therefore $Q^*(d\omega) = 0$. Since Q is a submersion, every tuple of tangent vectors on the orbit lifts to G. This forces $d\omega = 0$. ∎

A nondegenerate alternating form exists only in even dimension. To see this, let A be the matrix of such a form in a basis. Then $A^T = -A$. If the size d of A were odd, then transposition invariance of the determinant would give $\det A = \det(-A) = -\det A$. Hence $\det A = 0$, and the form would be degenerate. In particular, every coadjoint orbit has even dimension. This is a useful first check on a proposed calculation.

Reading the form through linear functions

There is a useful way to remember the sign in the defining formula (19.2). For $X \in \mathfrak{g}$, restrict the linear function $\mu_X(\eta) = \eta(X)$ to an orbit. Its derivative in a tangent direction is

$$(d\mu_X)_\eta(\xi_Y) = (\operatorname{ad}^*(Y)\eta)(X) = -\eta([Y,X]) = \eta([X,Y]).$$

Thus $d\mu_X = \iota_{\xi_X}\omega$, where contraction means $(\iota_V\omega)(W) = \omega(V, W)$. This identifies the infinitesimal action using only the orbit form and the linear functions.

For a smooth real function f, nondegeneracy determines a unique tangent vector field X_f by $\iota_{X_f}\omega = df$. It is called the Hamiltonian vector field in this convention. The preceding calculation says $X_{\mu_X} = \xi_X$. Consequently the bracket

$$\{f, h\} = \omega(X_f, X_h)$$

satisfies $\{\mu_X, \mu_Y\} = \mu_{[X,Y]}$. We only need this identity for linear functions. It says that the Lie bracket is still visible after passing to an orbit.

A central element $W \in \mathfrak{g}$ has $\xi_W = 0$. Thus $d\mu_W = \iota_{\xi_W}\omega = 0$, so μ_W is locally constant on each orbit. It is constant when G is connected. Such a constant can still be the value of a bracket of linear functions. For the Heisenberg basis X, Y and Z of the next section, the identity above gives $\{\mu_X, \mu_Y\} = \mu_Z$, and μ_Z takes one constant value on each orbit.

A change of convention for Hamiltonian vector fields would change some of these signs. Here the definitions, rather than a remembered physics formula, determine them. This calculation concerns finite-dimensional smooth manifolds. It does not differentiate an unbounded operator or assert a quantisation rule.

19.3 Heisenberg planes and points

Let H be the Heisenberg group of Lecture 18, with Lie algebra $\mathfrak{h}$ spanned by X, Y and Z. We use exactly its matrix coordinates:

$$(a, b, c)(a', b', c') = (a + a', b + b', c + c' + ab'), \qquad [X, Y] = Z.$$

In particular, c is a matrix entry. It is not the central coefficient in the logarithm, which is $c - ab/2$.

Write $\ell = (p, q, \lambda)$ when $\ell(X) = p$, $\ell(Y) = q$ and $\ell(Z) = \lambda$. The adjoint formulas of Lecture 18 are

$$\mathrm{Ad}(a, b, c)X = X - bZ, \qquad \mathrm{Ad}(a, b, c)Y = Y + aZ, \qquad \mathrm{Ad}(a, b, c)Z = Z.$$

The inverse of $g = (a, b, c)$ has first coordinates $(-a, -b)$. Hence $\mathrm{Ad}(g^{-1})X = X + bZ$ and $\mathrm{Ad}(g^{-1})Y = Y - aZ$. Applying ℓ to these vectors gives

$$(a, b, c) \cdot (p, q, \lambda) = (p + \lambda b, q - \lambda a, \lambda). \tag{19.3}$$

For $\lambda \neq 0$, this reaches every point of the plane with fixed last coordinate. Its stabiliser is the centre. For $\lambda = 0$, every point is fixed. Thus the orbit list consists of planes for nonzero λ and individual points $(p, q, 0)$.

On a plane, p and q are coordinates. Differentiating the action formula (19.3) gives

$$\xi_X = -\lambda\partial_q, \qquad \xi_Y = \lambda\partial_p, \qquad \xi_Z = 0.$$

The defining value of the form is $\omega(\xi_X, \xi_Y) = \ell(Z) = \lambda$. Since $(dp \wedge dq)(\xi_X, \xi_Y) = \lambda^2$, we obtain

$$\omega_\lambda = \frac{1}{\lambda}\, dp \wedge dq. \tag{19.4}$$

The coefficient is $1/\lambda$, not λ, because the vectors ξ_X and ξ_Y already carry a factor λ.

Point orbits have zero-dimensional tangent spaces. The zero form on such a tangent space is nondegenerate, since its kernel is the zero space. The orbit method will associate $(p, q, 0)$ with the one-dimensional character

$$\chi_{p,q}(a, b, c) = e^{i(pa+qb)}.$$

These are different representations for different (p, q). The trivial central character, with $\lambda = 0$, does not mean that the representation is trivial. Only $(0, 0, 0)$ gives the trivial character.

19.4 Polarisations and the orbit theorem

A Lie algebra is *nilpotent* if its lower central series terminates:

$$\mathfrak{g}_1 = \mathfrak{g}, \qquad \mathfrak{g}_{r+1} = [\mathfrak{g}, \mathfrak{g}_r], \qquad \mathfrak{g}_s = 0 \text{ for some } s.$$

For the Heisenberg algebra, $\mathfrak{g}_2 = \mathbb{R}Z$ and $\mathfrak{g}_3 = 0$. Nilpotence is stronger than solvability.

We import a global fact: for a connected simply connected nilpotent group, $\exp : \mathfrak{g} \to G$ is a diffeomorphism, and BCH is a finite polynomial. Every Lie subalgebra $\mathfrak{m}$ gives a closed subgroup $M = \exp \mathfrak{m}$. Indeed, finite BCH keeps products in $\mathfrak{m}$, and the exponential diffeomorphism takes the closed vector subspace $\mathfrak{m}$ to a closed subset. See Knapp, *Lie Groups Beyond an Introduction*, Section I.16.

Definition 19.3. A *real polarisation* for $\ell \in \mathfrak{g}^*$ is a Lie subalgebra $\mathfrak{m} \subset \mathfrak{g}$ such that

$$\ell([\mathfrak{m}, \mathfrak{m}]) = 0, \qquad \dim \mathfrak{m} = \frac{\dim \mathfrak{g} + \dim \mathfrak{g}_\ell}{2}.$$

A subspace $A \subseteq \mathfrak{g}$ is called *isotropic* for ℓ when

$$\ell([A, A]) = 0.$$

A real polarisation is thus an isotropic Lie subalgebra of the displayed dimension. This dimension says that the image of $\mathfrak{m}$ in $\mathfrak{g}/\mathfrak{g}_\ell$ is as large as the image of an isotropic subspace can be. In fact, the two displayed conditions force $\mathfrak{g}_\ell \subset \mathfrak{m}$, because an isotropic subspace of maximal dimension contains $\mathfrak{g}_\ell$. The subsection on the dimension of a polarisation below proves both statements.

For such a subalgebra, put

$$\chi_\ell(\exp W) = e^{i\ell(W)}, \qquad W \in \mathfrak{m}.$$

This is well-defined because exp is bijective. It is a character of M. Every BCH correction is a bracket of elements of $\mathfrak{m}$, so ℓ kills it and $\chi_\ell(\exp U \exp V) = \chi_\ell(\exp U)\chi_\ell(\exp V)$.

Theorem 19.4 (Nilpotent orbit theorem; cited)**.** *Let G be connected, simply connected and nilpotent. Every $\ell \in \mathfrak{g}^*$ has a real polarisation. The unitary induced representation*

$$\Pi_\ell = \operatorname{Ind}_{\exp \mathfrak{m}}^{G} \chi_\ell$$

is irreducible. Its equivalence class does not depend on the polarisation. Two such represen-

> *tations are equivalent exactly when their functionals lie on the same coadjoint orbit. Every irreducible strongly continuous unitary representation of G is unitarily equivalent to some Π_ℓ.*

This is Kirillov's theorem, imported from *Lectures on the Orbit Method*, Chapter 3. Induction means unitary induction with the quotient measure convention of Lecture 17. The statement concerns complex Hilbert representations, not just finite-dimensional modules.

The global assumptions matter. A central quotient can restrict which representations descend. A solvable group need not be nilpotent. Apart from the trivial group, no compact group and no semisimple group satisfies this theorem's hypotheses. We will meet related orbit constructions in Lecture 27, with a different integrality condition and a different representation model.

The dimension of a polarisation

We now isolate the linear algebra behind a polarisation. Define the alternating form

$$B_\ell(X, Y) = \ell([X, Y])$$

on $\mathfrak{g}$. Its kernel is $\mathfrak{g}_\ell$. Suppose $\dim O_\ell = 2d$ and $\dim \mathfrak{g}_\ell = k$. Then $\dim \mathfrak{g} = 2d + k$. The form B_ℓ induces a nondegenerate alternating form on the quotient $\mathfrak{g}/\mathfrak{g}_\ell$, which has dimension $2d$. A vector space with a nondegenerate alternating form is called a *symplectic vector space*.

Let L be a subspace of this quotient. Its *orthogonal* $L^\perp$ consists of the vectors whose induced pairing with every vector of L is zero. The subspace L is *isotropic* when the induced form vanishes on pairs of vectors of L, that is, when $L \subseteq L^\perp$. Nondegeneracy gives $\dim L^\perp = 2d - \dim L$, so an isotropic L has $\dim L \leq d$. Let $A \subset \mathfrak{g}$ be an isotropic subspace for ℓ. Its image in the quotient is isotropic, so this image has dimension at most d. Therefore

$$\dim A = \dim(A \cap \mathfrak{g}_\ell) + \dim\big((A + \mathfrak{g}_\ell)/\mathfrak{g}_\ell\big) \leq k + d.$$

Equality forces $\mathfrak{g}_\ell \subset A$. It also forces the image of A to be a d-dimensional isotropic subspace. The dimension in the definition of a polarisation is exactly $k + d$.

There is one extra demand: A must be a Lie subalgebra. A maximal isotropic vector subspace need not have this property. The existence of a suitable subalgebra is a substantial part of the nilpotent orbit theorem. The definition alone does not prove existence.

This count also predicts the number of variables in an induced model. If $\mathfrak{m}$ is a polarisation, then

$$\dim(G/M) = \dim \mathfrak{g} - \dim \mathfrak{m} = d = \tfrac{1}{2} \dim O_\ell.$$

The Heisenberg plane is two-dimensional, but its representation acts on functions of one real variable. The number of these variables is half the orbit dimension. The orbit does not single out which variables to use. A polarisation makes that choice.

19.5 Induction recovers the Schrödinger formula

Take $\ell = (0, 0, \lambda)$ with $\lambda \neq 0$. Its stabiliser algebra is $\mathbb{R}Z$, so a polarisation must have dimension two. Choose

$$\mathfrak{m} = \operatorname{span}\{Y, Z\}, \qquad M = \{(0, b, c)\}, \qquad \chi_\ell(0, b, c) = e^{i\lambda c}.$$

The subalgebra $\mathfrak{m}$ is abelian and two-dimensional, so it satisfies both polarisation conditions. A section of $H/M \cong \mathbb{R}$ is $s(t) = (t, 0, 0)$. The quotient measure is dt, invariant under the induced translation action.

An induced function satisfies $f(gm) = \chi_\ell(m)^{-1} f(g)$. Write $\varphi(t) = f(s(t))$. Since

$$(a, b, c) = s(a)(0, b, c - ab),$$

its values are

$$f(a, b, c) = e^{-i\lambda(c-ab)}\varphi(a).$$

Left translation acts by $(\Psi_\lambda(g)f)(h) = f(g^{-1}h)$. For $g = (a, b, c)$,

$$g^{-1}s(t) = (t - a, -b, -c + ab).$$

Insert this into the preceding covariance formula. Its central expression becomes $-c + ab - (t - a)(-b) = -c + bt$. Therefore

$$(\Psi_\lambda(a, b, c)\varphi)(t) = e^{i\lambda(c-bt)}\varphi(t - a). \tag{19.5}$$

The operator is unitary: the phase has absolute value one and translation preserves dt. The general unitary induction construction of Lecture 17 supplies the representation law and strong continuity. Both can also be checked from this formula.

The sign of the translation differs from the Schrödinger formula of Lecture 18 because our section uses the left action on H/M. It gives an equivalent model. Let $R\varphi(x) = \varphi(-x)$. Then $R^{-1} = R$, and

$$(R\Psi_\lambda(a, b, c)R^{-1}u)(x) = e^{i\lambda(c+bx)}u(x + a) = \Pi_\lambda(a, b, c)u(x).$$

Thus the orbit theorem reproduces the Schrödinger representation with the precise convention already used. The plane's parameter λ determines its nonzero central character.

For a point $(p, q, 0)$, the polarisation is all of $\mathfrak{h}$. Induction from H to itself gives $\chi_{p,q}$. The plane and point calculations therefore recover the complete Heisenberg classification from Lecture 18. The general orbit theorem extends that pattern to every connected simply connected nilpotent group.

What the Heisenberg orbit list explains

The orbit calculation separates two phenomena. For $\lambda \neq 0$, changing p or q stays within one plane. It cannot change the equivalence class of the representation. For $\lambda = 0$, each point is an orbit of its own, so p and q both survive as representation labels. The discontinuous change in orbit dimension matches the change from infinite-dimensional representations to characters.

One can check the zero-charge conclusion without the general orbit theorem. In an irreducible unitary representation, every central group element acts by a scalar, by the unitary Schur lemma used in Lecture 18. If that scalar character is trivial, the representation factors through $H/Z(H) \cong \mathbb{R}^2$. An irreducible unitary representation of this abelian group is one-dimensional. Its continuous characters are $e^{i(pa+qb)}$, obtained by applying the one-parameter character classification to each coordinate.

For nonzero central character, the Stone–von Neumann theorem gives uniqueness directly. The orbit theorem recovers that result and explains its geometric label. It also shows what to try for a larger nilpotent group: calculate ad^*, find orbit invariants, determine the stabiliser dimension, choose a real polarisation for ℓ, and induce its character. The last exercise carries out the geometric steps in one four-dimensional example.

Self-checks

1. Why must the stabiliser algebra contain the centre?

Answer. If X is central, then $[X, Y] = 0$ for every Y. The stabiliser formula (19.1) puts X in $\mathfrak{g}_\ell$ for every ℓ.

2. Does the formula $\omega_\lambda = \lambda\, dp \wedge dq$ have the right normalisation?

Answer. No. Its value on $(-\lambda\partial_q, \lambda\partial_p)$ would be λ^3. The required value is λ, so the coefficient is $1/\lambda$.

3. Can one apply the orbit theorem directly to a Heisenberg group with circle centre?

Answer. No. That group is not simply connected. Apply the theorem to its simply connected cover. Then use the Descent theorem of Lecture 4. Only representations trivial on the covering kernel descend.

Summary

The coadjoint action turns $\mathfrak{g}^*$ into a union of even-dimensional symplectic orbits. For connected simply connected nilpotent groups, a polarisation converts an orbit into an irreducible induced representation. For the Heisenberg group, planes give Schrödinger representations and points give characters. The cited classification is exact in this setting. Its hypotheses are part of the result.

Reading

- Kirillov, *Lectures on the Orbit Method*, Chapters 2–3: Heisenberg representations, polarisations and the nilpotent correspondence.
- Knapp, *Lie Groups Beyond an Introduction*, Section I.16: simply connected nilpotent groups.

- Warner, *Foundations of Differentiable Manifolds and Lie Groups*, Chapter 2: differential forms and exterior differentiation. The proof above uses only the two differential-form identities stated in that proof.

Lecture 20

Real Forms and Noncompact Structure

Objectives. *A connected real semisimple group with finite centre has a product decomposition into a compact subgroup, an abelian subgroup and a nilpotent subgroup. This lecture compares compact and split real forms and states the Cartan and Iwasawa decompositions. It proves both decompositions for* $\mathrm{SL}(2,\mathbb{R})$*, where they give coordinates on the upper half-plane.*

By the end of this lecture you will be able to:

- Distinguish the compact and split examples of real forms using their Killing forms and Cartan involutions.
- State the Cartan and Iwasawa decompositions with all their hypotheses.
- Compute $g = kan$ and $g = k \exp X$ for an explicit matrix in $\mathrm{SL}(2,\mathbb{R})$.
- Read both decompositions as coordinates on the upper half-plane.

20.1 Real forms and Cartan involutions

Throughout this section $\mathfrak{g}_{\mathbb{C}}$ is a complex semisimple Lie algebra with Killing form

$$\kappa(X,Y) = \operatorname{tr}(\operatorname{ad} X \operatorname{ad} Y).$$

We cite Knapp, *Lie Groups Beyond an Introduction*, digital second edition 2023, as Knapp Beyond.

Definition 20.1. A *real form* of $\mathfrak{g}_{\mathbb{C}}$ is a real Lie subalgebra $\mathfrak{g} \subset \mathfrak{g}_{\mathbb{C}}$ with $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \oplus i\mathfrak{g}$ as real vector spaces.

A real basis of $\mathfrak{g}$ is then a complex basis of $\mathfrak{g}_{\mathbb{C}}$. For $X \in \mathfrak{g}$, the matrix of $\operatorname{ad} X$ in this basis is real. It is the same over $\mathbb{R}$ and over $\mathbb{C}$. Hence the Killing form of $\mathfrak{g}$ is the restriction $\kappa|_{\mathfrak{g}}$. It is real and nondegenerate. Its signature is unchanged by real isomorphisms.

Definition 20.2. A real form $\mathfrak{g}$ is *compact* if $\kappa|_{\mathfrak{g}}$ is negative definite. It is *split* if it contains an abelian subalgebra $\mathfrak{h}_0$ with the following two properties. Every $\operatorname{ad} H$ with $H \in \mathfrak{h}_0$ is diagonalisable over $\mathbb{R}$. The sum $\mathfrak{h}_0 \oplus i\mathfrak{h}_0$ is a Cartan subalgebra of $\mathfrak{g}_{\mathbb{C}}$.

The Lie Algebras course supplies both types [I]. The real span of a Chevalley basis is a split form. Weyl's unitary trick turns the same basis into a compact form; see Knapp Beyond, Section VI.1.

The name “compact” refers to compact groups. By Lecture 6, the Lie algebra of a compact group has an Ad-invariant inner product, so each $\operatorname{ad} X$ is skew and

$$\kappa(X,X) = \operatorname{tr}\left((\operatorname{ad} X)^2\right) \le 0.$$

So a real form whose Killing form takes a positive value is not the Lie algebra of a compact group.

Example 20.3 (The two forms of $\mathfrak{sl}(2,\mathbb{C})$)**.** On $\mathfrak{sl}(2,\mathbb{C})$ one has $\kappa(X,Y) = 4\operatorname{tr}(XY)$. Take

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Then $H^2 = I$, $(E+F)^2 = I$ and $(E-F)^2 = -I$. The products

$$H(E+F) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad H(E-F) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad (E-F)(E+F) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

are traceless. In the basis $(H, E+F, E-F)$ of $\mathfrak{sl}(2,\mathbb{R})$ the Gram matrix of κ is therefore $\operatorname{diag}(8,8,-8)$. The signature is $(2,1)$. The real form $\mathfrak{sl}(2,\mathbb{R})$ is split with $\mathfrak{h}_0 = \mathbb{R}H$, because $\operatorname{ad} H$ has eigenvalues $2, 0, -2$ on E, H, F.

The course basis of $\mathfrak{su}(2)$ is

$$A = iH, \qquad B = E - F, \qquad C = i(E+F).$$

Thus $\kappa(A,A) = i^2 \cdot 8 = -8$, $\kappa(C,C) = -8$ and $\kappa(B,B) = -8$. The cross terms are multiples of the vanishing ones above. The Gram matrix is $-8I$, so $\mathfrak{su}(2)$ is compact. The signatures $(2,1)$ and $(0,3)$ differ, so the two real forms are not isomorphic.

The two real forms in the example share the line $\mathbb{R}B$. The split form adds the plane $\operatorname{span}(H, E+F)$. The compact form adds the plane $i\operatorname{span}(H, E+F)$, spanned by A and C. Thus

$$\mathfrak{sl}(2,\mathbb{R}) = \mathbb{R}B \oplus \operatorname{span}(H, E+F), \qquad \mathfrak{su}(2) = \mathbb{R}B \oplus i\operatorname{span}(H, E+F).$$

A Cartan involution makes this relation precise. Let $\mathfrak{g}$ now be any real semisimple Lie algebra. Write $\kappa(X,Y) = \operatorname{tr}(\operatorname{ad} X \operatorname{ad} Y)$ for the Killing form of $\mathfrak{g}$ itself. For a real form of $\mathfrak{g}_{\mathbb{C}}$, this is the restriction $\kappa|_{\mathfrak{g}}$ described above.

Definition 20.4. A *Cartan involution* of $\mathfrak{g}$ is a Lie algebra automorphism θ with $\theta^2 = \mathrm{id}$ such that

$$\kappa_\theta(X,Y) = -\kappa(X,\theta Y)$$

is positive definite on $\mathfrak{g}$.

The form κ_θ is symmetric. Since θ is an automorphism, $[\theta X, Z] = \theta[X, \theta^{-1}Z]$ for all $Z \in \mathfrak{g}$. So $\operatorname{ad}(\theta X) = \theta \operatorname{ad} X\, \theta^{-1}$. Hence $\kappa(\theta X, \theta Y) = \operatorname{tr}(\theta \operatorname{ad} X \operatorname{ad} Y\, \theta^{-1}) = \kappa(X,Y)$. Using $\theta^2 = \mathrm{id}$ and the symmetry of κ, we get

$$\kappa(X,\theta Y) = \kappa(\theta X, \theta^2 Y) = \kappa(\theta X, Y) = \kappa(Y, \theta X).$$

Thus $\kappa_\theta(X,Y) = \kappa_\theta(Y,X)$. So κ_θ is an inner product on $\mathfrak{g}$, as the spectral theorem in § 20.2 requires.

Write $\mathfrak{k}$ and $\mathfrak{p}$ for the $(+1)$- and (-1)-eigenspaces of θ. Since $\theta^2 = \mathrm{id}$, we have $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$. Because θ is an automorphism,

$$[\mathfrak{k},\mathfrak{k}] \subset \mathfrak{k}, \qquad [\mathfrak{k},\mathfrak{p}] \subset \mathfrak{p}, \qquad [\mathfrak{p},\mathfrak{p}] \subset \mathfrak{k}.$$

For instance, $X, Y \in \mathfrak{p}$ give $\theta[X,Y] = [-X,-Y] = [X,Y]$. Every real semisimple Lie algebra has a Cartan involution, unique up to inner automorphism [C: Knapp Beyond, Section VI.2].

Proposition 20.5. *Let $\mathfrak{g} = \mathfrak{sl}(n,\mathbb{R})$ and $\theta(X) = -X^T$.*

(a) θ is a Cartan involution. Here $\mathfrak{k} = \mathfrak{so}(n)$, and $\mathfrak{p}$ consists of the real symmetric traceless matrices.

(b) $\mathfrak{k} \oplus i\mathfrak{p} = \mathfrak{su}(n)$, and $\mathfrak{su}(n)$ is a compact real form of $\mathfrak{sl}(n,\mathbb{C})$.

(c) The traceless diagonal matrices form a maximal abelian subspace $\mathfrak{a} \subset \mathfrak{p}$. Each $\operatorname{ad} H$, $H \in \mathfrak{a}$, is diagonalisable over $\mathbb{R}$. Hence $\mathfrak{sl}(n,\mathbb{R})$ is split.

Proof. (a) Clearly $\theta^2 = \mathrm{id}$. It is an automorphism because

$$[\theta X, \theta Y] = X^T Y^T - Y^T X^T = (YX - XY)^T = \theta[X,Y].$$

On $\mathfrak{sl}(n,\mathbb{C})$ we import $\kappa(X,Y) = 2n \operatorname{tr}(XY)$. Then

$$\kappa_\theta(X,X) = 2n \operatorname{tr}(XX^T) = 2n \sum_{i,j} X_{ij}^2 > 0 \qquad (X \neq 0).$$

The eigenspace descriptions are immediate.

(b) Write $Z \in \mathfrak{gl}(n,\mathbb{C})$ as $Z = X + iY$ with X, Y real. Then $Z^* = X^T - iY^T$. So $Z^* = -Z$ exactly when $X^T = -X$ and $Y^T = Y$. Also $\operatorname{tr} Z = 0$ exactly when $\operatorname{tr} Y = 0$, since $\operatorname{tr} X = 0$ already. Hence $\mathfrak{su}(n) = \mathfrak{k} \oplus i\mathfrak{p}$. It is a real form because

$$\mathfrak{sl}(n,\mathbb{C}) = (\mathfrak{k} \oplus \mathfrak{p}) \oplus i(\mathfrak{k} \oplus \mathfrak{p}) = (\mathfrak{k} \oplus i\mathfrak{p}) \oplus i(\mathfrak{k} \oplus i\mathfrak{p}).$$

For nonzero $Z \in \mathfrak{su}(n)$, we get $\kappa(Z,Z) = 2n \operatorname{tr}(Z^2) = -2n \operatorname{tr}(Z^*Z) < 0$.

(c) Let $H = \operatorname{diag}(h_1, \dots, h_n)$ and let E_{ij} be the matrix units. Then

$$[H, E_{ij}] = (h_i - h_j) E_{ij}.$$

So $\operatorname{ad} H$ is diagonal, with real eigenvalues, in the basis of all E_{ij} with $i \neq j$ together with a basis of $\mathfrak{a}$. For maximality, choose $H \in \mathfrak{a}$ with distinct entries. If $Y \in \mathfrak{p}$ commutes with H, then $(h_i - h_j) Y_{ij} = 0$, so Y is diagonal. Finally, $\mathfrak{a} \oplus i\mathfrak{a}$ is the diagonal Cartan subalgebra of $\mathfrak{sl}(n,\mathbb{C})$. ∎

The recipe $\mathfrak{k} \oplus i\mathfrak{p}$ from the compact-form statement in Proposition 20.5 works in general. If θ is a Cartan involution of $\mathfrak{g}$, then $\mathfrak{k} \oplus i\mathfrak{p}$ is a compact real form of the complexification $\mathfrak{g}_{\mathbb{C}}$ of $\mathfrak{g}$, the complex Lie algebra in which $\mathfrak{g}$ is a real form [C: Knapp Beyond, Sections VI.1–VI.2]. For $\mathfrak{g} = \mathfrak{sl}(2,\mathbb{R})$ this recipe returns the decomposition of $\mathfrak{su}(2)$ displayed above.

20.2 Restricted roots and the global decompositions

Fix a real semisimple Lie algebra $\mathfrak{g}$ with Cartan involution θ, and write $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$.

Lemma 20.6. *For $X \in \mathfrak{p}$, the operator $\operatorname{ad} X$ is symmetric for κ_θ.*

Proof. Since θ is an automorphism and $\theta X = -X$, we have $[X, \theta Z] = -\theta[X, Z]$. Invariance of κ gives $\kappa([X,Y],W) = -\kappa(Y,[X,W])$. Hence

$$\kappa_\theta([X,Y],Z) = \kappa(Y,[X,\theta Z]) = -\kappa(Y,\theta[X,Z]) = \kappa_\theta(Y,[X,Z]).$$

∎

Choose a maximal abelian subspace $\mathfrak{a} \subset \mathfrak{p}$. The operators $\operatorname{ad} H$, $H \in \mathfrak{a}$, commute and are κ_θ-symmetric. By the spectral theorem they are simultaneously diagonalisable over $\mathbb{R}$.

Definition 20.7. For $\lambda \in \mathfrak{a}^*$ put

$$\mathfrak{g}_\lambda = \{X \in \mathfrak{g} : [H,X] = \lambda(H)X \text{ for all } H \in \mathfrak{a}\}.$$

A *restricted root* is a nonzero λ with $\mathfrak{g}_\lambda \neq 0$. The *real rank* of $\mathfrak{g}$ is $\dim \mathfrak{a}$.

The real rank does not depend on the choices of θ and $\mathfrak{a}$ [C: Knapp Beyond, Chapter VI]. First fix θ. Any two maximal abelian subspaces of $\mathfrak{p}$ are then conjugate by a finite product of automorphisms $e^{\operatorname{ad} Z}$ with $Z \in \mathfrak{k}$. This conjugacy is a statement about the Lie algebra $\mathfrak{g}$ alone.

Next, let θ' be a second Cartan involution of $\mathfrak{g}$, with (-1)-eigenspace $\mathfrak{p}'$. By the uniqueness statement in § 20.1, $\theta' = \varphi\theta\varphi^{-1}$ for an inner automorphism φ. If $\theta X = -X$, then $\theta'(\varphi X) = \varphi\theta X = -\varphi X$. Symmetrically, $\theta = \varphi^{-1}\theta'\varphi$, so $\varphi^{-1}(\mathfrak{p}') \subseteq \mathfrak{p}$. Thus φ maps $\mathfrak{p}$ onto $\mathfrak{p}'$. Since φ is a Lie algebra automorphism, it maps maximal abelian subspaces of $\mathfrak{p}$ to maximal abelian subspaces of $\mathfrak{p}'$. Hence $\dim \mathfrak{a}$ is unchanged.

Next fix $H_0 \in \mathfrak{a}$ with $\lambda(H_0) \neq 0$ for the finitely many restricted roots. Such an H_0 exists, because $\mathfrak{a}$ is not a union of finitely many hyperplanes $\ker \lambda$. Call λ positive if $\lambda(H_0) > 0$, and set

$$\mathfrak{n} = \bigoplus_{\lambda(H_0)>0} \mathfrak{g}_\lambda.$$

The Jacobi identity gives $[\mathfrak{g}_\lambda, \mathfrak{g}_\mu] \subset \mathfrak{g}_{\lambda+\mu}$. Hence $\mathfrak{n}$ is a subalgebra. Each $\operatorname{ad} X$ with $X \in \mathfrak{n}$ is nilpotent, because it strictly raises the value at H_0.

For $\mathfrak{su}(n)$ the identity is a Cartan involution, since $\kappa_{\mathrm{id}} = -\kappa$ is positive definite. The two forms of $\mathfrak{sl}(n,\mathbb{C})$ compare as follows.

Real form	θ	$\mathfrak{k}$	$\mathfrak{p}$	real rank
$\mathfrak{su}(n)$, compact	id	$\mathfrak{su}(n)$	0	0
$\mathfrak{sl}(n,\mathbb{R})$, split	$X \mapsto -X^T$	$\mathfrak{so}(n)$	symmetric traceless	$n-1$

For $\mathfrak{sl}(n,\mathbb{R})$, with $H_0 = \operatorname{diag}(n-1, n-3, \dots, 1-n)$, the algebra $\mathfrak{n}$ is the strictly upper triangular matrices (Exercise 20.2).

Theorem 20.8 (Cartan and Iwasawa decompositions; cited)**.** *Let G be a connected semisimple Lie group with finite centre. Let θ be a Cartan involution of $\mathfrak{g}$, with $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$, and let K be the connected subgroup with Lie algebra $\mathfrak{k}$. Choose $\mathfrak{a}$ and $\mathfrak{n}$ as above, and put $A = \exp \mathfrak{a}$ and $N = \exp \mathfrak{n}$. Then:*

(a) K is compact, and $K \times \mathfrak{p} \to G$, $(k, X) \mapsto k \exp X$, is a diffeomorphism;

(b) A and N are closed subgroups, and $K \times A \times N \to G$, $(k,a,n) \mapsto kan$, is a diffeomorphism.

For proofs see Knapp Beyond, Sections VI.3 and VI.4, or P. Etingof, *Lie Groups and Lie Algebras*, MIT 18.755 lecture notes, Lectures 15 and 18. The finite-centre hypothesis ensures that K is compact.

20.3 The rank-one model $\mathrm{SL}(2,\mathbb{R})$

Let $G = \mathrm{SL}(2,\mathbb{R})$ and $\theta(X) = -X^{T}$. By the description of $\mathfrak{sl}(n,\mathbb{R})$ in Proposition 20.5, $\mathfrak{k} = \mathbb{R}(E-F)$ and $\mathfrak{p} = \operatorname{span}(H, E+F)$. Take $\mathfrak{a} = \mathbb{R}H$. The restricted roots are $\pm\lambda_1$ with $\lambda_1(H) = 2$, and $\mathfrak{g}_{\lambda_1} = \mathbb{R}E$ and $\mathfrak{g}_{-\lambda_1} = \mathbb{R}F$. Hence the real rank is one. With $H_0 = H$ we get $\mathfrak{n} = \mathbb{R}E$. The subgroups K, A and N consist of the matrices

$$k_\varphi = \begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix}, \qquad a_t = \begin{pmatrix} e^{t} & 0 \\ 0 & e^{-t} \end{pmatrix}, \qquad n_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix},$$

with $K = \mathrm{SO}(2) = \{k_\varphi\}$, $A = \{a_t\}$ and $N = \{n_x\}$. We now prove the conclusions of the Cartan and Iwasawa theorem, Theorem 20.8, directly for G, without using that theorem.

Proposition 20.9 (Iwasawa decomposition of $\mathrm{SL}(2,\mathbb{R})$). *The map $K\times A\times N \to G$, $(k,a,n)\mapsto kan$, is a diffeomorphism. Explicitly, for*

$$g = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in G, \qquad r = \sqrt{\alpha^2+\gamma^2},$$

one has $g = k_\varphi a_t n_x$ with

$$\cos\varphi = \frac{\alpha}{r}, \qquad \sin\varphi = \frac{\gamma}{r}, \qquad e^{t} = r, \qquad x = \frac{\alpha\beta+\gamma\delta}{r^{2}}.$$

Proof. The first column of g is nonzero, so $r > 0$. Let $k = k_\varphi$ with φ as stated. This is Gram–Schmidt applied to the columns of g. Using $\alpha\delta - \beta\gamma = 1$,

$$k^{-1}g = \frac{1}{r}\begin{pmatrix} \alpha & \gamma \\ -\gamma & \alpha \end{pmatrix}\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} r & \dfrac{\alpha\beta+\gamma\delta}{r} \\ 0 & \dfrac{1}{r} \end{pmatrix}.$$

On the other hand,

$$a_t n_x = \begin{pmatrix} e^{t} & e^{t}x \\ 0 & e^{-t} \end{pmatrix}.$$

These agree for $e^{t} = r$ and $x = (\alpha\beta+\gamma\delta)/r^{2}$. This proves existence.

For uniqueness, suppose $kan = k'a'n'$. Then $k'^{-1}k = a'n'(an)^{-1}$ is orthogonal and upper triangular with positive diagonal. Its first column is $(p,0)^{T}$ with $p>0$ and unit length, so $p = 1$. The second column is a unit vector orthogonal to $(1,0)^{T}$ with positive second entry, so it is $(0,1)^{T}$. Hence $k = k'$ and $an = a'n'$. Comparing diagonals gives $a = a'$, and then $n = n'$.

The multiplication map is smooth. Its inverse is given by the formulas above, which are smooth in the entries of g because $r > 0$. $\blacksquare$

In particular G is connected and diffeomorphic to a circle times $\mathbb{R}^2$. So $\mathrm{SL}(2,\mathbb{R})$ satisfies the hypotheses of the Cartan and Iwasawa theorem, Theorem 20.8, with centre $\{\pm I\}$.

For the Cartan decomposition we need the exponential on $\mathfrak{p}$. Define the entire functions

$$C(u)=\sum_{k\ge 0}\frac{u^k}{(2k)!},\qquad S(u)=\sum_{k\ge 0}\frac{u^k}{(2k+1)!}.$$

For $u=s^2\ge 0$ they are $\cosh s$ and $\sinh s/s$, with the latter interpreted as 1 at $s=0$. Let $\mathcal{P}$ be the set of symmetric positive definite 2×2 matrices of determinant 1.

Lemma 20.10. *For $Y\in\mathfrak{p}$, set $u=-\det Y\ge 0$. Then $Y^2=uI$ and*

$$\exp Y=C(u)I+S(u)Y.$$

The map $\exp\colon\mathfrak{p}\to\mathcal{P}$ is a bijection. Its inverse is

$$L(Q)=\frac{Q-cI}{S(C^{-1}(c))},\qquad c=\tfrac12\operatorname{tr}Q,$$

which is smooth on a neighbourhood of $\mathcal{P}$ among symmetric matrices.

Proof. Write $Y=\begin{pmatrix}p & q\\ q & -p\end{pmatrix}$. Then $Y^2=(p^2+q^2)I=uI$. Splitting the exponential series into even and odd powers gives the formula. The matrix $\exp Y=(\exp\frac{Y}{2})^2$ is the square of an invertible symmetric matrix. Hence $\exp Y$ is positive definite. Moreover, $\det\exp Y=e^{\operatorname{tr}Y}=1$.

Every coefficient of the series for C' is positive, so $C'>0$ on $(0,\infty)$. Also, $C'(0)=\frac12$. So $C'>0$ on some interval $(-\varepsilon,\infty)$. Thus C is a strictly increasing bijection $[0,\infty)\to[1,\infty)$, with smooth inverse near $[1,\infty)$. Also $S>0$ near $[0,\infty)$.

For injectivity, let $Q=\exp Y$. Then $\operatorname{tr}Q=2C(u)$, which determines u because C is strictly increasing on $[0,\infty)$. The formula for $\exp Y$ then gives $Y=(Q-C(u)I)/S(u)=L(Q)$.

For surjectivity, let $Q\in\mathcal{P}$ have eigenvalues $\mu,\mu^{-1}>0$. Then $c=\frac12(\mu+\mu^{-1})\ge 1$. Put $u=C^{-1}(c)$ and $Y=L(Q)$. This Y is symmetric and traceless. Cayley–Hamilton gives $Q^2=2cQ-I$, so $(Q-cI)^2=(c^2-1)I$. Since $u\,S(u)^2=C(u)^2-1=c^2-1$, we get $Y^2=uI$. Therefore

$$\exp Y=C(u)I+S(u)Y=cI+(Q-cI)=Q.$$

∎

Proposition 20.11 (Cartan decomposition of $\mathrm{SL}(2,\mathbb{R})$)**.** *The map $K\times\mathfrak{p}\to G$, $(k,X)\mapsto k\exp X$, is a diffeomorphism. Its inverse is*

$$g\mapsto\bigl(g\exp(-X),\,X\bigr),\qquad X=\tfrac12 L(g^{T}g).$$

Proof. Suppose first that $g=k\exp X$ with $k\in K$. Since X is symmetric, so is $\exp X$. Also k is orthogonal. Hence

$$g^{T}g=\exp(X)k^{T}k\exp(X)=\exp(2X).$$

By the exponential lemma, Lemma 20.10, $2X=L(g^{T}g)$, so X and then k are forced. Conversely, given g, the matrix $g^{T}g$ lies in $\mathcal{P}$. Define X and $k=g\exp(-X)$ as stated. Then

$$k^{T}k=\exp(-X)\exp(2X)\exp(-X)=I,\qquad\det k=1,$$

so $k \in K$. The map and its inverse are smooth by the exponential lemma, Lemma 20.10. ∎

Corollary 20.12 ($G = KAK$). *Every $g \in G$ can be written as $g = k_1 a_t k_2$ with $k_1, k_2 \in K$ and $t \geq 0$, and t is unique. The number e^{2t} is the larger eigenvalue of $g^{T} g$.*

Proof. Write $g = k \exp X$. The symmetric traceless matrix X has eigenvalues $\pm t$ with $t \geq 0$. Take an orthonormal eigenbasis, ordered $(+t, -t)$, and negate one vector if needed. This gives $k' \in K$ with $X = k'(tH)k'^{-1}$. Then $\exp X = k' a_t k'^{-1}$ and $g = (kk')a_t k'^{-1}$. If $g = k_1 a_t k_2$, then $g^{T} g = k_2^{-1} a_{2t} k_2$ has eigenvalues $e^{\pm 2t}$, which fixes t. ∎

Example 20.13. We compute the Iwasawa, Cartan and KAK data of the lower triangular unipotent matrix

$$g = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

The Cartan data involve the golden ratio $\tau = \frac{1+\sqrt{5}}{2}$. Recall $\tau + \tau^{-1} = \sqrt{5}$ and $\tau - \tau^{-1} = 1$.

Iwasawa. Here $r = \sqrt{2}$, so $k = k_{\pi/4}$, $e^{t} = \sqrt{2}$ and $x = \tfrac{1}{2}$. Check:

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \sqrt{2} & \frac{\sqrt{2}}{2} \\ 0 & \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Cartan. Here

$$Q = g^{T} g = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

The half-trace of Q is $c = \frac{3}{2}$. The equation $C(u) = \cosh \sqrt{u} = \frac{3}{2}$ gives $\sqrt{u} = 2 \log \tau$, since $\tau^{2} + \tau^{-2} = 3$. Also $\sinh \sqrt{u} = \frac{\sqrt{5}}{2}$. Hence

$$X = \frac{1}{2} L(Q) = \frac{2 \log \tau}{\sqrt{5}} \begin{pmatrix} \frac{1}{2} & 1 \\ 1 & -\frac{1}{2} \end{pmatrix} = \frac{\log \tau}{\sqrt{5}} \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}.$$

Now $-\det X = (\log \tau)^{2}$, so the exponential lemma, Lemma 20.10, gives

$$\exp X = \frac{\sqrt{5}}{2} I + \frac{1}{2\sqrt{5}} \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}.$$

Then

$$k = g \exp(-X) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix} \in \mathrm{SO}(2).$$

The eigenvalues of X are $\pm \log \tau$. Hence, as in the proof of the KAK corollary, Corollary 20.12, the KAK parameter of g is $t = \log \tau$.

20.4 The upper half-plane

Let $\mathbb{H} = \{ z \in \mathbb{C} : \operatorname{Im} z > 0 \}$. For g with entries $\alpha, \beta, \gamma, \delta$ as in the Iwasawa decomposition, Proposition 20.9, put

$$g \cdot z = \frac{\alpha z + \beta}{\gamma z + \delta}.$$

Lemma 20.14. *For $z \in \mathbb{H}$ one has $\gamma z + \delta \neq 0$ and*

$$\operatorname{Im}(g \cdot z) = \frac{\operatorname{Im} z}{|\gamma z + \delta|^2}.$$

This defines an action of G on $\mathbb{H}$. The stabiliser of i is K, and $-I$ acts trivially.

Proof. If $\gamma \neq 0$, the root of $\gamma z + \delta$ is real. If $\gamma = 0$, then $\delta \neq 0$. Next,

$$g \cdot z = \frac{(\alpha z + \beta)(\gamma \bar{z} + \delta)}{|\gamma z + \delta|^2},$$

and the numerator has imaginary part $(\alpha\delta - \beta\gamma) \operatorname{Im} z = \operatorname{Im} z$.

For the action law, write $[v] = v_1/v_2$ for $v \in \mathbb{C}^2$ with $v_2 \neq 0$. Then $g \cdot z = [g(z,1)^T]$, and $[\lambda v] = [v]$ for $\lambda \neq 0$. If h has bottom row (γ', δ'), then $h(z,1)^T = (\gamma' z + \delta')(h \cdot z, 1)^T$. Hence

$$(gh) \cdot z = [gh(z,1)^T] = [g(h \cdot z, 1)^T] = g \cdot (h \cdot z).$$

If $g \cdot i = i$, then $\alpha i + \beta = -\gamma + \delta i$. So $\alpha = \delta$ and $\beta = -\gamma$, with $\alpha^2 + \gamma^2 = 1$. That is, $g = k_\varphi$. Conversely, $k_\varphi \cdot i = ie^{i\varphi}/e^{i\varphi} = i$. Finally, $(-I) \cdot z = z$. ∎

Inverting $g^{-1} = kan$ gives $g = n^{-1}a^{-1}k^{-1}$. So the Iwasawa decomposition, Proposition 20.9, also yields a unique decomposition $G = NAK$.

Proposition 20.15. *Write $g = n_x a_t k$. Then*

$$g \cdot i = x + ie^{2t}, \qquad x = \frac{\alpha\gamma + \beta\delta}{\gamma^2 + \delta^2}, \qquad e^{2t} = \frac{1}{\gamma^2 + \delta^2}.$$

The map $n_x a_t \mapsto n_x a_t \cdot i$ is a diffeomorphism $NA \to \mathbb{H}$. The map $gK \mapsto g \cdot i$ is a bijection $G/K \to \mathbb{H}$.

Proof. We have $a_t \cdot z = e^{2t} z$ and $n_x \cdot z = z + x$. Since $k \cdot i = i$, this gives $n_x a_t k \cdot i = x + ie^{2t}$. The stated formulas for x and e^{2t} follow by comparing real and imaginary parts in the direct computation

$$g \cdot i = \frac{(\alpha i + \beta)(\delta - \gamma i)}{\gamma^2 + \delta^2} = \frac{\alpha\gamma + \beta\delta + i}{\gamma^2 + \delta^2}.$$

The map $(x,t) \mapsto x + ie^{2t}$ has smooth inverse $z \mapsto (\operatorname{Re} z, \tfrac{1}{2} \log \operatorname{Im} z)$. Finally, $g \cdot i = g' \cdot i$ exactly when $g^{-1}g' \in K$, by the stabiliser statement of Lemma 20.14. ∎

So Iwasawa coordinates are Cartesian coordinates (x,y) with $y = e^{2t}$. Here N translates horizontally and A dilates from 0. The subgroup NA acts by the affine maps $z \mapsto e^{2t} z + x$; Exercise 20.6 identifies it with the affine group of Lecture 16.

The KAK form describes $\mathbb{H}$ by K-orbits. If $g \cdot i = z$ and $g = k_1 a_t k_2$, then $z = k_1 \cdot (ie^{2t})$. The number $t \geq 0$ depends only on z, because replacing g by gk does not change the eigenvalues of $g^T g$. So $\mathbb{H}$ is the disjoint union of the K-orbits of the points ie^{2t}, $t \geq 0$. Exercise 20.5 shows that these orbits are Euclidean circles.

20.5 Self-checks

1. What are the real ranks of $\mathfrak{sl}(n,\mathbb{R})$ and $\mathfrak{su}(n)$? What do A and N become for $\mathrm{SU}(n)$?
   Answer. The real rank of $\mathfrak{sl}(n,\mathbb{R})$ is $n-1$, and the real rank of $\mathfrak{su}(n)$ is 0. For $\mathrm{SU}(n)$, $\mathfrak{p}=0$, so $A=N=\{I\}$ and the Iwasawa decomposition reads $G=K$.
2. Find the Iwasawa decomposition of $g=\begin{pmatrix}0&1\\-1&2\end{pmatrix}$.
   Answer. Here $r=1$, $\cos\varphi=0$ and $\sin\varphi=-1$, so $k=k_{-\pi/2}$ and $a=I$. Also $x=(0\cdot 1+(-1)\cdot 2)/1=-2$. Indeed
   $$\begin{pmatrix}0&1\\-1&0\end{pmatrix}\begin{pmatrix}1&-2\\0&1\end{pmatrix}=\begin{pmatrix}0&1\\-1&2\end{pmatrix}.$$
3. Where does $-I$ sit in $G=KAN$, and what does it do on $\mathbb{H}$?
   Answer. In $G=KAN$, the element $-I$ has factors $k=k_{\pi}$ and $a=n=I$. The element $-I$ acts trivially on $\mathbb{H}$. Hence the action factors through $G/\{\pm I\}$.

Summary

A Cartan involution splits $\mathfrak{g}=\mathfrak{k}\oplus\mathfrak{p}$. The compact form $\mathfrak{k}\oplus i\mathfrak{p}$ of $\mathfrak{g}_{\mathbb{C}}$ has real rank zero. The split form $\mathfrak{sl}(n,\mathbb{R})$ has real rank $n-1$. For connected semisimple groups with finite centre, the cited Cartan and Iwasawa theorem gives $G=K\exp\mathfrak{p}$ and $G=KAN$. For $\mathrm{SL}(2,\mathbb{R})$ we proved both decompositions, with explicit inverses. On $\mathbb{H}$, identified with G/K, the subgroup NA gives Cartesian coordinates. The KAK form describes $\mathbb{H}$ as the disjoint union of the K-orbits of the points ie^{2t} with $t\geq 0$.

Reading

- Knapp Beyond, Chapter VI, Sections 1–4: compact real forms, the Cartan decomposition on the Lie algebra and group levels, and the Iwasawa decomposition.
- P. Etingof, *Lie Groups and Lie Algebras*, MIT 18.755 lecture notes, Lectures 15 and 18: a second route through real forms and these decompositions.
- Lecture 22 uses $G=KAN$ to construct the principal series.

Lecture 21

Smooth Vectors, K-Types and $(\mathfrak{g}, K)$-Modules

Objectives. *A representation of a noncompact group usually acts on an infinite-dimensional Hilbert space. We use its derivatives on smooth vectors. This lecture extracts an algebraic object from such a representation. The object consists of the smooth vectors that are finite under a compact subgroup K. On these vectors $\mathfrak{g}$ and K act together, subject to compatibility conditions. The model is $\mathrm{SL}(2,\mathbb{R})$ with $K = \mathrm{SO}(2)$.*

By the end of this lecture you will be able to:

- Define smooth vectors, K-finite vectors, K-types and admissibility.
- Prove that the K-finite smooth vectors of a strongly continuous representation on a Hilbert space form a $(\mathfrak{g}, K)$-module.
- Use the compatibility conditions to show that E and F shift the circle K-types of $\mathrm{SL}(2,\mathbb{R})$ by ± 2.
- State which global questions the $(\mathfrak{g}, K)$-module leaves open, and which cited theorems address them.

21.1 Smooth vectors and K-finite vectors

Throughout, G is a Lie group with Lie algebra $\mathfrak{g}$. The subgroup $K \subseteq G$ is compact, with Lie algebra $\mathfrak{k} \subseteq \mathfrak{g}$. The representation Π of G on a Hilbert space $\mathcal{H}$ is strongly continuous. Here each operator $\Pi(g)$ is bounded, linear and invertible. The map $g \mapsto \Pi(g)v$ is norm continuous for every $v \in \mathcal{H}$. Unitarity is assumed only where stated.

Definition 21.1. A vector $v \in \mathcal{H}$ is *smooth* if the orbit map $G \to \mathcal{H}$, $g \mapsto \Pi(g)v$, is smooth. This means that in every chart all iterated partial derivatives exist as norm limits and are continuous. We write $\mathcal{H}^\infty$ for the set of smooth vectors. For $v \in \mathcal{H}^\infty$ and $X \in \mathfrak{g}$ we set

$$d\Pi(X)v = \left.\frac{d}{dt}\right|_{t=0} \Pi(\exp tX)v.$$

The set $\mathcal{H}^\infty$ is a subspace of $\mathcal{H}$, because a linear combination of smooth orbit maps is smooth.

For unitary representations, the following facts were imported in Lecture 15. We now also import the density and invariance facts (F1)–(F2) for strongly continuous representations by bounded operators. A source for this more general smooth-vector theorem is Knapp RT,

Sections III.3–4. Here Knapp RT is A. W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples* (2001).

(F1) *Density* (Gårding). The space $\mathcal{H}^\infty$ is dense in $\mathcal{H}$.

(F2) *Invariance.* Each operator $d\Pi(X)$ maps $\mathcal{H}^\infty$ into itself, and $X \mapsto d\Pi(X)$ is a Lie algebra representation of $\mathfrak{g}$ on $\mathcal{H}^\infty$.

(F3) *Skew-symmetry.* If Π is unitary, then $\langle d\Pi(X)u, w\rangle = -\langle u, d\Pi(X)w\rangle$ for all $X \in \mathfrak{g}$ and $u, w \in \mathcal{H}^\infty$.

We extend $d\Pi$ complex-linearly to $\mathfrak{g}_{\mathbb{C}}$ by $d\Pi(X + iY) = d\Pi(X) + i\, d\Pi(Y)$ for $X, Y \in \mathfrak{g}$. By the invariance property (F2), this is a representation of $\mathfrak{g}_{\mathbb{C}}$ on $\mathcal{H}^\infty$.

One elementary fact is proved here. If $v \in \mathcal{H}^\infty$ and $h \in G$, then $\Pi(h)v \in \mathcal{H}^\infty$. Indeed, the orbit map of $\Pi(h)v$ is $g \mapsto \Pi(gh)v$. This is the orbit map of v composed with the smooth right translation $g \mapsto gh$.

Definition 21.2. A vector $v \in \mathcal{H}$ is *K-finite* if $\operatorname{span}\{\Pi(k)v : k \in K\}$ is finite-dimensional. We write $\mathcal{H}_K$ for the set of K-finite vectors.

The set $\mathcal{H}_K$ is a subspace, because for $a, b \in \mathbb{C}$ the span of the vectors $\Pi(k)(av + bw)$, $k \in K$, lies in $\operatorname{span}\Pi(K)v + \operatorname{span}\Pi(K)w$.

Let $\widehat{K}$ be the set of equivalence classes of irreducible continuous representations of K. Each class δ has a finite-dimensional representative space V_δ (Lecture 10). Let $\operatorname{Hom}_K(V_\delta, \mathcal{H})$ be the space of linear maps $V_\delta \to \mathcal{H}$ that commute with K.

Definition 21.3. The *multiplicity* of $\delta \in \widehat{K}$ in Π is $m_\delta(\Pi) = \dim \operatorname{Hom}_K(V_\delta, \mathcal{H})$. The class δ is a *K-type* of Π if $m_\delta(\Pi) \geq 1$. The representation Π is *admissible* if $m_\delta(\Pi) < \infty$ for every $\delta \in \widehat{K}$.

Let $v \in \mathcal{H}_K$ and $W_v = \operatorname{span}\Pi(K)v$. Then W_v is finite-dimensional and K-invariant. All norms on W_v are equivalent, so strong continuity makes K act continuously on W_v. By complete reducibility (Lecture 5), W_v is a direct sum of irreducible subspaces. Hence every K-finite vector is a finite sum of vectors from K-types.

For the circle we write k_θ, $\theta \in \mathbb{R}/2\pi\mathbb{Z}$, with $k_\theta k_\phi = k_{\theta+\phi}$. Its irreducible continuous representations are the characters $\chi_n(k_\theta) = e^{in\theta}$, $n \in \mathbb{Z}$ (Lecture 7). We label the K-type χ_n by n.

Example 21.4. Let $G = K = \mathrm{SO}(2)$ and $\mathcal{H} = L^2(\mathbb{R}/2\pi\mathbb{Z})$, with

$$\langle f, h\rangle = \frac{1}{2\pi}\int_0^{2\pi} \overline{f(x)}\, h(x)\, dx, \qquad (\Pi(k_\theta)f)(x) = f(x + \theta).$$

This representation is unitary and strongly continuous. The functions $e_n(x) = e^{inx}$ form an orthonormal basis (Lecture 10). Put $\mathcal{H}(n) = \{v : \Pi(k_\theta)v = e^{in\theta}v \text{ for all } \theta\}$. Then $e_n \in \mathcal{H}(n)$.

We show that $\mathcal{H}(n) = \mathbb{C}e_n$. Let $a_m(v) = \langle e_m, v\rangle$. For continuous v, the substitution $y = x + \theta$ and periodicity give

$$a_m(\Pi(k_\theta)v) = \frac{1}{2\pi}\int_0^{2\pi} e^{-im(y-\theta)}v(y)\, dy = e^{im\theta}a_m(v).$$

Both sides are continuous linear functionals of v, so density extends the identity to all L^2 vectors. If $v \in \mathcal{H}(n)$, the left side is also $e^{in\theta}a_m(v)$. For $m \neq n$ take $\theta = \pi/(m-n)$. Then $e^{im\theta} = -e^{in\theta}$, so $a_m(v) = 0$. Hence $v = a_n(v)e_n$.

So $\mathcal{H}_K$ is the space of trigonometric polynomials $f = \sum_{|n|\le N} c_n e_n$. Each such f is smooth, because its orbit map

$$\theta \mapsto \Pi(k_\theta)f = \sum_{|n|\le N} c_n\, e^{in\theta}\, e_n$$

is a finite sum of smooth scalar functions times fixed vectors. Thus every K-finite vector is smooth. Every multiplicity is 1, so Π is admissible. The space $\mathcal{H}_K$ is dense but proper. For example, $\sum_n 2^{-|n|}e_n$ is not K-finite.

A K-finite vector need not be smooth. Let $G = \mathbb{R}$ act on $L^2(\mathbb{R})$ by translation, and let $K = \{0\}$. Every vector is K-finite. The indicator function of $[0,1]$ is not smooth (Exercise 21.5). So we impose both conditions. We write $\mathcal{H}_K^\infty$ for the space of smooth K-finite vectors:

$$\mathcal{H}_K^\infty = \mathcal{H}^\infty \cap \mathcal{H}_K.$$

21.2 The compatibility conditions

Definition 21.5. A $(\mathfrak{g}, K)$-*module* is a complex vector space V with a Lie algebra representation of $\mathfrak{g}$ on V, written $v \mapsto X \cdot v$, and a group homomorphism $K \to \mathrm{GL}(V)$, written $v \mapsto k \cdot v$, such that:

(a) (*finiteness condition*) every $v \in V$ lies in a finite-dimensional K-invariant subspace W on which K acts continuously;

(b) (*derivative condition*) for $X \in \mathfrak{k}$ and $v \in V$, with W as in the finiteness condition (a),

$$X \cdot v = \frac{d}{dt}\Big|_{t=0} \exp(tX) \cdot v \qquad \text{(derivative computed in } W\text{)};$$

(c) (*covariance condition*) for $k \in K$, $X \in \mathfrak{g}$ and $v \in V$,

$$k \cdot (X \cdot v) = (\mathrm{Ad}(k)X) \cdot (k \cdot v).$$

A *morphism* of $(\mathfrak{g}, K)$-modules is a linear map that commutes with both actions. A $(\mathfrak{g}, K)$-module V is *admissible* if $\dim \mathrm{Hom}_K(V_\delta, V) < \infty$ for every $\delta \in \widehat{K}$.

The derivative in the derivative condition (b) does not depend on the choice of W. Indeed, two choices W and W' both lie in the finite-dimensional space $W + W'$, and all norms on $W + W'$ are equivalent.

The $\mathfrak{g}$-action extends complex-linearly to $\mathfrak{g}_{\mathbb{C}}$. Both sides of the covariance condition (c) are complex-linear in X, so the covariance condition also holds for $X \in \mathfrak{g}_{\mathbb{C}}$.

The derivative condition (b) involves only elements $\exp(tX)$, which lie in the identity component of K. Hence, if K is disconnected, the K-action carries information that $\mathfrak{k}$ cannot detect (Exercise 21.7).

The covariance condition (c) describes how the K-action transforms the $\mathfrak{g}$-action. In § 21.3 it gives the shifts of the circle K-types for $\mathrm{SL}(2,\mathbb{R})$.

Proposition 21.6. *Let Π be a strongly continuous representation of G on $\mathcal{H}$, and let $K \subseteq G$ be compact. Then $\mathcal{H}_K^\infty$ is invariant under $\Pi(k)$ for $k \in K$ and under $d\Pi(X)$ for $X \in \mathfrak{g}_{\mathbb{C}}$. With these actions $\mathcal{H}_K^\infty$ is a $(\mathfrak{g}, K)$-module. Let Π' be a strongly continuous representation of G on a Hilbert space $\mathcal{H}'$. Let $T : \mathcal{H} \to \mathcal{H}'$ be bounded and linear, with $T\Pi(g) = \Pi'(g)T$ for all $g \in G$. Then T maps $\mathcal{H}_K^\infty$ into $\mathcal{H}_K'^\infty$, and its restriction is a morphism.*

Proof. Step 1: the K-action. Let $v \in \mathcal{H}_K^\infty$ and $k \in K$. Then $\Pi(k)v$ is smooth by the right-translation argument in § 21.1. Since $Kk = K$,

$$\operatorname{span} \Pi(K)\Pi(k)v = \operatorname{span} \Pi(Kk)v = \operatorname{span} \Pi(K)v,$$

which is finite-dimensional. So $\Pi(k)v \in \mathcal{H}_K^\infty$. In particular, $W_v = \operatorname{span} \Pi(K)v$ is contained in $\mathcal{H}_K^\infty$.

Step 2: the covariance identity. Let $k \in K$, $X \in \mathfrak{g}$ and $w \in \mathcal{H}^\infty$. We claim

$$\Pi(k)\, d\Pi(X)w = d\Pi(\operatorname{Ad}(k)X)\, \Pi(k)w. \tag{21.1}$$

By the conjugation formula for exponentials (Lecture 2), $k \exp(tX)k^{-1} = \exp(t \operatorname{Ad}(k)X)$. Hence, for $t \neq 0$,

$$\Pi(k)\Big[\frac{\Pi(\exp tX)w - w}{t}\Big] = \frac{\Pi(\exp(t \operatorname{Ad}(k)X))\, \Pi(k)w - \Pi(k)w}{t}.$$

Let $t \to 0$. On the left, the bracket tends to $d\Pi(X)w$, and $\Pi(k)$ is bounded. So the left side tends to $\Pi(k)\, d\Pi(X)w$. On the right, $\Pi(k)w$ is smooth, so the quotient tends to $d\Pi(\operatorname{Ad}(k)X)\Pi(k)w$.

Step 3: $\mathfrak{g}$ preserves K-finiteness. This is the main step. The difficulty is that $d\Pi(X)$ does not commute with $\Pi(k)$. So the span of $\Pi(K)\, d\Pi(X)v$ is not obviously finite-dimensional. The covariance identity (21.1) replaces commutation. It moves X to $\operatorname{Ad}(k)X$, which still lies in the finite-dimensional space $\mathfrak{g}$.

Let $v \in \mathcal{H}_K^\infty$ and $W = W_v$. By Step 1, $W \subseteq \mathcal{H}^\infty$, so $d\Pi(X)w$ is defined for $w \in W$. Choose bases $X_1, \dots, X_d$ of $\mathfrak{g}$ and $w_1, \dots, w_r$ of W. Put

$$U = \operatorname{span}\{d\Pi(X_i)w_j : 1 \le i \le d,\ 1 \le j \le r\}.$$

Then $\dim U \le dr$. Let $X \in \mathfrak{g}$, $w \in W$ and $k \in K$. Expand

$$\operatorname{Ad}(k)X = \sum_{i=1}^{d} a_i(k)X_i, \qquad \Pi(k)w = \sum_{j=1}^{r} b_j(k)w_j,$$

with scalars $a_i(k)$ and $b_j(k)$. The second expansion uses $\Pi(k)w \in W$. By the covariance identity (21.1) and linearity of $d\Pi$ in both arguments,

$$\Pi(k)\, d\Pi(X)w = d\Pi(\operatorname{Ad}(k)X)\, \Pi(k)w = \sum_{i=1}^{d}\sum_{j=1}^{r} a_i(k)\, b_j(k)\, d\Pi(X_i)w_j \in U.$$

In particular $\Pi(K)\, d\Pi(X)v \subseteq U$. So $d\Pi(X)v$ is K-finite. It is smooth by the invariance property (F2). Complex combinations stay in the subspace $\mathcal{H}_K^\infty$, so the same holds for $X \in \mathfrak{g}_{\mathbb{C}}$.

Step 4: the axioms. By the invariance property (F2) and Step 3, $d\Pi$ is a representation of $\mathfrak{g}$ on $\mathcal{H}_K^\infty$. The finiteness condition (a) holds with $W = W_v$, by Step 1 and the continuity of

the K-action on W_v shown in § 21.1. For the derivative condition (b), let $X \in \mathfrak{k}$. The curve $t \mapsto \Pi(\exp tX)v$ lies in W_v. Its derivative in $\mathcal{H}$ is $d\Pi(X)v$. A finite-dimensional subspace is closed, and W_v carries the restricted norm. So the difference quotients converge in W_v to $d\Pi(X)v$. The covariance condition (c) is the covariance identity (21.1).

Step 5: intertwiners. Let $v \in \mathcal{H}_K^\infty$. The orbit map $g \mapsto \Pi'(g)Tv = T\Pi(g)v$ is smooth, because T is bounded and linear. Also $\operatorname{span}\Pi'(K)Tv = T(\operatorname{span}\Pi(K)v)$ is finite-dimensional. Differentiating $T\Pi(\exp tX)v = \Pi'(\exp tX)Tv$ at $t = 0$ gives $T\,d\Pi(X)v = d\Pi'(X)Tv$. Commutation with K is the hypothesis. ∎

Definition 21.7. The $(\mathfrak{g}, K)$-*module of* Π is $\mathcal{H}_K^\infty$ with the actions given in the $(\mathfrak{g}, K)$-module proposition, Proposition 21.6. Two representations are *infinitesimally equivalent* if their $(\mathfrak{g}, K)$-modules are isomorphic.

21.3 Circle K-types for $\mathrm{SL}(2,\mathbb{R})$

Now $G = \mathrm{SL}(2,\mathbb{R})$ and $K = \mathrm{SO}(2)$. Put

$$W = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad k_\theta = \exp(\theta W) = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}.$$

The matrix k_θ here is $k_{-\theta}$ in the notation of Lecture 20. Then $\mathfrak{k} = \mathbb{R}W$, and $\theta \mapsto k_\theta$ identifies $\mathbb{R}/2\pi\mathbb{Z}$ with K. So K is connected.

Let V be a $(\mathfrak{g}, K)$-module. For $n \in \mathbb{Z}$ put

$$V(n) = \{v \in V : k_\theta \cdot v = e^{in\theta}v \text{ for all } \theta\}.$$

By the finiteness condition (a) and complete reducibility (Lecture 5), every $v \in V$ is a finite sum of vectors from the spaces $V(n)$. The sum is direct, because the functions $e^{in\theta}$ are linearly independent. So $V = \bigoplus_{n\in\mathbb{Z}} V(n)$, and $\dim V(n)$ is the multiplicity of the K-type n.

Let $H = \operatorname{diag}(1,-1)$ and $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Then H, S, W is a basis of $\mathfrak{g} = \mathfrak{sl}(2,\mathbb{R})$. Multiplying matrices gives

$$[H,S] = 2W, \qquad [W,H] = -2S, \qquad [W,S] = 2H.$$

In $\mathfrak{g}_\mathbb{C} = \mathfrak{sl}(2,\mathbb{C})$ we set

$$H' = -iW = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad E = \frac{1}{2}(H + iS) = \frac{1}{2}\begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}, \quad F = \frac{1}{2}(H - iS) = \frac{1}{2}\begin{pmatrix} 1 & -i \\ -i & -1 \end{pmatrix}.$$

The real brackets give

$$\begin{aligned} [H', E] &= -\tfrac{i}{2}\big([W,H] + i[W,S]\big) = -\tfrac{i}{2}(-2S + 2iH) = H + iS = 2E, \\ [H', F] &= -\tfrac{i}{2}\big([W,H] - i[W,S]\big) = -\tfrac{i}{2}(-2S - 2iH) = -(H - iS) = -2F, \\ [E, F] &= \tfrac{1}{4}\big(-i[H,S] + i[S,H]\big) = -\tfrac{i}{2}[H,S] = -iW = H'. \end{aligned}$$

Thus (H', E, F) is an $\mathfrak{sl}_2$-triple. The $\mathfrak{sl}_2$ theory of the prerequisite Lie Algebras course applies to it. The element H' spans $\mathfrak{k}_\mathbb{C}$, whereas $H \notin \mathfrak{k}$. Also $F = \overline{E}$, where $\overline{X + iY} = X - iY$ for $X, Y \in \mathfrak{g}$.

Lemma 21.8. *For every $\theta \in \mathbb{R}$, the complex-linear extension of $\mathrm{Ad}(k_\theta)$ to $\mathfrak{g}_{\mathbb{C}}$ satisfies $\mathrm{Ad}(k_\theta)H' = H'$, $\mathrm{Ad}(k_\theta)E = e^{2i\theta}E$ and $\mathrm{Ad}(k_\theta)F = e^{-2i\theta}F$.*

Proof. Since $W = iH'$, we get

$$\mathrm{ad}(W)E = i[H', E] = 2iE, \qquad \mathrm{ad}(W)F = i[H', F] = -2iF, \qquad \mathrm{ad}(W)H' = 0.$$

By the relation between Ad and ad in Lecture 2, $\mathrm{Ad}(\exp \theta W) = e^{\theta \,\mathrm{ad}\, W}$, extended complex-linearly to $\mathfrak{g}_{\mathbb{C}}$. On an eigenvector of $\mathrm{ad}\, W$ with eigenvalue μ this operator is multiplication by $e^{\theta\mu}$. The eigenvalues 0, $2i$ and $-2i$ of $\mathrm{ad}\, W$ on H', E and F give the three formulas. ∎

Exercise 21.1 checks the adjoint lemma by matrix multiplication.

Proposition 21.9. *Let V be a $(\mathfrak{g}, K)$-module for $G = \mathrm{SL}(2, \mathbb{R})$ and $K = \mathrm{SO}(2)$. If $v \in V(n)$, then*

$$H' \cdot v = nv, \qquad E \cdot v \in V(n+2), \qquad F \cdot v \in V(n-2).$$

Proof. Let $v \in V(n)$. Since $\exp(\theta W) \cdot v = e^{in\theta}v$, the derivative condition (b) with $X = W$ gives

$$W \cdot v = \frac{d}{d\theta}\Big|_{\theta=0} e^{in\theta}v = in\, v, \qquad \text{so} \qquad H' \cdot v = -i\, W \cdot v = nv.$$

By the covariance condition (c), extended to $X \in \mathfrak{g}_{\mathbb{C}}$, and the adjoint lemma, Lemma 21.8,

$$k_\theta \cdot (E \cdot v) = (\mathrm{Ad}(k_\theta)E) \cdot (k_\theta \cdot v) = e^{2i\theta}E \cdot (e^{in\theta}v) = e^{i(n+2)\theta}\, E \cdot v.$$

So $E \cdot v \in V(n+2)$. The argument for F is the same with $e^{-2i\theta}$, and gives $F \cdot v \in V(n-2)$. ∎

In this proof the shift by 2 comes from the covariance condition (c). For this connected group K, the shift also follows from the derivative condition (b) and the bracket $[H', E] = 2E$. Indeed, this bracket and $H' \cdot v = nv$ show that H' multiplies $E \cdot v$ by $n+2$. The operator H' acts on each summand $V(m)$ of $V = \bigoplus_{m \in \mathbb{Z}} V(m)$ by m, so $E \cdot v \in V(n+2)$. The bracket $[H', F] = -2F$ gives $F \cdot v \in V(n-2)$ in the same way.

Let Ω be the element

$$\Omega = H'^2 + 2EF + 2FE$$

of the universal enveloping algebra $U(\mathfrak{g}_{\mathbb{C}})$. It is the Casimir element of the triple. By the prerequisite Lie Algebras course, it is central in $U(\mathfrak{g}_{\mathbb{C}})$. Substituting $EF = FE + H'$ gives

$$\Omega = H'^2 + 2H' + 4FE, \qquad \Omega \cdot v = n(n+2)v + 4F \cdot E \cdot v \quad (v \in V(n)). \tag{21.2}$$

By the shift proposition, Proposition 21.9, $F \cdot E \cdot v \in V(n)$. So Ω preserves each $V(n)$.

Example 21.10. The group $\mathrm{SL}(2, \mathbb{R})$ acts on $\mathbb{C}^2$ by matrix multiplication, and hence on $V_m = \mathrm{Sym}^m \mathbb{C}^2$, of dimension $m+1$. The orbit maps are polynomial in the matrix entries, so every vector is smooth. Since V_m is finite-dimensional, every vector is also K-finite. Hence V_m is its own $(\mathfrak{g}, K)$-module. The Lie algebra acts on products by the Leibniz rule. Put $u_\pm = (1, \pm i)^T$. Then

$$k_\theta u_+ = \begin{pmatrix} \cos\theta + i\sin\theta \\ -\sin\theta + i\cos\theta \end{pmatrix} = e^{i\theta}u_+, \qquad k_\theta u_- = e^{-i\theta}u_-.$$

Direct multiplication (Exercise 21.2) gives $Eu_- = u_+$, $Eu_+ = 0$, $Fu_+ = u_-$ and $Fu_- = 0$. Let $w_j = u_+^j u_-^{m-j}$ for $0 \le j \le m$. The group acts on each factor, so

$$k_\theta w_j = (e^{i\theta}u_+)^j (e^{-i\theta}u_-)^{m-j} = e^{i(2j-m)\theta} w_j.$$

Hence the K-types of V_m are $-m, -m+2, \dots, m$, each with multiplicity one. The Leibniz rule gives

$$E \cdot w_j = (m-j) w_{j+1} \quad (0 \le j < m), \qquad F \cdot w_j = j w_{j-1} \quad (0 < j \le m).$$

The endpoint actions are $Ew_m = 0$ and $Fw_0 = 0$, in agreement with the shift proposition, Proposition 21.9. Since $E \cdot w_m = 0$ and $w_m \in V(m)$, the Casimir formula (21.2) gives $\Omega \cdot w_m = m(m+2) w_m$. The K-types are the familiar $\mathfrak{sl}_2$-weights, read with respect to the compact element H' rather than the diagonal element H.

If Π is unitary and Ω acts on $\mathcal{H}_K^\infty$ by a scalar λ, the skew-symmetry (F3) turns the Casimir formula (21.2) into formulas for $\|d\Pi(E)v\|^2$ and $\|d\Pi(F)v\|^2$. Exercise 21.4 derives them. Exercise 21.3 builds infinite-dimensional $(\mathfrak{g}, K)$-modules from the same raising and lowering pattern.

21.4 What the $(\mathfrak{g}, K)$-module does not record

A $(\mathfrak{g}, K)$-module has no topology. It does not record the norm or the action of elements of G outside K. We cite theorems that relate it back to representations. None of them is proved or used later in this lecture.

Admissibility. Let G be a connected linear reductive group with maximal compact subgroup K (Lecture 20). The following two statements are in Knapp RT, Chapter VIII.

(i) If a strongly continuous Hilbert representation Π is admissible and $\Pi|_K$ is unitary, then every K-finite vector is smooth, so $\mathcal{H}_K^\infty = \mathcal{H}_K$.

(ii) (Harish-Chandra) Every irreducible unitary representation of G is admissible.

The circle example, Example 21.4, shows the conclusion of the smoothness statement (i) directly in the case $G = K$.

Globalization. Passing from a $(\mathfrak{g}, K)$-module to a continuous group representation is called globalization. A Harish-Chandra module is an admissible $(\mathfrak{g}, K)$-module finitely generated over $U(\mathfrak{g}_{\mathbb{C}})$. A positive definite Hermitian form on a Harish-Chandra module is invariant if the K-action is unitary and the $\mathfrak{g}$-action is skew-symmetric for real $X \in \mathfrak{g}$.

For connected linear reductive G with maximal compact K, an irreducible Harish-Chandra module with an invariant positive definite Hermitian form has an irreducible unitary globalization. It is unique up to unitary equivalence. This theorem is cited from P. Etingof's MIT 18.757 notes. The globalization is Theorem 7.5, printed p. 38, and its uniqueness is Proposition 7.1, printed p. 37. The theorem supplies an action of G on a Hilbert space completing the module, together with its strong continuity. A completion of the vector space alone supplies neither.

One $(\mathfrak{g}, K)$-module can arise from completions of different kinds. Take $G = K = \mathrm{SO}(2)$ as in the circle example, Example 21.4. Translation also acts continuously on the Banach

space $C(\mathbb{R}/2\pi\mathbb{Z})$ with the supremum norm, which is not a Hilbert space. With the same definitions, the smooth K-finite vectors of $C(\mathbb{R}/2\pi\mathbb{Z})$ are again the trigonometric polynomials. So $L^2(\mathbb{R}/2\pi\mathbb{Z})$ and $C(\mathbb{R}/2\pi\mathbb{Z})$ have the same $(\mathfrak{g}, K)$-module.

This common module has infinitely many distinct circle types and is not finitely generated over $U(\mathfrak{k}_{\mathbb{C}})$. Thus this example is outside the irreducible Harish-Chandra theorem. The more general Casselman–Wallach globalization theorem belongs to further study.

Integration. The identities among $d\Pi(E)$, $d\Pi(F)$ and $d\Pi(H')$ were valid because Π was given. Formal commutation relations among unbounded operators do not by themselves produce a group representation. For the modules of Exercise 21.3, the question of which ones come from unitary representations is taken up in Lecture 22 and Lecture 23.

21.5 Self-checks

Check 1. In Step 3 of the proof of the $(\mathfrak{g}, K)$-module proposition, Proposition 21.6, why must all of W_v consist of smooth vectors?

Answer. The space U is spanned by vectors $d\Pi(X_i)w_j$ with $w_j \in W_v$. These are defined only for smooth w_j. Step 1 shows that each $\Pi(k)v$ is smooth.

Check 2. What are the K-types of Ad on $\mathfrak{g}_{\mathbb{C}}$? Which V_m has the same K-types?

Answer. By the adjoint lemma, Lemma 21.8, the elements E, H' and F have K-types 2, 0, -2, each with multiplicity one. The module V_2 has the same K-types.

Check 3. Could the eigenvalues of $H = \operatorname{diag}(1, -1)$ serve as K-type labels?

Answer. No. The derivative condition (b) relates the K-action only to elements of $\mathfrak{k}_{\mathbb{C}} = \mathbb{C}H'$. Since $H \notin \mathfrak{k}_{\mathbb{C}}$, the action of H is not determined by the K-action. In the V_m example, Example 21.10, the element $H = E + F$ moves $w_j \in V(n)$, where $n = 2j - m$, into $V(n+2) \oplus V(n-2)$.

Summary

Let K be a compact subgroup of G. For a strongly continuous representation of G on a Hilbert space, the smooth K-finite vectors carry an action of $\mathfrak{g}$ and of K. The derivative condition (b) relates the two actions on $\mathfrak{k}$. The covariance condition (c) describes how the K-action transforms the $\mathfrak{g}$-action. The proof that $\mathfrak{g}$ preserves K-finiteness uses the covariance condition.

For the circle, the K-finite vectors are the trigonometric polynomials. For $\mathrm{SL}(2, \mathbb{R})$ with $K = \mathrm{SO}(2)$, the covariance condition (c) shows that E and F shift the circle K-types by 2 and -2, respectively. The derivative condition (b) shows that H' acts on the K-type n by multiplication by n. For this connected K, the brackets $[H', E] = 2E$ and $[H', F] = -2F$ then give the same shifts.

For connected linear reductive G with maximal compact subgroup K, three results are cited. Irreducible unitary representations are admissible. In an admissible strongly continuous

Hilbert representation Π with $\Pi|_K$ unitary, K-finite vectors are smooth. An irreducible Harish-Chandra module with an invariant positive definite Hermitian form has an irreducible unitary globalization, unique up to unitary equivalence.

Reading

- Knapp RT, Sections III.3–4: smooth vectors and the facts (F1)–(F3).
- Knapp RT, Chapter VIII: admissible representations and Harish-Chandra's admissibility theorem.
- P. Etingof, MIT 18.757 *Representations of Lie Groups*, lecture notes, Lectures 2 and 5–7: a second account of this material.
- Etingof, Lecture 7, Theorem 7.5 and Proposition 7.1: the unitary globalization result stated here and its uniqueness. For further study, Wallach, *Real Reductive Groups II*, develops Casselman–Wallach globalization.

Lecture 22

The principal series of $\mathrm{SL}(2,\mathbb{R})$

Objectives. *This lecture builds the principal series of $G = \mathrm{SL}(2,\mathbb{R})$. The principal series is induced from characters of the upper triangular subgroup. We realise it on functions on a circle. We prove unitarity when the parameter is purely imaginary. Finally we compute how the Lie algebra moves vectors between K-types.*

By the end of this lecture you will be able to:

- State the inducing data (ε, ν) and the role of the modular function.
- Pass between induced functions, homogeneous functions on the plane and the circle model.
- Prove that $\Pi_{\varepsilon,\nu}$ is unitary for $\nu \in i\mathbb{R}$.
- Compute the ladder of K-types and locate the reducibility points.

22.1 Inducing data

Throughout, $G = \mathrm{SL}(2,\mathbb{R})$. We use the matrices

$$k_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad a_t = \begin{pmatrix} e^{t} & 0 \\ 0 & e^{-t} \end{pmatrix}, \qquad n_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}.$$

Let $K = \{k_\theta\}$, $A = \{a_t\}$, $N = \{n_x\}$ and $M = \{\pm I\}$. The group M is the centre of G.

Definition 22.1. The *minimal parabolic subgroup* of G is the group $P = MAN$ of upper triangular matrices in G.

Each $p \in P$ has a unique form $p = m a_t n_x$. Its diagonal determines m and t; its upper-right entry then determines x. The subgroup N is normal in P, so $m a_t n_x \mapsto m a_t$ is a homomorphism. Note that $K \cap P = M$.

The tangent algebra of P has basis $H = \mathrm{diag}(1,-1)$ and $E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Direct multiplication gives

$$\mathrm{Ad}(a_t)H = H, \qquad \mathrm{Ad}(a_t)E = e^{2t}E, \qquad \mathrm{Ad}(n_x)H = H - 2xE, \qquad \mathrm{Ad}(n_x)E = E,$$

and $\mathrm{Ad}(-I) = \mathrm{id}$. Since Ad and det are multiplicative, these formulas give $|\det \mathrm{Ad}(m a_t n_x)| = e^{2t}$. We use the modular convention of Lecture 16, namely

$$\int f(xg)\,dx = \Delta(g)^{-1} \int f(x)\,dx.$$

With this convention $\Delta(g) = |\det \mathrm{Ad}(g)|^{-1}$. Therefore

$$\Delta_P(ma_tn_x) = e^{-2t}. \tag{22.1}$$

The group G itself is unimodular because it is connected with semisimple Lie algebra, by Lecture 16.

Definition 22.2. Let $\varepsilon \in \{0,1\}$ and $\nu \in \mathbb{C}$. The *inducing character* $\chi_{\varepsilon,\nu} \colon P \to \mathbb{C}^\times$ is

$$\chi_{\varepsilon,\nu}(ma_tn_x) = \sigma_\varepsilon(m)\, e^{\nu t}, \qquad \sigma_\varepsilon(\pm I) = (\pm 1)^\varepsilon.$$

It is a continuous character, since it factors through $P \to MA$. It is unitary exactly when $\nu \in i\mathbb{R}$.

Definition 22.3. The space $I_{\varepsilon,\nu}$ consists of the continuous $f \colon G \to \mathbb{C}$ with

$$f(gp) = \Delta_P(p)^{1/2}\chi_{\varepsilon,\nu}(p)^{-1} f(g) \qquad (g \in G,\ p \in P).$$

For $g \in G$, the operator $\Pi_{\varepsilon,\nu}(g)$ on $I_{\varepsilon,\nu}$ is given by $(\Pi_{\varepsilon,\nu}(g)f)(h) = f(g^{-1}h)$ for $h \in G$.

By the modular formula (22.1), the defining condition reads

$$f(g\, ma_tn_x) = \sigma_\varepsilon(m)\, e^{-(\nu+1)t} f(g). \tag{22.2}$$

Left and right translations commute, so $\Pi_{\varepsilon,\nu}(g)$ preserves $I_{\varepsilon,\nu}$. Also $\Pi(g_1g_2) = \Pi(g_1)\Pi(g_2)$. We drop subscripts when (ε,ν) is fixed.

Remark 22.4. The factor $\Delta_P^{1/2}$ gives another form of the normalisation in Lecture 17. In Lecture 17 the square root appears in the action.

Let q be the rho-function of Lecture 17. It is a positive continuous function on G with $q(gp) = \Delta_P(p)q(g)$ for $g \in G$ and $p \in P$. Let f satisfy the unitary covariance condition of Lecture 17, and set $F(g) = \sqrt{q(g)}f(g)$. By the transformation law of q, the square root moves into the covariance condition. The action becomes plain left translation. For $\nu \in i\mathbb{R}$ the resulting condition is

$$|F(gp)|^2 = \Delta_P(p)|F(g)|^2.$$

This condition allows integration over G/P. Below we give the circle integral and prove its invariance directly. Nothing else from Lecture 17 is used.

The shift $e^{-t} = \Delta_P(a_t)^{1/2}$ is often written $e^{-\rho(tH)}$. Here ρ is the linear functional on $\mathbb{R}H$ with $\rho(H) = 1$, half the eigenvalue 2 of $\mathrm{ad}\, H$ on E.

22.2 The plane model and the circle model

Let $\mathbf{e}_1 = (1,0)^{\mathsf{T}}$ and $u_\theta = (\cos\theta, \sin\theta)^{\mathsf{T}} = k_\theta \mathbf{e}_1$.

Lemma 22.5. *The map $g \mapsto g\mathbf{e}_1$ induces a homeomorphism $G/N \to \mathbb{R}^2 \setminus \{0\}$.*

Proof. If $g\mathbf{e}_1 = \mathbf{e}_1$, the first column of g is $\mathbf{e}_1$, and $\det g = 1$ forces $g \in N$. So the fibres of $g \mapsto g\mathbf{e}_1$ are the cosets gN. The map is onto, with continuous section

$$s(v) = \begin{pmatrix} v_1 & -v_2/|v|^2 \\ v_2 & v_1/|v|^2 \end{pmatrix}, \qquad s(v)\mathbf{e}_1 = v, \qquad \det s(v) = 1.$$

Thus $v \mapsto s(v)N$ is a continuous inverse. ∎

Proposition 22.6. *The rule $F(g\mathbf{e}_1) = f(g)$ is a bijection from $I_{\varepsilon,\nu}$ onto the continuous functions F on $\mathbb{R}^2 \setminus \{0\}$ satisfying*

$$F(cv) = \operatorname{sgn}(c)^{\varepsilon}\,|c|^{-(\nu+1)}F(v) \qquad (c \in \mathbb{R}^{\times}). \tag{22.3}$$

Under it, $\Pi_{\varepsilon,\nu}(g)$ becomes $F \mapsto F(g^{-1}\cdot)$.

Proof. Taking $p = n_x$ in the covariance law (22.2) shows $f(gn_x) = f(g)$. By the plane identification (Lemma 22.5), F is well defined, and $F = f \circ s$ is continuous. For $c = \pm e^t$ we have $g(\pm a_t)\mathbf{e}_1 = c\,g\mathbf{e}_1$. So the covariance law with $p = \pm a_t$ gives the homogeneity rule (22.3).

Conversely, given F, put $f(g) = F(g\mathbf{e}_1)$. Since $g\,ma_tn_x\mathbf{e}_1 = \pm e^t g\mathbf{e}_1$ for $m = \pm I$, the homogeneity rule gives the covariance law. Finally, $(\Pi(g)f)(h) = f(g^{-1}h) = F(g^{-1}h\mathbf{e}_1)$ for $h \in G$. So $\Pi(g)$ becomes $F \mapsto F(g^{-1}\cdot)$. ∎

A homogeneous function F is determined by its values on the unit circle. Put $\varphi(\theta) = F(u_\theta) = f(k_\theta)$. Then

$$F(ru_\theta) = r^{-(\nu+1)}\varphi(\theta) \quad (r > 0), \qquad \varphi(\theta+\pi) = (-1)^{\varepsilon}\varphi(\theta).$$

The second rule is the homogeneity rule (22.3) with $c = -1$. Let C_ε be the space of continuous 2π-periodic functions with this parity. For every ν, restriction to K identifies $I_{\varepsilon,\nu}$ with C_ε. This identification is the *compact picture*. In the compact picture the space C_ε does not depend on ν.

Definition 22.7. For $h \in G$, the map τ_h of $\mathbb{R}/2\pi\mathbb{Z}$ is defined by $hu_\theta = |hu_\theta|\,u_{\tau_h(\theta)}$.

Apply the plane model (Proposition 22.6) with $g^{-1}u_\theta = |g^{-1}u_\theta|\,u_{\tau_{g^{-1}}(\theta)}$. This gives the circle model:

$$(\Pi_{\varepsilon,\nu}(g)\varphi)(\theta) = |g^{-1}u_\theta|^{-(\nu+1)}\,\varphi\big(\tau_{g^{-1}}(\theta)\big). \tag{22.4}$$

The parameter ν enters only through the scalar factor.

Example 22.8. For rotations $k_\alpha^{-1}u_\theta = u_{\theta-\alpha}$, so

$$(\Pi(k_\alpha)\varphi)(\theta) = \varphi(\theta-\alpha), \qquad \Pi(-I)\varphi = (-1)^{\varepsilon}\varphi.$$

The second formula uses $-I = k_\pi$. Thus the centre M acts by σ_ε for every ν. We call ε the central parity.

For the diagonal subgroup, $a_s^{-1}u_\theta = (e^{-s}\cos\theta,\ e^{s}\sin\theta)^{\mathsf{T}}$. Hence

$$r_s(\theta)^2 := |a_s^{-1}u_\theta|^2 = e^{-2s}\cos^2\theta + e^{2s}\sin^2\theta, \qquad \tan\tau_{a_s^{-1}}(\theta) = e^{2s}\tan\theta.$$

When $\varepsilon = 0$ the constant function $e_0 = 1$ lies in C_0, and $\Pi_{0,\nu}(a_s)e_0 = r_s^{-(\nu+1)}$.

22.3 Unitarity on the imaginary axis

Let $\mathcal{H}_\varepsilon$ be the closure of C_ε in $L^2(\mathbb{R}/2\pi\mathbb{Z})$, with inner product

$$\langle \varphi, \psi \rangle = \frac{1}{2\pi} \int_0^{2\pi} \overline{\varphi(\theta)}\, \psi(\theta)\, d\theta.$$

By Fourier analysis on the circle (Lecture 10), $\mathcal{H}_\varepsilon$ is the closed span of the functions $e_n(\theta) = e^{in\theta}$ with $n \equiv \varepsilon \pmod 2$. The key step towards unitarity is a change of variables on the circle.

Lemma 22.9. *Each τ_h is an orientation-preserving diffeomorphism. Moreover $\tau_{h_1 h_2} = \tau_{h_1} \circ \tau_{h_2}$ and*

$$\tau_h'(\theta) = |h u_\theta|^{-2}.$$

Proof. Linear maps commute with positive scaling, so $h_1 h_2 u_\theta$ is a positive multiple of $h_1 u_{\tau_{h_2}(\theta)}$. This gives the composition law, and $\tau_{h^{-1}}$ inverts τ_h. Locally $\tau_h(\theta)$ is a smooth branch of the argument of the nonvanishing smooth curve $h u_\theta$. So τ_h is smooth.

For the derivative, write $r = |h u_\theta|$ and $\tau = \tau_h(\theta)$. Differentiate $h u_\theta = r\, u_\tau$ in θ:

$$h u_\theta' = r' u_\tau + r \tau' u_\tau', \qquad u_\theta' = (-\sin\theta, \cos\theta)^{\mathsf{T}}.$$

Now compute the determinant of the pair $(h u_\theta, h u_\theta')$ in two ways. By multiplicativity of the determinant it equals $\det h \cdot \det(u_\theta, u_\theta') = 1$. By the first display it also equals

$$\det(r u_\tau,\ r' u_\tau + r\tau' u_\tau') = r^2 \tau' \det(u_\tau, u_\tau') = r^2 \tau'.$$

Hence $\tau' = r^{-2} > 0$. The inverse $\tau_{h^{-1}}$ is smooth by the same argument. ∎

As a check, take $h = a_s^{-1}$ and differentiate $\tan\tau = e^{2s}\tan\theta$ from the diagonal example (Example 22.8):

$$\tau' = \frac{e^{2s}\sec^2\theta}{1 + e^{4s}\tan^2\theta} = \frac{e^{2s}}{\cos^2\theta + e^{4s}\sin^2\theta} = \frac{1}{e^{-2s}\cos^2\theta + e^{2s}\sin^2\theta} = \frac{1}{r_s(\theta)^2}.$$

Proposition 22.10. *For $\nu \in \mathbb{C}$, $\varphi, \psi \in C_\varepsilon$ and $g \in G$,*

$$\langle \Pi_{\varepsilon,-\bar\nu}(g)\varphi,\ \Pi_{\varepsilon,\nu}(g)\psi \rangle = \langle \varphi, \psi \rangle.$$

Proof. Put $h = g^{-1}$, $r(\theta) = |h u_\theta|$ and $\theta' = \tau_h(\theta)$. Since $r > 0$, we have $\overline{r^{-(-\bar\nu+1)}} = r^{\nu-1}$. By the circle-model formula (22.4), the integrand is

$$\overline{(\Pi_{\varepsilon,-\bar\nu}(g)\varphi)(\theta)}\,(\Pi_{\varepsilon,\nu}(g)\psi)(\theta) = r^{\nu-1} r^{-\nu-1}\,\overline{\varphi(\theta')}\psi(\theta') = \overline{\varphi(\theta')}\psi(\theta')\, r(\theta)^{-2}.$$

By the Jacobian lemma (Lemma 22.9), $r(\theta)^{-2}\, d\theta = d\theta'$. As θ runs once around the circle, so does θ', in the same direction. So the integral equals $\int_0^{2\pi} \overline{\varphi(\theta')}\psi(\theta')\, d\theta'$. After division by 2π, this is $\langle \varphi, \psi \rangle$. ∎

Theorem 22.11. *Let $\nu \in i\mathbb{R}$. Each $\Pi_{\varepsilon,\nu}(g)$ extends uniquely from C_ε to a unitary operator on $\mathcal{H}_\varepsilon$. The map $g \mapsto \Pi_{\varepsilon,\nu}(g)$ is a strongly continuous unitary representation of G.*

Proof. For $\nu \in i\mathbb{R}$ we have $-\bar{\nu} = \nu$. So the pairing identity (Proposition 22.10) says that $\Pi(g)$ preserves norms on C_ε. As C_ε is dense, $\Pi(g)$ extends uniquely to an isometry of $\mathcal{H}_\varepsilon$.

The relation $\Pi(g_1 g_2) = \Pi(g_1)\Pi(g_2)$ holds on C_ε by the plane model (Proposition 22.6). By continuity it holds on $\mathcal{H}_\varepsilon$. So $\Pi(g^{-1})$ inverts $\Pi(g)$. An invertible isometry is unitary.

For strong continuity, fix $\varphi \in C_\varepsilon$ and $g_0 \in G$. By the circle-model formula (22.4), the function $(g,\theta) \mapsto (\Pi(g)\varphi)(\theta)$ is continuous. It is uniformly continuous on $U \times [0, 2\pi]$ for a compact neighbourhood U of g_0. Hence $\sup_\theta |(\Pi(g)\varphi - \Pi(g_0)\varphi)(\theta)| \to 0$ as $g \to g_0$. The L^2 norm is at most this supremum.

For general $\psi \in \mathcal{H}_\varepsilon$ and $\delta > 0$, choose $\varphi \in C_\varepsilon$ with $\|\psi - \varphi\| < \delta$. Since every $\Pi(g)$ is unitary,

$$\|\Pi(g)\psi - \Pi(g_0)\psi\| \le 2\delta + \|\Pi(g)\varphi - \Pi(g_0)\varphi\|,$$

and the right side is below 3δ for g near g_0. ∎

Example 22.12. Unitarity can fail off the imaginary axis. For example, take $\varepsilon = 0$ and $\nu \in \mathbb{R}$. By the diagonal example (Example 22.8),

$$\|\Pi_{0,\nu}(a_s)e_0\|^2 = \frac{1}{2\pi}\int_0^{2\pi} \frac{d\theta}{(A\cos^2\theta + B\sin^2\theta)^{\nu+1}}, \qquad A = e^{-2s},\ B = e^{2s}.$$

For $A, B > 0$, the substitution $u = \tan\theta$ on each quarter period gives

$$\int_0^{2\pi} \frac{d\theta}{A\cos^2\theta + B\sin^2\theta} = 4\int_0^{\infty} \frac{du}{A + Bu^2} = \frac{2\pi}{\sqrt{AB}}.$$

Applying $-\partial_A - \partial_B$ to this identity gives

$$\int_0^{2\pi} \frac{d\theta}{(A\cos^2\theta + B\sin^2\theta)^2} = \frac{\pi(A+B)}{(AB)^{3/2}}.$$

Here $AB = 1$. For $\nu = 0$ the norm is 1, as the unitarity theorem (Theorem 22.11) predicts. For $\nu = 1$ the squared norm is $\frac{1}{2}(e^{2s} + e^{-2s}) = \cosh 2s$, which exceeds 1 for $s \neq 0$. Exercise 22.6 gives the general skew-symmetry test.

For general $\nu \in \mathbb{C}$ the substitution in the proof of the pairing identity (Proposition 22.10) still gives

$$\|\Pi_{\varepsilon,\nu}(g)\varphi\|^2 \le \max_\theta |g^{-1}u_\theta|^{-2\operatorname{Re}\nu}\,\|\varphi\|^2.$$

So each $\Pi_{\varepsilon,\nu}(g)$ is bounded, locally uniformly in g. The proof of the unitarity theorem (Theorem 22.11) then gives a strongly continuous representation on $\mathcal{H}_\varepsilon$. In its density step, 2δ is replaced by $2C\delta$, where C bounds $\|\Pi_{\varepsilon,\nu}(g)\|$ on a compact neighbourhood of g_0. By the pairing identity (Proposition 22.10), the adjoint is $\Pi_{\varepsilon,\nu}(g)^* = \Pi_{\varepsilon,-\bar{\nu}}(g^{-1})$.

22.4 K-types and the Lie algebra action

By the rotation example (Example 22.8), $\Pi(k_\alpha)e_n = e^{-in\alpha}e_n$. So the character $k_\alpha \mapsto e^{-in\alpha}$ of K occurs exactly once when $n \equiv \varepsilon \pmod 2$, and not at all otherwise. This matches the centre: $k_\pi = -I$ acts on e_n by $(-1)^n = (-1)^\varepsilon$. Let

$$V_{\varepsilon,\nu} = \operatorname{span}\{e_n : n \equiv \varepsilon \pmod 2\} \qquad \text{(finite sums)}.$$

For $\nu \in i\mathbb{R}$ this is the space of K-finite vectors of Lecture 21.

For smooth $\varphi \in C_\varepsilon$ and $X \in \mathfrak{g} = \mathfrak{sl}(2,\mathbb{R})$ put

$$d\Pi(X)\varphi = \frac{d}{ds}\Big|_{s=0} \Pi_{\varepsilon,\nu}(\exp sX)\varphi.$$

By the circle-model formula (22.4), the function $(s,\theta) \mapsto (\Pi(\exp sX)\varphi)(\theta)$ is smooth. So the difference quotients converge uniformly in θ. Hence they converge in $\mathcal{H}_\varepsilon$.

More generally, the function $(g,\theta) \mapsto (\Pi(g)\varphi)(\theta)$ is jointly smooth. On compact sets in a group chart, the difference quotients for its derivatives converge uniformly in θ. Hence φ is a smooth vector in the Hilbert representation, in the sense used in Lecture 21.

For $\nu \in i\mathbb{R}$ this derivative agrees with the derivative on smooth vectors in Lecture 15. We extend $d\Pi$ complex linearly to $\mathfrak{g}_{\mathbb{C}}$.

Lemma 22.13. *For smooth $\varphi \in C_\varepsilon$ and $X \in \mathfrak{g}$, using the dot product of $\mathbb{R}^2$,*

$$(d\Pi(X)\varphi)(\theta) = (\nu+1)\,(u_\theta \cdot Xu_\theta)\,\varphi(\theta) - \det(u_\theta, Xu_\theta)\,\varphi'(\theta).$$

Proof. Let $v(s) = \exp(-sX)u_\theta = r(s)u_{\tau(s)}$. Then $r(0)=1$, $\tau(0)=\theta$ and $v'(0) = -Xu_\theta$. As in the proof of the Jacobian lemma (Lemma 22.9), $\det(v,v') = r^2\tau'$, so $\tau'(0) = -\det(u_\theta, Xu_\theta)$. Also $(r^2)' = 2\,v\cdot v'$, so $r'(0) = -u_\theta \cdot Xu_\theta$. Now differentiate $r(s)^{-(\nu+1)}\varphi(\tau(s))$ at $s=0$. The product rule with these values of $r'(0)$ and $\tau'(0)$ gives the stated formula. ∎

We use the basis

$$W = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

with $\exp(\alpha W) = k_\alpha$. We have $Wu_\theta = u_\theta'$, $Hu_\theta = (\cos\theta, -\sin\theta)^{\mathsf{T}}$ and $Qu_\theta = (\sin\theta, \cos\theta)^{\mathsf{T}}$. The derivative formula (Lemma 22.13) gives the table:

X	$u_\theta \cdot Xu_\theta$	$\det(u_\theta, Xu_\theta)$	$d\Pi(X)\varphi$
W	0	1	$-\varphi'$
H	$\cos 2\theta$	$-\sin 2\theta$	$(\nu+1)\cos 2\theta\,\varphi + \sin 2\theta\,\varphi'$
Q	$\sin 2\theta$	$\cos 2\theta$	$(\nu+1)\sin 2\theta\,\varphi - \cos 2\theta\,\varphi'$

Set $X_\pm = H \pm iQ \in \mathfrak{g}_{\mathbb{C}}$. Since $\sin 2\theta \mp i\cos 2\theta = \mp i e^{\pm 2i\theta}$,

$$d\Pi(X_\pm)\varphi = e^{\pm 2i\theta}\big((\nu+1)\varphi \mp i\varphi'\big).$$

The element W used here is the negative of the W in Lecture 21, so k_α here is $k_{-\alpha}$ there. In the notation of Lecture 21, $E = X_+/2$, $F = X_-/2$ and $H' = iW$. Thus the Fourier vector e_n has type n in that lecture's parametrisation, even though its character here is $e^{-in\alpha}$.

Proposition 22.14. *For $n \equiv \varepsilon \pmod 2$,*

$$d\Pi(W)e_n = -in\,e_n, \qquad d\Pi(X_+)e_n = (\nu+1+n)\,e_{n+2}, \qquad d\Pi(X_-)e_n = (\nu+1-n)\,e_{n-2}.$$

Proof. Insert $e_n' = in\,e_n$ into the table row for W and into the display for $d\Pi(X_\pm)$. For example, $(\nu+1) - i(in) = \nu+1+n$. ∎

The ladder formulas also show that $d\Pi$ respects brackets, for every $\nu \in \mathbb{C}$. On e_n the eigenvalues of $d\Pi(W)$ give

$$[d\Pi(W), d\Pi(X_\pm)]e_n = \big(-i(n\pm 2) + in\big)(\nu+1\pm n)\,e_{n\pm 2} = \mp 2i\,d\Pi(X_\pm)e_n.$$

The products of the two ladder operators are

$$d\Pi(X_\mp)d\Pi(X_\pm)e_n = (\nu+1\pm n)(\nu-1\mp n)\,e_n = \big(\nu^2 - (1\pm n)^2\big)e_n.$$

Subtracting the two cases gives

$$[d\Pi(X_+), d\Pi(X_-)]e_n = \big((1+n)^2 - (1-n)^2\big)e_n = 4n\,e_n = 4i\,d\Pi(W)e_n.$$

In $\mathfrak{g}_{\mathbb{C}}$ the brackets $[W,H] = 2Q$, $[W,Q] = -2H$ and $[H,Q] = -2W$ give $[W, X_\pm] = \mp 2iX_\pm$ and $[X_+, X_-] = 4iW$ (Exercise 22.4). The elements W, X_+ and X_- span $\mathfrak{g}_{\mathbb{C}}$, and $d\Pi$ is complex linear. Hence $d\Pi([Y,Z]) = [d\Pi(Y), d\Pi(Z)]$ for all $Y, Z \in \mathfrak{g}_{\mathbb{C}}$. So $d\Pi$ is a representation of $\mathfrak{g}_{\mathbb{C}}$ on $V_{\varepsilon,\nu}$ for every $\nu \in \mathbb{C}$.

Proposition 22.15. *A subspace $U \subseteq V_{\varepsilon,\nu}$ is stable under $\Pi(K)$ and $d\Pi(\mathfrak{g})$ exactly when $U = \operatorname{span}\{e_n : n \in S\}$ for a set S such that the following two conditions hold. If $n \in S$ and $\nu+1+n \neq 0$, then $n+2 \in S$. If $n \in S$ and $\nu+1-n \neq 0$, then $n-2 \in S$. Hence the $(\mathfrak{g}, K)$-module $V_{\varepsilon,\nu}$ is irreducible if and only if ν is not an integer with $\nu \equiv \varepsilon + 1 \pmod 2$.*

Proof. Let U be K-stable and let $u = \sum c_n e_n \in U$ be a finite sum. The e_n with $c_n \neq 0$ span a finite-dimensional K-stable space F. The continuous curve $\alpha \mapsto e^{in\alpha}\Pi(k_\alpha)u$ lies in the closed subspace $U \cap F$ of F. Hence

$$\frac{1}{2\pi}\int_0^{2\pi} e^{in\alpha}\,\Pi(k_\alpha)u\,d\alpha = c_n e_n$$

lies in U. Thus U is spanned by the e_n it contains. Since H and Q are linear combinations of $X_\pm$, and $d\Pi(W)$ preserves every span of the e_n, stability under $d\Pi(\mathfrak{g})$ means stability under $d\Pi(X_\pm)$. By the ladder formulas (Proposition 22.14), this is the stated property of S.

If no coefficient $\nu+1\pm n$ vanishes, every nonempty admissible S contains all $n \equiv \varepsilon$. If $\nu+1+n_0 = 0$, then $S = \{n \le n_0\}$ is admissible and proper. If $\nu+1-n_0 = 0$, use $S = \{n \ge n_0\}$. Finally, $\nu = -(n_0+1)$ or $\nu = n_0 - 1$ with $n_0 \equiv \varepsilon$ means exactly that $\nu \in \mathbb{Z}$ and $\nu \equiv \varepsilon + 1 \pmod 2$. ∎

The passage from $(\mathfrak{g}, K)$-submodules to closed invariant subspaces is Harish-Chandra's correspondence for admissible representations. We do not prove it. We cite the outcome.

Theorem 22.16 (cited). *Let $\nu \in \mathbb{C}$. The representation $\Pi_{\varepsilon,\nu}$ on $\mathcal{H}_\varepsilon$ has no closed invariant subspace other than 0 and $\mathcal{H}_\varepsilon$ exactly when $\nu \notin \varepsilon + 1 + 2\mathbb{Z}$. In that case $V_{\varepsilon,\nu} \cong V_{\varepsilon,-\nu}$ as*

> $(\mathfrak{g}, K)$*-modules. For* $\nu \in i\mathbb{R}$*, the representations* $\Pi_{\varepsilon,\nu}$ *and* $\Pi_{\varepsilon,-\nu}$ *are unitarily equivalent, and all* $\Pi_{\varepsilon,\nu}$ *are irreducible except* $\Pi_{1,0}$*. The space* $\mathcal{H}_1$ *of* $\Pi_{1,0}$ *is the orthogonal sum of the closed spans of* $\{e_n : n > 0\}$ *and* $\{e_n : n < 0\}$*, and each summand is irreducible. See Knapp,* Representation Theory of Semisimple Groups*, §II.5, and Etingof's 18.757 notes, Lecture 19.*

The pieces at the reducibility points return in Lecture 23.

The reducible module at $(\varepsilon, \nu) = (1, 0)$

The simplest reducibility point is $(\varepsilon, \nu) = (1, 0)$. The Fourier labels are the odd integers, and the ladder formulas become

$$d\Pi(X_+)e_n = (1+n)e_{n+2}, \qquad d\Pi(X_-)e_n = (1-n)e_{n-2}.$$

At $n = -1$, the upward coefficient vanishes. At $n = 1$, the downward coefficient vanishes. Thus the space $V_{1,0}$ is the direct sum

$$V_{1,0} = \operatorname{span}\{e_1, e_3, e_5, \ldots\} \;\oplus\; \operatorname{span}\{e_{-1}, e_{-3}, e_{-5}, \ldots\}.$$

Each summand is preserved by both ladder operators and by K. Within either chain, the remaining connecting coefficients are nonzero. The weight-projection argument in the proof of the submodule criterion (Proposition 22.15) shows that each chain is algebraically irreducible.

The cited irreducibility theorem (Theorem 22.16) gives the two closed irreducible summands. Their group invariance is part of that cited theorem, beyond the algebraic calculation above. Lecture 23 identifies them as limits of discrete series and explains their norms. The full odd ladder breaks at the gap between -1 and 1.

For comparison, $(\varepsilon, \nu) = (0, 0)$ has only even labels. Neither equation $1 + n = 0$ nor $1 - n = 0$ has an even solution. Its algebraic module is therefore irreducible.

We compare the two modules through the Casimir element $\Omega = H^2 + Q^2 - W^2$ of $U(\mathfrak{g}_{\mathbb{C}})$. Exercise 22.4 shows $\Omega = X_+X_- - 2iW - W^2$ in $U(\mathfrak{g}_{\mathbb{C}})$. Since $d\Pi$ is a representation of $\mathfrak{g}_{\mathbb{C}}$, it extends to $U(\mathfrak{g}_{\mathbb{C}})$, and this identity holds for the operators. By the ladder formulas (Proposition 22.14), the three terms act on e_n by $\nu^2 - (1-n)^2$, $-2n$ and n^2. Hence Ω acts on $V_{\varepsilon,\nu}$ by the scalar

$$\nu^2 - (1-n)^2 - 2n + n^2 = \nu^2 - 1.$$

At $(\varepsilon, \nu) = (1, 0)$ and at $(0, 0)$ this scalar is -1. The two modules nevertheless have different central parity and different reducibility. A Casimir value alone is not a classification label.

22.5 Self-checks

1. What is $\Pi_{\varepsilon,\nu}(-I)$?
   Answer. $(-I)^{-1}u_\theta = u_{\theta+\pi}$ has length 1. So $(\Pi(-I)\varphi)(\theta) = \varphi(\theta+\pi) = (-1)^\varepsilon\varphi(\theta)$. The centre acts by σ_ε for every ν.

2. For which (ε,ν) is e_0 fixed by G?
   Answer. We need $e_0 \in C_\varepsilon$, so $\varepsilon = 0$. We also need $d\Pi(X_+)e_0 = (\nu+1)e_2 = 0$, so $\nu = -1$. Conversely, for $\nu = -1$ the factor in the circle-model formula (22.4) is 1, so $\Pi(g)e_0 = e_0$.
3. Why is $V_{1,0}$ reducible?
   Answer. Here $\nu + 1 + n = 0$ at $n = -1$ and $\nu + 1 - n = 0$ at $n = 1$. So $d\Pi(X_+)e_{-1} = 0$ and $d\Pi(X_-)e_1 = 0$. The sets $\{n \geq 1\}$ and $\{n \leq -1\}$ are admissible, and $V_{1,0}$ is the direct sum of their spans.

22.6 Summary

The principal series $\Pi_{\varepsilon,\nu}$ is induced from the character $\sigma_\varepsilon \otimes e^{\nu t}$ of $P = MAN$, shifted by $\Delta_P^{1/2}$. Restriction to K identifies every $I_{\varepsilon,\nu}$ with the continuous parity-ε functions on the circle. The group acts on these functions by the circle-model formula (22.4). The identity $\tau'_h = |hu_\theta|^{-2}$ makes the pairing of ν with $-\bar{\nu}$ invariant. Hence, for $\nu \in i\mathbb{R}$, the operators $\Pi_{\varepsilon,\nu}(g)$ extend to a strongly continuous unitary representation on $\mathcal{H}_\varepsilon$.

Each K-type occurring in $V_{\varepsilon,\nu}$ occurs once and is spanned by one e_n with $n \equiv \varepsilon \pmod 2$. The elements $X_\pm$ move n by ± 2 with coefficients $\nu + 1 \pm n$. We proved that the $(\mathfrak{g}, K)$-module $V_{\varepsilon,\nu}$ is reducible exactly at integers ν of parity opposite to ε. The cited irreducibility theorem gives the same criterion for closed invariant subspaces of $\mathcal{H}_\varepsilon$.

22.7 Further reading

- A. W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples* (Princeton, 2001), §II.5 for $\mathrm{SL}(2,\mathbb{R})$, and §§VII.1–2 for induction from parabolic subgroups in general.
- P. Etingof, lecture notes for MIT 18.757, *Representations of Lie Groups*, Lectures 9 and 19.

Lecture 23

Discrete series and the rank one unitary picture

Objectives. *We realise the holomorphic discrete series of* $\mathrm{SL}(2,\mathbb{R})$ *on weighted spaces of holomorphic functions on the disk. We then compare it with the finite-dimensional and principal series modules. Finally we state the full list of irreducible unitary representations.*

By the end of this lecture you will be able to:

- Prove that Π_n is an irreducible unitary representation on the weighted disk space $\mathcal{A}_n$ for each integer $n \geq 2$.
- Compute its K-types $n, n+2, \dots$ and the action of H', E, F.
- Decide which lowest weights λ occur in irreducible unitary representations of $\mathrm{SL}(2,\mathbb{R})$, and when such a representation factors through $\mathrm{PSL}(2,\mathbb{R})$.
- State Bargmann's classification and derive the positivity bounds behind it.

23.1 The disk model

Let $G = \mathrm{SL}(2,\mathbb{R})$. We keep the conventions of Lecture 21:

$$W = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad k_\theta = \exp(\theta W) = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}, \qquad K = \{k_\theta\}.$$

Thus $k_{2\pi} = I$ and $k_\pi = -I$. With $H_0 = \mathrm{diag}(1,-1)$ and $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ put

$$H' = -iW, \qquad E = \tfrac{1}{2}(H_0 + iS), \qquad F = \tfrac{1}{2}(H_0 - iS) = \overline{E}.$$

Then $[H', E] = 2E$, $[H', F] = -2F$ and $[E, F] = H'$. In a representation Π, a vector v has *K-type k* if $\Pi(k_\theta)v = e^{ik\theta}v$ for all θ. If such a v is smooth, then $d\Pi(H')v = kv$.

The disk carries an action of

$$\mathrm{SU}(1,1) = \left\{ \begin{pmatrix} \alpha & \beta \\ \bar\beta & \bar\alpha \end{pmatrix} : |\alpha|^2 - |\beta|^2 = 1 \right\}.$$

Fix the Cayley matrix $\gamma = \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$ and set $\varphi(g) = \gamma g \gamma^{-1}$. For $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G$, multiplying out gives

$$\varphi(g) = \begin{pmatrix} \alpha & \beta \\ \bar\beta & \bar\alpha \end{pmatrix}, \qquad \alpha = \tfrac{1}{2}\big(a + d + i(c - b)\big), \qquad \beta = \tfrac{1}{2}\big(a - d + i(b + c)\big). \tag{23.1}$$

Then $|\alpha|^2 - |\beta|^2 = ad - bc = 1$. Solving (23.1) for a, b, c, d shows that φ is an isomorphism of Lie groups from G onto $\mathrm{SU}(1,1)$. Also

$$\varphi(k_\theta) = \mathrm{diag}(e^{-i\theta}, e^{i\theta}), \qquad \varphi(-I) = -I. \tag{23.2}$$

Conjugation by γ is complex linear on all 2×2 matrices. A direct multiplication gives

$$\gamma H' \gamma^{-1} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \gamma E \gamma^{-1} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad \gamma F \gamma^{-1} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}. \tag{23.3}$$

A function on an open subset of the complex plane is *holomorphic* if it is complex differentiable at every point.

We import three results from elementary complex analysis. The first is the local power-series theorem. A holomorphic function on a disk has a Taylor series converging uniformly on every smaller closed disk. The second concerns power series. A power series about the centre of a disk that converges at every point of the disk defines a holomorphic function there. The third follows from the Cauchy–Riemann equations. The real Jacobian of a holomorphic map is the squared modulus of its complex derivative. These are the analytic inputs used below. See MIT 18.112, the lectures on Cauchy's formula, power series and the Cauchy–Riemann equations.

Let $D = \{z \in \mathbb{C} : |z| < 1\}$ with closure $\overline{D}$. For $h \in \mathrm{SU}(1,1)$ as above and $z \in \overline{D}$ put

$$h \cdot z = \frac{\alpha z + \beta}{\bar\beta z + \bar\alpha}, \qquad j(h, z) = \bar\beta z + \bar\alpha.$$

Lemma 23.1. *Let $h, h_1, h_2 \in \mathrm{SU}(1,1)$ and $z \in \overline{D}$.*

(a) $|j(h,z)| \geq |\alpha| - |\beta| > 0$ *and* $|j(h,z)|^2 - |\alpha z + \beta|^2 = 1 - |z|^2$.

(b) h *maps* D *into* D *and* $\overline{D}$ *into* $\overline{D}$, $(h_1 h_2) \cdot z = h_1 \cdot (h_2 \cdot z)$ *and* $j(h_1 h_2, z) = j(h_1, h_2 \cdot z)\, j(h_2, z)$.

(c) $d(h \cdot z)/dz = j(h,z)^{-2}$.

Proof. (a) Use $|z| \leq 1$ and $|\alpha|^2 = 1 + |\beta|^2$. Expanding both squares, the cross terms cancel and leave $(|\alpha|^2 - |\beta|^2)(1 - |z|^2)$.

(b) Dividing (a) by $|j(h,z)|^2$ gives

$$1 - |h \cdot z|^2 = \frac{1 - |z|^2}{|j(h,z)|^2}. \tag{23.4}$$

Hence $|h \cdot z| < 1$ whenever $|z| < 1$, so h maps D into D. Also, $1 - |h \cdot z|^2 \geq 0$ when $|z| \leq 1$, with equality exactly when $|z| = 1$. So h maps $\overline{D}$ into $\overline{D}$.

Next,

$$j(h_1, h_2 \cdot z)\, j(h_2, z) = \bar\beta_1(\alpha_2 z + \beta_2) + \bar\alpha_1(\bar\beta_2 z + \bar\alpha_2).$$

The right side is the bottom row of $h_1 h_2$ applied to $(z, 1)^T$, so it equals $j(h_1 h_2, z)$. In the same way, the top row of $h_1 h_2$ applied to $(z, 1)^T$ is the numerator of $(h_1 h_2) \cdot z$. It equals $j(h_2, z)$ times the numerator of $h_1 \cdot (h_2 \cdot z)$. Dividing the two rows cancels $j(h_2, z)$ and gives $(h_1 h_2) \cdot z = h_1 \cdot (h_2 \cdot z)$.

(c) By the quotient rule the numerator is $\alpha(\bar\beta z + \bar\alpha) - \bar\beta(\alpha z + \beta) = 1$. ∎

Definition 23.2. Let $n \geq 2$ be an integer and $dA = dx\,dy$. For f holomorphic on D put

$$\|f\|_n^2 = \frac{n-1}{\pi} \lim_{r \to 1} \int_{|z| \leq r} |f(z)|^2 (1-|z|^2)^{n-2}\, dA(z) \in [0, \infty].$$

The *weighted disk space* $\mathcal{A}_n$ is the space of holomorphic f on D with $\|f\|_n < \infty$.

The limit exists because the integrand is nonnegative. For the same reason it equals the supremum of the integrals over compact regions in D with smooth boundary.

Lemma 23.3. *If $f(z) = \sum_{m \geq 0} a_m z^m$ is holomorphic on D, then*

$$\|f\|_n^2 = \sum_{m \geq 0} c_m |a_m|^2, \qquad c_m = \|z^m\|_n^2 = \frac{m!\,(n-1)!}{(n+m-1)!} = \binom{n+m-1}{m}^{-1}.$$

So $\mathcal{A}_n$ is a Hilbert space with orthonormal basis $e_m = c_m^{-1/2} z^m$. Moreover $\|g\|_n \leq \sup_D |g|$ for bounded holomorphic g.

Proof. On $|z| = s < 1$ the series converges uniformly. Orthogonality of the functions $e^{im\theta}$ gives

$$\int_0^{2\pi} |f(se^{i\theta})|^2 d\theta = 2\pi \sum_m |a_m|^2 s^{2m}.$$

Integration in polar coordinates then gives

$$\frac{n-1}{\pi} \int_{|z| \leq r} |f|^2 (1-|z|^2)^{n-2} dA = \sum_m |a_m|^2\, (n-1) \int_0^r 2s^{2m+1} (1-s^2)^{n-2} ds.$$

The terms are nonnegative and increase with r, so we may let $r \to 1$ termwise. With $t = s^2$ and repeated integration by parts,

$$(n-1) \int_0^1 t^m (1-t)^{n-2} dt = (n-1) \frac{m!\,(n-2)!}{(n+m-1)!} = c_m.$$

Conversely, let $(a_m)_{m \geq 0}$ be complex numbers with $C = \sum_m c_m |a_m|^2 < \infty$. Then $|a_m|^2 \leq C \binom{n+m-1}{m} \leq C(m+1)^{n-1}$, so $\sum a_m z^m$ converges on D. By the power-series import, its sum f is holomorphic on D. The formula above then gives $\|f\|_n^2 = C$. Hence $f \mapsto (c_m^{1/2} a_m)_m$ is an isometry of $\mathcal{A}_n$ onto ℓ^2. This gives completeness, the inner product by polarisation, and the basis. Finally $\|g\|_n^2 \leq \sup |g|^2\, \|1\|_n^2 = \sup |g|^2$, since $c_0 = 1$. ∎

Theorem 23.4. *For $h \in \mathrm{SU}(1,1)$ and $f \in \mathcal{A}_n$ define*

$$(\Pi_n(h) f)(z) = j(h^{-1}, z)^{-n}\, f(h^{-1} \cdot z). \tag{23.5}$$

Then $g \mapsto \Pi_n(\varphi(g))$ is a strongly continuous unitary representation of G on $\mathcal{A}_n$. We write $\Pi_n(g)$ for $\Pi_n(\varphi(g))$.

Proof. Homomorphism. Apply the cocycle identity of Lemma 23.1(b) to h_2^{-1} and h_1^{-1}:

$$(\Pi_n(h_1)\Pi_n(h_2) f)(z) = j(h_1^{-1}, z)^{-n}\, j(h_2^{-1}, h_1^{-1} \cdot z)^{-n} f(h_2^{-1} h_1^{-1} \cdot z) = (\Pi_n(h_1 h_2) f)(z).$$

Isometry. Substitute $z = h\cdot w$. The cocycle identity with $h^{-1}h = I$ gives $j(h^{-1}, h\cdot w) = j(h,w)^{-1}$. With the identity (23.4) for $1-|h\cdot w|^2$ and the derivative formula of Lemma 23.1(c) we get

$$\begin{aligned}&|(\Pi_n(h)f)(h\cdot w)|^2(1-|h\cdot w|^2)^{n-2}\left|\frac{d(h\cdot w)}{dw}\right|^2\\ &\quad= |j(h,w)|^{2n}|f(w)|^2\frac{(1-|w|^2)^{n-2}}{|j(h,w)|^{2n-4}}\frac{1}{|j(h,w)|^4}.\end{aligned}$$

The powers of $|j(h,w)|$ cancel, leaving $|f(w)|^2(1-|w|^2)^{n-2}$.

For a holomorphic map the real Jacobian determinant is $|d(h\cdot w)/dw|^2$, by the Cauchy–Riemann equations. So on each compact region $R \subset D$, the change of variables formula equates the weighted integral of $|\Pi_n(h)f|^2$ over $h\cdot R$ with that of $|f|^2$ over R.

The map $w \mapsto h\cdot w$ is a bijection of D with inverse $w \mapsto h^{-1}\cdot w$. So the regions $h\cdot R$ also run over all compact regions. Taking suprema gives $\|\Pi_n(h)f\|_n = \|f\|_n$. Thus $\Pi_n(h)$ is an isometry with inverse $\Pi_n(h^{-1})$, hence unitary.

Continuity. Let f, g be polynomials. By the lower bound for $|j(h,z)|$ in Lemma 23.1(a), the function $\overline{g(z)}(\Pi_n(h)f)(z)(1-|z|^2)^{n-2}$ is continuous on $\mathrm{SU}(1,1)\times\overline{D}$. Here $n \geq 2$ is used at the boundary. It is uniformly continuous on $U\times\overline{D}$ for a compact neighbourhood U of I. So $h \mapsto \langle g, \Pi_n(h)f\rangle_n$ is continuous. Polynomials are dense and $\|\Pi_n(h)\| = 1$, so an $\varepsilon/3$ argument extends this to all f, g. Then

$$\|\Pi_n(h)f - f\|_n^2 = 2\|f\|_n^2 - 2\,\mathrm{Re}\langle f, \Pi_n(h)f\rangle_n \to 0 \qquad (h \to I).$$

Unitarity moves this to every other point. ∎

Remark 23.5. Put $\delta = \mathrm{diag}(1,-1)$. Since $\bar\gamma = \gamma\delta$, the matrix $\bar\gamma$ in place of γ gives $\Pi_n^-(g) = \Pi_n(\delta g\delta^{-1})$. Conjugation by δ preserves G, although $\delta \notin G$. Since $\delta k_\theta \delta^{-1} = k_{-\theta}$, the K-types of Π_n^- are the negatives of those of Π_n. Which one is called holomorphic depends on the Cayley matrix and the sign of the K-angle. We label them by K-types.

23.2 K-types and Lie algebra operators

Proposition 23.6. *For all θ and $m \geq 0$,*

$$\Pi_n(k_\theta)z^m = e^{i(n+2m)\theta}z^m.$$

Hence the K-types of Π_n are $n, n+2, n+4, \dots$, each with multiplicity one, and the K-finite vectors are the polynomials.

Proof. By the formula (23.2), the element $h = \varphi(k_\theta)$ has $h^{-1} = \mathrm{diag}(e^{i\theta}, e^{-i\theta})$. So $h^{-1}\cdot z = e^{2i\theta}z$ and $j(h^{-1}, z) = e^{-i\theta}$. Now (23.5) gives $e^{in\theta}(e^{2i\theta}z)^m$.

For an integer k define

$$P_k f = \frac{1}{2\pi}\int_0^{2\pi} e^{-ik\theta}\,\Pi_n(k_\theta)f\,d\theta,$$

a Riemann integral of a continuous $\mathcal{A}_n$-valued function. Then $P_k e_m = e_m$ if $k = n + 2m$, and $P_k e_m = 0$ otherwise. Being bounded, P_k is the orthogonal projection onto $\mathbb{C}e_m$, or zero. Its range is the space of vectors of K-type k.

If v is K-finite, its orbit spans a finite-dimensional space U. Every $P_k v$ lies in U, as a limit of Riemann sums there. The nonzero $P_k v$ are orthogonal, so only finitely many occur. The basis expansion of v is therefore a polynomial. Conversely, every polynomial is a finite sum of K-types. ∎

Lemma 23.7. *Polynomials are smooth vectors for Π_n. For $Y = \begin{pmatrix} p & q \\ r & s \end{pmatrix} \in \mathfrak{su}(1,1)$ and a polynomial f,*

$$d\Pi_n(Y) f = n(rz + s) f + \bigl(rz^2 + (s - p)z - q\bigr) f'. \tag{23.6}$$

Proof. By the lower bound for $|j(h, z)|$ in Lemma 23.1(a), $(h, z) \mapsto (\Pi_n(h) f)(z)$ is smooth on $\mathrm{SU}(1,1) \times \overline{D}$. By Taylor's theorem on compact sets, difference quotients in h of every order converge uniformly on $\overline{D}$. By the supremum bound in Lemma 23.3, they converge in $\mathcal{A}_n$. So f is a smooth vector in the sense of Lecture 21. Moreover $d\Pi_n(Y) f$ is the pointwise t-derivative at 0 of $(\Pi_n(h_t) f)(z)$, where $h_t = \exp(tY)$. Since $h_t^{-1} = I - tY + O(t^2)$, uniformly in $z \in \overline{D}$,

$$j(h_t^{-1}, z) = 1 - t(rz + s) + O(t^2), \qquad h_t^{-1} \cdot z = \frac{(1 - tp)z - tq}{1 - t(rz + s)} = z + t\bigl(rz^2 + (s - p)z - q\bigr) + O(t^2).$$

So $j(h_t^{-1}, z)^{-n} = 1 + nt(rz + s) + O(t^2)$, and the product and chain rules give (23.6). ∎

Proposition 23.8. *Extend $d\Pi_n$ complex-linearly to $\mathfrak{g}_{\mathbb{C}}$. Then*

$$\begin{aligned} d\Pi_n(H') z^m &= (n + 2m) z^m, \\ d\Pi_n(E) z^m &= (n + m) z^{m+1}, \\ d\Pi_n(F) z^m &= -m\, z^{m-1} \quad (m \geq 1), \qquad d\Pi_n(F) 1 = 0. \end{aligned}$$

The Casimir element $\Omega = H'^2 + 2EF + 2FE$ acts on every polynomial by $n(n - 2)$.

Proof. The right side of (23.6) is complex linear in (p, q, r, s). So it also computes $d\Pi_n$ on the matrices in (23.3). For H' we have $p = -1$, $s = 1$, giving $nf + 2zf'$. For E we have $r = 1$, giving $nzf + z^2 f'$. For F we have $q = 1$, giving $-f'$. Apply these to z^m. Since $[E, F] = H'$, we have $\Omega = H'^2 + 2H' + 4FE$. On z^m its eigenvalue is

$$(n + 2m)^2 + 2(n + 2m) - 4(n + m)(m + 1) = n^2 - 2n.$$ ∎

For a unitary representation Π and $X \in \mathfrak{g}$, the operator $d\Pi(X)$ is skew-symmetric on smooth vectors. Extend $d\Pi$ complex-linearly to $\mathfrak{g}_{\mathbb{C}}$. The matrices H_0, S and W are real, and the inner product is antilinear in the first slot. Moving $d\Pi(E)$ to the second slot therefore negates the real generators H_0 and S. It also turns the coefficient i of S into $-i$, which produces $-d\Pi(F)$. For $H' = -iW$ the two sign changes cancel. Hence smooth vectors u and v satisfy

$$\langle d\Pi(E) u, v \rangle = -\langle u, d\Pi(F) v \rangle, \qquad \langle d\Pi(H') u, v \rangle = \langle u, d\Pi(H') v \rangle. \tag{23.7}$$

In the model, $u = z^m$ and $v = z^{m+1}$ give $(n+m)c_{m+1}$ on the left and $(m+1)c_m$ on the right. These agree by the formula for c_m in Lemma 23.3. Conversely, distinct K-types are orthogonal, so (23.7) forces $c_{m+1}/c_m = (m+1)/(n+m)$. Up to a scalar, the weighted norm is the only such inner product on polynomials.

Theorem 23.9. *Π_n is irreducible for every integer $n \geq 2$.*

Proof. Let $V \neq 0$ be a closed invariant subspace and $0 \neq v \in V$. Each P_k maps V into V, since its Riemann sums do and V is closed. As $P_{n+2m}v = \langle e_m, v\rangle e_m$, some e_m lies in V. For $X \in \mathfrak{g}$, $d\Pi_n(X)e_m$ is the norm limit of $t^{-1}(\Pi_n(\exp tX)e_m - e_m) \in V$, so it lies in V. Hence $d\Pi_n(E)e_m$ and $d\Pi_n(F)e_m$ lie in V. By the operator formulas of Proposition 23.8, E sends z^m to $(n+m)z^{m+1}$ with $n+m \neq 0$. For $m \geq 1$, F sends z^m to $-mz^{m-1}$ with $m \neq 0$. So every z^k lies in V. Polynomials are dense, so $V = \mathcal{A}_n$. ∎

An irreducible unitary representation belongs to the *discrete series* if its matrix coefficients are square integrable on G. The family Π_n, $n \geq 2$, is the holomorphic discrete series in our labelling. Its square integrability is cited from Knapp's §VI.1; Exercise 23.6 checks the coefficient of e_0.

23.3 Descent conditions

Proposition 23.10. *Let Π be a strongly continuous representation of G on a Hilbert space. Let $v \neq 0$ have K-type k.*

(a) *$k \in \mathbb{Z}$. In particular, for $\lambda \notin \mathbb{Z}$ no such Π has a lowest-weight module with weights $\lambda, \lambda+2, \dots$ as its K-finite part.*

(b) *$\Pi(-I)v = (-1)^k v$.*

(c) *$\Pi_n(-I) = (-1)^n \mathrm{id}$. Hence Π_n factors through $\mathrm{PSL}(2,\mathbb{R}) = G/\{\pm I\}$ if and only if n is even.*

Proof. (a) Since $k_{2\pi} = I$, we have $v = \Pi(k_{2\pi})v = e^{2\pi i k}v$. As $v \neq 0$, this forces $e^{2\pi i k} = 1$, so $k \in \mathbb{Z}$. A lowest-weight K-finite part with weights $\lambda, \lambda+2, \dots$ contains a vector of K-type λ, so it requires $\lambda \in \mathbb{Z}$.

(b) Since $k_\pi = -I$, the definition of K-type at $\theta = \pi$ gives $\Pi(-I)v = e^{i\pi k}v$. This equals $(-1)^k v$ because $k \in \mathbb{Z}$.

(c) By the formula (23.2) for $\varphi(-I)$, $(-I)\cdot z = z$ and $j(-I, z) = -1$, so $\Pi_n(-I)f = (-1)^{-n}f$. The quotient map $G \to \mathrm{PSL}(2,\mathbb{R})$ is onto with kernel $\{\pm I\}$. So Π_n factors exactly when $\Pi_n(-I) = \mathrm{id}$. ∎

The value of $\Pi_n(-I)$ in Proposition 23.10(c) agrees with the sign rule of Proposition 23.10(b). Indeed, every K-type $n+2m$ of Π_n has the parity of n. The integrality argument of Proposition 23.10(a) uses the period 2π of K in G. On the universal covering group the lift of K is a copy of $\mathbb{R}$, so (a) gives no condition there. We do not study that group.

Remark 23.11. The ratio forced by (23.7) still makes sense for $n = 1$. It gives $c_m = 1$, so the norm is $\sum |a_m|^2$. No positive measure μ on D gives this norm by area integration. It would satisfy $\int_D 1\, d\mu = \int_D |z|^2 d\mu = 1$, although $1 - |z|^2$ is strictly positive on D. This is impossible. So lowest weight 1 lies outside the family $\mathcal{A}_n$. It appears below as a limit of discrete series.

23.4 Comparison and the unitary picture

Let V be a $(\mathfrak{g}, K)$-module of G (Lecture 21) on which Ω acts by a scalar c. Write v_k for a vector of K-type k. Since $\Omega = H'^2 + 2H' + 4FE = H'^2 - 2H' + 4EF$,

$$4FEv_k = \big(c - k(k+2)\big)v_k, \qquad 4EFv_k = \big(c - k(k-2)\big)v_k. \tag{23.8}$$

Example 23.12. Fix $n \geq 2$ and $c = n(n-2) = (n-1)^2 - 1$. By (23.8), $FEv_k = 0$ exactly when $k \in \{-n, n-2\}$.

(i) In Π_n the K-types are $n, n+2, \ldots$. F kills z^0, and E is injective.

(ii) Put $d = n - 2$ and replace n by $-d$ in (23.5). Call the resulting operator $\Pi_{-d}(h)$. It sends f to $j(h^{-1}, z)^d f(h^{-1} \cdot z)$ and preserves the polynomials of degree at most d. The computation of (23.6) gives $d\Pi(E)z^m = (m-d)z^{m+1}$, which kills z^d. The K-types are $-d, \ldots, d$. On the lowest vector $v_{-d} = z^0$, we have $Fv_{-d} = 0$. Thus (23.8) gives $c = (-d)(-d-2) = d(d+2) = n(n-2)$. Exercise 23.3 gives the polynomial matrices.

(iii) The principal series of Lecture 22 has every integer of one parity as a K-type, once each. Take the principal series module $V_{\varepsilon,\nu}$ of Lecture 22 with $\nu = n - 1$, and with ε chosen so that its K-types have the parity of n. By Lecture 22, Exercise 22.4, Ω acts on it by $\nu^2 - 1 = n(n-2)$. At $k = n-2$, (23.8) says that the coefficient of $E\colon v_{n-2} \mapsto v_n$ times that of $F\colon v_n \mapsto v_{n-2}$ vanishes. The same holds between $-n$ and $2-n$. So the module is reducible. It has a filtration whose subquotients have K-types $k \leq -n$, $2 - n \leq k \leq n - 2$ and $k \geq n$. These subquotients have the K-types and Casimir value of Π_n^-, of the finite-dimensional module Π_{-d} with $d = n - 2$, and of Π_n.

Proposition 23.13. *Let V be an irreducible $(\mathfrak{g}, K)$-module of G. Assume each K-type has multiplicity at most one and Ω acts by c. Suppose V has a positive definite inner product satisfying (23.7) for the module action. If k and $k+2$ are K-types, then*

$$\|Ev_k\|^2 = \tfrac{1}{4}\big(k(k+2) - c\big)\|v_k\|^2 > 0. \tag{23.9}$$

Consequently:

(i) if the K-types are $\lambda, \lambda+2, \lambda+4, \ldots$, then $\lambda \geq 1$ is an integer and $c = \lambda(\lambda-2)$;

(ii) if the K-types are $2\mathbb{Z}$, then $c < 0$;

(iii) if the K-types are $2\mathbb{Z}+1$, then $c < -1$.

Proof. By (23.7) and (23.8), $\|Ev_k\|^2 = -\langle v_k, FEv_k\rangle = \frac{1}{4}(k(k+2) - c)\|v_k\|^2$. Since $\|Ev_k\|^2$ and $\|v_k\|^2$ are real and $v_k \neq 0$, this identity shows that c is real.

Next we show that $Ev_k \neq 0$, arguing by contradiction. Suppose $Ev_k = 0$. For each parity, the sum of the K-type spaces of that parity is a submodule. By irreducibility, all K-types have

the parity of k. Since E raises and F lowers K-types by 2, the span of the v_j with $j \le k$ is invariant. It is nonzero and misses v_{k+2}, which contradicts irreducibility. So (23.9) is strict.

For the lowest-weight pattern (i), $Fv_\lambda = 0$, so (23.8) gives $c = \lambda(\lambda - 2)$. Then (23.9) at $k = \lambda$ reads $\|Ev_\lambda\|^2 = \lambda\|v_\lambda\|^2$, so $\lambda > 0$. The circle action on the $(\mathfrak{g}, K)$-module satisfies $k_{2\pi} = I$. Thus $v_\lambda = e^{2\pi i\lambda} v_\lambda$, which makes λ an integer. For the two-sided patterns (ii) and (iii), write $k(k+2) = (k+1)^2 - 1$. Its minimum is 0 over even k and -1 over odd k. ∎

Twisting by δ from Remark 23.5 interchanges E, F, sends H' to $-H'$ and preserves (23.7). Thus the lowest-weight case (i) also gives $c = \lambda(\lambda - 2)$ with integer $\lambda \ge 1$ for highest-weight patterns $-\lambda, -\lambda - 2, \ldots$.

Theorem 23.14 (Bargmann; cited). *Every irreducible unitary representation of G is equivalent to exactly one entry of the following list. Each entry is irreducible and unitary, and is determined by its K-types and its Casimir value c. Here Π_1 means formula (23.5) at $n = 1$ on the completion of polynomials in the norm $\sum |a_m|^2$. The unitarity and irreducibility of Π_1 are part of this cited theorem.*

Family	*K-types*	c
trivial	$\{0\}$	0
even principal series	$2\mathbb{Z}$	$c \le -1$
odd principal series	$2\mathbb{Z} + 1$	$c < -1$
complementary series	$2\mathbb{Z}$	$-1 < c < 0$
Π_n, $n \ge 1$	$n, n+2, \ldots$	$n(n-2)$
Π_n^-, $n \ge 1$	$-n, -n-2, \ldots$	$n(n-2)$

For $n \ge 2$ the last two rows are the discrete series. For $n = 1$ they are the limits of discrete series, realised with the norm of Remark 23.11. Their sum is the odd principal series at $c = -1$.

Proposition 23.13 proves the necessity of the c-ranges for these K-type patterns. Call a $(\mathfrak{g}, K)$-module *even* if its K-types are $2\mathbb{Z}$, and *odd* if they are $2\mathbb{Z} + 1$. An irreducible even module with a positive definite inner product satisfying (23.7) needs $c < 0$. The part $c \le -1$ is the unitary principal series of Lecture 22. The interval $-1 < c < 0$ is the complementary series. The inner product of the complementary series is fixed by (23.9), not by the L^2 norm of Lecture 22 (Exercise 23.5). An irreducible odd module with such an inner product needs $c < -1$, so there is no odd complementary series. At $c = -1$ the odd principal series module has $FEv_{-1} = 0$ and splits.

The cited parts are the converse and the exhaustion. They use two results of Harish-Chandra (compare Lecture 21). First, every irreducible unitary representation is admissible. Its K-finite vectors form an irreducible $(\mathfrak{g}, K)$-module with multiplicity-one K-types on which Ω is scalar. Second, each module above with an invariant positive definite Hermitian form integrates to an irreducible unitary representation of G. Etingof's MIT 18.757 notes treat admissibility in Lecture 5 and unitary globalisation in Theorem 7.5.

Since G is simple and noncompact, its only finite-dimensional unitary irreducible representation is trivial. Exercise 23.3 checks this directly for the finite-dimensional modules Π_{-d} of Example 23.12(ii).

23.5 Self-checks

1. For $n = 3$, compute c_1 and c_2. Check (23.7) for $u = z$ and $v = z^2$.

   Answer. $c_1 = 1!\,2!/3! = 1/3$ and $c_2 = 2!\,2!/4! = 1/6$. The left side is $(3+1)c_2 = 2/3$. The right side is $-\langle z, -2z\rangle = 2c_1 = 2/3$.

2. Does Π_5 factor through $\mathrm{PSL}(2,\mathbb{R})$? Compute $\Pi_5(k_{\pi/2})z^m$.

   Answer. No, since $\Pi_5(-I) = -\mathrm{id}$. By the K-type formula of Proposition 23.6, $\Pi_5(k_{\pi/2})z^m = i^{5+2m}z^m = (-1)^m\, i\, z^m$.

3. Is there an irreducible unitary representation with K-types $1, 3, 5, \dots$ and $c = -\frac{1}{2}$?

   Answer. No. By the cited Harish-Chandra facts after Theorem 23.14, the K-finite vectors of such a representation satisfy the hypotheses of Proposition 23.13. Its lowest-weight case (i) shows that the lowest K-type 1 forces $c = 1 \cdot (1-2) = -1$.

23.6 Summary

For each integer $n \geq 2$, formula (23.5) defines an irreducible unitary representation Π_n on $\mathcal{A}_n$. Its K-types are $n, n+2, \dots$, each once. The operator E raises the K-type by 2 and F lowers it by 2. The Casimir element Ω acts by $n(n-2)$. The relation $k_{2\pi} = I$ forces integral K-types. The representation Π_n descends to $\mathrm{PSL}(2,\mathbb{R})$ exactly for even n. In an irreducible unitary module with K-types $\lambda, \lambda+2, \dots$, the lowest weight is an integer $\lambda \geq 1$, and Ω acts by $\lambda(\lambda-2)$. Positivity of $\|Ev_k\|^2$ explains the shape of Bargmann's cited list, including the complementary series.

23.7 Reading

- A. W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples*, Princeton University Press, 2001 edition. §VI.1 treats the holomorphic discrete series of $\mathrm{SU}(1,1)$. §§II.5–II.6 treat the unitary representations of $\mathrm{SL}(2,\mathbb{R})$.
- MIT 18.112, *Functions of a Complex Variable*, Fall 2008: Cauchy's formula, power series and the Cauchy–Riemann equations.
- P. Etingof, *Representations of Lie Groups*, MIT 18.757 lecture notes, Fall 2023. Lecture 5 treats admissibility, and Theorem 7.5 is the unitary globalisation theorem.
- V. Bargmann, Irreducible unitary representations of the Lorentz group, *Ann. of Math.* 48 (1947), 568–640, for the original classification.

Lecture 24

Lorentz groups and finite-dimensional representations

Objectives. *We build the double cover* $\mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ *from Hermitian* 2×2 *matrices. We then classify the finite-dimensional irreducible continuous complex representations of both groups by complexifying the real Lie algebra.*

By the end of this lecture you will be able to:

- Prove that $\mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ is a surjective covering homomorphism with kernel $\{\pm I\}$.
- Identify $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ with $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$.
- Label the irreducible finite-dimensional continuous complex representations of $\mathrm{SL}(2,\mathbb{C})$ by (j_L, j_R) and decide when they descend to $\mathrm{SO}^+(1,3)$.
- Recognise Weyl spinors and four-vectors in this list.

In special relativity an event has coordinates $x = (x_0, \vec{x})$, with time x_0 and position $\vec{x} \in \mathbb{R}^3$. We use units in which light has speed 1. Inertial observers are related by linear maps that preserve the Minkowski form below. Relativistic fields take values in finite-dimensional representations of the Lorentz group or its double cover $\mathrm{SL}(2,\mathbb{C})$.

24.1 Minkowski space and the Lorentz group

Let $e_0, \dots, e_3$ be the standard basis of $\mathbb{R}^4$. The *Minkowski form* is

$$\langle x, y\rangle = x_0 y_0 - \vec{x}\cdot\vec{y} = x^{T}\eta y, \qquad \eta = \mathrm{diag}(1,-1,-1,-1).$$

Definition 24.1. The *Lorentz group* is $\mathrm{O}(1,3) = \{\Lambda \in \mathrm{GL}(4,\mathbb{R}) : \Lambda^{T}\eta\Lambda = \eta\}$. The *future cone* is $T^+ = \{x \in \mathbb{R}^4 : x_0 > |\vec{x}|\}$.

Let $\Lambda \in \mathrm{O}(1,3)$. Then $\det\Lambda = \pm 1$ and $\Lambda^{-1} = \eta\Lambda^{T}\eta$, so $(\Lambda^{-1})_{00} = \Lambda_{00}$. Since $\langle \Lambda e_0, \Lambda e_0\rangle = 1$,

$$\Lambda_{00}^2 = 1 + \Lambda_{10}^2 + \Lambda_{20}^2 + \Lambda_{30}^2 \geq 1.$$

So either $\Lambda_{00} \geq 1$ or $\Lambda_{00} \leq -1$.

Lemma 24.2. *Let* $\Lambda \in \mathrm{O}(1,3)$. *Then* $\Lambda_{00} \geq 1$ *if and only if* $\Lambda T^+ \subseteq T^+$. *Consequently the elements with* $\Lambda_{00} \geq 1$ *form a subgroup* $\mathrm{O}^+(1,3)$.

Proof. Suppose $\Lambda_{00} \geq 1$ and let $x \in T^+$. Put $w = \Lambda^{-1} e_0$. Then $\langle w, w \rangle = 1$ and $w_0 = (\Lambda^{-1})_{00} = \Lambda_{00} \geq 1$, so $w_0 > |\vec{w}|$. Since Λ preserves the form,

$$(\Lambda x)_0 = \langle e_0, \Lambda x \rangle = \langle w, x \rangle = w_0 x_0 - \vec{w} \cdot \vec{x} \geq w_0 x_0 - |\vec{w}|\,|\vec{x}| > 0.$$

The last step uses $w_0 > |\vec{w}|$ and $x_0 > |\vec{x}|$. Also $\langle \Lambda x, \Lambda x \rangle = \langle x, x \rangle > 0$. These two facts give $\Lambda x \in T^+$. Conversely, if $\Lambda T^+ \subseteq T^+$, then $\Lambda e_0 \in T^+$. So $\Lambda_{00} > 0$, hence $\Lambda_{00} \geq 1$. The cone condition is preserved by products, and $(\Lambda^{-1})_{00} = \Lambda_{00}$. So these elements form a subgroup. ∎

Definition 24.3. The *proper orthochronous Lorentz group* is

$$\mathrm{SO}^+(1,3) = \{\Lambda \in \mathrm{O}(1,3) : \det \Lambda = 1,\ \Lambda_{00} \geq 1\}.$$

By the cone lemma, Lemma 24.2, it is a group. It is closed in $\mathrm{GL}(4,\mathbb{R})$, hence a Lie group by the closed-subgroup theorem (Lecture 3). By the two dichotomies above, for $\det \Lambda$ and for Λ_{00}, it is also open in $\mathrm{O}(1,3)$. So it has the Lie algebra of $\mathrm{O}(1,3)$:

$$\mathfrak{so}(1,3) = \{Y \in \mathfrak{gl}(4,\mathbb{R}) : Y^T \eta + \eta Y = 0\}.$$

The map $Y \mapsto \eta Y$ is a bijection onto the antisymmetric matrices, so $\dim \mathfrak{so}(1,3) = 6$.

A basis consists of the *rotation generators* J_k and the *boost generators* K_k, for $k = 1, 2, 3$. Here $J_k e_0 = 0$, and J_k acts on $\vec{x}$ by $e_k \times \vec{x}$. Moreover $K_k e_0 = e_k$ and $K_k e_k = e_0$, and K_k maps the other two basis vectors to zero. For example,

$$J_3 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad K_3 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Then $R_3(\theta) = \exp(\theta J_3)$ rotates the (x_1, x_2)-plane by the positive angle θ. The *boost* $B_3(t) = \exp(t K_3)$ fixes x_1, x_2 and acts by

$$x_0 \pm x_3 \mapsto e^{\pm t}(x_0 \pm x_3).$$

Parity $\mathrm{diag}(1,-1,-1,-1)$ and time reversal $\mathrm{diag}(-1,1,1,1)$ lie in $\mathrm{O}(1,3)$ but outside the open subgroup $\mathrm{SO}^+(1,3)$. The Lie algebra $\mathfrak{so}(1,3)$ carries no information about these elements. We study only $\mathrm{SO}^+(1,3)$.

24.2 The double cover

Let $\sigma_0 = I$ and

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

These form a real basis of the space $\mathcal{H}$ of Hermitian 2×2 matrices. For $x \in \mathbb{R}^4$ put

$$X(x) = \sum_{\mu=0}^{3} x_\mu \sigma_\mu = \begin{pmatrix} x_0 + x_3 & x_1 - i x_2 \\ x_1 + i x_2 & x_0 - x_3 \end{pmatrix}.$$

Direct expansion gives

$$\det X(x) = \langle x, x\rangle, \qquad x_0 = \tfrac{1}{2}\operatorname{tr} X(x).$$

For $A \in \mathrm{SL}(2,\mathbb{C})$ the matrix $AX(x)A^*$ is again Hermitian. Define $\varphi(A) \in \mathrm{GL}(4,\mathbb{R})$ by

$$X(\varphi(A)x) = A\,X(x)\,A^*. \tag{24.1}$$

The entries of $\varphi(A)$ are real polynomials in the real and imaginary parts of the entries of A. Hence φ is smooth.

Example 24.4 (Diagonal elements)**.** Let $A = \operatorname{diag}(a, a^{-1})$ with $a \in \mathbb{C}^\times$. Then

$$AX(x)A^* = \begin{pmatrix} |a|^2(x_0+x_3) & (a/\bar a)(x_1 - ix_2) \\ (\bar a/a)(x_1+ix_2) & |a|^{-2}(x_0-x_3)\end{pmatrix}.$$

For $a = e^{-i\theta/2}$, the coordinates x_0, x_3 are fixed and $x_1 + ix_2 \mapsto e^{i\theta}(x_1+ix_2)$. For $a = e^{t/2}$, we get $x_0 \pm x_3 \mapsto e^{\pm t}(x_0 \pm x_3)$, with x_1, x_2 fixed. Put $X_k = -\frac{i}{2}\sigma_k$. Then

$$\varphi(\exp \theta X_3) = R_3(\theta), \qquad \varphi(\exp tiX_3) = B_3(t).$$

Differentiating at 0 gives $d\varphi(X_3) = J_3$ and $d\varphi(iX_3) = K_3$. Also $\operatorname{diag}(e^{i\theta}, e^{-i\theta}) = \exp(-2\theta X_3)$ maps to $R_3(-2\theta)$, as in the convention for $\mathrm{SU}(2) \to \mathrm{SO}(3)$ in Lecture 4. Finally, $\exp(2\pi X_3) = -I$ maps to $R_3(2\pi) = I$.

Let $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}$ denote $\mathfrak{sl}(2,\mathbb{C})$ regarded as a six-dimensional real Lie algebra.

Theorem 24.5. *The map φ is a surjective homomorphism $\mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ with kernel $\{\pm I\}$. Its differential $d\varphi : \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \to \mathfrak{so}(1,3)$ is an isomorphism of real Lie algebras, and φ is a two-sheeted covering map.*

Proof. Step 1: a homomorphism into $\mathrm{O}^+(1,3)$. Since $(AB)X(AB)^* = A(BXB^*)A^*$, we have $\varphi(AB) = \varphi(A)\varphi(B)$. Since $\det A = 1$, $\det(AXA^*) = \det X$. So $\varphi(A)$ preserves $\langle x, x\rangle$. By polarisation it preserves $\langle x, y\rangle$, so $\varphi(A) \in \mathrm{O}(1,3)$. Its 00 entry is the time component of AIA^*:

$$\varphi(A)_{00} = \tfrac{1}{2}\operatorname{tr}(AA^*) = \tfrac{1}{2}\textstyle\sum_{i,j}|A_{ij}|^2 > 0.$$

By the dichotomy above, $\varphi(A)_{00} \geq 1$.

Step 2: determinant one. We show that $\mathrm{SL}(2,\mathbb{C})$ is path connected. Let $A \in \mathrm{SL}(2,\mathbb{C})$. Let $P = (A^*A)^{1/2}$ be the positive definite square root, and $U = AP^{-1}$. Then $U^*U = I$ and $\det P = |\det A| = 1$, so $U \in \mathrm{SU}(2)$. Diagonalise U and P by the spectral theorem. Since both have determinant 1, we can write $U = \exp Y_1$ and $P = \exp Y_2$ with Y_1 skew-Hermitian, Y_2 Hermitian, and both traceless. Then $t \mapsto \exp(tY_1)\exp(tY_2)$, for $t \in [0,1]$, is a path in $\mathrm{SL}(2,\mathbb{C})$ from I to A. Along it, $\det\varphi$ is continuous with values in $\{\pm 1\}$, and it starts at 1. So $\det\varphi(A) = 1$.

Step 3: the kernel. Suppose $\varphi(A) = I$. Then $AXA^* = X$ for all $X \in \mathcal{H}$. Taking $X = I$ gives $AA^* = I$, so $AXA^{-1} = X$ for all $X \in \mathcal{H}$. Every complex matrix is $X_1 + iX_2$ with $X_1, X_2 \in \mathcal{H}$. So A commutes with all of $M_2(\mathbb{C})$, hence $A = cI$. Then $\det A = 1$ gives $c = \pm 1$. Conversely, $\pm I$ lie in the kernel.

Step 4: surjectivity. This is the main step. Let $\Lambda \in \mathrm{SO}^+(1,3)$ and $v = \Lambda e_0$. First we remove the boost part. The matrix $V = X(v)$ satisfies

$$\det V = \langle v, v\rangle = 1, \qquad \operatorname{tr} V = 2\Lambda_{00} > 0.$$

Its real eigenvalues have product 1 and positive sum, so both are positive. Thus $V = W\,\mathrm{diag}(\lambda, \lambda^{-1})W^*$ with W unitary and $\lambda > 0$. Put $P = W\,\mathrm{diag}(\lambda^{1/2}, \lambda^{-1/2})W^*$. Then $P \in \mathrm{SL}(2,\mathbb{C})$, $P^* = P$ and $P^2 = V$. Hence $X(\varphi(P)e_0) = PIP^* = V$, that is, $\varphi(P)e_0 = v$.

Next we treat the rotation part. Put

$$\Lambda' = \varphi(P)^{-1}\Lambda.$$

This element lies in $\mathrm{SO}^+(1,3)$ and fixes e_0. So it preserves $e_0^{\perp} = \mathrm{span}(e_1, e_2, e_3)$, where $-\langle\cdot,\cdot\rangle$ is the Euclidean inner product. Hence $\Lambda' = \mathrm{diag}(1, R)$ with $R \in \mathrm{SO}(3)$. Let $n = Re_3$, a unit vector, and set $N = \sum_k n_k\sigma_k$. This Hermitian matrix satisfies

$$\operatorname{tr} N = 0, \qquad \det N = -|n|^2 = -1,$$

so the eigenvalues of N are ± 1. By the spectral theorem, $N = W'\sigma_3 W'^*$ with W' unitary. We correct the determinant of W' by replacing W' with

$$W'\,\mathrm{diag}(1, \overline{\det W'}).$$

Diagonal matrices commute with σ_3, so this replacement preserves the diagonalisation of N. The corrected W' belongs to $\mathrm{SU}(2)$.

Then $\varphi(W')e_0 = e_0$, since $W'W'^* = I$. Also $\varphi(W')e_3 = n$, since $W'\sigma_3 W'^* = N$. So $\varphi(W')^{-1}\Lambda'$ fixes e_0 and e_3. It acts on $\mathrm{span}(e_1, e_2)$ by an element of $\mathrm{SO}(2)$, so it equals $R_3(\theta)$ for some θ. By the rotation formula of Example 24.4,

$$\Lambda = \varphi(P)\,\varphi(W')\,R_3(\theta) = \varphi\big(PW'\exp(\theta X_3)\big).$$

In particular, φ maps $\mathrm{SU}(2)$ onto the rotations $\mathrm{diag}(1, R)$.

Step 5: the differential and the covering. Differentiate (24.1) along $\exp(tY)$, using $\exp(tY)^* = \exp(tY^*)$:

$$X(d\varphi(Y)x) = YX(x) + X(x)Y^*, \qquad Y \in \mathfrak{sl}(2,\mathbb{C}). \tag{24.2}$$

Suppose $d\varphi(Y) = 0$. Taking $X = I$ gives $Y^* = -Y$. Then $YX - XY = 0$ for all $X \in \mathcal{H}$, so Y is scalar, as in Step 3. Since $\operatorname{tr} Y = 0$, we get $Y = 0$. Thus $d\varphi$ is injective. Both algebras have real dimension 6, so $d\varphi$ is a linear bijection. The differential of a Lie group homomorphism preserves brackets, so $d\varphi$ is an isomorphism of real Lie algebras.

By Steps 2 and 4, $\mathrm{SO}^+(1,3) = \varphi(\mathrm{SL}(2,\mathbb{C}))$ is connected. A continuous surjective homomorphism of connected Lie groups with bijective differential is a covering map (Lecture 4). Its fibres are cosets of $\ker\varphi = \{\pm I\}$, so it has two sheets. ∎

Formula (24.2) gives $d\varphi(X_k) = J_k$ and $d\varphi(iX_k) = K_k$ for every k. The case $k = 3$ is the diagonal computation of Example 24.4, Exercise 24.1 treats $k = 1$, and the case $k = 2$ is identical. The brackets $[X_i, X_j] = \epsilon_{ijk}X_k$ then transfer through $d\varphi$:

$$[J_i, J_j] = \epsilon_{ijk}J_k, \qquad [J_i, K_j] = \epsilon_{ijk}K_k, \qquad [K_i, K_j] = -\epsilon_{ijk}J_k.$$

24.3 Complexifying the Lorentz algebra

By the double-cover theorem, Theorem 24.5, $\mathfrak{so}(1,3) \cong \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}$. A complex representation of this real algebra need not be complex linear. For example, $Y \mapsto \bar{Y}$ is a real Lie algebra automorphism that is not complex linear. Complexification treats both kinds together.

Let $\mathfrak{g}$ be a real Lie algebra and V a finite-dimensional complex space. A real Lie algebra map $\rho : \mathfrak{g} \to \mathfrak{gl}(V)$ extends uniquely to a complex-linear Lie algebra map $\rho_{\mathbb{C}}(Y \otimes z) = z\rho(Y)$ on $\mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}$. A complex subspace is ρ-invariant if and only if it is $\rho_{\mathbb{C}}$-invariant. A linear map intertwines two such maps ρ if and only if it intertwines their extensions.

Proposition 24.6. *The complex-linear map*

$$\iota : \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C} \to \mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C}), \qquad Y \otimes z \mapsto (zY,\, z\bar{Y}),$$

is an isomorphism of complex Lie algebras.

Proof. The pair $(zY, z\bar{Y})$ is real bilinear in (Y, z), so ι is well defined and complex linear. Complex conjugation of matrices preserves brackets, so

$$\iota([Y \otimes z, Y' \otimes z']) = \big(zz'[Y,Y'],\, zz'[\bar{Y},\bar{Y}']\big) = [\iota(Y \otimes z), \iota(Y' \otimes z')].$$

Both Lie algebras have complex dimension 6, so it suffices to prove that ι is injective. A general element $Y \otimes 1 + Y' \otimes i$ has image $(Y + iY',\, \bar{Y} + i\bar{Y}')$. If this is zero, then $Y = -iY'$ and $\bar{Y} = -i\bar{Y}'$. Conjugating the first equation gives $\bar{Y} = i\bar{Y}'$. Hence $\bar{Y}' = 0$, so $Y' = 0$ and $Y = 0$. ∎

Since $\overline{iX_k} = -i\bar{X}_k$, we have $\iota(X_k \otimes 1) = (X_k, \bar{X}_k)$ and $\iota(iX_k \otimes i) = (-X_k, \bar{X}_k)$. Hence the elements

$$X_k^L = \tfrac{1}{2}(X_k \otimes 1 - iX_k \otimes i), \qquad X_k^R = \tfrac{1}{2}(X_k \otimes 1 + iX_k \otimes i)$$

satisfy $\iota(X_k^L) = (X_k, 0)$ and $\iota(X_k^R) = (0, \bar{X}_k)$. We identify $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ with $\mathfrak{so}(1,3)_{\mathbb{C}} = \mathfrak{so}(1,3) \otimes_{\mathbb{R}} \mathbb{C}$ through the complex-linear extension of $d\varphi$. Then $X_k^L = \frac{1}{2}(J_k - iK_k)$ and $X_k^R = \frac{1}{2}(J_k + iK_k)$. These elements satisfy

$$[X_i^L, X_j^L] = \epsilon_{ijk} X_k^L, \qquad [X_i^R, X_j^R] = \epsilon_{ijk} X_k^R, \qquad [X_i^L, X_j^R] = 0.$$

Moreover $J_k = X_k^L + X_k^R$ and $K_k = i(X_k^L - X_k^R)$.

24.4 Classification of finite-dimensional representations

For $m \geq 0$ let $V_m = \mathrm{Sym}^m(\mathbb{C}^2)$, and let $S^m(A)$ be the restriction of $A^{\otimes m}$ to V_m. The entries of $S^m(A)$ are polynomials in the entries of A, so dS^m is complex linear on $\mathfrak{sl}(2,\mathbb{C})$. From the Lie algebra background, dS^m is the irreducible module of highest weight m. The element $H = \mathrm{diag}(1,-1)$ acts with eigenvalues $m, m-2, \dots, -m$. On $\mathrm{SU}(2)$, V_m is the spin-$m/2$ representation.

Definition 24.7. For $j_L, j_R \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, the representation (j_L, j_R) of $\mathrm{SL}(2,\mathbb{C})$ is

$$D^{(j_L,j_R)}(A) = S^{2j_L}(A) \otimes S^{2j_R}(\bar{A}) \quad \text{on} \quad V_{2j_L} \otimes V_{2j_R}.$$

Since $A \mapsto \bar{A}$ is a continuous automorphism of $\mathrm{SL}(2,\mathbb{C})$, this is a continuous representation. Its dimension is $(2j_L+1)(2j_R+1)$.

A representation Π of $\mathrm{SL}(2,\mathbb{C})$ *descends* to $\mathrm{SO}^+(1,3)$ if there is a continuous representation Π' of $\mathrm{SO}^+(1,3)$ with

$$\Pi = \Pi' \circ \varphi.$$

This is the descent of Lecture 4, now through the covering φ.

Theorem 24.8. *(a) Each (j_L, j_R) is irreducible, and distinct pairs give inequivalent representations. Every irreducible finite-dimensional continuous complex representation of $\mathrm{SL}(2,\mathbb{C})$ is equivalent to some (j_L, j_R).*

(b) $D^{(j_L,j_R)}(-I) = (-1)^{2j_L+2j_R}$. Hence (j_L, j_R) descends to $\mathrm{SO}^+(1,3)$ if and only if $2j_L+2j_R$ is even. Up to equivalence, these descended representations are all the irreducible finite-dimensional continuous complex representations of $\mathrm{SO}^+(1,3)$.

(c) The restriction of (j_L, j_R) to $\mathrm{SU}(2)$ is equivalent to $\bigoplus_j V_{2j}$, where $j = |j_L - j_R|, |j_L - j_R| + 1, \dots, j_L + j_R$.

Proof. (a) Write $D = D^{(j_L,j_R)}$, $m = 2j_L$ and $n = 2j_R$. Since $\overline{\exp tY} = \exp t\bar{Y}$, differentiation gives

$$dD(Y) = dS^m(Y) \otimes 1 + 1 \otimes dS^n(\bar{Y}), \qquad Y \in \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}.$$

So the complex-linear extension is

$$dD_{\mathbb{C}}(Y \otimes z) = dS^m(zY) \otimes 1 + 1 \otimes dS^n(z\bar{Y}) = (dS^m \boxtimes dS^n)\big(\iota(Y \otimes z)\big).$$

Thus, through ι, $dD_{\mathbb{C}}$ is the module $V_m \boxtimes V_n$ of $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$. We import from the Lie algebra background: these modules are irreducible and pairwise non-isomorphic, since their highest weights (m,n) differ. Every finite-dimensional irreducible module of $\mathfrak{sl}(2,\mathbb{C})\oplus\mathfrak{sl}(2,\mathbb{C})$ is one of them.

Now we pass from algebra to group. Every continuous finite-dimensional representation Π is smooth (Lecture 3), with $\Pi(\exp Y) = \exp d\Pi(Y)$ (Lecture 2). By Step 2 of the proof of Theorem 24.5, $\mathrm{SL}(2,\mathbb{C})$ is connected, so it is generated by $\exp \mathfrak{sl}(2,\mathbb{C})$ (Lecture 2). Hence a subspace is Π-invariant if and only if it is $d\Pi$-invariant. A linear map intertwines two group representations if and only if it intertwines their differentials. By the extension facts in § 24.3, both statements pass to $d\Pi_{\mathbb{C}}$. So D is irreducible, and distinct pairs are inequivalent.

Finally, let Π be an irreducible finite-dimensional continuous complex representation. Then $d\Pi_{\mathbb{C}} \circ \iota^{-1} \cong V_m \boxtimes V_n$ for some (m,n). This is the complexified differential of $D^{(m/2,n/2)}$, so $\Pi \cong D^{(m/2,n/2)}$.

(b) We have $S^m(-I) = (-1)^m$ and $\overline{-I} = -I$, so $D(-I) = (-1)^{m+n}$. Covering maps are open. So by the double-cover theorem, Theorem 24.5, φ identifies $\mathrm{SO}^+(1,3)$ with $\mathrm{SL}(2,\mathbb{C})/\{\pm I\}$ as topological groups. By Lecture 4, continuous representations of $\mathrm{SO}^+(1,3)$ correspond to continuous representations of $\mathrm{SL}(2,\mathbb{C})$ that are trivial on $-I$. The correspondence preserves irreducibility and equivalence. Now combine this with (a).

(c) Let $\varepsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. For $U \in \mathrm{SU}(2)$ one has $\bar{U} = \varepsilon U \varepsilon^{-1}$ (Exercise 24.3). So $S^n(\bar{U}) = S^n(\varepsilon)S^n(U)S^n(\varepsilon)^{-1}$, and the restriction is equivalent to $V_m \otimes V_n$. The Clebsch–Gordan rule from the background gives the decomposition. ∎

Every finite-dimensional continuous complex representation of $\mathrm{SL}(2,\mathbb{C})$ is a direct sum of representations (j_L, j_R). This follows from Weyl's complete reducibility theorem for the semisimple algebra $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$, transferred as in the proof.

Only $(0,0)$ admits an invariant inner product. The reason is that a boost generator has the nonzero real eigenvalue $j_L + j_R$ on every $(j_L, j_R) \neq (0,0)$. Exercise 24.6 gives the proof.

In physics, let a field ψ take values in a representation space V with matrices $D(A)$. Its transformation is

$$\psi'(x) = D(A)\psi(\varphi(A)^{-1}x).$$

The label (j_L, j_R) classifies the components of ψ. It does not describe the unitary Hilbert space of particle states, which is the subject of Lecture 25.

24.5 Weyl spinors and four-vectors

Example 24.9 (Weyl spinors). The representation $(\frac{1}{2}, 0)$ is $\mathbb{C}^2$ with $A \mapsto A$. The representation $(0, \frac{1}{2})$ is $\mathbb{C}^2$ with $A \mapsto \bar{A}$. Physicists call their vectors left-handed and right-handed Weyl spinors. The choice of hands is a convention. By the classification theorem, Theorem 24.8, the two representations are inequivalent. A direct check uses traces. For $A = \mathrm{diag}(2i, -\frac{i}{2}) \in \mathrm{SL}(2,\mathbb{C})$,

$$\operatorname{tr} A = \tfrac{3i}{2}, \qquad \operatorname{tr} \bar{A} = -\tfrac{3i}{2}.$$

Equivalent representations have equal traces. These traces differ, so the trace check confirms that the two representations are inequivalent. Both representations restrict to V_1 on $\mathrm{SU}(2)$. Both send $-I$ to -1, so neither descends to $\mathrm{SO}^+(1,3)$. The Dirac representation $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ also sends $-I$ to -1, so it does not descend either.

Example 24.10 (Four-vectors). Let $\Psi : \mathbb{C}^2 \otimes \mathbb{C}^2 \to M_2(\mathbb{C})$ be the linear isomorphism given by $\Psi(\xi \otimes \zeta) = \xi\zeta^T$. The map Ψ sends $e_i \otimes e_j$ to the matrix unit E_{ij}. For $A \in \mathrm{SL}(2,\mathbb{C})$,

$$\Psi(A\xi \otimes \bar{A}\zeta) = A\xi(\bar{A}\zeta)^T = A\xi\zeta^T\bar{A}^T = A\,\Psi(\xi \otimes \zeta)\,A^*.$$

So $D^{(\frac{1}{2},\frac{1}{2})}(A)$ becomes $Z \mapsto AZA^*$ on $M_2(\mathbb{C}) = \mathcal{H} \oplus i\mathcal{H}$. This action preserves $\mathcal{H}$ and acts there by $\varphi(A)$. Hence $(\frac{1}{2}, \frac{1}{2})$ is equivalent to $\mathbb{C}^4 = \mathbb{R}^4 \otimes_{\mathbb{R}} \mathbb{C}$ with $\varphi(A)$ extended complex-linearly. Since $2 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2} = 2$ is even, the representation $(\frac{1}{2}, \frac{1}{2})$ descends by the descent criterion of Theorem 24.8. The descended representation is the defining representation of $\mathrm{SO}^+(1,3)$ on $\mathbb{C}^4$.

Two checks are consistent with this identification. First, the restriction rule of Theorem 24.8(c) predicts $V_0 \oplus V_2$ on $\mathrm{SU}(2)$. Indeed $U \in \mathrm{SU}(2)$ acts by $Z \mapsto UZU^{-1}$. It fixes I, which spans the time direction. It preserves the traceless matrices, which are the spatial vectors and carry spin 1. Second, compare boost eigenvalues. In $(\frac{1}{2}, \frac{1}{2})$, the element $iX_3 = \frac{1}{2}\sigma_3$ acts by $\frac{1}{2}(H \otimes 1 + 1 \otimes H)$. Its eigenvalues are $\frac{1}{2}(q + q')$ with $q, q' \in \{\pm 1\}$:

$$1, \quad 0, \quad 0, \quad -1.$$

On $\mathbb{R}^4$, the matrix $d\varphi(iX_3) = K_3$ has eigenvectors $e_0 + e_3$, e_1, e_2, $e_0 - e_3$ with eigenvalues $1, 0, 0, -1$. The lists agree.

The table lists small cases. Exercise 24.4 identifies $(1,0)\oplus(0,1)$ with $\Lambda^2\mathbb{C}^4$, the complexified antisymmetric tensors.

(j_L, j_R)	dimension	on SU(2)	descends	name
$(0,0)$	1	V_0	yes	scalar
$(\frac12,0)$, $(0,\frac12)$	2	V_1	no	Weyl spinors
$(1,0)$, $(0,1)$	3	V_2	yes	summands of $\Lambda^2\mathbb{C}^4$
$(\frac12,\frac12)$	4	$V_0\oplus V_2$	yes	four-vector
$(1,\frac12)$	6	$V_1\oplus V_3$	no	–

24.6 Self-checks

(1) Compute $\varphi(\operatorname{diag}(2,\frac12))$.

(2) Find all $A\in \mathrm{SL}(2,\mathbb{C})$ with $\varphi(A)=R_3(\pi)$.

(3) Does $(\frac32,\frac12)$ descend to $\mathrm{SO}^+(1,3)$? Give its dimension and its restriction to SU(2).

Answer. (1) By the diagonal formula of Example 24.4 with $a=2$, we get $x_0\pm x_3\mapsto 4^{\pm1}(x_0\pm x_3)$. Since $a/\bar a=1$, the coordinates x_1 and x_2 are fixed. This is $B_3(t)$ with $e^t=4$, so $\cosh t=\frac{17}{8}$ and $\sinh t=\frac{15}{8}$. Thus

$$\varphi\big(\operatorname{diag}(2,\tfrac12)\big)=\begin{pmatrix}\frac{17}{8}&0&0&\frac{15}{8}\\0&1&0&0\\0&0&1&0\\\frac{15}{8}&0&0&\frac{17}{8}\end{pmatrix}.$$

Check: $17^2-15^2=64$.

(2) The element $\exp(\pi X_3)=\operatorname{diag}(-i,i)$ maps to $R_3(\pi)$. The fibre is a coset of $\{\pm I\}$, so the answer is $A=\pm\operatorname{diag}(-i,i)$.

(3) Here $2j_L+2j_R=4$ is even, so it descends. Its dimension is $4\cdot 2=8$. The restriction is $V_2\oplus V_4$, of dimension $3+5=8$.

24.7 Summary

The action $X\mapsto AXA^*$ on Hermitian matrices preserves $\det X=\langle x,x\rangle$. It gives a two-sheeted covering $\varphi:\mathrm{SL}(2,\mathbb{C})\to\mathrm{SO}^+(1,3)$ with kernel $\{\pm I\}$. The map ι identifies the complexified Lorentz algebra with $\mathfrak{sl}(2,\mathbb{C})\oplus\mathfrak{sl}(2,\mathbb{C})$. Up to equivalence, the irreducible finite-dimensional continuous representations of $\mathrm{SL}(2,\mathbb{C})$ are the (j_L,j_R), given by $S^{2j_L}(A)\otimes S^{2j_R}(\bar A)$. Exactly those with $2j_L+2j_R$ even descend to $\mathrm{SO}^+(1,3)$. Weyl spinors are $(\frac12,0)$ and $(0,\frac12)$, and four-vectors are $(\frac12,\frac12)$.

24.8 Reading

- A. W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples*, Princeton University Press, 2001, Chapter II, Section 3, on finite-dimensional representations of $\mathrm{SL}(2,\mathbb{C})$.

Lecture 25

Poincaré representations and little groups

Objectives. *This lecture reduces the classification of positive-energy irreducible unitary representations of the Poincaré group to little groups. It makes the massive and finite-helicity sectors explicit. Translations are diagonalised by momenta. Lorentz transformations move momenta along orbits. The orbit and an irreducible unitary representation of the stabiliser of one of its points determine the representation up to unitary equivalence.*

By the end of this lecture you will be able to:

- Construct the group $\mathbb{R}^4 \rtimes \mathrm{SL}(2,\mathbb{C})$ and its action on momenta.
- Classify the momentum orbits and compute the massive and massless little groups.
- State Mackey's classification with its hypotheses, and separate the finite-helicity, continuous-spin and zero-momentum sectors.
- Verify the massive spin model and name the steps it omits.

25.1 The Poincaré group as a semidirect product

Spacetime is $\mathbb{R}^4$ with points $x = (x_0, x_1, x_2, x_3)$, where x_0 is time. The *Minkowski pairing* is

$$x \cdot y = x_0 y_0 - x_1 y_1 - x_2 y_2 - x_3 y_3.$$

Lorentz transformations are the linear maps preserving this pairing. The proper orthochronous ones form the group $\mathrm{SO}^+(1,3)$. Adding translations gives the Poincaré group. Quantum symmetries act projectively, and Lecture 13 explains why one then passes to a cover. We prove no lifting theorem. We study unitary representations of the cover built below.

Identify $\mathbb{R}^4$ with the 2×2 Hermitian matrices by

$$\Phi(x) = \begin{pmatrix} x_0 + x_3 & x_1 - ix_2 \\ x_1 + ix_2 & x_0 - x_3 \end{pmatrix}, \qquad \det \Phi(x) = x \cdot x, \qquad \operatorname{tr} \Phi(x) = 2x_0. \tag{25.1}$$

For $g \in \mathrm{SL}(2,\mathbb{C})$ define the map $\Lambda(g) \colon \mathbb{R}^4 \to \mathbb{R}^4$ by

$$\Phi(\Lambda(g)x) = g\, \Phi(x)\, g^*.$$

Lemma 25.1. *The map $\Lambda(g)$ is linear, and its entries are real polynomials in the real and imaginary parts of the entries of g. Moreover $\Lambda(gg') = \Lambda(g)\Lambda(g')$ and $\Lambda(g)x \cdot \Lambda(g)y = x \cdot y$.*

Proof. The matrix $g\Phi(x)g^{*}$ is Hermitian and depends linearly on x. For a Hermitian matrix $h = (h_{ij})$, the vector $\Phi^{-1}(h)$ depends linearly on h. For example, its time coordinate is $x_0 = \tfrac{1}{2}(h_{11} + h_{22})$. Hence $\Lambda(g)$ is linear. Each entry of $g\Phi(x)g^{*}$ is a sum of products of an entry of g, an entry of $\Phi(x)$ and the complex conjugate of an entry of g. So the entries of $\Lambda(g)$ are real polynomials in the real and imaginary parts of the entries of g.

The second claim follows from $gg'\Phi(x)(gg')^{*} = g\big(g'\Phi(x)g'^{*}\big)g^{*}$. For the third, $\det(g\Phi(x)g^{*}) = |\det g|^{2}\det\Phi(x) = x\cdot x$. Polarisation, $2x\cdot y = (x+y)\cdot(x+y) - x\cdot x - y\cdot y$, gives the pairing. ∎

By Lecture 24, Λ maps $\mathrm{SL}(2,\mathbb{C})$ onto $\mathrm{SO}^{+}(1,3)$ with kernel $\{\pm I\}$. The final claim of Proposition 25.3 and the descent statements use this fact. The classification itself does not depend on it.

Definition 25.2. The group $G = \mathbb{R}^{4} \rtimes \mathrm{SL}(2,\mathbb{C})$ is the set $\mathbb{R}^{4} \times \mathrm{SL}(2,\mathbb{C})$ with the product

$$(a,g)(a',g') = \big(a + \Lambda(g)a',\; gg'\big). \tag{25.2}$$

Proposition 25.3. *G is a Lie group of dimension 10. Its identity is $(0,I)$ and $(a,g)^{-1} = \big(-\Lambda(g)^{-1}a,\; g^{-1}\big)$. The translations $A = \{(a,I)\} \cong \mathbb{R}^{4}$ form a closed normal subgroup, and*

$$(0,g)(a,I)(0,g)^{-1} = \big(\Lambda(g)a,\; I\big). \tag{25.3}$$

The map $(a,g) \mapsto (a,\Lambda(g))$ is a homomorphism onto $\mathbb{R}^{4} \rtimes \mathrm{SO}^{+}(1,3)$ with kernel $\{(0,\pm I)\}$.

Proof. Associativity is the main point. Both bracketings of a triple product have second entry $gg'g''$. Their first entries are

$$a + \Lambda(g)a' + \Lambda(gg')a'' \qquad \text{and} \qquad a + \Lambda(g)\big(a' + \Lambda(g')a''\big).$$

These agree because $\Lambda(g)$ is linear and Λ is multiplicative, by Lemma 25.1. The stated inverse is an inverse on both sides, since

$$(a,g)\big(-\Lambda(g)^{-1}a,\, g^{-1}\big) = (a-a,\; I), \qquad \big(-\Lambda(g)^{-1}a,\, g^{-1}\big)(a,g) = \big(-\Lambda(g)^{-1}a + \Lambda(g)^{-1}a,\; I\big).$$

The manifold $\mathbb{R}^{4} \times \mathrm{SL}(2,\mathbb{C})$ has dimension $4 + 6$. Multiplication is polynomial in real coordinates by Lemma 25.1. So is inversion, since g^{-1} is the adjugate of g. Hence G is a Lie group. The projection $(a,g) \mapsto g$ is a continuous homomorphism with kernel A, so A is closed and normal. Next, $(0,g)(a,I) = (\Lambda(g)a, g)$, and multiplying on the right by $(0,g^{-1})$ gives (25.3). The final map respects (25.2), because only $\Lambda(g)$ acts on translations. Its image and kernel come from those of Λ. ∎

From now on, the Poincaré group means G. The central element $z = (0,-I)$ maps to the identity.

25.2 Momentum characters and mass shells

Let U be a strongly continuous unitary representation of G on a Hilbert space. On A it gives commuting unitary operators. Their building blocks are characters. Exercise 25.2 proves the next lemma.

Lemma 25.4. *Every continuous homomorphism $\mathbb{R}^4 \to \mathrm{U}(1)$ equals $\chi_p(a) = e^{-ip\cdot a}$ for a unique $p \in \mathbb{R}^4$.*

The sign follows the generator convention of Lecture 15. For the one-dimensional representation χ_p and the translation generator $X = e_0$, we have $\chi_p(te_0) = e^{-itp_0}$. Thus $A_X = p_0$ in that convention, and p_0 is the energy.

A *projection-valued measure* E on $\mathbb{R}^4$ assigns an orthogonal projection $E(S)$ to every Borel set $S \subseteq \mathbb{R}^4$, with $E(\mathbb{R}^4) = I$ and countable additivity in the strong operator topology for disjoint sets. We say that E is *concentrated* on a Borel set S when $E(S) = I$.

We cite the *Stone–Naimark–Ambrose–Godement (SNAG) theorem* for vector groups; see Berndt, *Representations of Linear Groups* (2007), Section 5.2, Theorem 5.2. For a strongly continuous unitary representation of $\mathbb{R}^4$, it gives a unique projection-valued measure E on $\mathbb{R}^4$ with

$$U(a, I) = \int_{\mathbb{R}^4} \chi_p(a)\, dE(p).$$

The *support* of E is the complement of the union of open sets O with $E(O) = 0$. Since $\mathbb{R}^4$ has a countable basis, this union is a countable union of such null sets. Thus the support is the smallest closed set F with $E(F) = I$. We call it the *translation spectrum*.

We use the spectral calculus for projection-valued measures as a cited analytic input (Hall, *Quantum Theory for Mathematicians*, §10.1). It defines the self-adjoint operator

$$P_0 = \int_{\mathbb{R}^4} p_0\, dE(p), \qquad \mathrm{Dom}(P_0) = \left\{ v : \int_{\mathbb{R}^4} p_0^2\, d\langle v, E(p)v\rangle < \infty \right\}.$$

By Stone's theorem with the convention of Lecture 15, this is the generator of $t \mapsto U(te_0, I)$. Positive energy means $P_0 \geq 0$. By the spectral calculus this is equivalent to concentration of E on $\{p_0 \geq 0\}$. Momentum eigenvectors need not exist in the Hilbert space.

Still, a formal computation shows the mechanism. If $U(a, I)\psi = \chi_p(a)\psi$ for all a, then (25.3) gives

$$U(a, I)U(0, g)\psi = U(0, g)\, U(\Lambda(g)^{-1}a, I)\psi = \chi_{\Lambda(g)p}(a)\, U(0, g)\psi. \tag{25.4}$$

Here $p \cdot \Lambda(g)^{-1}a = \Lambda(g)p \cdot a$ by the pairing invariance in Lemma 25.1. The same calculation with E, together with the uniqueness of E in the SNAG theorem, gives

$$U(0, g)E(S)U(0, g)^{-1} = E(\Lambda(g)S).$$

Hence the translation spectrum is stable under the following action.

Definition 25.5. The *mass-shell action* of $\mathrm{SL}(2, \mathbb{C})$ on momentum space $\mathbb{R}^4$ is $g \cdot p = \Lambda(g)p$. For $m, \kappa > 0$ set

$$X_m^{\pm} = \{p \cdot p = m^2,\ \pm p_0 > 0\}, \qquad C^{\pm} = \{p \cdot p = 0,\ \pm p_0 > 0\}, \qquad Y_\kappa = \{p \cdot p = -\kappa^2\}.$$

Theorem 25.6. *The orbits of the mass-shell action are exactly the sets $X_m^{\pm}$, $C^{\pm}$, Y_κ and $\{0\}$.*

Proof. Invariants. The number $p \cdot p$ is invariant, because $\Lambda(g)$ preserves the pairing by Lemma 25.1. Write $\vec{p} = (p_1, p_2, p_3)$, with Euclidean norm $|\vec{p}|$. Suppose $p \cdot p \geq 0$ and $p \neq 0$. Then $p_0^2 \geq |\vec{p}|^2$, so $p_0 \neq 0$. The eigenvalues $\mu_1 \geq \mu_2$ of $\Phi(p)$ satisfy

$$\mu_1 \mu_2 = p \cdot p \geq 0, \qquad \mu_1 + \mu_2 = 2p_0.$$

So $p_0 > 0$ exactly when $\Phi(p)$ is positive semidefinite and nonzero. The map $\Phi \mapsto g\Phi g^*$ preserves both properties, because $v^* g \Phi g^* v = (g^* v)^* \Phi (g^* v)$ and g is invertible. Hence each listed set is a union of orbits. The listed sets partition $\mathbb{R}^4$.

Transitivity. Write $\Phi(p) = u \operatorname{diag}(\mu_1, \mu_2)\, u^*$ with $u \in \mathrm{U}(2)$. A scalar factor $e^{i\phi}$ with $e^{2i\phi} \det u = 1$ puts u in $\mathrm{SU}(2)$ and changes nothing. Choose the base points

$$k_m = (m, 0, 0, 0), \qquad k = (1, 0, 0, 1), \qquad \ell_\kappa = (0, 0, 0, \kappa).$$

Their Φ-images are $\operatorname{diag}(m, m)$, $\operatorname{diag}(2, 0)$ and $\operatorname{diag}(\kappa, -\kappa)$. For $s > 0$ let $d_s = \operatorname{diag}(s, s^{-1})$. Then

$$d_s \operatorname{diag}(\alpha, \beta)\, d_s^* = \operatorname{diag}(s^2 \alpha,\ s^{-2} \beta). \tag{25.5}$$

On X_m^+ we have $\mu_1, \mu_2 > 0$ and $\mu_1 \mu_2 = m^2$. Take $s^2 = \mu_1 / m$. Then $s^{-2} m = \mu_2$, so $p = \Lambda(u d_s) k_m$. On C^+ we have $\mu_1 = 2p_0$ and $\mu_2 = 0$; take $s^2 = \mu_1 / 2$. On Y_κ we have $\mu_1 > 0 > \mu_2$ and $\mu_1 \mu_2 = -\kappa^2$; take $s^2 = \mu_1 / \kappa$. Since $\Lambda(g)$ is linear, X_m^- and C^- are the orbits of $-k_m$ and $-k$. Finally 0 is fixed. ∎

By (25.5), $\Lambda(d_s)$ multiplies $p_0 + p_3$ by s^2 and $p_0 - p_3$ by s^{-2}. It is a boost along the x_3-axis.

A subset of $\mathbb{R}^4$ is *locally closed* when it is the intersection of a closed set and an open set. Each orbit is locally closed. For example, C^+ is the closed set $\{p \cdot p = 0\}$ intersected with the open set $\{p_0 > 0\}$.

The invariant Borel sets $\{p \cdot p < r\}$ with r rational, together with $\{p \cdot p \geq 0,\ p_0 > 0\}$, $\{p \cdot p \geq 0,\ p_0 < 0\}$ and $\{0\}$, separate the orbits. Local closedness and this separation property form the regularity used below.

25.3 Little groups and Mackey's classification

Definition 25.7. The *little group* of $p \in \mathbb{R}^4$ is its stabiliser $H_p = \{g \in \mathrm{SL}(2, \mathbb{C}) : \Lambda(g) p = p\}$.

For $b \in \mathbb{C}$ and $|\lambda| = 1$ write

$$n(b) = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}, \qquad r(\lambda) = \begin{pmatrix} \lambda & 0 \\ 0 & \bar{\lambda} \end{pmatrix}.$$

Proposition 25.8. *(a)* $H_{k_m} = \mathrm{SU}(2)$.

(b) $H_k = \{n(b) r(\lambda) : b \in \mathbb{C},\ |\lambda| = 1\}$, *and* $r(\lambda) n(b) r(\lambda)^{-1} = n(\lambda^2 b)$.

(c) $\Lambda(r(e^{-i\theta/2}))$ *rotates the* (x_1, x_2)*-plane by the angle* $+\theta$ *and fixes* x_0 *and* x_3.

(d) $H_{\ell_\kappa} = \{g : g \sigma_3 g^* = \sigma_3\} = \mathrm{SU}(1, 1)$, *where* $\sigma_3 = \operatorname{diag}(1, -1)$. *Also* $H_0 = \mathrm{SL}(2, \mathbb{C})$.

Proof. (a) The condition is $g(mI)g^{*} = mI$, that is, $gg^{*} = I$.

(b) Since $\Phi(k) = 2e_1e_1^{*}$, the condition is $vv^{*} = e_1e_1^{*}$ for $v = ge_1$. In entries,

$$\begin{pmatrix} |v_1|^2 & v_1\bar{v}_2 \\ v_2\bar{v}_1 & |v_2|^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

So $v_2 = 0$ and $v_1 = \lambda$ with $|\lambda| = 1$. Since $\det g = 1$,

$$g = \begin{pmatrix} \lambda & w \\ 0 & \bar{\lambda} \end{pmatrix} = n(\lambda w)\, r(\lambda).$$

Every such g fixes k. The conjugation formula is a product of three matrices:

$$\begin{pmatrix} \lambda & 0 \\ 0 & \bar{\lambda} \end{pmatrix} \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \bar{\lambda} & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} 1 & \lambda^2 b \\ 0 & 1 \end{pmatrix}.$$

(c) Conjugation by $r(\lambda)$ fixes the diagonal of $\Phi(p)$. It multiplies the entry $p_1 + ip_2$ by $\bar{\lambda}^2$, which is $e^{i\theta}$ for $\lambda = e^{-i\theta/2}$.

(d) This restates the definition, since $\Phi(\ell_\kappa) = \kappa\sigma_3$. ∎

Write $N = \{n(b)\} \cong \mathbb{C}$ and $T = \{r(\lambda)\} \cong \mathrm{U}(1)$. By the factorisation in Proposition 25.8(b), $H_k = N \rtimes T$. The map $n(b)r(\lambda) \mapsto (b, \lambda^2)$ is two-to-one onto the group of orientation-preserving Euclidean motions of the plane. So we call N the *little-group translations*. The maps $\Lambda(n(b))$ are Lorentz transformations fixing k, not spacetime translations.

We have $r(e^{-i\theta/2}) = \exp(\theta X_3)$ with $X_3 = -i\sigma_3/2$. By the rotation formula in Proposition 25.8(c), this generates positive-angle rotations about x_3, as in Lecture 24.

The next theorem uses unitary induction from Lecture 17.

Theorem 25.9 (Mackey, cited). *Let $G = \mathbb{R}^n \rtimes H$, where H is a second-countable Lie group acting linearly and continuously on $\mathbb{R}^n$. For $p \in (\mathbb{R}^n)^{*}$ put $\chi_p(a) = e^{-ip(a)}$. Let H act on these momenta so that $\chi_{h\cdot p}(a) = \chi_p(h^{-1}\cdot a)$. Assume the action is* regular*: orbits are locally closed, and countably many invariant Borel sets separate them. Let H_p be the stabiliser of p. For a strongly continuous unitary representation U of G, let E be the projection-valued measure on $(\mathbb{R}^n)^{*}$ that the SNAG theorem attaches to the translations $a \mapsto U(a, e)$.*

(a) If U is irreducible, then E is concentrated on one orbit Ω.

(b) Let $p \in \Omega$ and let σ be a strongly continuous irreducible unitary representation of H_p. Put $(\chi_p \otimes \sigma)(a, h) = \chi_p(a)\sigma(h)$. Then $\operatorname{Ind}_{\mathbb{R}^n \rtimes H_p}^{G}(\chi_p \otimes \sigma)$ is irreducible, and its measure E is concentrated on Ω.

(c) Every strongly continuous irreducible unitary representation whose measure E is concentrated on Ω is unitarily equivalent to one of the induced representations in (b). For a fixed $p \in \Omega$, two of these induced representations are unitarily equivalent exactly when their representations σ are unitarily equivalent.

This is Mackey's method for semidirect products, as presented in A. A. Kirillov, *Elements of the Theory of Representations* (1976), §13. For our G it is Wigner's classification of 1939. For our spacetime group, the Minkowski pairing identifies the functionals $p \in (\mathbb{R}^4)^{*}$ of the theorem with the vectors p used above. The remarks after the orbit theorem, Theorem 25.6, verify regularity. The dual action is $g \cdot p = \Lambda(g)p$ by (25.4).

Corollary 25.10. *(a) For $m > 0$, up to unitary equivalence, the strongly continuous irreducible unitary representations of G with E concentrated on X_m^+ are labelled by $n \geq 0$. The label is the $\mathrm{SU}(2)$-representation V_n, of dimension $n+1$ and spin $n/2$.*

(b) Up to unitary equivalence, those with E concentrated on C^+ are labelled by the equivalence classes of strongly continuous irreducible unitary representations of $H_k = N \rtimes T$.

Proof. Combine Mackey's theorem, Theorem 25.9, with the little groups computed in Proposition 25.8. In the massive case, strongly continuous irreducible unitary representations of the compact group $\mathrm{SU}(2)$ are finite-dimensional by Lecture 10. Hence they are the V_n of Lecture 8. ∎

25.4 Massless sectors: helicity and continuous spin

Proposition 25.11. *Let σ be a finite-dimensional continuous unitary representation of H_k on V. Then $\sigma(n(b)) = I$ for all b. If σ is irreducible, then $\sigma(n(b)r(\lambda)) = \lambda^\nu$ for a unique $\nu \in \mathbb{Z}$, and*

$$\sigma(\exp \theta X_3) = e^{-i\theta\nu/2}.$$

Proof. The operators $\sigma(n(b))$ are unitary and commute. So V is the direct sum of their joint eigenspaces. On each, $b \mapsto \sigma(n(b))$ is a continuous character of $\mathbb{C} \cong \mathbb{R}^2$. As in the character lemma, Lemma 25.4, it has the form $b \mapsto e^{-i\,\mathrm{Re}(\bar\zeta b)}$ for a unique $\zeta \in \mathbb{C}$. Let V_ζ be the joint eigenspace for ζ. Let S be the finite set of ζ with $V_\zeta \neq 0$.

This is the key step. By the conjugation formula in Proposition 25.8(b), $n(b)r(\lambda) = r(\lambda)n(\bar\lambda^2 b)$. Take $v \in V_\zeta$. Then

$$\sigma(n(b))\sigma(r(\lambda))v = \sigma(r(\lambda))\sigma(n(\bar\lambda^2 b))v = e^{-i\,\mathrm{Re}(\overline{\lambda^2\zeta}\, b)}\,\sigma(r(\lambda))v.$$

So $\sigma(r(\lambda))V_\zeta = V_{\lambda^2\zeta}$. Every unit complex number is some λ^2. If some $\zeta \neq 0$ lay in S, the whole circle $|\zeta'| = |\zeta|$ would lie in S. This contradicts finiteness. So $S = \{0\}$, and N acts trivially.

Then σ factors through $n(b)r(\lambda) \mapsto \lambda$. This is a homomorphism onto $\mathrm{U}(1)$, since $n(b)r(\lambda)n(b')r(\lambda') = n(b + \lambda^2 b')r(\lambda\lambda')$. The operators $\sigma(r(\lambda))$ commute. If σ is irreducible, a common eigenline is invariant, so $\dim V = 1$. By the one-variable character argument of Exercise 25.2(a), $\theta \mapsto \sigma(r(e^{i\theta}))$ equals $e^{i\nu\theta}$ for some real ν. Periodicity in θ forces $\nu \in \mathbb{Z}$. Finally $\exp \theta X_3 = r(e^{-i\theta/2})$ maps to $e^{-i\nu\theta/2}$. ∎

Compare with the generator convention $\Pi(\exp \theta X) = \exp(-i\theta A_X)$ of Lecture 15. The self-adjoint generator of rotations about the momentum direction is $A_{X_3} = \nu/2$. This number is the *helicity*.

The *finite-helicity* massless representations are the representations induced from $\chi_k \otimes \sigma$, where σ is a finite-dimensional continuous irreducible unitary representation of H_k. By Proposition 25.11, these are the representations $\mathrm{Ind}(\chi_k \otimes \sigma_\nu)$ with $\nu \in \mathbb{Z}$ and $\sigma_\nu(n(b)r(\lambda)) = \lambda^\nu$. Each has the single helicity $\nu/2$. In these representations the little-group translations act trivially.

Remark 25.12. Mackey's theorem, Theorem 25.9, also applies to $H_k = N \rtimes T$. The T-orbits on the characters of N are $\{0\}$ and the circles $|\zeta| = \rho > 0$. These orbits are closed. The sets $\{|\zeta| < r\}$ with $r \in \mathbb{Q}$, together with $\{0\}$, separate them. Hence the action is regular.

The orbit $\{0\}$ gives the σ_ν. A circle has stabiliser $\{r(\pm 1)\}$ with two characters. These give infinite-dimensional irreducible representations of H_k that are nontrivial on N (Exercise 25.7). Inducing them to G gives the *continuous-spin* representations. Their spectral measure is concentrated on C^+, so their energy is positive. Positive energy alone does not exclude them. They form a sector separate from finite helicity.

Now suppose a nonzero representation has E concentrated on C^+. Its closed support meets C^+ and is Lorentz-invariant by covariance of E. Transitivity then puts all of C^+ in the support. Closedness adds its limit point 0. So concentration on an orbit differs from equality of closed supports.

The orbit $\{0\}$ is also separate. Translations act trivially there. Any strongly continuous irreducible unitary representation of $\mathrm{SL}(2,\mathbb{C})$ gives a representation of this zero-momentum sector by composition with the projection $G \to \mathrm{SL}(2,\mathbb{C})$. The trivial one is the usual vacuum choice. Positive energy alone does not select it.

Positive energy excludes X_m^- and C^-, where $p_0 < 0$. It also excludes Y_κ. Indeed, the set $S = Y_\kappa \cap \{p_0 < 0\}$ is nonempty and relatively open. By transitivity, the Lorentz translates of S cover Y_κ. By second countability, countably many of these translates already cover Y_κ. Positive energy gives $E(S) = 0$, since $S \subset \{p_0 < 0\}$. Then covariance of E makes every translate of S an E-null set. Disjointising this countable cover and using countable additivity would give $E(Y_\kappa) = 0$. This contradicts concentration of E on that orbit in a nonzero representation.

25.5 The massive induced model

Fix $m > 0$ and $n \geq 0$. Let D_n be the irreducible unitary representation of $\mathrm{SU}(2)$ on V_n. Put $\omega(\vec{p}) = \sqrt{m^2 + |\vec{p}|^2}$. Define a measure μ_m on X_m^+ by

$$\int_{X_m^+} f \, d\mu_m = \int_{\mathbb{R}^3} f(\omega(\vec{p}), \vec{p}) \, \frac{d^3\vec{p}}{2\,\omega(\vec{p})}.$$

For $p \in X_m^+$ the matrix $\Phi(p)$ is positive definite. Let $h_p = (\Phi(p)/m)^{1/2}$ be its positive square root. Then $\det h_p = 1$ and $h_p(mI)h_p^* = \Phi(p)$, so $\Lambda(h_p)k_m = p$. The *Wigner rotation* is

$$W(g,p) = h_p^{-1}\, g\, h_{\Lambda(g)^{-1}p}.$$

The model space is $\mathcal{H}_{m,n} = L^2(X_m^+, \mu_m; V_n)$. The operators are

$$(U(a,g)\psi)(p) = e^{-ip\cdot a}\, D_n\big(W(g,p)\big)\, \psi(\Lambda(g)^{-1}p). \tag{25.6}$$

Proposition 25.13. *(a)* $W(g,p) \in \mathrm{SU}(2)$.
(b) $W(gg',p) = W(g,p)\,W(g',\Lambda(g)^{-1}p)$.
(c) $U((a,g)(a',g')) = U(a,g)\,U(a',g')$.
(d) If μ_m is Λ-invariant, then each $U(a,g)$ is unitary.
(e) If $g \in \mathrm{SU}(2)$, then $W(g,p) = g$.

Proof. (a) Put $q = \Lambda(g)^{-1}p$. Then h_q maps k_m to q, g maps q to p, and h_p^{-1} maps p to k_m. So $W(g,p) \in H_{k_m} = \mathrm{SU}(2)$.

(b) With q as above, $W(g,p)W(g',q) = h_p^{-1} g h_q \, h_q^{-1} g' h_{\Lambda(g')^{-1}q}$. The middle factors cancel, and $\Lambda(g')^{-1}q = \Lambda(gg')^{-1}p$.

(c) Apply (25.6) twice:

$$U(a,g)U(a',g')\psi(p) = e^{-ip\cdot a}\, e^{-iq\cdot a'}\, D_n(W(g,p))\, D_n(W(g',q))\, \psi(\Lambda(g')^{-1}q).$$

By the pairing invariance in Lemma 25.1, $q \cdot a' = p \cdot \Lambda(g)a'$, so the phase is $e^{-ip\cdot(a+\Lambda(g)a')}$. By the cocycle identity (b), the operator is $D_n(W(gg',p))$. The argument is $\Lambda(gg')^{-1}p$. By (25.2) this is $U(a+\Lambda(g)a', gg')\psi(p)$.

(d) The phase has modulus one and D_n is unitary. Therefore

$$\|U(a,g)\psi\|^2 = \int \|\psi(\Lambda(g)^{-1}p)\|^2 \, d\mu_m(p) = \|\psi\|^2.$$

By the homomorphism property (c), $U(a,g)$ has inverse $U((a,g)^{-1})$.

(e) For unitary g, the matrix gh_qg^{-1} is positive definite, and its square is $g\Phi(q)g^*/m = \Phi(p)/m$. By uniqueness of the positive square root, $gh_qg^{-1} = h_p$. Hence $W(g,p) = g$. ∎

By the rotation case of the Wigner rotation, Proposition 25.13(e), rotations act by $(U(0,u)\psi)(p) = D_n(u)\psi(\Lambda(u)^{-1}p)$. At the rest momentum k_m the little group acts through D_n, which is spin $n/2$. The central element acts by

$$U(z) = D_n(-I) = (-1)^n I.$$

Hence half-integer spin does not descend to $\mathbb{R}^4 \rtimes \mathrm{SO}^+(1,3)$. Time translation by t acts by multiplication by e^{-itp_0}. Its self-adjoint generator P_0 is multiplication by p_0, with domain

$$\mathrm{Dom}(P_0) = \{\psi \in \mathcal{H}_{m,n} : p_0\psi \in \mathcal{H}_{m,n}\}.$$

Since $p_0 \geq m$ on X_m^+, we get $P_0 \geq m$. The energy is positive.

Three steps complete the identification with a unitary induced representation. The measure and intertwining calculations are assigned to the sheet. Continuity is explained here.

1. *Invariance of μ_m.* Exercise 25.4 writes $g = u\, d_s\, v$ with $u, v \in \mathrm{SU}(2)$. Rotations fix p_0 and act orthogonally on $\vec{p}$. The boost d_s needs a Jacobian computation.
2. *Strong continuity.* For continuous compactly supported ψ, the formula is jointly continuous in (a,g,p). The positive square root h_p is continuous by matrix functional calculus. For (a,g) in a compact neighbourhood, transformed supports lie in one compact set and the integrands are uniformly bounded there. Dominated convergence gives norm continuity. Density and unitarity extend it to all ψ.
3. *Identification with induction.* The section map is

   $$f \longmapsto \psi_f, \qquad \psi_f(p) = f(0, h_p).$$

   It carries the induced representation from $A \rtimes \mathrm{SU}(2)$, with inducing data $\chi_{k_m} \otimes D_n$, onto (25.6). Exercise 25.5 checks the intertwining. Then the irreducibility of induced representations in Mackey's theorem, Theorem 25.9(b), shows that the model is irreducible.

Remark 25.14. The finite-dimensional representations of $\mathrm{SL}(2,\mathbb{C})$ admit no invariant inner product unless trivial, as proved in Lecture 24. In physics they label the components of fields. They do not classify the particle Hilbert space. For a massive particle of spin $n/2$, that space is $\mathcal{H}_{m,n}$, built from unitary $\mathrm{SU}(2)$ data on an orbit.

Self-checks

1. Is $(-1,0,0,0)$ in the orbit of k_1?
   Answer. No. $\Phi(k_1) = I$ is positive definite, but $\Phi(-1,0,0,0) = -I$ is not. Conjugation $\Phi \mapsto g\Phi g^*$ preserves positivity.
2. Do the defining matrices of H_k on $\mathbb{C}^2$ preserve some inner product?
   Answer. No. By Proposition 25.11, every finite-dimensional continuous unitary representation of H_k is trivial on N. An invariant inner product would make the defining representation such a representation. But $n(1) \neq I$.
3. In the model with $n = 2$, what are $U(z)$ and $W(u,p)$ for $u \in \mathrm{SU}(2)$?
   Answer. $U(z) = D_2(-I) = 1$, and $W(u,p) = u$ by the rotation case of the Wigner rotation, Proposition 25.13(e).

Summary

The Poincaré cover is $G = \mathbb{R}^4 \rtimes \mathrm{SL}(2,\mathbb{C})$. Translations act through momentum characters $e^{-ip\cdot a}$, and $\mathrm{SL}(2,\mathbb{C})$ moves momenta by Λ. The orbits are mass shells, light cones, spacelike hyperboloids and $\{0\}$. Mackey's theorem reduces the classification to little groups. Up to unitary equivalence, irreducible unitary representations with E concentrated on X_m^+ are labelled by an $\mathrm{SU}(2)$ spin $n/2$ with $n \geq 0$. Finite-helicity massless representations have trivial little-group translations. Continuous-spin sectors also have positive energy and are separate. The zero-momentum orbit $\{0\}$ gives a further separate sector.

Reading

- R. Berndt, *Representations of Linear Groups* (2007), Section 5.2, Theorem 5.2: the spectral-measure theorem for vector groups.
- E. P. Wigner, *On Unitary Representations of the Inhomogeneous Lorentz Group*, Annals of Mathematics 40 (1939), 149–204. Read the classification of momentum orbits and little groups.
- A. A. Kirillov, *Elements of the Theory of Representations* (1976), §13. Read the treatment of induced representations and Mackey's method for semidirect products.

Lecture 26

Equivariant line bundles and Borel–Weil

Objectives. *Induced representations act on equivariant functions. The same functions can be read as sections of a vector bundle. In the Borel–Weil setting, holomorphic sections of an associated line bundle form a finite-dimensional representation. We compute these holomorphic sections completely for the projective line. The general Borel–Weil theorem is a cited extension.*

By the end of this lecture you will be able to:

- Compute all holomorphic sections of $\mathcal{O}(m)$ on $\mathbb{CP}^1$.
- Identify the resulting SU(2) representation and its central action.
- State Borel–Weil with a specified bundle character and dual convention.

26.1 From equivariant functions to sections

Let G be a Lie group, H a closed subgroup and $\sigma : H \to \mathrm{GL}(V)$ a continuous finite-dimensional complex representation. The quotient projection $G \to G/H$ is a submersion by Lecture 3. The local-section theorem [C: Lee, *Introduction to Smooth Manifolds*, second edition, Theorem 4.26] therefore gives a smooth local section near every point of G/H. We now attach a copy of V above each coset.

Definition 26.1. The *associated vector bundle* is

$$E = G \times_H V = (G \times V)/\!\sim, \qquad (gh, v) \sim (g, \sigma(h)v).$$

Its projection is $[g, v] \mapsto gH$. A *section* is a map $s : G/H \to E$ whose value at x lies in the fibre above x. If $\dim V = 1$, it is a *line bundle*.

Each fibre is a vector space: choose g over the coset and use $v \mapsto [g, v]$. Changing g changes this identification by an invertible linear map.

Choose smooth local sections $s_i : U_i \to G$ whose domains cover G/H. The map

$$U_i \times V \longrightarrow E|_{U_i}, \qquad (x, v) \longmapsto [s_i(x), v]$$

is a bijection, because every g over x has the form $s_i(x)h$ with $h \in H$. It is linear on each fibre. On $U_i \cap U_j$, write $s_j(x) = s_i(x)h_{ij}(x)$. The same bundle vector then has coordinates

$$v_i = \sigma(h_{ij})v_j.$$

The identity $h_{ij}h_{jk} = h_{ik}$ ensures consistency on triple overlaps.

A *smooth complex vector bundle* of rank k over a manifold M is a smooth manifold E with a smooth surjection onto M and a complex vector-space structure on each fibre E_x, the preimage of x. It is required to be locally a product. Each point of M has an open neighbourhood U and a diffeomorphism $E|_U \to U \times \mathbb{C}^k$ mapping each fibre E_x complex-linearly onto $\{x\} \times \mathbb{C}^k$. These diffeomorphisms are the *smooth local trivialisations*.

The bundle $E = G \times_H V$ has this structure. The local sections s_i are smooth, so the maps $h_{ij} = s_i^{-1}s_j$ are smooth into G. They take values in the embedded subgroup H, so they are smooth into H by the codomain-restriction result for embedded submanifolds [C: Lee, Corollary 5.30]. The representation σ is smooth by the automatic-smoothness theorem of Lecture 3. Hence the transition functions $\sigma(h_{ij})$ are smooth. Let $k = \dim V$, choose a basis of V, and regard $\mathbb{C}^k$ as $\mathbb{R}^{2k}$. Each fibre of E is then a real vector space of dimension $2k$, and the real matrices of $\sigma(h_{ij})$ are smooth maps into $\mathrm{GL}(2k, \mathbb{R})$. The vector bundle chart lemma [C: Lee, Lemma 10.6] applies to these real data. It gives E a unique topology and smooth structure for which the inverses of the maps above are smooth local trivialisations. These maps are complex-linear on fibres, so E is a smooth complex vector bundle.

This structure does not depend on the chosen family of local sections. For a second family (s'_j), the maps $s_i^{-1}s'_j$ are smooth into H by the same codomain-restriction argument. So the mixed transition functions $\sigma(s_i^{-1}s'_j)$ are smooth, and the chart lemma applies to the union of the two families. By its uniqueness clause, the union gives the same structure as each family alone.

A section is equivalently a function $f : G \to V$ with

$$f(gh) = \sigma(h)^{-1} f(g). \tag{26.1}$$

The correspondence is $s(gH) = [g, f(g)]$. It is well-defined because

$$[gh, f(gh)] = [g, \sigma(h) f(gh)] = [g, f(g)].$$

Conversely, a section determines a unique $f(g)$ in the coordinates supplied by g. Over U_i, the section is $x \mapsto [s_i(x), f(s_i(x))]$. The local trivialisation is a diffeomorphism, so s is smooth over U_i exactly when the coefficient $f \circ s_i$ is smooth. If f is smooth, then each $f \circ s_i$ is smooth. Conversely, suppose that each $f \circ s_i$ is smooth. Write $\pi : G \to G/H$ for the projection, and put

$$q(g) = s_i(\pi(g))^{-1} g \qquad (g \in \pi^{-1}(U_i)).$$

This map is smooth into G and takes values in H, so it is smooth into the embedded subgroup H by the same codomain-restriction result. Since $g = s_i(\pi(g))q(g)$, the covariance (26.1) gives

$$f(g) = \sigma\big(q(g)\big)^{-1} f\big(s_i(\pi(g))\big).$$

Hence f is smooth on each open set $\pi^{-1}(U_i)$, and therefore on G. The same argument with continuity in place of smoothness shows that s is continuous exactly when f is continuous.

We also need a holomorphic version of this construction. A function on an open subset of $\mathbb{C}^d$ is *holomorphic* if it is real C^∞ and its differential is $\mathbb{C}$-linear at every point. Polynomials are holomorphic, and so are quotients of polynomials where the denominator is nonzero. Composition of holomorphic maps is holomorphic by the chain rule.

Let Φ be a map between open subsets of $\mathbb{C}^d$ with holomorphic components and invertible differential at a point. The real inverse function theorem gives a smooth local inverse. Its differential is the inverse of a $\mathbb{C}$-linear map, hence $\mathbb{C}$-linear. So the local inverse is holomorphic. This is the *holomorphic inverse function theorem*. A *complex manifold* is a Hausdorff, second-countable topological space covered by charts. These charts are homeomorphisms of open sets onto open subsets of $\mathbb{C}^d$. The transition maps between them have holomorphic components. A map between complex manifolds is *holomorphic* if it is continuous and its expressions in charts have holomorphic components.

A *complex submanifold* of a complex manifold M is a subset N with the following property. Each point of N lies in a chart of M in which N is given by the vanishing of some coordinates. The remaining coordinates give charts of N. Hence a holomorphic map into M with values in N is also holomorphic as a map into N. We call this the *slice-chart property*.

A *complex Lie group* is a group with a complex manifold structure for which multiplication and inversion are holomorphic. The groups $K_{\mathbb{C}}$ of Lecture 14 are examples. A *closed complex subgroup* of a complex Lie group G is a closed subgroup that is also a complex submanifold.

Suppose that G is a complex Lie group, H is a closed complex subgroup and G/H is a complex manifold. Suppose also that the projection $\pi : G \to G/H$ is holomorphic, that σ is holomorphic and that the local sections s_i are holomorphic. Assume the U_i cover G/H.

If $g = s_i(x)h$, then $R_h \circ s_i$ is a holomorphic, hence smooth, local section through g, because $\pi \circ R_h = \pi$. By the local-section theorem of Lee, Theorem 4.26, π is a smooth submersion.

The topology of G/H is the quotient topology. Indeed, π is continuous, since it is holomorphic. Conversely, let $U \subset G/H$ be a set such that $\pi^{-1}(U)$ is open. Then $U \cap U_i = s_i^{-1}(\pi^{-1}(U))$ is open for each i, because $\pi \circ s_i = \mathrm{id}$. The sets U_i cover G/H, so U is open. By the uniqueness of the quotient structure in Lecture 3, the smooth structure underlying the complex manifold G/H is the quotient smooth structure.

In this case $h_{ij} = s_i^{-1} s_j$ is holomorphic as a map into G, because the group operations of G are holomorphic. By the slice-chart property, h_{ij} is holomorphic as a map into H. Then $\sigma(h_{ij})$ is holomorphic, as a composition of holomorphic maps.

Over the domain of a holomorphic chart φ of G/H inside U_i, the map $[s_i(x), v] \mapsto (\varphi(x), v)$ is a chart of E, with V identified with $\mathbb{C}^k$ by a basis. The change between two such charts has the form $(y, u) \mapsto (\psi(y), A(y)u)$. Here ψ is a transition map of G/H, and $A(y)$ is the matrix of $\sigma(h_{ij})$ at the point with coordinate y. The entries of A are holomorphic, so this change is holomorphic. Hence these charts make E a complex manifold. In this case E is a *holomorphic* vector bundle. The local trivialisations given by the sections s_i are its *holomorphic bundle charts*. For a second family (s'_j) of holomorphic local sections, the mixed changes of chart have the same form with $\sigma(s_i^{-1} s'_j)$, which is holomorphic by the slice-chart property. Hence both families give one holomorphic atlas, and the complex structure of E does not depend on the family. A section is *holomorphic* when each coefficient $f \circ s_i$ is holomorphic.

Left multiplication defines $a[g, v] = [ag, v]$. On sections its action is

$$(\Pi(a)s)(x) = a\, s(a^{-1}x).$$

In the function description this becomes $(\Pi(a)f)(g) = f(a^{-1}g)$. This is precisely the compact induction convention of Lecture 12.

26.2 The projective line and its two charts

Definition 26.2. The *complex projective line* $\mathbb{CP}^1$ is the set of one-dimensional complex subspaces of $\mathbb{C}^2$. A nonzero vector (v_0, v_1) determines the point $[v_0 : v_1]$, with $[cv_0 : cv_1] = [v_0 : v_1]$ for $c \neq 0$.

There are two charts. On $U_0 = \{v_0 \neq 0\}$ put $z = v_1/v_0$. On $U_1 = \{v_1 \neq 0\}$ put $w = v_0/v_1$. On the overlap, the transition between the charts is $w = 1/z$.

The charts give $\mathbb{CP}^1$ its topology. The points $[1 : 0]$ and $[0 : 1]$ have disjoint neighbourhoods $\{|z| < 1\}$ and $\{|w| < 1\}$. Every other pair lies in a common chart, so the space is Hausdorff. The two charts give a countable base. Each chart is a copy of $\mathbb{C}$, and the transition $w = 1/z$ is holomorphic on the overlap. Hence the two charts make $\mathbb{CP}^1$ a complex manifold, in the sense of the first section.

The group $\mathrm{SU}(2)$ acts on lines by matrix multiplication. In the z chart, the action of $g = \left(\begin{smallmatrix} a & -\bar{b} \\ b & \bar{a} \end{smallmatrix}\right)$ is

$$z \longmapsto \frac{b + \bar{a}z}{a - \bar{b}z}$$

where the denominator is nonzero. The other chart expressions are similar. These rational expressions show that the action is smooth. It acts transitively. Indeed, a unit vector $v = (a, b)$ is the first column of

$$g_v = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix} \in \mathrm{SU}(2).$$

The stabiliser of the line $\mathbb{C}(1, 0)$ is

$$T = \{t_\theta = \mathrm{diag}(e^{i\theta}, e^{-i\theta})\}.$$

By the homogeneous-space theorem of Lecture 3, $\mathbb{CP}^1 \cong \mathrm{SU}(2)/T$ as smooth manifolds. It is compact, as the image of the compact group.

The *tautological line bundle* has fibre the line itself:

$$\mathcal{O}(-1) = \{(\ell, v) : \ell \in \mathbb{CP}^1,\ v \in \ell\}.$$

Its dual is $\mathcal{O}(1)$. For $m \geq 0$, let $\mathcal{O}(m) = \mathcal{O}(1)^{\otimes m}$. Set $\mathcal{O}(0)$ equal to the trivial line bundle. For $m < 0$, use the $(-m)$th tensor power of $\mathcal{O}(-1)$.

The local frames of $\mathcal{O}(-1)$ are $e_0(z) = (1, z)$ and $e_1(w) = (w, 1)$. On the overlap, $e_1 = z^{-1}e_0$. The dual frames satisfy $e_1^* = ze_0^*$. For $m < 0$, $(e_i^*)^m$ means $e_i^{\otimes(-m)}$. Since $e_1^{\otimes(-m)} = z^m e_0^{\otimes(-m)}$, the same transition rule holds for $m < 0$. Hence a section of $\mathcal{O}(m)$ is a pair of functions f_0, f_1 with

$$f_0(z) = z^m f_1(1/z), \qquad f_1(w) = w^m f_0(1/w). \tag{26.2}$$

The frames $(e_0^*)^m$ and $(e_1^*)^m$ give fibrewise-linear bijections $\mathcal{O}(m)|_{U_i} \to U_i \times \mathbb{C}$. Their transition is multiplication by z^m, which is smooth and holomorphic on $U_0 \cap U_1$. The vector bundle chart lemma of Lee, Lemma 10.6, therefore makes $\mathcal{O}(m)$ a smooth line bundle with these trivialisations. We therefore call the section *holomorphic* when both functions are *entire*, that is, holomorphic on all of $\mathbb{C}$. The Borel–Weil discussion below shows that this definition agrees with the notion for associated bundles. We denote this space by $H^0(\mathbb{CP}^1, \mathcal{O}(m))$. The superscript 0 means ordinary global sections.

26.3 Computing the holomorphic sections

We import two results from complex analysis, the power-series theorem and Cauchy's estimate. If f is entire, then

$$f(z) = \sum_{k \geq 0} a_k z^k, \qquad |a_k| \leq R^{-k} \max_{|z|=R} |f(z)| \quad (R > 0).$$

Both are imported from [C: Stein and Shakarchi, *Complex Analysis*, 2003, Chapter 2, Section 4]. Theorem 4.4 and the paragraph after it state that an entire function has a power series about 0 that converges on all of $\mathbb{C}$, with $a_k = f^{(k)}(0)/k!$. The Cauchy inequalities, Corollary 4.3, give

$$|f^{(k)}(0)| \leq k! R^{-k} \max_{|z|=R} |f(z)|.$$

Dividing by $k!$ gives the estimate for a_k. These results apply to our entire functions, because a $\mathbb{C}$-linear differential makes a function complex differentiable.

Theorem 26.3. *For $m \geq 0$, the holomorphic sections of $O(m)$ are naturally the homogeneous polynomials of degree m on $\mathbb{C}^2$. In particular,*

$$\dim H^0(\mathbb{CP}^1, O(m)) = m + 1.$$

For $m < 0$, the space of holomorphic sections is zero.

Proof. Let (f_0, f_1) satisfy the transition rule (26.2). Since f_1 is bounded on $|w| \leq 1$, there is C with

$$|f_0(z)| \leq C|z|^m \qquad (|z| \geq 1).$$

If $f_0(z) = \sum a_k z^k$, Cauchy's estimate on a circle of radius $R \geq 1$ gives $|a_k| \leq CR^{m-k}$. Let $R \to \infty$. For $k > m$ this forces $a_k = 0$.

If $m \geq 0$, it follows that f_0 is a polynomial of degree at most m. Conversely every such polynomial works, because

$$f_0(z) = \sum_{k=0}^{m} a_k z^k \quad \Longrightarrow \quad f_1(w) = \sum_{k=0}^{m} a_k w^{m-k}$$

is entire and satisfies the transition. The corresponding homogeneous polynomial is

$$F(v_0, v_1) = \sum_{k=0}^{m} a_k v_0^{m-k} v_1^k.$$

Its chart coefficients are $F(1, z) = f_0(z)$ and $F(w, 1) = f_1(w)$. It obeys $F(cv) = c^m F(v)$ for $c \in \mathbb{C}$, so its restriction to each line is an element of that line's mth dual tensor power. Hence F is the coordinate-free description of the section.

If $m < 0$, every $k \geq 0$ satisfies $k > m$, so all a_k vanish. The transition then makes $f_1 = 0$ too. ∎

For $m = 2$, take $f_0(z) = a_0 + a_1 z + a_2 z^2$. Then $f_1(w) = a_0 w^2 + a_1 w + a_2$. Thus $F = a_0 v_0^2 + a_1 v_0 v_1 + a_2 v_1^2$.

26.4 The group action and its dual

A matrix $g \in \mathrm{SL}(2,\mathbb{C})$ sends a tautological fibre ℓ linearly to $g\ell$. The inverse transpose of this linear map sends the dual line ℓ^* to $(g\ell)^*$. Hence g acts on $O(1)$ and on all its tensor powers. Under the polynomial description, the section action is

$$(\Pi_m(g)F)(v) = F(g^{-1}v). \tag{26.3}$$

Indeed, at the line ℓ, the transformed functional evaluates a vector $v \in \ell$ by first pulling it back to $g^{-1}v \in g^{-1}\ell$. This gives the inverse in the formula.

The representation is $\mathrm{Sym}^m(\mathbb{C}^2)^*$, the dual of the symmetric tensor representation. For $\mathrm{SU}(2)$ this is isomorphic to V_m, by the prerequisite highest-weight theory. Here is also a direct weight check. On the basis $F_k = v_0^{m-k}v_1^k$,

$$\Pi_m(t_\theta)F_k = e^{i(2k-m)\theta}F_k, \qquad 0 \le k \le m.$$

The weights are $-m, -m+2, \dots, m$. They match V_m, with highest weight m and dimension $m+1$. Irreducibility is the symmetric-power result from Lie Algebras.

For a concrete matrix computation, put

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad g^{-1} = \begin{pmatrix} \bar{a} & \bar{b} \\ -b & a \end{pmatrix}.$$

For $m = 2$, in the ordered basis (v_0^2, v_0v_1, v_1^2), expansion gives

$$D^{(2)}(g) = \begin{pmatrix} \bar{a}^2 & -\bar{a}b & b^2 \\ 2\bar{a}\bar{b} & |a|^2 - |b|^2 & -2ab \\ \bar{b}^2 & a\bar{b} & a^2 \end{pmatrix}.$$

For example, the middle column comes from

$$(\bar{a}v_0 + \bar{b}v_1)(-bv_0 + av_1) = -\bar{a}b\,v_0^2 + (|a|^2 - |b|^2)v_0v_1 + a\bar{b}\,v_1^2.$$

The centre acts by $\Pi_m(-I) = (-1)^m I$. Thus the action on sections descends to $\mathrm{SO}(3)$ precisely for even m.

The same sections inside compact induction

For a homogeneous polynomial F of degree $m \ge 0$, a corresponding equivariant function $f_F : \mathrm{SU}(2) \to \mathbb{C}$ is given by $f_F(g) = F(g(1,0)^T)$. In the proof of Theorem 26.3, the section defined by F assigns to each line the restriction of F to that line. So $f_F(g)$ is the value of this restriction at the spanning vector $g(1,0)^T$. Since $gt_\theta(1,0)^T = e^{i\theta}g(1,0)^T$, homogeneity gives

$$f_F(gt_\theta) = e^{im\theta}f_F(g).$$

This is the covariance (26.1) for $H = T$ and $\sigma(t_\theta) = e^{-im\theta}$. Also

$$f_{\Pi_m(a)F}(g) = F(a^{-1}g(1,0)^T) = f_F(a^{-1}g),$$

so $F \mapsto f_F$ intertwines Π_m with the left-translation action $(\Pi(a)f)(g) = f(a^{-1}g)$ of the first section. A supplementary example in the solution of Exercise 26.5 identifies $O(m)$ with $\mathrm{SU}(2) \times_T \mathbb{C}_{-m}$, where T acts on $\mathbb{C}_{-m}$ by $e^{-im\theta}$.

26.5 The general Borel–Weil theorem

Let K be compact, connected, simply connected and semisimple. Its complexification $G = K_{\mathbb{C}}$ was introduced in Lecture 14. Sepanski, *Compact Lie Groups*, states Borel–Weil for a matrix model of G, so we first identify the two groups.

A faithful representation ρ of K exists [C: Bröcker and tom Dieck (BD), Section III.4, Theorem 4.1]. Averaging makes ρ unitary (Lecture 5). The entries of $r = \rho \oplus \overline{\rho} : K \to \mathrm{U}(N)$ generate $\mathscr{R}(K)$ [C: BD, Section III.4, Proposition 4.3].

Recall from Lecture 14 that each $\psi \in K_{\mathbb{C}} = \mathcal{X}(K)$ is a unital algebra homomorphism from $\mathscr{R}(K)$ to $\mathbb{C}$. The *coefficient map* $r_{\mathbb{C}}$ evaluates ψ on the matrix entries r_{ij} of r:

$$r_{\mathbb{C}}(\psi) = \big(\psi(r_{ij})\big)_{i,j=1}^{N}.$$

By [C: BD, Section III.8, Proposition 8.2 and the next paragraph], the map $r_{\mathbb{C}}$ sends $K_{\mathbb{C}}$ biholomorphically onto a closed subgroup and analytic submanifold of $\mathrm{GL}(N,\mathbb{C})$.

The resulting complex structure on $K_{\mathbb{C}}$ does not depend on r. Let r' be a second faithful unitary representation of K whose entries also generate $\mathscr{R}(K)$. Let $r'_{\mathbb{C}}$ be its coefficient map, defined by the same construction. Each entry of r' is a polynomial in the entries of r, and conversely. Every $\psi \in K_{\mathbb{C}}$ is a unital algebra homomorphism, so it respects these polynomial relations. Hence $r'_{\mathbb{C}} \circ r_{\mathbb{C}}^{-1}$ and its inverse are restrictions of polynomial maps. By the slice-chart property, they are holomorphic between the two analytic submanifolds. This common structure is the complex structure of Lecture 14.

The Lie algebra $\mathfrak{g}$ of $r_{\mathbb{C}}(K_{\mathbb{C}})$ is the complex span of that of $r(K)$ [C: BD, Proposition 8.4 and the preceding paragraph].

Since K is connected, so is $K_{\mathbb{C}}$ (Lecture 14). A connected Lie subgroup of $\mathrm{GL}(N,\mathbb{C})$ is determined by its Lie algebra [C: Kirillov Jr., *An Introduction to Lie Groups and Lie Algebras*, Theorem 3.40]. Hence $r_{\mathbb{C}}(K_{\mathbb{C}})$ is the complexification of $r(K)$ in Sepanski, Definition 7.48. We identify G with $r_{\mathbb{C}}(K_{\mathbb{C}})$ and K with $r(K)$, since $r_{\mathbb{C}}(\mathrm{ev}_g) = r(g)$.

Sepanski's construction applies to every faithful unitary r.

Fix a maximal torus T of K and positive roots. The *positive Borel subalgebra* $\mathfrak{b}$ is the complex Cartan subalgebra plus the positive root spaces. Let B be the connected Lie subgroup of G with this Lie algebra (Sepanski, Definition 7.48). By Sepanski, Lemma 7.49, B is closed. By Sepanski, Theorem 7.50, the inclusion induces a diffeomorphism from K/T onto the *flag manifold* G/B.

Since B is closed, it is an embedded Lie subgroup of G by the closed-subgroup theorem of Lecture 3. The inclusion of B with Sepanski's Lie subgroup structure is smooth into this embedded submanifold by Lee, Corollary 5.30. It is a bijective Lie group homomorphism, so it is a Lie group isomorphism by Lee, Corollary 7.6. Hence the embedded group B has Lie algebra $\mathfrak{b}$, and exp maps some neighbourhood O of 0 in $\mathfrak{b}$ diffeomorphically onto an open subset of B.

Choose a complex complement $\mathfrak{m}$ of $\mathfrak{b}$ in $\mathfrak{g}$. For $b \in B$, define $\Phi : \mathfrak{b} \times \mathfrak{m} \to G$ by

$$\Phi(X,Y) = b \exp X \exp Y.$$

The matrix exponential is holomorphic and maps $\mathfrak{g}$ into G, so Φ is holomorphic into G by the slice-chart property. Its differential at $(0,0)$ is the complex-linear isomorphism $(X,Y) \mapsto b(X+Y)$. By the holomorphic inverse function theorem, after shrinking O, there is a neighbourhood O' of 0 in $\mathfrak{m}$ such that Φ maps $O \times O'$ biholomorphically onto an open set $W' \subset G$. The set $\Phi(O \times \{0\}) = b \exp O$ is open in B. Because B has the subspace topology, it equals $B \cap W''$ for some open set W''. On $W = W' \cap W''$, the set B is exactly the slice $Y = 0$. Hence B is a closed complex subgroup.

The complex quotient theorem [C: Kirillov Jr., Theorem 2.11 and its proof] makes G/B a complex manifold, with holomorphic projection $\pi : G \to G/B$ and holomorphic local sections. Thus the first section applies to G, B and G/B.

Let λ be an integral weight. Since K is simply connected, $X^*(T) = P$ (Lecture 7). Let N_+ be the connected subgroup of G whose Lie algebra is the sum of the positive root spaces. By Sepanski, Lemma 7.49(d), each $b \in B$ is uniquely a product $t \exp(iH)n$ with $t \in T$, $H \in \mathfrak{t}$ and $n \in N_+$. These factors depend smoothly on b.

Our character of weight λ is $\exp H \mapsto e^{i\lambda(H)}$, so Sepanski's weight is our $i\lambda$ on $\mathfrak{t}$ (Sepanski, Definitions 6.23 and 6.26, Theorem 6.27 and formula (6.28)).

In our labels, Sepanski, Definition 7.52, defines $\chi_\lambda : B \to \mathbb{C}^\times$ by

$$\chi_\lambda\big(\exp(H')\exp(iH)n\big) = e^{i\lambda(H')-\lambda(H)} \qquad (H, H' \in \mathfrak{t},\ n \in N_+).$$

We import from Sepanski, Definition 7.52, the fact that this formula defines a homomorphism. The formula is well defined on $T = \exp \mathfrak{t}$ because $\lambda \in P = X^*(T)$. By the smooth dependence of the factors on b, the homomorphism χ_λ is smooth. It is trivial on N_+. Its differential at e is the complex-linear extension of $i\lambda$ on $\mathfrak{t}_{\mathbb{C}}$ and zero on the positive root spaces, so it is $\mathbb{C}$-linear. Left translations of B are holomorphic, and χ_λ is multiplicative. So its differential is $\mathbb{C}$-linear at every point, and χ_λ is holomorphic.

Define

$$L_\lambda = G \times_B \mathbb{C}_{-\lambda}, \qquad (gb, u) \sim (g, \chi_\lambda(b)^{-1}u).$$

Sections correspond to functions with $f(gb) = \chi_\lambda(b) f(g)$. Let $s_i : U_i \to G$ be holomorphic local sections of $\pi : G \to G/B$ whose domains cover G/B. By the first section, they and the holomorphic character χ_λ give holomorphic bundle charts of L_λ. A section is holomorphic in that sense exactly when its function $f : G \to \mathbb{C}$ is holomorphic. The proof is the argument of the first section, with holomorphic in place of smooth. In it, the map $q(g) = s_i(\pi(g))^{-1}g$ on $\pi^{-1}(U_i)$ is now holomorphic into B by the slice-chart property. The covariance writes $f(g)$ on $\pi^{-1}(U_i)$ as the product of $\chi_\lambda(q(g))$ and $f(s_i(\pi(g)))$, and both factors are holomorphic. Hence f is holomorphic on each set $\pi^{-1}(U_i)$, and therefore on G. Left translation acts as before. Write $H^0(G/B, L_\lambda)$ for the space of holomorphic sections of L_λ.

Theorem 26.4 (Borel–Weil; cited). *With the preceding positive Borel and bundle convention, if λ is dominant integral, then*

$$H^0(G/B, L_\lambda) \cong V_\lambda^*$$

as a K-representation. This is finite-dimensional and irreducible. If an integral λ is not dominant, then $H^0(G/B, L_\lambda) = 0$.

We import the theorem from Sepanski, Theorem 7.58, for the groups G and B above. Sepanski's Definition 7.54 calls a section holomorphic when its function on G is holomorphic.

By the equivalence above, his holomorphic sections are ours. Our L_λ is his line bundle of index $-\lambda$, and this change of index produces the dual. For $\mathrm{SU}(2)$, take r to be the standard representation. Its four entries generate the representative functions, and $r_{\mathbb{C}}(K_{\mathbb{C}}) = \mathrm{SL}(2,\mathbb{C})$ (Lecture 14). Thus $G = \mathrm{SL}(2,\mathbb{C})$, and B is the subgroup of upper triangular matrices in G. The character is

$$\chi_m(b) = c^m, \qquad b = \begin{pmatrix} c & \beta \\ 0 & c^{-1} \end{pmatrix} \in B.$$

It is trivial on the unipotent matrices, where $c = 1$. On the diagonal matrices it is $\mathrm{diag}(c, c^{-1}) \mapsto c^m$, the character of weight m. The fibre at $\mathbb{C}(1,0)$ transforms by c^{-m}.

The map $gB \mapsto [g(1,0)^T]$ identifies G/B with $\mathbb{CP}^1$. Indeed, G acts transitively on lines, since $\mathrm{SU}(2)$ already does, and B is the stabiliser of the line $\mathbb{C}(1,0)$.

The preliminary note in the solution of Exercise 26.5 compares the two section spaces. With Theorem 26.3, it follows that $F \mapsto f_F$, with $f_F(g) = F(g(1,0)^T)$ for $g \in G$, is an isomorphism $H^0(\mathbb{CP}^1, \mathcal{O}(m)) \cong H^0(G/B, L_m)$ of $\mathrm{SU}(2)$-representations for $m \geq 0$. For $m < 0$, both spaces are zero. Hence our explicit calculation is exactly the case $\mathrm{SU}(2)$ of Borel–Weil.

Self-checks

1. What is wrong with choosing an arbitrary entire f_0 as a section of $\mathcal{O}(1)$?

Answer. The coefficient $f_1(w) = w f_0(1/w)$ must also extend holomorphically to $w = 0$. This forces f_0 to have degree at most one.

2. Which representation occurs in $H^0(\mathbb{CP}^1, \mathcal{O}(3))$?

Answer. It is $\mathrm{Sym}^3(\mathbb{C}^2)^* \cong V_3$, of dimension four and spin $3/2$. The central element $-I$ acts by the scalar $(-1)^3 = -1$, so this representation does not descend to $\mathrm{SO}(3)$.

3. Does L_λ always produce highest weight λ in our general convention?

Answer. No. Its section space is V_λ^*, of highest weight $-w_0\lambda$. For $\mathrm{SU}(2)$ these weights coincide; for general groups they need not.

Summary

Equivariant functions correspond to sections of associated bundles. On $\mathbb{CP}^1$ and for $m \geq 0$, the transition rule for $\mathcal{O}(m)$ identifies holomorphic sections with homogeneous polynomials of degree m. Borel–Weil extends this construction to flag manifolds.

Lecture 27

Compact orbits and geometric quantisation

Objectives. *A compact coadjoint orbit has a symplectic form. Not every such orbit supplies a representation of the chosen global group. We calculate the obstruction on a sphere. The symplectic area of the sphere divided by $2\pi\hbar$ must be integral. An equivariant lift imposes a further condition when the group is* $\mathrm{SO}(3)$. *Holomorphic sections then connect the orbit to the representations of the previous lecture.*

By the end of this lecture you will be able to:

- Compute the coadjoint-orbit form of $\mathrm{SU}(2)$ with its radius normalisation.
- Define a unitary connection locally and prove the sphere integrality condition.
- Distinguish an ordinary prequantum bundle from an equivariant one.
- Recover spin $m/2$ by holomorphic sections and keep this construction separate from shifted character formulas.

27.1 The orbit sphere and its area

Use the basis of Lecture 4,

$$X_j = -\frac{i}{2}\sigma_j, \qquad [X_1, X_2] = X_3$$

with cyclic versions of the bracket. The Pauli matrices satisfy

$$\sigma_i\sigma_j = \delta_{ij}I + i\sum_k \epsilon_{ijk}\sigma_k.$$

Identify $\mathfrak{su}(2)^*$ with $\mathbb{R}^3$ by $x_j = \ell(X_j)$. For $u = (u_1, u_2, u_3)$ put $X_u = \sum_j u_j X_j$. Then $[X_u, X_v] = X_{u\times v}$.

The adjoint action is the rotation cover from Lecture 4. Its dual action is the same rotation action in these orthonormal coordinates. Thus its orbits are the spheres

$$S_r^2 = \{x : |x| = r\} \quad (r > 0)$$

and the origin. The infinitesimal action is $\xi_u(x) = u \times x$. Indeed, pairing it with v gives $(u \times x)\cdot v = -x \cdot (u \times v)$, which is exactly the dual derivative from Lecture 19.

Proposition 27.1. *Let $r > 0$. The orbit form at $x \in S^2_r$ is*

$$\omega_x(v, w) = \frac{x \cdot (v \times w)}{r^2}, \qquad v, w \in T_x S^2_r.$$

If Ω is the outward-oriented area form of the sphere, then $\omega = \Omega/r$. In spherical coordinates,

$$\omega = r \sin\theta \, d\theta \wedge d\phi, \qquad \int_{S^2_r} \omega = 4\pi r. \tag{27.1}$$

Proof. The KKS formula of Lecture 19 requires

$$\omega_x(u \times x, v \times x) = x \cdot (u \times v).$$

At $x = re_3$, the two tangent vectors are $r(u_2, -u_1, 0)$ and $r(v_2, -v_1, 0)$. Substitution in the stated formula gives $r(u_1 v_2 - u_2 v_1)$, the required value. Rotation invariance gives the formula at every x.

The outward area form is $\Omega_x(v, w) = x \cdot (v \times w)/r$. Thus $\omega = \Omega/r$. The usual sphere parametrisation has area form $r^2 \sin\theta \, d\theta \wedge d\phi$, giving the spherical formula (27.1). Integration over $0 \le \theta \le \pi$ and $0 \le \phi \le 2\pi$ gives $4\pi r$. ∎

Write $\xi_j = \xi_{e_j}$ for the standard basis vectors e_j. At the north pole, $\xi_1 = -r\partial_{x_2}$ and $\xi_2 = r\partial_{x_1}$, and $\omega(\xi_1, \xi_2) = r$. This is a convenient sign check. The origin must be treated separately: it is a point orbit, and none of the formulas dividing by r applies there.

Fix a positive constant $\hbar$. It sets the scale at which we compare symplectic area with phases. Keeping it explicit will make the integrality condition easier to track.

27.2 Connections described in local frames

A Hermitian line bundle is a complex line bundle with a smoothly varying positive inner product on its fibres. A local *unitary frame* is a section of norm one that never vanishes. In such a frame, a section is described by a complex coefficient function u.

Definition 27.2. A *unitary connection* differentiates local coefficients by

$$\nabla u = du + Au,$$

where A is an imaginary-valued 1-form. Let unitary frames e_i on U_i and e_j on U_j have connection forms A_i and A_j. If $e_j = g_{ij} e_i$, with $g_{ij} : U_i \cap U_j \to \mathrm{U}(1)$, the connection forms are required to satisfy

$$A_j = A_i + g_{ij}^{-1} dg_{ij}.$$

The *curvature* F_∇ is the imaginary-valued 2-form given on U_i by $F_\nabla = dA_i$.

Here du is the ordinary differential, evaluated on tangent vectors. A section s has coefficients u_i and u_j in the two frames. Since $e_j = g_{ij} e_i$, these coefficients satisfy $u_i = g_{ij} u_j$, or equivalently $u_j = g_{ij}^{-1} u_i$. Differentiating $u_i = g_{ij} u_j$ gives

$$du_i + A_i u_i = g_{ij}\big(du_j + (A_i + g_{ij}^{-1} dg_{ij}) u_j\big).$$

The left side is the coefficient of ∇s in the frame e_i. In the frame e_j this coefficient is $du_j + A_j u_j$, and coefficients change by the factor g_{ij}. Hence frame independence of ∇s for every section is exactly the rule for A_j in the definition.

Since g_{ij} is scalar, $d(g_{ij}^{-1} dg_{ij}) = 0$ locally. Indeed, write $g_{ij} = e^{i\alpha}$ with real α, so this form is $i\,d\alpha$. Hence $dA_i = dA_j$ and the curvature is global.

The same connection can be written in any nonvanishing frame. The change-of-frame formulas remain valid for $g_{ij} \in \mathbb{C}^\times$, with complex-valued A_i. Only unitary frames require imaginary-valued forms.

Along a path, the equation for parallel transport is $du + Au = 0$. In one frame its solution multiplies u by $\exp(-\int A)$. This is a phase because A is imaginary. Curvature measures the resulting phase around small loops.

Definition 27.3. A *prequantum line bundle* for (M, ω) at scale $\hbar$ is a Hermitian line bundle with unitary connection satisfying

$$F_\nabla = -\frac{i}{\hbar}\omega. \tag{27.2}$$

The sign is part of our convention. A prequantum bundle is an initial geometric input. Its space of all smooth sections is usually far larger than the desired representation. A polarisation, such as holomorphicity, supplies a further selection.

We cite the general prequantisation theorem. A closed real 2-form ω on a smooth manifold admits a prequantum line bundle at scale $\hbar$ exactly when its periods divided by $2\pi\hbar$ are integers. A *period* of ω is the integral of ω over a closed oriented surface mapped smoothly into the manifold, or over an integer combination of such surfaces. Equivalently, $[\omega/(2\pi\hbar)]$ is an integral cohomology class. See Kirillov, Chapter 1, Section 2.4.

We prove the sphere case directly.

27.3 The integral sphere bundle

Put

$$m = \frac{2r}{\hbar}.$$

Take a northern chart excluding the south pole and a southern chart excluding the north pole. Define

$$A_N = -\frac{im}{2}(1 - \cos\theta)d\phi, \qquad A_S = \frac{im}{2}(1 + \cos\theta)d\phi. \tag{27.3}$$

Although $d\phi$ is singular at each pole, A_N is smooth at the north pole and A_S at the south pole. For example, in a northern complex coordinate $z = \tan(\theta/2)e^{i\phi}$,

$$A_N = \frac{m}{2}\frac{z\,d\bar{z} - \bar{z}\,dz}{1 + |z|^2}.$$

This expression is smooth at $z = 0$. The southern formula has the same property in its own coordinate.

Differentiation gives

$$dA_N = dA_S = -\frac{im}{2}\sin\theta\, d\theta \wedge d\phi = -\frac{i}{\hbar}\omega.$$

On the overlap, $A_S - A_N = im\, d\phi$. Thus A_N and A_S satisfy the change-of-frame formula if the frame transition is

$$e_S = e^{im\phi} e_N.$$

This is a single-valued function on the overlap exactly when $m \in \mathbb{Z}$. For positive radius this means m is a positive integer.

Proposition 27.4. *The sphere S^2_r, $r > 0$, admits a prequantum line bundle if and only if $2r/\hbar \in \mathbb{Z}$. For $m = 2r/\hbar$, the preceding construction supplies one.*

Proof. Sufficiency is the explicit construction. For necessity, suppose that S^2_r carries a prequantum line bundle.

Step 1 (unitary frames on the two disks). Use the two open disk charts obtained by deleting the opposite poles. On each chart, parallel transport a unit vector at its origin along radial segments. Cover each radial segment by finitely many local unitary frames. The change-of-frame law makes transport agree on overlaps. The resulting finite composition of ODE solutions depends smoothly on the endpoint and gives a smooth unitary frame. Each chart contains a neighbourhood of its closed hemisphere, so these frames are valid up to and beyond the equator. Call them e_N and e_S, with connection forms A_N and A_S.

Step 2 (the transition function and its lift). Write $e_S = g e_N$ near the equator, with $g : S^1 \to \mathrm{U}(1)$. Orient the equator by increasing ϕ, as the boundary of the northern disk.

Lift $g(e^{i\phi})$ to a real phase $\alpha(\phi)$ for $0 \le \phi \le 2\pi$, using the circle covering from Lecture 4. Since the endpoints give the same value, $\alpha(2\pi) - \alpha(0) = 2\pi k$ for an integer k.

Step 3 (Stokes). By the change-of-frame law, $A_S = A_N + g^{-1}dg$. Moreover $g^{-1}dg = i\, d\alpha$, since $g = e^{i\alpha}$ along the equator. Let D_N and D_S be the closed northern and southern hemispheres. We apply the calculus Stokes formula for a 1-form on each oriented disk. The southern disk induces the opposite orientation on the equator. Hence

$$\int_{S^2} F_\nabla = \int_{D_N} dA_N + \int_{D_S} dA_S = \int_{S^1} (A_N - A_S) = -\int_{S^1} g^{-1}dg = -2\pi i k.$$

By the prequantum condition (27.2) and the spherical formula (27.1),

$$k = \frac{i}{2\pi}\int F_\nabla = \frac{1}{2\pi\hbar}\int \omega = \frac{2r}{\hbar}.$$

Hence $2r/\hbar \in \mathbb{Z}$. ∎

Since $m > 0$, the curvature has no global primitive on the sphere. If $F = dA$ globally, Stokes' theorem would give $\int_{S^2} F = 0$. The period is nonzero because $m \neq 0$. The two local potentials are therefore essential.

27.4 Holomorphic sections and equivariance

Identify the sphere with $\mathbb{CP}^1$ by

$$z \longmapsto \frac{r}{1+|z|^2}\bigl(2\operatorname{Re} z, 2\operatorname{Im} z, 1-|z|^2\bigr).$$

The north pole is $z=0$, and $z=\tan(\theta/2)e^{i\phi}$ is the northern coordinate used above. Recall from Lecture 26 that $\mathrm{SU}(2)$ acts on lines by $g\cdot[v]=[gv]$. The identification map is equivariant for this action and the coadjoint action. Indeed, the line of $v=(1,z)$ corresponds to the traceless Hermitian matrix proportional to

$$vv^*/\|v\|^2 - I/2.$$

Conjugation by g sends this matrix to the matrix for the line of gv, and it rotates the Pauli coordinates of the matrix. With $|z|=\tan(\theta/2)$, we have $d|z| = \frac{1}{2}(1+|z|^2)\,d\theta$ and $r\sin\theta = 2r|z|/(1+|z|^2)$. Substitution in the spherical formula (27.1) gives the sphere form

$$\omega = \frac{2ir\,dz\wedge d\bar z}{(1+|z|^2)^2}.$$

Since $i\,dz\wedge d\bar z = 2\,d(\operatorname{Re} z)\wedge d(\operatorname{Im} z)$, this form has the same positive orientation as the spherical formula (27.1).

The holomorphic frame transition of $O(m)$ in Lecture 26 is $e_S^{\mathrm{hol}} = z^m e_N^{\mathrm{hol}}$. The Hermitian metric with

$$\|e_N^{\mathrm{hol}}\|^2 = (1+|z|^2)^{-m}$$

gives unitary frames $e_N = (1+|z|^2)^{m/2}e_N^{\mathrm{hol}}$ and e_S whose transition is $(z/|z|)^m = e^{im\phi}$. This is the transition of the integral sphere bundle, so the potentials (27.3) define a unitary connection on $O(m)$.

The metric is $\mathrm{SU}(2)$-invariant. To see this, we describe it fibre by fibre. A point f of the fibre over a line $[v]$ is a function on that line with $f(\lambda v)=\lambda^m f(v)$ for $\lambda\in\mathbb{C}^\times$. In this model, the holomorphic sections of Lecture 26 are the homogeneous polynomials of degree m, restricted to each line. On the fibre over $[v]$ put

$$\|f\|^2 = \frac{|f(v)|^2}{\|v\|^{2m}}.$$

Replacing v by λv multiplies the numerator and the denominator by $|\lambda|^{2m}$. Hence the norm does not depend on the chosen vector v. An element $g\in\mathrm{SU}(2)$ sends f to the point gf over $[gv]$ with $(gf)(gv)=f(v)$. Since g is unitary, $\|gv\|=\|v\|$, and so $\|gf\|=\|f\|$. The northern frame e_N^{hol} takes the value 1 at $v=(1,z)$, as the polynomial v_0^m does. Since $\|(1,z)\|^2 = 1+|z|^2$, this norm agrees with the metric displayed above.

A connection is *compatible* with the holomorphic structure when its form in a holomorphic frame has no $d\bar z$ term. Apply the change-of-frame law to $e_N^{\mathrm{hol}} = (1+|z|^2)^{-m/2}e_N$ and use the coordinate formula for A_N above. The form of this connection in the frame e_N^{hol} is then

$$A_N^{\mathrm{hol}} = A_N - \frac{m}{2}\,d\log(1+|z|^2) = -m\frac{\bar z\,dz}{1+|z|^2}.$$

Exercise 27.4 carries out this frame change in coordinates. The form A_N^{hol} has only a dz term, and its curvature is

$$dA_N^{\mathrm{hol}} = m\frac{dz \wedge d\bar{z}}{(1+|z|^2)^2} = -\frac{i}{\hbar}\omega.$$

Thus the prequantum bundle can be taken to be $O(m)$, with its usual holomorphic structure.

For a given metric, the unitary connection compatible with the holomorphic structure is unique. Let e be a holomorphic frame and put $h = \|e\|^2$. Let A be the connection form in the frame e. The unitary frame $h^{-1/2}e$ has an imaginary-valued connection form A_u. The frame e is $h^{1/2}$ times this unitary frame, so the change-of-frame law gives $A = A_u + \frac{1}{2}d\log h$. Moreover $\bar{A}_u = -A_u$, and $\log h$ is real. Hence

$$A + \bar{A} = d\log h.$$

For a function f, write $\partial f = (\partial f/\partial z)\,dz$ and $\bar{\partial} f = (\partial f/\partial \bar{z})\,d\bar{z}$, so that $df = \partial f + \bar{\partial} f$. Suppose that A has no $d\bar{z}$ term. Then $\bar{A}$ has no dz term. Comparing the dz terms on the two sides gives

$$A = \partial \log h.$$

Thus the metric determines the compatible unitary connection. For $h = (1+|z|^2)^{-m}$ this formula gives $-m\bar{z}\,dz/(1+|z|^2)$, which is A_N^{hol} again.

When $m = 2r/\hbar$ is a positive integer, the *holomorphic quantisation* of S_r^2 is the section space

$$Q(S_r^2) = H^0(\mathbb{CP}^1, O(m)) \cong V_m.$$

By the section computation of Lecture 26, this space has dimension $m+1$ and spin $j = m/2 = r/\hbar$. An invariant positive inner product comes from compact averaging. The point orbit uses the trivial line and gives the trivial representation, without a radius denominator.

Definition 27.5. Let K be a Lie group with Lie algebra $\mathfrak{k}$, and let $M \subset \mathfrak{k}^*$ be a coadjoint orbit with its KKS form ω. An *equivariant prequantum bundle* for K is a prequantum line bundle for (M,ω) at scale $\hbar$ with a smooth K-action on its total space. The action must cover the coadjoint action on M, be linear and unitary on fibres, preserve the connection, and satisfy the *orbit normalisation*. This normalisation requires that, at each $\ell \in M$, the stabiliser K_ℓ act on the fibre over ℓ with differential

$$X \mapsto i\ell(X)/\hbar, \qquad X \in \mathrm{Lie}(K_\ell).$$

The orbit normalisation fixes the infinitesimal lift. Without it, merely asking for a bundle action would not specify its relation to the orbit functional.

At the north pole, the stabiliser in SU(2) is generated by $\exp(tX_3)$, of period 4π. In an equivariant prequantum bundle, this stabiliser acts on the fibre by the character

$$\exp(tX_3) \longmapsto e^{irt/\hbar} = e^{imt/2}.$$

This is a character exactly when $m \in \mathbb{Z}$. It is the action on the fibre of $O(m)$ from Lecture 26: $t_\theta = \exp(-2\theta X_3)$ acts there by $e^{-im\theta}$. The natural SU(2) action is linear on fibres and preserves the invariant metric. Each $g \in \mathrm{SU}(2)$ acts by a holomorphic bundle map. It sends holomorphic frames to holomorphic frames. Its holomorphic coordinate change sends forms with only a dz term to forms of the same kind. Hence g transports the connection to a unitary

connection compatible with the holomorphic structure. By uniqueness, the transported connection is the original one. Therefore the natural action supplies the required lift.

For SO(3) the same stabiliser rotation has period 2π. Its required character is single-valued only when $e^{2\pi i r/\hbar} = 1$, or m is even. Conversely, for even m the centre $-I$ acts trivially on every fibre of $O(m)$, so the action descends. Therefore

ordinary prequantum bundle	$m \in \mathbb{Z}$
SU(2)-equivariant prequantum bundle	$m \in \mathbb{Z}$
SO(3)-equivariant prequantum bundle	$m \in 2\mathbb{Z}$

for the positive-radius sphere, with $m > 0$. An ordinary integral sphere with odd m does exist. It does not quantise equivariantly for the global rotation group SO(3).

Phase changes and the odd-degree obstruction

One might try to change the SU(2) lift by phases so that $-I$ acts trivially. We show that, for this sphere and its fixed connection, every such phase change is trivial.

Throughout this subsection, an action on $O(m)$ means a smooth action that is linear and unitary on fibres, as in the definition of an equivariant prequantum bundle. Suppose two such SU(2) actions cover the same sphere action and preserve the connection. For fixed g, compare their fibre maps. Their ratio is a bundle automorphism covering the identity. It is linear and unitary on each fibre, and each fibre is a complex line. Hence the ratio is multiplication by a smooth function $a_g : S^2 \to \mathrm{U}(1)$. Since both actions preserve the connection,

$$a_g^{-1} da_g = 0.$$

The sphere is connected, so a_g is constant. Call this constant $\chi(g)$. The group-action law then implies

$$\chi(gh) = \chi(g)\chi(h).$$

Thus changing the lift while preserving the connection can only twist it by a continuous character of SU(2).

Every such character is trivial. Its differential is a real Lie algebra map from $\mathfrak{su}(2)$ to the abelian algebra $i\mathbb{R}$, so it vanishes on brackets. Since

$$[\mathfrak{su}(2), \mathfrak{su}(2)] = \mathfrak{su}(2),$$

the differential vanishes. Since the connected group SU(2) is generated by exponentials (Lecture 2), the character is identically one. Consequently a phase change of this kind cannot alter the action of the centre. Hence the natural action of Lecture 26 is the only such SU(2) action on $O(m)$ that covers the sphere action and preserves the connection.

Now suppose that SO(3) acted on $O(m)$ in this way. Composition with the covering $\mathrm{SU}(2) \to \mathrm{SO}(3)$ would give such an SU(2) action on which $-I$ acts trivially. By uniqueness, that action would be the natural one, where $-I$ acts on every fibre by $(-1)^m$. Hence for odd m no SO(3) action on $O(m)$ that is linear and unitary on fibres covers the rotation action and preserves the connection. This conclusion does not use the orbit normalisation.

This phase calculation explains the relation of the stabiliser-character test to projective representations in Lecture 13: a cover acts on the bundle and on sections, while its central kernel may remain visible as a phase.

A spin-one comparison with functions on the sphere

For $r = \hbar$, the integer is $m = 2$. The quantisation is the three-dimensional space of homogeneous quadratic polynomials in (v_0, v_1)

$$F(v_0, v_1) = Av_0^2 + Bv_0v_1 + Cv_1^2.$$

There is another spin-one model in Lecture 12: the linear coordinate functions on the sphere. The two models are isomorphic as representations, although a bundle section and a scalar function are different geometric objects.

We now identify the two models explicitly. Write $F(v) = v^T Qv$ and set

$$Q = \begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix}, \qquad \epsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad T(Q) = \epsilon^{-1}Q = \begin{pmatrix} -B/2 & -C \\ A & B/2 \end{pmatrix}.$$

This is an invertible linear map from symmetric matrices to traceless matrices. The polynomial action sends Q to $g^{-T}Qg^{-1}$. Since $g^T\epsilon g = \epsilon$ for $g \in \mathrm{SL}(2,\mathbb{C})$,

$$T(g^{-T}Qg^{-1}) = gT(Q)g^{-1}.$$

Thus T intertwines the quadratic section representation with the complexified adjoint representation.

For $Y \in \mathfrak{su}(2)_{\mathbb{C}}$, let ℓ_x be the functional with coordinates $x_j = \ell_x(X_j)$ and define the linear function $f_Y(x) = \ell_x(Y)$ on the sphere, extending ℓ_x complex-linearly. Under pullback by g^{-1},

$$f_Y(g^{-1} \cdot x) = \ell_x(\mathrm{Ad}(g)Y) = f_{\mathrm{Ad}(g)Y}(x).$$

Hence the map sending Y to f_Y is another intertwiner. The three basis elements X_j give the independent functions x_j, so this map is an isomorphism onto their span. Since $\mathfrak{su}(2)_{\mathbb{C}} = \mathfrak{sl}(2,\mathbb{C})$, the map $Q \mapsto f_{T(Q)}$ identifies the two spin-one models explicitly. The normalised period is still $m = 2$, the dimension is three, and the action descends to $\mathrm{SO}(3)$.

27.5 The compact construction and the shift

The sphere illustrates the general compact picture. Let K be compact, connected, simply connected and semisimple. Let λ be a dominant integral weight in the real convention $d\chi = i\lambda$. Recall the positive Borel subalgebra fixed in Lecture 26. Our sign conventions follow that choice. Extend λ by zero on the root directions, and take the orbit through $\ell = -\hbar\lambda$. By the orbit normalisation, the stabiliser of ℓ acts on the fibre over ℓ by a character with differential $-i\lambda$.

For strictly dominant λ, the stabiliser is a maximal torus T. The orbit is K/T and the bundle is L_λ of Lecture 26. Choose its invariant complex structure from the positive Borel subalgebra. The Borel–Weil theorem of Lecture 26 then gives V_λ^*. For a weight on a chamber wall, the stabiliser K_ℓ is larger. The orbit is then the partial flag manifold K/K_ℓ. The analogous line bundle and holomorphic-section result are cited from Kirillov, Chapter 5, Sections 2–3. We do not prove that general geometric construction here.

For a different global form of K, the stabiliser character must exist for the actual stabiliser group. This is the equivariant integrality test. It reproduces the character-lattice restriction of Lecture 8. An ordinary integral period condition alone does not encode the global form.

No ρ shift was used in this construction. On the sphere, the orbit label is m, and the uncorrected section space has dimension $m+1$. The character nevertheless contains $m+1$:

$$\chi_m(t_\theta) = \sum_{k=0}^{m} e^{i(m-2k)\theta} = \frac{\sin((m+1)\theta)}{\sin\theta} \qquad (\sin\theta \neq 0).$$

This is the Weyl character formula, with its familiar $\lambda+\rho$ numerator. It does not say that the prequantum orbit above had label $m+1$. Half-form corrections are another construction and also require a convention. Optional Exercise 27.6 shows one such choice on the sphere, and nothing in this lecture depends on it.

Self-checks

1. If $r = 3\hbar/2$, find the normalised period and the quantised representation.

Answer. The period divided by $2\pi\hbar$ is $2r/\hbar = 3$. The bundle is $O(3)$ and its sections give V_3, of dimension four and spin $3/2$.

2. Does that sphere admit an SO(3)-equivariant prequantum lift with the orbit normalisation?

Answer. No. Its stabiliser character would be $e^{3it/2}$ on a rotation of period 2π. It takes the value -1 at $t = 2\pi$.

3. Why can the connection have local potentials although no global potential exists?

Answer. The difference of two local potentials is the logarithmic differential of a transition function. A nonzero curvature period prevents a single global primitive but is consistent with these local potentials.

Summary

The SU(2) orbit of radius $r > 0$ has symplectic area $4\pi r$. Prequantisation requires $m = 2r/\hbar$ to be integral. Holomorphic sections of $O(m)$ give spin $m/2$. Equivariance for SO(3) further requires even m. The ordinary period, the global equivariant lift and the holomorphic selection are separate steps.

Reading

Kirillov, *Lectures on the Orbit Method*, Chapter 1, Section 2.4, for integrality, and Chapter 5, Sections 2–3, for compact orbits, holomorphic constructions and character formulas. Sepanski, *Compact Lie Groups*, Section 7.4, supplies the Borel–Weil statement used in Lecture 26.

Lecture 28

Reconstruction and the boundary of quantum groups

Objectives. *The group, its coefficient algebra and its tensor representations encode the same global information in different forms. Differentiation connects these descriptions to the enveloping algebra. It loses information, however, unless the allowed group representations are retained. We make that connection explicit and identify the structures that a later course on quantum groups will deform.*

By the end of this lecture you will be able to:

- Construct the pairing between $U(\mathfrak{k}_{\mathbb{C}})$ and representative functions, with the correct multiplication order.
- For connected K, explain why representative functions form a restricted part of an algebraic dual.
- Recover compact and complex groups from the appropriate tensor automorphisms.
- Use the circle and the SU(2)/SO(3) pair to identify what global data survive differentiation.

28.1 The two algebras

Let K be a compact Lie group and $\mathfrak{k}$ its real Lie algebra. As in Lecture 14, $\mathcal{R}(K)$ is the span of the coefficients

$$c^V_{\lambda,v}(g) = \lambda(\Pi_V(g)v), \qquad \lambda \in V^*,\ v \in V,$$

of finite-dimensional continuous complex representations. Every sum of coefficients is one coefficient of a direct sum. Its Hopf structure is

$$\Delta_{\mathcal{R}} f(g,h) = f(gh), \qquad \varepsilon_{\mathcal{R}} f = f(e), \qquad S_{\mathcal{R}} f(g) = f(g^{-1}), \qquad f^*(g) = \overline{f(g)}.$$

These formulas and their targets were proved in Lecture 14. We use the ordinary algebraic tensor product. Thus $\Delta_{\mathcal{R}} f$ is a finite sum of tensors, not an infinite expansion.

On the other side, put $U = U(\mathfrak{k}_{\mathbb{C}})$, the universal enveloping algebra from Lie Algebras. It is generated by $\mathfrak{k}_{\mathbb{C}}$ with $XY - YX = [X,Y]$. Its standard Hopf maps are determined by

$$\Delta_U X = X \otimes 1 + 1 \otimes X, \qquad \varepsilon_U X = 0, \qquad S_U X = -X. \tag{28.1}$$

Here Δ_U and ε_U extend as algebra homomorphisms. The antipode S_U extends so that it reverses products. This extension is well defined because reversing the defining relation

sends $XY - YX - [X,Y]$ to $YX - XY + [X,Y]$, which is again zero in U. We check that Δ_U respects the defining relation:

$$[\Delta_U X, \Delta_U Y] = [X,Y] \otimes 1 + 1 \otimes [X,Y] = \Delta_U [X,Y].$$

Mixed tensor factors commute, which is why the cross terms vanish. Coassociativity and the counit identities follow by checking the generators, since the maps are algebra homomorphisms.

The antipode identities hold on X because $-X + X = 0$. They extend to products. Suppose that they hold on u and v. Then

$$\begin{aligned} \sum S_U(u_{(1)}v_{(1)})u_{(2)}v_{(2)} &= \sum S_U(v_{(1)})S_U(u_{(1)})u_{(2)}v_{(2)} \\ &= \varepsilon_U(u) \sum S_U(v_{(1)})v_{(2)} = \varepsilon_U(u)\varepsilon_U(v)1. \end{aligned}$$

The first equality uses that S_U reverses products. The second uses the identity for u, and the third uses the identity for v. The other antipode identity is proved the same way.

The two algebras have opposite kinds of commutativity. The coefficient algebra is commutative. It is cocommutative exactly when K is abelian, because coefficients separate points. The enveloping algebra is cocommutative, as the generator formulas (28.1) show, but is generally noncommutative.

The compact real form also defines an involution on U by $X^* = -X$ for real $X \in \mathfrak{k}$, extended antilinearly with $(uv)^* = v^*u^*$. For real X, Y, applying $*$ to the defining relation gives $YX - XY + [X,Y]$, which is again zero. Thus $*$ is well defined. Applying it twice fixes the generators, so it is an involution.

In every finite-dimensional unitary representation,

$$d\Pi(u^*) = d\Pi(u)^*.$$

This follows first for real generators, whose operators are skew-adjoint, then for sums and products. The involution remembers the compact real form inside its complexification.

28.2 Differentiating coefficients

For $X \in \mathfrak{k}$, let

$$(R_X f)(g) = \left.\frac{d}{dt}\right|_0 f(g \exp(tX)).$$

This is the left-invariant vector field generated by X, acting on functions. The letter R records multiplication on the right in the argument. It is distinct from the right regular group action, although differentiating that action gives this operator. By Lecture 1, $[R_X, R_Y] = R_{[X,Y]}$. Hence, by the universal property of U, these operators extend to R_u for $u \in U$.

Definition 28.1. The *coefficient pairing* is

$$\langle u, f \rangle = (R_u f)(e), \qquad u \in U,\ f \in \mathcal{R}(K).$$

The coefficient pairing is complex bilinear. Its angle brackets denote a pairing, not a Hermitian inner product. For a coefficient, direct differentiation gives

$$R_X c^V_{\lambda,v}(g) = \lambda(\Pi_V(g)d\Pi_V(X)v), \qquad \langle u, c^V_{\lambda,v}\rangle = \lambda(d\Pi_V(u)v). \tag{28.2}$$

The second identity follows for products of generators and then by linearity. Since the pairing was defined on functions by differential operators, it does not depend on the chosen coefficient expression.

The order matters. For two real generators $X, Y \in \mathfrak{k}$,

$$\langle XY, f\rangle = \partial_t\partial_s|_0\, f(\exp(tX)\exp(sY)).$$

Indeed, for $f = c^V_{\lambda,v}$ we have $R_X R_Y f(g) = \lambda(\Pi_V(g)d\Pi_V(X)d\Pi_V(Y)v)$. Interchanging X and Y would reverse the matrix product. Subtracting the pairing of YX with f from the pairing of XY with f gives $\langle [X,Y], f\rangle$.

Proposition 28.2 (Hopf pairing identities). *Write finite coproducts as $\Delta_U u = \sum u_{(1)} \otimes u_{(2)}$ and $\Delta_{\mathcal{R}} f = \sum f_{(1)} \otimes f_{(2)}$. Then*

$$\begin{aligned}
\langle ab, f\rangle &= \sum \langle a, f_{(1)}\rangle\langle b, f_{(2)}\rangle,\\
\langle u, fh\rangle &= \sum \langle u_{(1)}, f\rangle\langle u_{(2)}, h\rangle,\\
\langle 1, f\rangle &= \varepsilon_{\mathcal{R}} f, \qquad \langle u, 1\rangle = \varepsilon_U u,\\
\langle S_U u, f\rangle &= \langle u, S_{\mathcal{R}} f\rangle.
\end{aligned}$$

Proof. Choose a basis (e_j) of V with dual (e^j). For $f = c^V_{\lambda,v}$,

$$\Delta_{\mathcal{R}} f = \sum_j c^V_{\lambda,e_j} \otimes c^V_{e^j,v}.$$

Applying the right side of the first identity gives

$$\sum_j \lambda(d\Pi(a)e_j)\, e^j(d\Pi(b)v) = \lambda(d\Pi(a)d\Pi(b)v).$$

This is $\langle ab, f\rangle$ by the coefficient formula (28.2).

For the second identity, $R_X(fh) = (R_X f)h + f(R_X h)$ is the ordinary Leibniz rule. Iterating it gives exactly the coproduct (28.1). For instance,

$$\Delta_U(XY) = XY \otimes 1 + X \otimes Y + Y \otimes X + 1 \otimes XY.$$

Thus evaluation at e proves the claim for every product and hence for every u. The identity $\langle 1, f\rangle = \varepsilon_{\mathcal{R}} f$ follows because R_1 is the identity. Applying derivatives to constants gives $\langle u, 1\rangle = \varepsilon_U u$.

Finally, write $\Pi^\vee$ for the dual representation on V^*, given by $\Pi^\vee(g)\mu = \mu \circ \Pi_V(g)^{-1}$. The function $S_{\mathcal{R}} c^V_{\lambda,v}(g) = \lambda(\Pi_V(g)^{-1}v)$ is the coefficient of $\Pi^\vee$ with vector λ and with the functional given by evaluation at v. On V^*, a real generator acts by $d\Pi^\vee(X)\mu = -\mu \circ d\Pi(X)$. For every $u \in U$,

$$d\Pi^\vee(u)\mu = \mu \circ d\Pi(S_U u).$$

This holds for generators and extends multiplicatively because S_U reverses products. Apply the coefficient formula (28.2) to this coefficient of $\Pi^\vee$, and then to $c^V_{\lambda,v}$. Both sides of the last identity equal $\lambda(d\Pi(S_U u)v)$, which proves it. ∎

Hence the pairing makes the product of each algebra dual to the coproduct of the other. In this algebraic form, it relates the infinitesimal action to group multiplication. The proposition does not assert that one of these algebras is the entire linear dual of the other.

A noncommutative calculation

Take $K = \mathrm{SU}(2)$, so $\mathfrak{k}_{\mathbb{C}} = \mathfrak{sl}(2,\mathbb{C})$. We test the order identity in its defining representation. Put

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

These are complexified generators. The derivative is extended complex-linearly; we are not claiming that $\exp(tE)$ belongs to $\mathrm{SU}(2)$ for real t. Let $D_{ij}(g) = g_{ij}$ be coefficients of its defining representation. Then

$$EF = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad FE = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

It follows that $\langle EF, D_{11}\rangle = 1$, whereas $\langle FE, D_{11}\rangle = 0$. Their difference is $\langle H, D_{11}\rangle = 1$. The pairing therefore distinguishes EF from FE in U, even though its other argument belongs to a commutative function algebra.

The coefficient coproduct gives the same answer:

$$\Delta_{\mathcal{R}} D_{11} = D_{11} \otimes D_{11} + D_{12} \otimes D_{21}.$$

Pairing the first tensor factor with E and the second with F yields

$$E_{11}F_{11} + E_{12}F_{21} = 0 + 1 = 1.$$

Pairing the first factor with F and the second with E yields zero. Thus a commutative algebra can have a coproduct that encodes a noncommutative group law. Commutativity of the product does not by itself decide cocommutativity of the coproduct.

The regular actions on coefficients

Let $L_g f(x) = f(g^{-1}x)$ and $R_g f(x) = f(xg)$, as in the compact harmonic-analysis lectures. Here R_g denotes the group action. The symbols R_X and R_u denote differential operators. The pairings of the translates of a coefficient are

$$\langle u, L_g c_{\lambda,v}\rangle = \lambda\big(\Pi(g^{-1})d\Pi(u)v\big),$$

$$\langle u, R_g c_{\lambda,v}\rangle = \lambda\big(d\Pi(u)\Pi(g)v\big).$$

For the first formula, $L_g c$ is the coefficient with functional $\lambda \circ \Pi(g^{-1})$. For the second, $R_g c$ has vector $\Pi(g)v$. Inserting these into the coefficient formula (28.2) proves both identities without any operator-domain question.

Notice the positions of the group matrix and the derivative matrix. Left translation changes the dual variable. Right translation changes the vector variable. This is the same tensor order $V^* \otimes V$ that appeared in the Peter–Weyl theorem. The enveloping-algebra pairing continues that convention rather than introducing a second one.

28.3 What the dual contains

For any unital algebra A, its *finite dual* A° consists of the functionals that vanish on a two-sided ideal of finite codimension. Equivalently they are matrix coefficients of finite-dimensional A-modules. Here is the elementary equivalence.

A finite-dimensional module gives a homomorphism $A \to \mathrm{End}(V)$ with finite-codimensional kernel; every coefficient vanishes on that kernel. Conversely, if φ vanishes on such an ideal I, use the finite-dimensional regular module A/I, vector $1 + I$ and functional $\bar{\varphi}$. Then $\varphi(a) = \bar{\varphi}(a(1 + I))$.

By the coefficient formula (28.2), the pairing therefore maps $\mathcal{R}(K)$ into U°. Only the modules that integrate to the chosen compact group occur in this image. A finite-dimensional representation of $\mathfrak{k}_{\mathbb{C}}$ need not respect the periods of K or the restrictions imposed by a central quotient.

> **Proposition 28.3** (Injectivity for connected groups). *If K is connected, the map $\mathcal{R}(K) \to U^\circ$ from the pairing is injective.*

Proof. Suppose $\langle u, f\rangle = 0$ for every u. Since every sum of coefficients is one coefficient of a direct sum, we may write $f = c^V_{\lambda,v}$. For any finite list $X_1, \dots, X_s \in \mathfrak{k}$, expand the finite-dimensional matrix exponentials in

$$f(\exp(t_1X_1)\cdots\exp(t_sX_s)) = \lambda\big(e^{t_1 d\Pi(X_1)}\cdots e^{t_s d\Pi(X_s)}v\big).$$

These power series converge absolutely for every real or complex tuple $(t_1, \dots, t_s)$. The Taylor coefficient of $t_1^{a_1}\cdots t_s^{a_s}$ is

$$\frac{\langle X_1^{a_1}\cdots X_s^{a_s}, f\rangle}{a_1!\cdots a_s!} = 0.$$

Thus the displayed function vanishes for all real tuples. A connected Lie group is generated by its exponentials (Lecture 2), so $f = 0$ on K. ∎

For disconnected K, the same argument detects only the identity component. A coefficient may vanish on that component and be nonzero elsewhere. Thus connectedness cannot simply be removed.

Example 28.4 (The circle). Let $K = \mathrm{U}(1)$ and choose X with $\exp(tX) = e^{it}$. Then

$$U = \mathbb{C}[X], \qquad \mathcal{R}(K) = \mathbb{C}[z, z^{-1}], \qquad \langle X^k, z^n\rangle = (in)^k.$$

The image consists of finite linear combinations of the evaluation functionals $p(X) \mapsto p(in)$, $n \in \mathbb{Z}$. The finite dual is larger. Evaluation at $X = 1$ belongs to U° but is not in this image. If it were a combination supported on $n_1, \dots, n_s$, the polynomial $p(X) = \prod_j (X - in_j)$ would have zero value under that combination, but $p(1) \neq 0$.

The full linear dual is larger still. It contains the functional L given by $L(X^k) = k!$. By the equivalence above, every element of U° is a matrix coefficient of a finite-dimensional module. Suppose that X acts on that module by the operator A and that the coefficient uses the vector v and the functional λ. Then the coefficient takes the value $\lambda(A^k v)$ on X^k. Choose a norm on the module, with the corresponding operator and dual norms. This value is then bounded in absolute value by $C\|A\|^k$, where $C = \|\lambda\|\|v\|$. The sequence $k!$ has no such exponential bound. Thus L is not even in U°.

28.4 Global forms and tensor reconstruction

The groups SU(2) and SO(3) have isomorphic real Lie algebras, hence the same complex enveloping algebra after this identification. Their coefficient algebras retain different representations. Pullback through $p : \mathrm{SU}(2) \to \mathrm{SO}(3)$ gives

$$p^*\mathcal{R}(\mathrm{SO}(3)) = \{f \in \mathcal{R}(\mathrm{SU}(2)) : f(-g) = f(g)\}. \tag{28.3}$$

Indeed, a representation descends exactly when $-I$ acts trivially. On V_m it acts by $(-1)^m$. The algebraic Peter–Weyl decomposition therefore identifies the right side with precisely the coefficient blocks for even m. Odd blocks have been excluded, not annihilated by differentiation. Both coefficient algebras inject into the same finite dual, with different images.

To recover the group, keep the category of finite-dimensional continuous complex representations of K, all its intertwiners, and its forgetful functor ω. This functor sends each representation to its underlying vector space and each intertwiner to its linear map.

A *tensor automorphism* of ω is a family of maps $\theta_V \in \mathrm{GL}(V)$, one for each representation V, such that

$$\theta_W T = T\theta_V \quad \text{for every intertwiner } T : V \to W, \qquad \theta_{V\otimes W} = \theta_V \otimes \theta_W, \qquad \theta_{\mathbb{C}} = 1.$$

Here $\mathbb{C}$ is the trivial representation. A tensor automorphism is *unitary* if every θ_V is unitary for every invariant inner product on V. Write $\mathrm{Aut}_\otimes(\omega)$ for the group of tensor automorphisms and $\mathrm{Aut}^u_\otimes(\omega)$ for its subgroup of unitary tensor automorphisms.

Each $g \in K$ supplies the unitary tensor automorphism $\theta_V = \Pi_V(g)$. A tensor automorphism θ corresponds to the algebra homomorphism $\mathcal{R}(K) \to \mathbb{C}$ given by $c^V_{\lambda,v} \mapsto \lambda(\theta_V v)$. Unitary tensor automorphisms correspond to $*$-homomorphisms. Exercises 14.4–14.5 prove this correspondence.

Let $K_\mathbb{C}$ be the complexification of K introduced in Lecture 14. It is a complex Lie group, and its identity component is reductive. The reconstruction theorem is cited from Bröcker–tom Dieck, Sections III.7–8. It identifies

$$K \cong \mathrm{Aut}^u_\otimes(\omega), \qquad K_\mathbb{C} \cong \mathrm{Aut}_\otimes(\omega).$$

The compact group has the topology of pointwise convergence on the representation spaces. In words, every unitary tensor automorphism is $\Pi_V(g)$ for a unique $g \in K$. Every tensor automorphism is given by a unique element of $K_\mathbb{C}$, acting through the holomorphic extensions of the representations Π_V to $K_\mathbb{C}$. The unitary condition distinguishes the compact target from its complexification.

For the circle, write $\mathbb{C}_n$ for the one-dimensional representation $z \mapsto z^n$ of U(1), where $n \in \mathbb{Z}$. Every finite-dimensional representation of U(1) is a direct sum of copies of these representations. Naturality makes θ act on every copy of $\mathbb{C}_n$ by one scalar $a_n \in \mathbb{C}^\times$. The tensor rule, the isomorphism $\mathbb{C}_m \otimes \mathbb{C}_n \cong \mathbb{C}_{m+n}$ and the unit rule give $a_{m+n} = a_m a_n$ and $a_0 = 1$. Therefore $a_n = a^n$ with $a = a_1 \in \mathbb{C}^\times$. Every such choice of a defines a tensor automorphism. This tensor automorphism is unitary exactly when $|a| = 1$. This recovers $\mathbb{C}^\times$ and U(1) respectively.

For SU(2), let $V = \mathbb{C}^2$ be the defining representation. The alternating tensor $e_1 \otimes e_2 - e_2 \otimes e_1$ is invariant. Since $\theta_{V\otimes V} = \theta_V \otimes \theta_V$ and $\theta_{\mathbb{C}} = 1$, naturality with the intertwiner from $\mathbb{C}$ given by this tensor forces $\theta_V \otimes \theta_V$ to preserve it. Because $\theta_V \otimes \theta_V$ multiplies this tensor by $\det \theta_V$, we get $\det \theta_V = 1$. Every irreducible representation is a symmetric power of V, and direct sums and intertwiners determine the rest of the tensor automorphism.

Conversely every $A \in \mathrm{SL}(2,\mathbb{C})$ acts on these tensor constructions and gives a tensor automorphism, using the complexification from Lecture 14. Hence the complex target is $\mathrm{SL}(2,\mathbb{C})$. Imposing unitarity on V gives SU(2). For the even-weight category the central element $-I$ acts as the identity tensor automorphism, so reconstruction instead yields the quotient global group.

A circle quotient and its representation lattice

There is a smaller example of global information in a coefficient subalgebra. Let $\mu_N \subset \mathrm{U}(1)$ be the Nth roots of unity. The quotient is again a circle, but keep the quotient map and the identification of Lie algebras fixed. A character z^n descends through this map exactly when

$$\zeta^n = 1 \quad \text{for every } \zeta \in \mu_N, \qquad \text{or equivalently } N \mid n.$$

Thus the pulled-back coefficient algebra is

$$\mathbb{C}[z^N, z^{-N}] \subset \mathbb{C}[z, z^{-1}].$$

Under the same infinitesimal pairing, its evaluation points are $iN\mathbb{Z}$, instead of $i\mathbb{Z}$. The real Lie algebras of these two circles are isomorphic. The quotient map, however, has changed which periods a representation must respect. Rescaling an abstract Lie algebra isomorphism does not erase the specified quotient map.

The subalgebra remains closed under the Hopf operations. For its generator $w = z^N$,

$$\Delta w = w \otimes w, \qquad \varepsilon w = 1, \qquad Sw = w^{-1}, \qquad w^* = w^{-1}.$$

Reconstruction from this algebra and its unitary structure recovers the quotient circle together with its representation lattice. The same calculation with even highest weights gives the SU(2) to SO(3) distinction. In both cases, keeping a Lie algebra while changing the allowed coefficient subalgebra changes the global representation problem.

28.5 What a quantum-group course changes

The structures above give a concrete starting point for deformation. The coefficient algebra encodes the multiplication of group elements in its coproduct. The enveloping algebra encodes the infinitesimal bracket in its associative product. The pairing between them relates these two operations. A deformation can change these operations while retaining the Hopf identities.

Classical object	Product	Coproduct
$\mathcal{R}(K)$	commutative	usually not cocommutative
$U(\mathfrak{k}_{\mathbb{C}})$	usually not commutative	cocommutative

A noncommutative coefficient algebra is no longer an algebra of scalar-valued functions on an ordinary group. A noncocommutative coproduct changes how tensor products of representations carry an action. The order of tensor factors can then matter. Associativity of the tensor product of representations still follows from coassociativity.

A braiding is additional structure, not an automatic property of a Hopf algebra. In suitable quasitriangular examples on the enveloping side, an R-matrix supplies an intertwiner between the two tensor orders. The coefficient-side formulation uses the corresponding coquasitriangular structure. The future course will define these notions and prove their compatibility identities. They are not needed for any theorem here.

The $*$-structure also remains substantive. An arbitrary Hopf $*$-algebra is not automatically a compact quantum group; positivity and analytic completion require further hypotheses. Similarly, a representation category does not automatically become a fusion category. For an infinite compact Lie group there are infinitely many irreducibles. Finite fusion categories in quantum-group examples require particular constructions and assumptions.

The useful preparation is therefore the classical bookkeeping developed in this course: distinguish an algebraic coefficient space from a completion, track coproduct order, keep the real or unitary structure, and retain the global representation lattice. These are the data a deformation must replace or preserve. The theory of deformation parameters, $U_q(\mathfrak{g})$ and roots of unity begins in the later course.

Self-checks

1. For the circle, compute $\langle X^2, z^{-3} \rangle$.

Answer. It is $(-3i)^2 = -9$. The pairing is bilinear, so there is no complex conjugation in this calculation.

2. Does $U(\mathfrak{sl}(2,\mathbb{C}))$ alone decide whether V_1 (spin $1/2$) is allowed?

Answer. No. It is allowed for $\mathrm{SU}(2)$ and excluded for $\mathrm{SO}(3)$. The chosen coefficient subalgebra, or equivalently the allowed global representations, records that distinction.

3. For the circle tensor automorphism $a = 2$, what are $\theta_{\mathbb{C}_2}$ and $\theta_{\mathbb{C}_{-1}}$? Is it unitary?

Answer. They are 4 and $1/2$. It is a complex tensor automorphism, but it is not unitary and does not represent an element of the compact circle.

Summary

Differentiation pairs representative functions with the enveloping algebra. For connected K, the pairing embeds them in the finite dual. There the global group selects a specific subalgebra. Tensor reconstruction recovers K when unitary structure is kept and $K_{\mathbb{C}}$ when it is forgotten. The coefficient and enveloping Hopf structures provide the classical starting point for quantum groups.

Reading

Bröcker–tom Dieck, *Representations of Compact Lie Groups*, Sections III.1 and III.7–8: coefficient algebras, reconstruction and complexification. The enveloping-algebra and tensor-product inputs are from the prerequisite Lie Algebras course. Lecture 14 contains the coefficient Hopf identities used here. The quantum-group paragraph is an outlook, not an additional classification theorem.